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New York State Household Travel Patterns: A Comprehensive Comparison Analysis



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June 2025

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A COMPREHENSIVE COMPARISON ANALYSIS**

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June 2025

Prepared for
New York State Department of Transportation

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ABBREVIATIONS AND ACRONYMS

ACS	American Community Survey
MOV	multioccupancy vehicle
MPO	Metropolitan Planning Organization
MSA	Metropolitan Statistical Area
Nass-Suff	Nassau and Suffolk
NHTS	National Household Travel Survey
NYC	New York City
NYMTC	New York Metropolitan Transportation Council
NYS	New York State
ORNL	Oak Ridge National Laboratory
PMT	person miles of travel
POV	privately owned vehicle
Putn-Rock-West	Putnam, Rockland, and Westchester
RDD	random digit dialing
SOV	single-occupancy vehicle
SUV	sport utility vehicle
VMT	vehicle miles of travel

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EXECUTIVE SUMMARY

In 1969, the U.S. Department of Transportation began an initiative to collect detailed data on household and personal travel, with the most recent surveys being the 2017 and 2022 National Household Travel Surveys. The primary goal of these surveys is to collect trip-based data on the nature and characteristics of personal travel, allowing for an analysis of the relationships between travel patterns and the socio-economic and demographic characteristics of travelers and their households. In addition to the sample households allocated to New York State (NYS) in the national survey, they procured an additional sample of households in the 1995, 2001, 2009, and 2017 surveys. This report presents a comprehensive analysis of travel patterns, vehicle ownership trends, and travel behaviors across NYS regions and the rest of the nation. Following is a summary of key findings:

Comparison Between NYS and the Rest of the United States

- NYS has a higher share of zero-vehicle households. In 2017, 29% of NYS households did not own a vehicle, compared with only 8% of households nationwide (Figure ES-1).
- In 2017, the share of privately owned vehicles was 59% in NYS and 84% in the rest of the United States. Walking trips in NYS were significantly higher than those in the rest of the United States across all years. For example, in 2017, walking trips accounted for 24% of person trips in NYS, compared with only 10% of person trips in the rest of the United States.

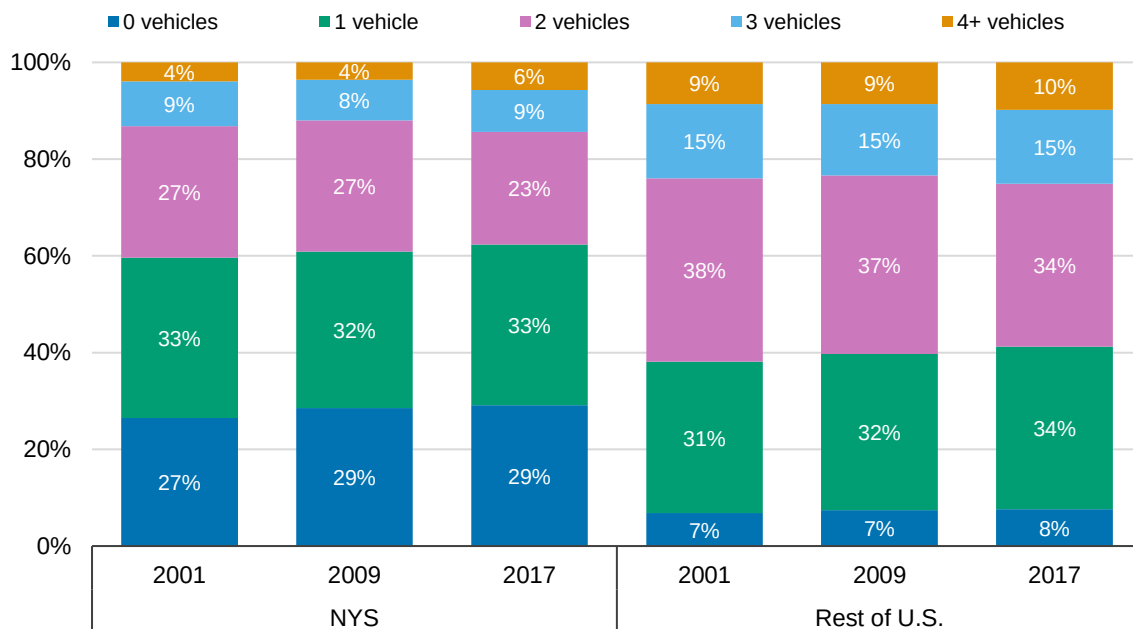


Figure ES-1. Distribution of household vehicle ownership. Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number.

New York City (NYC) vs. the Rest of NYS and the Rest of the United States

- Despite having 43% of NYS households, New York City (NYC) accounts for only 21% of the state's vehicles, with 54% of NYC households having no vehicle.

- NYC residents rely on public transit and walking more than those in the rest of NYS and the rest of the United States. For example, walking was the most common travel mode in NYC (41% of person trips), whereas single-occupancy vehicle person trips were 10% in 2017, significantly lower than those in the rest of NYS and the rest of the United States (Figure ES-2).
- Although average daily vehicle trips declined from 2001 to 2017 (Figure ES-3), trip lengths increased (Figure ES-4), reflecting longer commutes and increasing congestion.

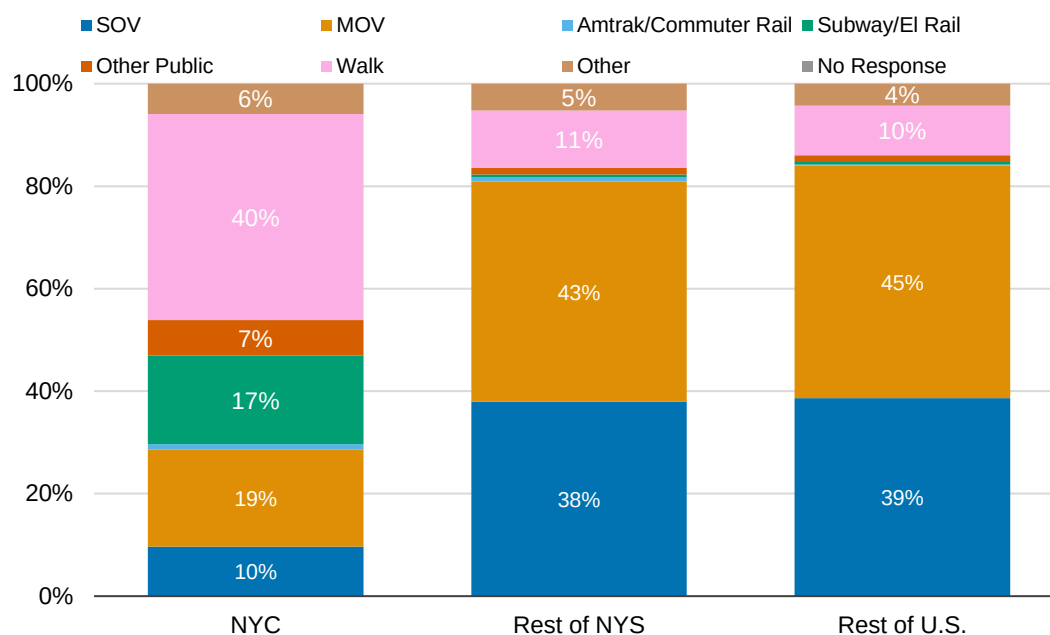


Figure ES-2. Person trip share by selected modes by region in 2017.

Notes: SOV = single-occupancy vehicle; MOV = multioccupancy vehicle; El = elevated rail. Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number. Unlabeled segments represent values lower than 4%.

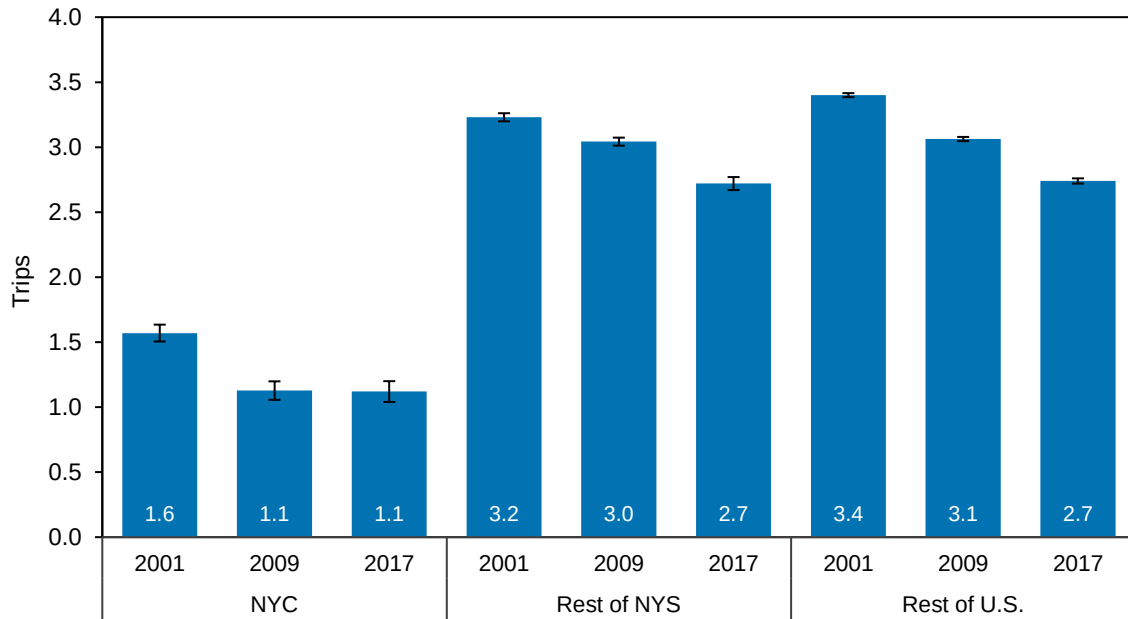


Figure ES-3. Average daily vehicle trips per driver by region and year.

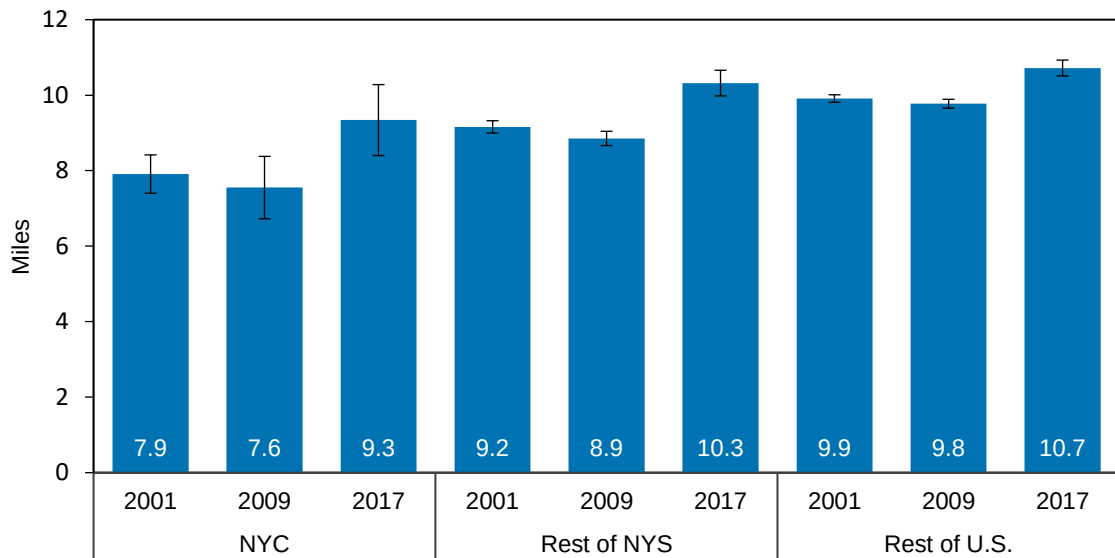


Figure ES-4. Average vehicle trip length by region and year.

Population Density Effect on Travel

- The number of vehicle trips per driver decreases as population density increases in NYS and the rest of the United States. For areas with fewer than 500 people per square mile, NYS drivers average 2.6 trips per day compared with 2.7 trips per day among drivers in the rest of the United States. In the highest-density areas (50,000+ people per square mile), NYS drivers take only 0.8 daily trips on average, which is statistically significantly lower than the 1.9 daily trips observed in the rest of the United States.

- Both NYS and the rest of the United States show an increase in average driving distances from 2001 to 2017 across all density categories. The trend is more pronounced in lower-density areas. For instance, in regions with population density lower than 500 people per square mile, NYS driving distances increased statistically significantly from 11 miles in 2001 to 13 miles in 2017 (Figure ES-5).
- A decrease in the share of drivers can be observed as population density increases. Furthermore, the gap between NYS and the rest of the United States metros widened with increasing population density (Figure ES-6).
- NYS commuters generally experienced longer-travel-time trips compared with those of their counterparts in the rest of the nation, especially for public transit users. The disparity is most pronounced in the least populated areas in 2017, where public transit times often exceed an hour (Figure ES-7).

Metropolitan Statistical Areas (MSAs) and Regional Variations

- Average daily person trips per person declined across most MSAs (Figure ES-8), and trip distances increased, particularly in suburban and midsized MSAs (Figure ES-9).
- Work-related vehicle trips have consistently declined in major metropolitan areas, particularly in NYC, where they accounted for over 25% of trips in 2001 but dropped to around 10% by 2017. Social and recreational trips have gained a more prominent share, particularly in urban areas, where they have offset declines in work-related travel.
- Urban centers generally exhibit longer driving times than those in smaller cities and towns. Albany, Syracuse, and Rochester consistently report driving times lower than those in NYC and Rockland–Westchester (Figure ES-10).

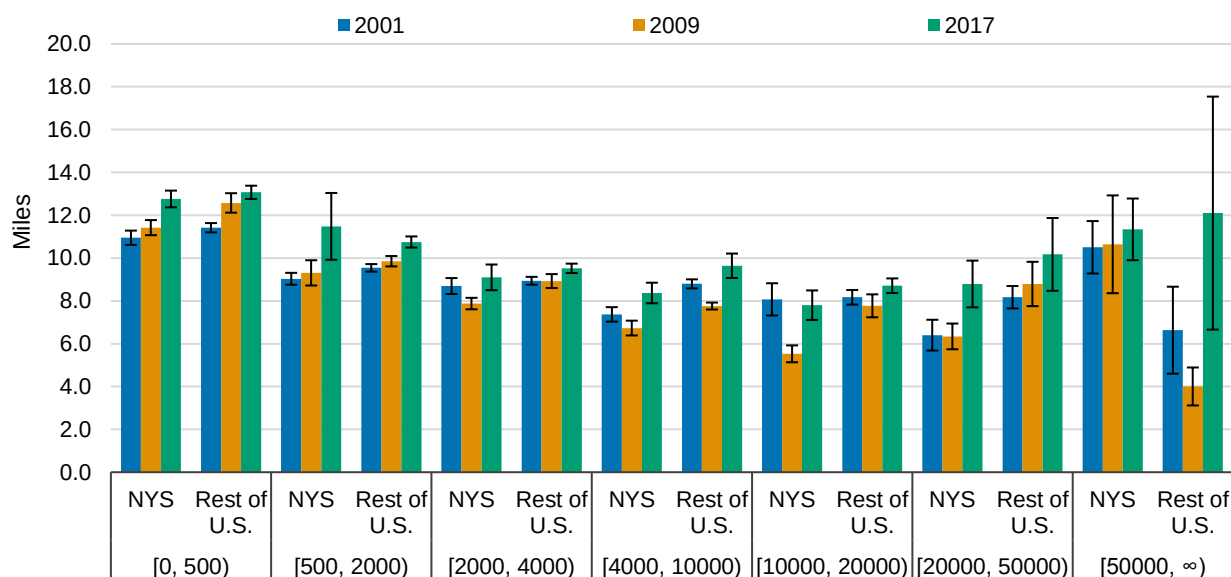


Figure ES-5. Average driving distances by population density, region, and year.

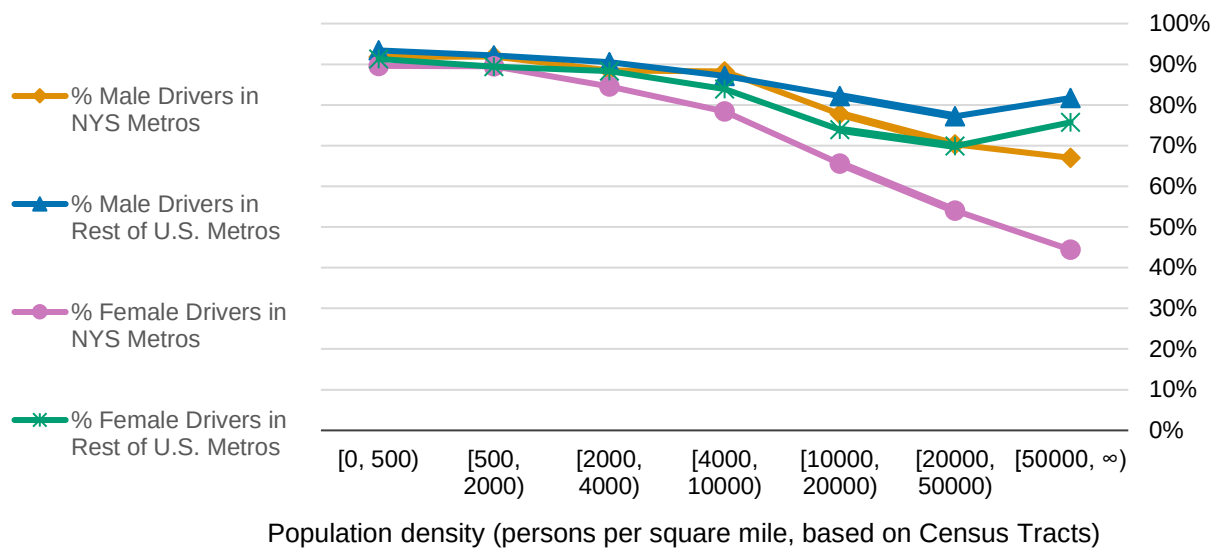


Figure ES-6. Driver rates by gender, population density, and region in 2017.

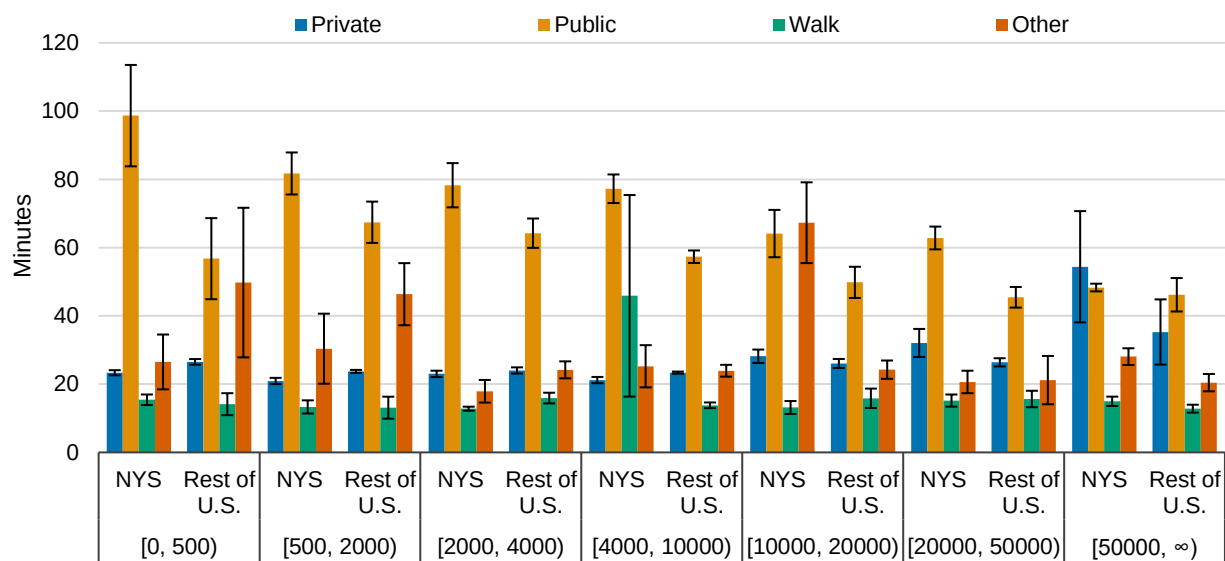


Figure ES-7. Commute travel time by mode, population density, and region in 2017.

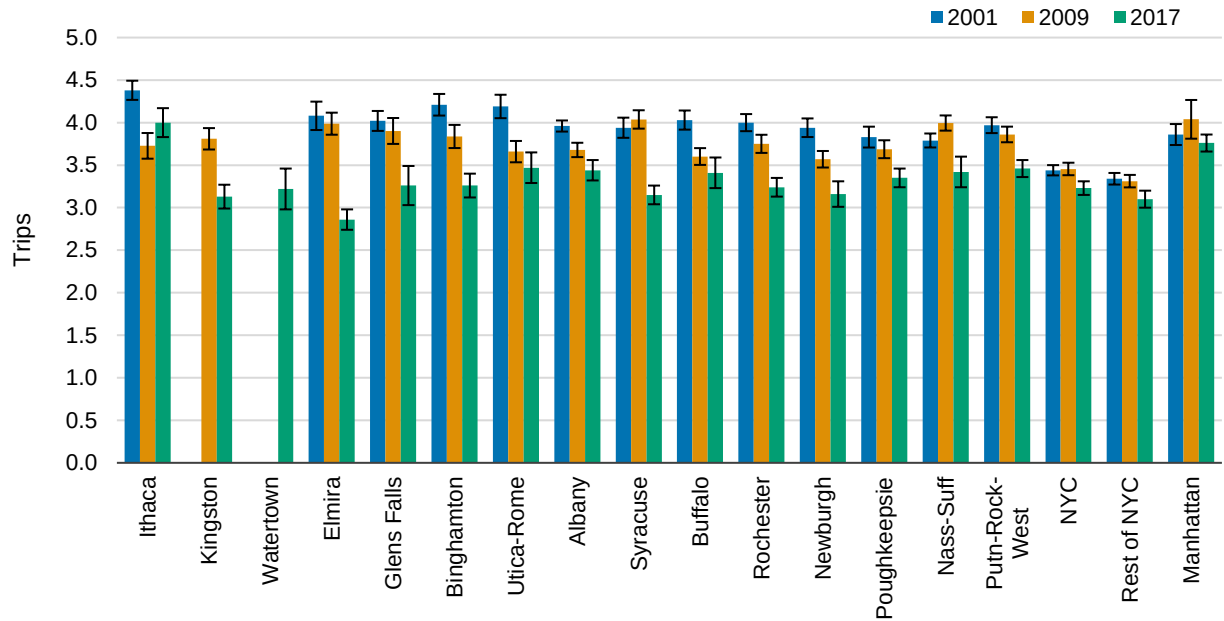


Figure ES-8. Average daily person trips by year and location. Notes: Nass-Suff = Nassau and Suffolk; Putn-Rock-West = Putnam, Rockland, and Westchester.

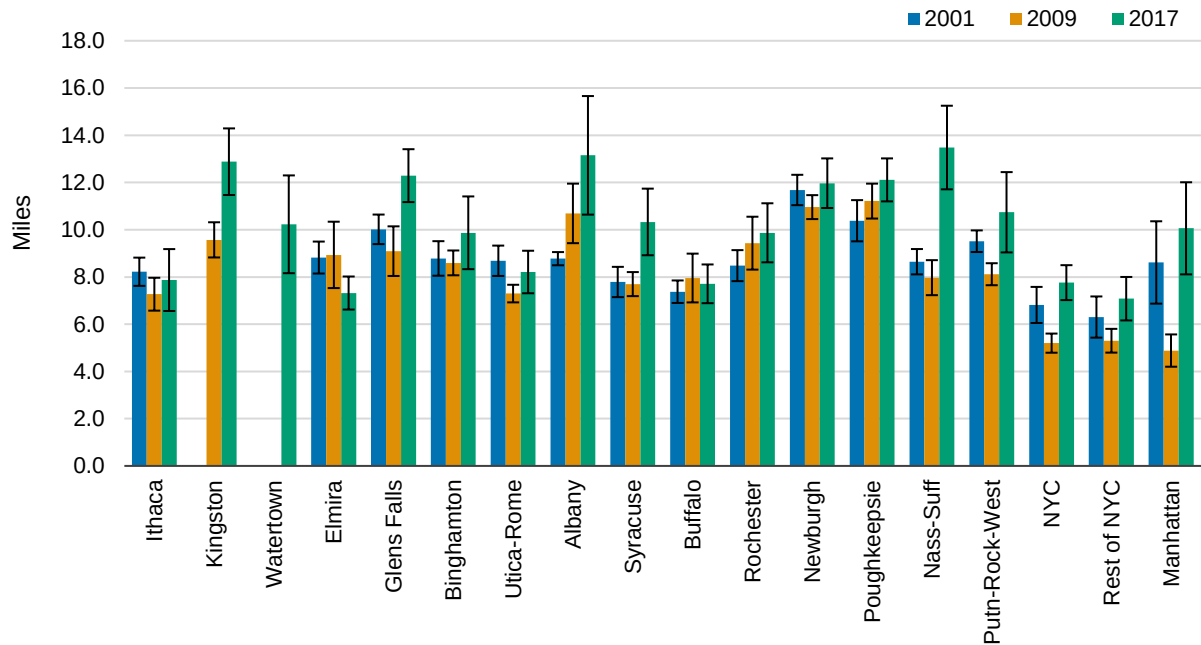


Figure ES-9. Average person trip distances by year and location. Notes: Nass-Suff = Nassau and Suffolk; Putn-Rock-West = Putnam, Rockland, and Westchester.

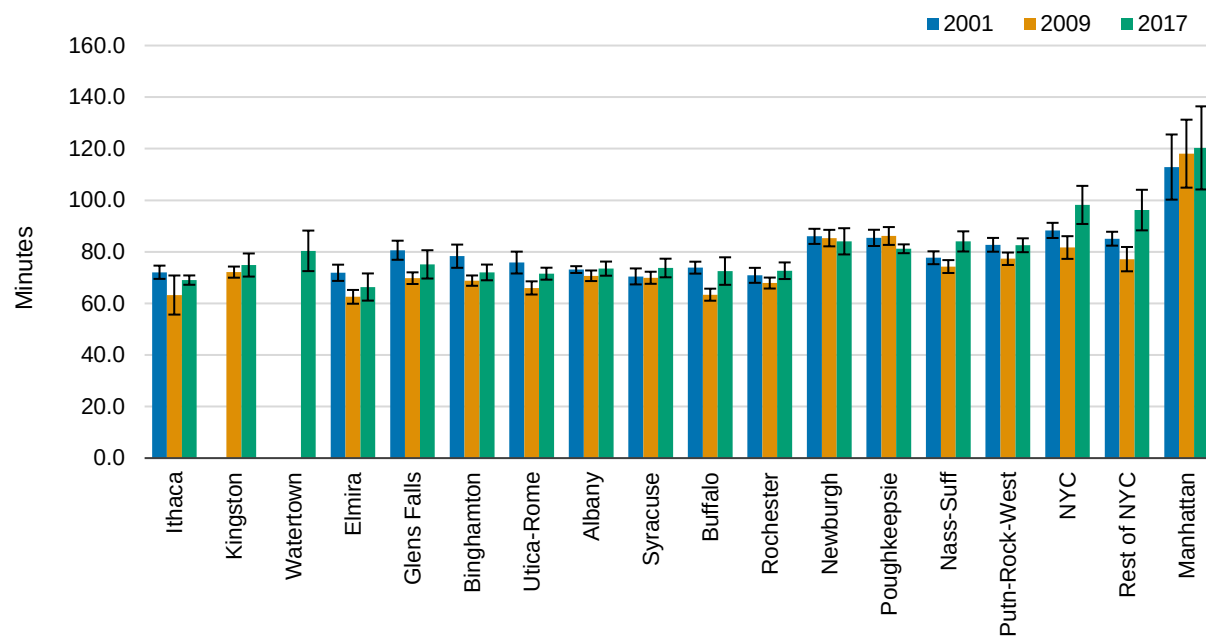


Figure ES-10. Average time driving in private vehicles by year and location.

Notes: Nass-Suff = Nassau and Suffolk; Putn-Rock-West = Putnam, Rockland, and Westchester.

1. INTRODUCTION

1.1 BACKGROUND AND OBJECTIVES

Transportation statistics, such as data on personal travel patterns, play a critical role in shaping infrastructure planning, investment strategies, and strategic decisions. Understanding travel behavior at the household and individual levels is essential for assessing the efficiency, capacity, and adaptability of transportation systems to current and future demand. Reliable data on personal travel trends help policymakers evaluate the feasibility and effectiveness of congestion-alleviating technologies, such as high-speed rail and intelligent transportation systems, as well as guide decisions on transportation investment programs. Additionally, these statistics are vital for quantifying the energy use and environmental impact of different transportation policies, supporting efforts to enhance sustainability, reduce emissions, and improve air quality. In a rapidly evolving mobility landscape, where factors such as urbanization, economic shifts, and emerging transportation technologies continuously reshape travel demand, comprehensive and accurate data are indispensable for long-term planning and operation strategies.

To meet the aforementioned data needs, the U.S. Department of Transportation launched a national travel survey in 1969 to systematically collect detailed information on household and personal travel. Initially known as the Nationwide Personal Transportation Survey, the survey focusing on daily travel patterns was conducted in 1977, 1983, 1990, and 1995. In 2001, the survey was renamed the National Household Travel Survey (NHTS) and expanded to include both daily and long-distance travel, capturing trip characteristics over a 4 week period. In 2009 and 2017, the methodology shifted back to a single-day travel diary approach, with limited data collection on long trips. The primary objective of these surveys has remained consistent: to analyze the relationships between personal travel choices and socio-economic and demographic characteristics, providing a data-driven foundation for transportation research, infrastructure development, and policymaking.

Over the years, the sponsorship and funding structure of NHTS have evolved. The Federal Highway Administration has been the primary sponsor of the survey, with additional contributions from the Federal Transit Administration, the American Automobile Association, and the Public Policy Institute of the American Association of Retired Persons in certain years. In 2009, the nationwide survey collected responses from 151,147 households, of which 25,510 were part of the core national sample funded by federal agencies. The remaining households were obtained through state and regional add-ons, funded by 14 state transportation agencies and 6 metropolitan planning organizations (MPOs), which opted to supplement their data collection for more detailed local insights. By 2017, the NHTS sample had been further expanded to include a national sample and additional regional add-ons funded by various agencies, enhancing its utility for state and local transportation planning efforts. The 2022 NHTS reduced the overall household sample size to 27,290 (7,893 from the national sample and 19,397 from add-ons). Additionally, unlike previous surveys, the 2022 NHTS did not report data at the state level but instead reported data at the Census Division level.

Recognizing the need for more granular state-specific data, New York State (NYS) has participated in the add-on program several times. In 2001, NYS purchased an add-on sample that expanded the state's dataset beyond what was available from the national sample alone. This practice continued in 2009, when the NYS add-on included responses from 14,102 additional households, bringing the total sample to 16,165 usable households. Similarly, in 2017, NYS procured an add-on sample that increased the dataset to 17,209 households. These add-ons enable more detailed analyses of travel behavior at the regional level, capturing variations among urban, suburban, and rural areas as well as differences across demographic groups. The larger sample size allows for robust estimates of travel trends, which are crucial for transportation planning and policy decisions tailored to the unique needs of NYS. However, NYS did

not participate as an add-on partner in 2022, meaning that the latest available state-specific data remain from the 2017 NHTS. Despite this, the 2022 NHTS data for the Middle Atlantic Census Division (i.e., New York, New Jersey, and Pennsylvania) still provide valuable insights into broader regional travel trends and can be used to track larger-scale shifts in mobility patterns.

This report is part of a series of analyses that explore household travel behavior within NYS (Liu et al., 2022; Pan et al., 2023, 2024; Patwary et al., 2025a, 2025b; Southworth et al., 2012; Uddin et al., 2024). The objective of this study is to examine travel patterns, vehicle ownership trends, and trip-making behaviors across different regions of NYS and compare them to national trends. The report aims to answer several key research questions: How do travel behaviors in NYS differ from those observed in the rest of the country? To what extent does the travel behavior of New York City (NYC) residents influence the overall trends observed in the state? Are there notable differences in travel patterns across NYS regions, particularly between urban and rural areas? And how have these patterns evolved over time, especially between the 2001, 2009, 2017, and 2022 survey periods? By addressing these questions, the report provides a comprehensive understanding of travel trends in NYS, supporting data-driven decision-making for future transportation planning initiatives.

1.2 OVERVIEW OF THE NATIONAL HOUSEHOLD TRAVEL SURVEY

As mentioned in the previous section, NHTS is the most comprehensive source of data on personal travel in the US. It provides critical insights into household and individual travel behavior, including trip frequency, mode choice, travel purpose, vehicle ownership, and travel time. This section, focusing on the 2009 and 2017 surveys, discusses survey participants, methods, survey periods, sample sizes, and sampling weights.

1.2.1 Who is Included

The NHTS surveys a representative sample of the civilian, noninstitutionalized US population to capture a wide range of travel behaviors. Individuals living in households are included, whereas people residing in group quarters such as college dormitories, military barracks, nursing homes, long-term medical facilities, prisons, and other institutional settings are excluded. Since 2001, NHTS has collected travel information for all household members, regardless of age. However, interviews are conducted directly only with individuals 5 years of age and older. For children between the ages of 5 and 13, an adult household member provides travel details on their behalf. This approach ensures that travel behaviors across all age groups, including school trips and child care-related travel, are adequately captured in the dataset.

1.2.2 How the Data Were Collected

The methodology for data collection in the NHTS has evolved over time to incorporate technological advancements and improve data accuracy. Whereas the 2009 NHTS relied primarily on telephone-based surveys, the 2017 NHTS transitioned to a mixed-method approach, utilizing a web-based survey along with telephone retrieval interviews for respondents unable or unwilling to participate online. This shift in methodology improved efficiency, reduced survey costs, and expanded participation by reaching households that may have been difficult to contact through traditional telephone methods.

2009 NHTS Data Collection

The 2009 NHTS used a random digit dialing (RDD) method to contact households via telephone, employing computer-assisted telephone interviewing technology. The sample was drawn from a list-assisted RDD sample frame, ensuring a representative cross section of United States households.

Households that agreed to participate were assigned a specific 24 h “travel day,” during which they recorded all trips made by household members.

To ensure comprehensive data collection, a diary package was mailed to participating households shortly after their recruitment interview. The package included the following:

- A letter of thanks from the U.S. Department of Transportation acknowledging the household’s participation.
- A survey brochure providing background on the study.
- A travel day diary personalized for each household member ages 5 years or older, with instructions and an example of a completed entry.
- A reminder card with the household’s assigned travel date.
- A \$2 cash incentive to encourage participation.
- An odometer mileage form for recording the make, model, and year of each household vehicle, along with their odometer readings.

The travel day diary played a crucial role in the 2009 NHTS because participants were required to record trip details manually and later report them during a follow-up telephone interview. The assignment of travel days was evenly distributed across the week, ensuring that data collection accounted for weekday and weekend variations in travel behavior.

2017 NHTS Data Collection

In 2017, the NHTS adopted a new address-based sampling method, replacing the previous RDD approach. This methodological shift improved coverage by reaching households without landline telephones and ensured better representation of younger populations, lower-income households, and rural areas.

The 2017 NHTS followed a two-stage collection process:

1. **Household Recruitment:** Selected households received an invitation letter with instructions to complete a web-based recruitment survey or to opt for a phone interview retrieval method if they preferred. Households that completed this initial step were assigned a specific 24 h travel day.
2. **Travel Diary Reporting:** Each household member recorded their entire day’s travel using a provided travel log in which they detailed trip purpose, mode, duration, and distance.

The 2017 diary package was similar to the 2009 version, except for incentives. In 2017, a \$2 cash incentive was awarded to each sampled address for the initial recruitment mailing. Then, the travel log package included a \$5 cash incentive to improve response rates.

1.2.3 When the Data Were Collected

The 2009 NHTS was conducted from March 2008 through May 2009, ensuring a 12 month data collection period to capture seasonal variations in travel behavior. Travel data were collected for all seven days of the week, including holidays, with each household assigned a specific 24 h travel day that started

at 4:00 a.m. and ended at 3:59 a.m. the following day. To correct for seasonal imbalances, sample weights were applied, ensuring an even distribution of household and person interviews across all months. The NYS add-on sample followed the same methodology, with data collection spanning March 2008 through April 2009.

The 2017 NHTS followed a similar structure, running from March 2016 through May 2017, with travel days assigned and recorded in the same 24 h format. The survey design maintained a seasonal balance using adjusted weights to ensure proportional representation across different months. The New York add-on sample was collected between April 2016 and April 2017, aligning with the national survey period.

1.2.4 Sample Sizes and Weighted Sums

The 2009 NHTS had a sample size of 151,147 households nationwide. This consisted of 25,510 households in the core national sample and 124,637 households from add-on participants, including 14,102 additional households from NYS. When combined with the national sample, NYS's total dataset included 16,165 usable households. The 2017 NHTS further expanded the dataset, collecting data from 129,696 households nationwide, representing over 264,000 individuals. NYS's add-on sample increased its dataset to 17,209 usable households.

Because NHTS employs a complex survey design, weighting factors are applied to adjust for sample design, selection probabilities, and nonresponse biases, ensuring the dataset accurately reflects the national and state populations. Without these adjustments, estimates could misrepresent actual travel behavior. For instance, in the unweighted sample in the 2017 NHTS, NYS households reported an average of 7.0 daily trips, whereas after applying appropriate weighting factors, the true estimate was 8.5 daily trips per household. These weighting adjustments allow for reliable comparisons across survey years and geographic regions.

1.2.5 Key Changes in 2017 NHTS Survey Methodology

The 2017 NHTS introduced several methodological changes, some of which affect direct comparisons with previous survey years. The changes included the following:

- **Shift from Self-Reported to System-Calculated Trip Distances:** The 2017 NHTS used Google Maps API to compute the shortest-path route distance between reported trip origins and destinations, replacing the previous method of relying on participants' self-reported distances. This has improved precision but resulted in systematically lower estimates of vehicle miles of travel (VMT) compared with estimates from previous years.
- **Reclassification of Loop Trips:** In prior surveys, "loop trips" (trips that begin and end at the same location) were counted as two separate trips. In 2017, these were recorded as single trips, leading to a reduction in reported trip counts. Notably, loop trips are made primarily via active transportation modes (e.g., bicycling or walking).
- **Transition to Online Data Collection:** Most respondents completed the survey online rather than by phone, leading to differences in response patterns and potential mode-specific reporting biases.

1.2.6 Adjustment of Trip Distances

The 2017 NHTS introduced a methodological change in how trip distances were calculated. Instead of relying on self-reported distances from survey respondents, the 2017 survey used system-calculated trip distances derived from Google Maps' shortest-path routing between geocoded origins and destinations.

Although this approach improved consistency and reduced reporting errors, it also led to systematically shorter trip lengths compared with those in previous survey years, affecting estimates of VMT and person miles of travel (PMT).

To correct for this discrepancy, an adjustment factor was developed by comparing self-reported distances from previous surveys with system-calculated distances in 2017. This analysis, detailed in the 2017 NHTS Summary of Travel Trends Report (Federal Highway Administration, 2018), revealed the following:

- Work-related trips showed a 7% difference between self-reported and system-calculated distances, meaning system-calculated values were slightly shorter.
- Nonwork trips had a larger 11% discrepancy, requiring greater adjustment.

These adjustments were applied to VMT and PMT estimates, ensuring that 2017 data remained comparable with data from previous survey years and consistent with external travel data sources. The adjusted estimates provided a more accurate representation of actual travel behavior, thereby helping policymakers and researchers avoid underestimating total travel distances when comparing trends over time.

1.3 ORGANIZATION OF REPORT

This report provides a comprehensive analysis of transportation patterns, vehicle ownership trends, and travel behaviors across NYS and the rest of the United States over multiple NHTS years (2001, 2009, 2017, and 2022). The study focuses on how travel patterns have evolved, highlighting differences by geographic characteristics, household income, and personal mobility choices. Chapter 2 examines statewide trends, comparing NYS with the rest of the nation in terms of trip frequency, travel mode choices, vehicle ownership, and vehicle characteristics. It also explores long-term shifts in mobility patterns, including declines in vehicle use, increased reliance on alternative transportation, and regional variations in travel behavior. Chapter 3 analyzes travel patterns within NYS, contrasting NYC with the rest of the state. The discussion highlights the effect of public transit accessibility, differences in vehicle ownership rates, and the divergence in travel behavior between urban and suburban areas. Chapter 4 focuses on how population density influences travel behavior, comparing mobility trends across low-, mid-, and high-density areas in both NYS and the rest of the United States. It assesses trip rates, mode choices, commute patterns, and evolving travel distances over time. Chapter 5 evaluates household and travel patterns by Metropolitan Statistical Areas (MSAs), exploring how travel demand, vehicle access, and commuting habits vary across different metro areas. It also investigates regional disparities in zero-vehicle households, public transit use, and VMT. Finally, Chapter 6 summarizes key findings and suggests directions for future research.

2. COMPARISON OF NEW YORK STATE DATA WITH THE REST OF THE UNITED STATES

This chapter investigates travel patterns of NYS residents in comparison with those across the United States. It focuses on various travel metrics, including personal trips, PMT, and VMT, analyzing these by demographic factors such as age, gender, and vehicle ownership. The primary objective is to analyze how travel behaviors in NYS are different from those in the rest of the United States and how these differences have evolved across NHTS survey years (2001, 2009, and 2017). Owing to the availability of 2022 NHTS data only at the Census Division level, comparisons are also made between the Middle Atlantic Census Division and the rest of the United States for 2017 and 2022.

Key questions addressed in the chapter include the following:

- Do households without any vehicles exhibit similar mobility patterns to those with one or more vehicles?
- How does the travel behavior of non-vehicle-owning residents in NYS compare with that of similar residents nationwide?
- For the NYS households that own and use private vehicles, what are the types, ages, and average number of these vehicles relative to households in other regions?

Additionally, the chapter explores the differences in walking and cycling habits among NYS residents compared with national trends in nonmotorized travel. It assesses how these activities align with contemporary profiles of active and healthy lifestyles.

The effects of vehicle ownership and usage patterns on fossil fuel consumption are also assessed. This analysis focused on whether NYS residents' multimodal travel activities result in higher or lower greenhouse gas emissions per person-mile compared with those in the rest of the country.

2.1 HOUSEHOLD PROFILE

Table 2-1 compares household characteristics and vehicle ownership patterns between NYS and the rest of the United States. In NYS, the share of households with incomes below \$10,000 increased significantly from 3.6% in 2001 to 4.6% in 2009. Although this trend continued from 2009 to 2017, with the share rising to 4.8%, the increase was not statistically significant. Regarding driver licensing, the percentage of female drivers in NYS declined significantly, dropping from 77.2% in 2009 to 67.7% in 2017. A similar trend was observed among male drivers, whose licensing rate fell from 86.4% in 2009 to 80.3% in 2017.

2.1.1 Number of Household Drivers, Licenses, Workers, and Vehicles

In NYS, the number of drivers increased by 15% from 2001 to 2009, but then decreased by 7% from 2009 to 2017 (Figure 2-1). In contrast, the United States saw an increase in drivers during both periods, though the growth was slower from 2009 to 2017 compared with that of 2001 to 2009 (6% vs. 11%). In NYS, the number of vehicles decreased by 4% between 2001 and 2009 but rose by 4% from 2009 to 2017. Meanwhile, the rest of the United States experienced steady vehicle growth of around 5% in both periods.

Table 2-1. Summary of household characteristics and vehicle licensing rates.

	2001		2009		2017	
	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.
Households	7,183	100,182	7,137	105,964	7,261	110,947
Drivers	11,167	179,258	12,838	199,471	11,949	211,347
Workers	9,645	135,627	9,759	141,614	9,842	147,147
Vehicles	9,545	193,041	9,144	202,357	9,528	213,051
Workers per Household	1.34	1.35	1.37	1.34	1.36	1.33
Vehicles per Household	1.33	1.93	1.28	1.91	1.31	1.92
Vehicles per Driver	0.85	1.08	0.71	1.01	0.80	1.01
Household Income % ^a (in U.S.\$[2022] ^b)						
Less than \$10,000	3.6%	3.5%	4.6%	4.1%	4.8%	4.7%
\$10,000 to \$14,999	4.6%	4.0%	4.1%	4.8%	6.2%	5.5%
\$15,000 to \$24,999	6.8%	6.6%	8.3%	8.9%	7.0%	8.4%
\$25,000 to \$34,999	7.5%	7.7%	8.1%	8.6%	6.6%	7.5%
\$35,000 to \$49,999	11.1%	12.3%	9.7%	11.2%	9.6%	11.7%
\$50,000 to \$74,999	14.4%	16.5%	15.2%	15.6%	12.5%	15.1%
\$75,000 to \$99,999	13.5%	14.8%	11.3%	12.1%	11.2%	11.7%
\$100,000 to \$124,999	7.3%	7.8%	8.4%	9.3%	10.0%	9.4%
\$125,000 to \$149,999	6.7%	6.7%	6.0%	5.8%	7.0%	7.1%
\$150,000 to \$199,999	6.2%	5.6%	5.9%	5.9%	8.2%	7.6%
\$200,000 or more	7.9%	6.3%	9.7%	6.9%	12.8%	8.3%
Unreported	10.3%	8.1%	8.8%	6.8%	4.0%	3.0%
Licensing Rate ^c						
Female Drivers/Female 16+	71.1%	87.7%	77.2%	86.8%	67.7%	86.4%
Male Drivers/Male 16+	83.5%	93.3%	86.4%	91.6%	80.3%	89.4%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

^b 2001, 2009, and 2017 income categories are adjusted to 2022 dollars.

^c Licensing rate is determined using the self-reported response to the question “Are you a driver?” in the NHTS survey. As such, this rate should not be compared with the official NYS Department of Transportation drivers’ licensing statistics.

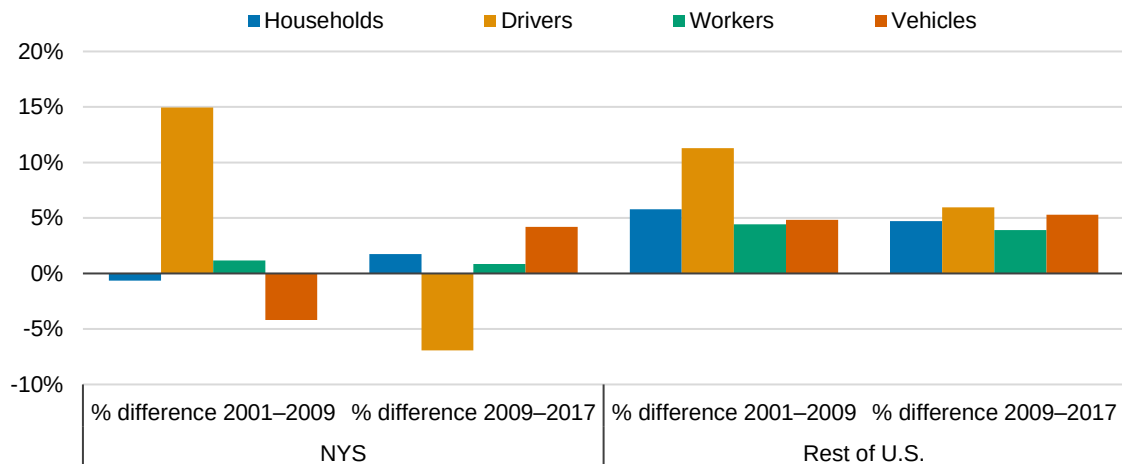


Figure 2-1. Percent differences in number of households, drivers, workers, and vehicles by location and across two time periods.

2.1.2 Household Incomes and Vehicle Ownership

The distribution of vehicle ownership in NYS and the rest of the United States is presented in Table 2-2. NYS shows an increasing trend in zero-vehicle households from 2009 to 2017, although the change is not statistically significant. However, a statistically significant increase was observed in the share of households with four or more vehicles, rising from 3.6% in 2009 to 5.7% in 2017. Whereas the rest of the United States had a higher share of households with four or more vehicles in 2017 (9.8%) compared with those of NYS, the absolute increase from 2009 to 2017 was smaller than the increase seen in NYS. Among households owning vehicles, NYS households were on average less likely to own two or more vehicles than households in the rest of the country.

Table 2-2. Distribution of household vehicle ownership by year and region.

Vehicle Ownership ^a	NYS			Rest of U.S.		
	2001	2009	2017	2001	2009	2017
0 vehicles	26.5%	28.5%	29.1%	6.8%	7.4%	7.6%
1 vehicle	33.2%	32.4%	33.2%	31.3%	32.3%	33.6%
2 vehicles	27.2%	27.1%	23.3%	37.9%	36.9%	33.7%
3 vehicles	9.3%	8.4%	8.7%	15.4%	14.8%	15.3%
4+ vehicles	3.9%	3.6%	5.7%	8.6%	8.6%	9.8%
2+ vehicles	40.4%	39.1%	37.7%	61.9%	60.3%	58.8%
2+ vehicles/1+ vehicles	54.9%	54.7%	53.2%	66.4%	65.1%	63.6%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Figure 2-2 illustrates the vehicle ownership patterns in NYS compared with those of the rest of the United States. Across all 3 years, the share of households with no vehicles was statistically significantly higher in NYS. For example, in 2017, 29% of households in NYS had no vehicles, compared with just 8% of households in the rest of the United States. NYS, particularly NYC, has a well-developed public transportation system that provides residents with viable alternatives to owning a vehicle. The extensive subway and bus networks enable residents to easily meet their travel needs without the necessity of a personal vehicle. This level of accessibility could lead to higher rates of households choosing not to own a car, especially in urban centers where public transport is more accessible.

Households with two or more vehicles are more common in the rest of the United States than in NYS for all years. In 2017, 34% of households in the rest of the United States owned two vehicles, whereas only 23% of households in NYS did. The share of households with just one vehicle was comparable between NYS and the rest of the United States. For instance, in 2017, this share was 34% in the rest of the United States and 33% in NYS.

2.2 PERSONAL TRAVEL COMPARISONS

2.2.1 Trip Frequency and Trip Length

The average number of person trips per person in NYS decreased significantly from 3.73 in 2001 to 3.64 in 2009, and further to 3.29 in 2017 (Table 2-3). In 2001, person trips were significantly higher in the rest of the United States (4.11) than those in NYS. However, by 2017, this difference had narrowed, with the rest of the United States at 3.38, which was not statistically different from the number in NYS. Females and males in NYS made about the same number of person trips, whereas in the rest of the United States, females made slightly more trips than males made.

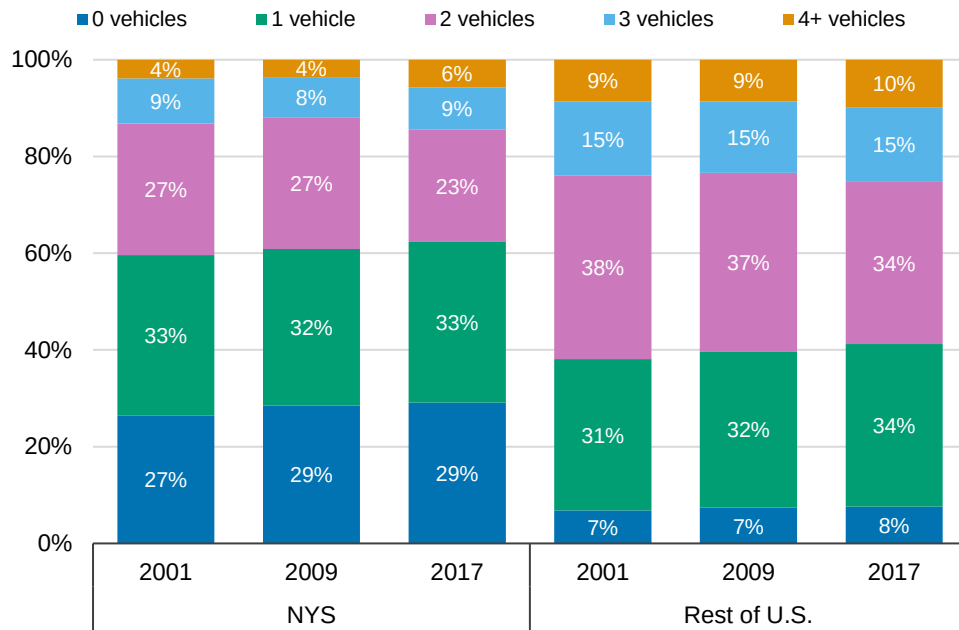


Figure 2-2. Distribution of household vehicle ownership by year and region.

Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number.

Table 2-3. Personal travel statistics by demographic characteristics, year, and region.

Travel Statistics	NYS			Rest of U.S.		
	2001	2009	2017	2001	2009	2017
Person Trips per Person	3.73	3.64	3.29	4.11	3.81	3.38
Gender						
Male	3.77	3.63	3.29	4.11	3.76	3.32
Female	3.70	3.64	3.28	4.12	3.85	3.44
Household Income (U.S.\$[2022])						
Less than \$10,000	2.74	2.52	2.45	2.98	2.85	2.82
\$10,000 to \$14,999	3.17	2.80	2.81	3.43	2.96	2.91
\$15,000 to \$24,999	3.18	3.27	3.03	3.40	3.26	3.06
\$25,000 to \$34,999	3.36	3.09	2.96	3.75	3.49	3.18
\$35,000 to \$49,999	3.52	3.49	3.16	3.92	3.68	3.34
\$50,000 to \$74,999	3.78	3.44	3.30	4.16	3.80	3.43
\$75,000 to \$99,999	3.91	3.89	3.28	4.34	3.94	3.40
\$100,000 to \$124,999	3.98	4.05	3.32	4.45	4.11	3.52
\$125,000 to \$149,999	4.12	4.12	3.46	4.44	4.33	3.54
\$150,000 to \$199,999	4.24	4.10	3.56	4.61	4.23	3.68
\$200,000 or more	4.22	4.30	3.82	4.58	4.31	3.66
Unreported	3.18	2.95	2.79	3.45	3.35	2.91
Weekday Person Trips per Person	3.66	3.74	3.28	4.00	3.92	3.35
Weekend Person Trips per Person	3.95	3.37	3.34	4.44	3.52	3.49
Average Person Trip Length (miles)	8.30	7.49	9.22	10.14	9.89	10.80

Figure 2-3 shows person trips by household income in NYS, revealing that higher-income groups generally take more trips. A declining trend is seen across all years in person trips from 2001 to 2009 and further to 2017. For example, in NYS, the number of trips in 2017 for households earning \$200K or more (3.8) was significantly lower than that in 2009 (4.3).

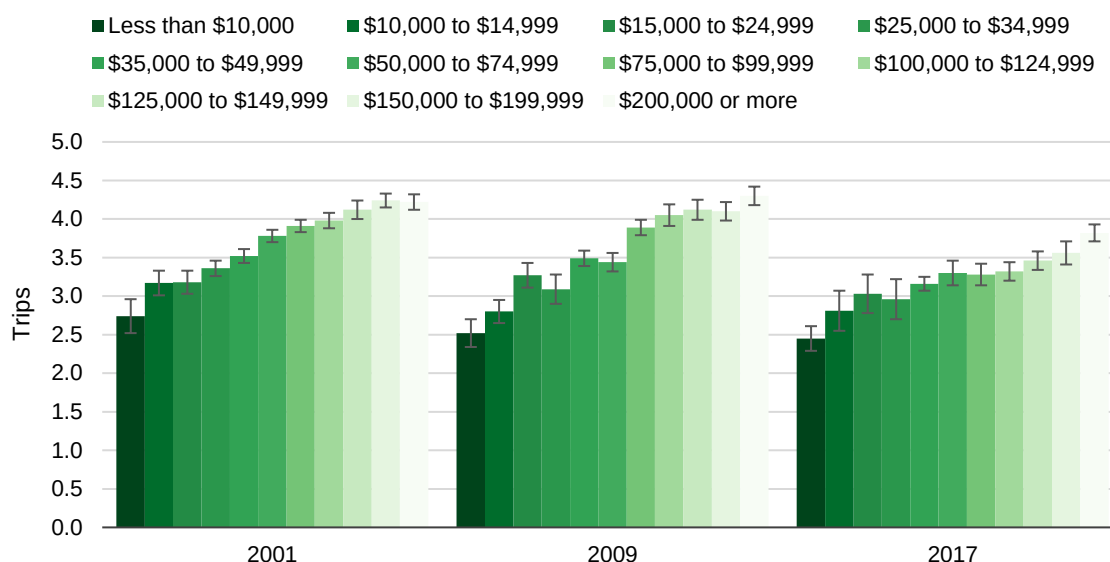


Figure 2-3. Person trips per person by household income and year for NYS.

The number of weekday person trips per person was significantly higher than the number of weekend trips in both 2001 and 2009 in NYS (Figure 2-4). However, by 2017, weekday and weekend trips became comparable. In the rest of the United States, the number of weekend trips was statistically significantly higher than weekday trips in 2001 (4.2 vs. 3.9), a pattern that persisted in 2009. In 2017, the number of weekend trips per person was statistically significantly higher than weekday trips in the rest of the United States, though the absolute difference is exceedingly small (0.1).

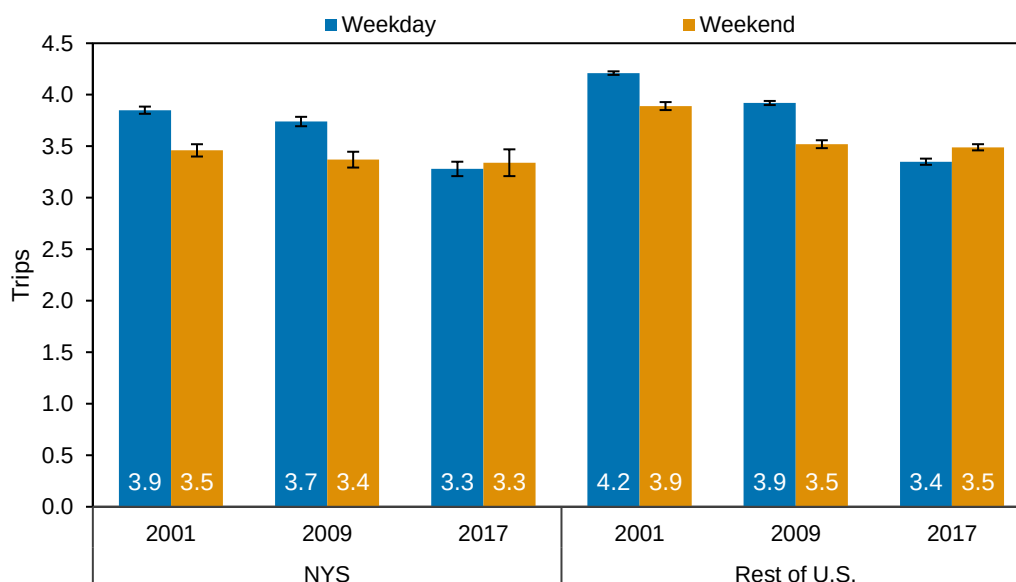


Figure 2-4. Weekday and weekend person trips per person by year and region.

Note: Weekend trips = Saturday and Sunday trips.

Figure 2-5 shows that the average person trip length in the rest of the United States remained relatively stable between 2001 and 2017. In contrast, NYS saw a statistically significant increase in average trip length in 2017 compared with 2009, rising from 7.5 to 9.2 miles.

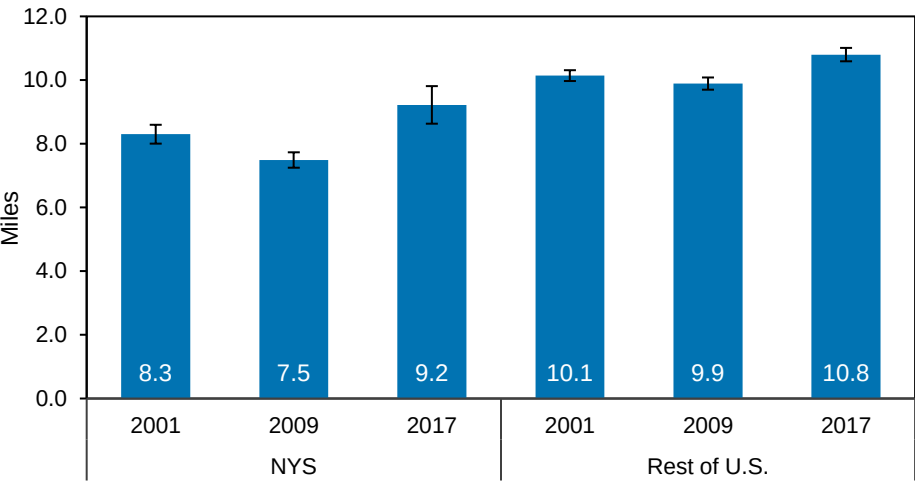


Figure 2-5. Average person trip length by year and region.

Comparing 2017 and 2022, person trips per person in the Middle Atlantic census division decreased significantly, from 3.32 to 2.29 (Table 2-4). A similar trend is observed in the rest of the United States. The group that experienced the largest reduction, as shown in Figure 2-6, is the lowest income group, those earning less than \$10,000. For example, 47.4% of the reduction in person trips was among this group. A similar pattern was also evident in the rest of the United States.

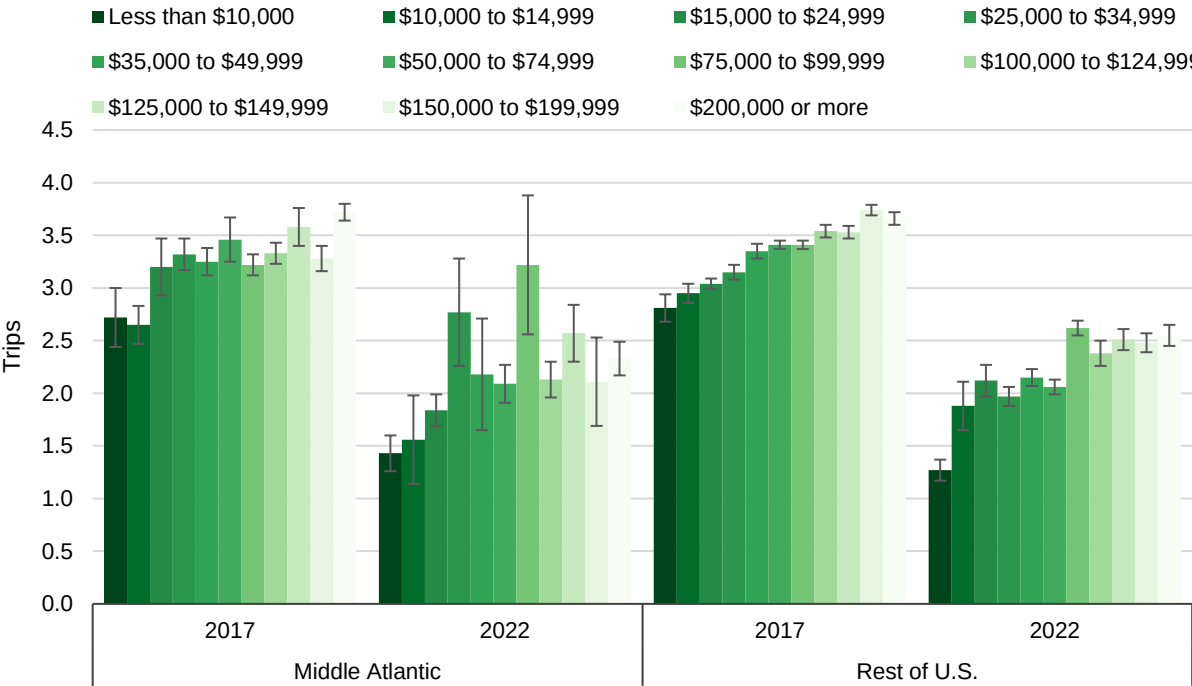


Figure 2-6. Person trips per person by household income and year for the Middle Atlantic Census Division and the rest of the United States.

Table 2-4. Personal travel statistics by household income and year for the Middle Atlantic Census Division and the rest of the United States.

Travel Statistics	Middle Atlantic		Rest of U.S.		2017–2022 % difference	
	2017	2022	2017	2022	Middle Atlantic	Rest of U.S.
Person Trips per Person	3.32	2.29	3.38	2.27	–31.0%	–32.8%
Household Income						
Less than \$10,000	2.72	1.43	2.81	1.27	–47.4%	–54.8%
\$10,000 to \$14,999	2.65	1.56	2.95	1.88	–41.1%	–36.3%
\$15,000 to \$24,999	3.20	1.84	3.04	2.12	–42.5%	–30.3%
\$25,000 to \$34,999	3.32	2.77	3.15	1.97	–16.6%	–37.5%
\$35,000 to \$49,999	3.25	2.18	3.35	2.15	–32.9%	–35.8%
\$50,000 to \$74,999	3.46	2.09	3.41	2.06	–39.6%	–39.6%
\$75,000 to \$99,999	3.22	3.22	3.41	2.62	0.0%	–23.2%
\$100,000 to \$124,999	3.33	2.13	3.54	2.38	–36.0%	–32.8%
\$125,000 to \$149,999	3.58	2.57	3.53	2.51	–28.2%	–28.9%
\$150,000 to \$199,999	3.28	2.11	3.74	2.48	–35.7%	–33.7%
\$200,000 or more	3.72	2.33	3.66	2.55	–37.4%	–30.3%
Unreported	2.50	2.35	2.98	1.75	–6.0%	–41.3%

2.2.2 Modes of Travel

Privately owned vehicles (POVs), including single-occupancy vehicles (SOVs) and multioccupancy vehicles (MOVs), account for over 50% of person trips in both NYS and the rest of the United States (Table 2-5). In 2017, the share of POVs was 59% in NYS and 84.1% in the rest of the United States. Walking trips were significantly higher in NYS than in the rest of the United States across all years. For example, in 2017, walking trips accounted for 23.4% of person trips in NYS, compared with only 9.7% in the rest of the United States. A similar pattern is observed for POVs in relation to PMT.

From 2001 to 2009, person trips decreased by 2.4%, and from 2009 to 2017, the decrease was 9.6% (Table 2-6). The decline was even more pronounced in the rest of the United States, where person trips decreased by 11.3% from 2009 to 2017. In terms of PMT, NYS experienced an increase of 17.5% from 2009 to 2017, whereas a 1.1% decrease was observed in the rest of the United States during the same period.

Table 2-7 reveals that average minutes spent walking and/or biking increased from 2001 to 2017 in NYS. However, the increase was not statistically significant. Figure 2-7 shows that the 21-to-35-year-old age group has the highest daily walking time in NYS. Younger adults may also be more physically active or have greater flexibility in their travel choices compared with older populations (Liu et al., 2022; Patwary et al., 2025). In 2017, walking time was statistically significantly lower in the rest of the United States than in NYS across all age groups.

Table 2-5. Distribution of person travel rates and mode shares^a by year and region.

	NYS			Rest of U.S.		
	2001	2009	2017	2001	2009	2017
Person Trips per Person (%)	3.73	3.64	3.29	4.11	3.81	3.38
SOV	30.3%	30.5%	26.0%	40.3%	40.5%	38.7%
MOV	35.3%	31.7%	33.0%	47.3%	44.4%	45.4%
Other POV	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Amtrak/Commuter Rail	0.6%	1.1%	0.9%	0.1%	0.1%	0.2%
Subway/Elevated Rail	5.3%	4.2%	7.5%	0.2%	0.1%	0.5%
Other Public	3.8%	4.8%	3.6%	0.8%	1.0%	1.3%
Walk	20.0%	22.0%	23.4%	7.9%	9.7%	9.7%
Other	4.6%	5.6%	5.5%	3.3%	4.1%	4.3%
Unreported	0.1%	0.1%	0.0%	0.1%	0.2%	0.0%
PMT per Person (%)	29.40	25.76	30.26	41.03	36.85	36.43
SOV	31.8%	34.5%	24.4%	37.6%	39.0%	32.3%
MOV	43.0%	44.5%	35.8%	51.3%	49.9%	44.9%
Other POV	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Amtrak/Commuter Rail	1.7%	1.8%	3.1%	0.3%	0.4%	0.7%
Subway/Elevated Rail	4.7%	3.7%	7.2%	0.1%	0.1%	0.5%
Other Public	2.0%	2.7%	1.8%	0.5%	0.6%	0.9%
Walk	1.8%	2.3%	2.7%	0.6%	0.7%	0.7%
Other	14.6%	10.5%	25.1%	9.3%	9.4%	19.9%
Unreported	0.6%	0.1%	0.0%	0.5%	0.1%	0.0%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 2-6. Difference of person trips and PMT from 2001 to 2009 and from 2009 to 2017 by region.

	NYS		Rest of U.S.	
	% difference 2001–2009	% difference 2009–2017	% difference 2001–2009	% difference 2009–2017
Person Trips per Person	–2.4%	–9.6%	–7.3%	–11.3%
PMT per Person	–12.4%	17.5%	–10.2%	–1.1%

Table 2-7. Average minutes spent walking and/or biking per day by age, year, and region.

	NYS			Rest of US		
	2001	2009	2017	2001	2009	2017
All Ages	10.74	12.34	12.40	5.50	6.24	5.96
5-to-10-year-olds	10.17	8.68	7.50	7.15	5.13	4.85
11-to-15-year-olds	12.92	10.06	8.98	8.23	9.02	5.49
16-to-20-year-olds	10.54	10.70	11.69	5.05	5.90	5.43
21-to-35-year-olds	12.47	15.38	13.63	4.85	6.25	6.64
36-to-64-year-olds	10.39	13.35	13.06	5.00	6.59	6.05
65+-year-olds	8.46	9.21	12.84	5.05	4.48	5.83

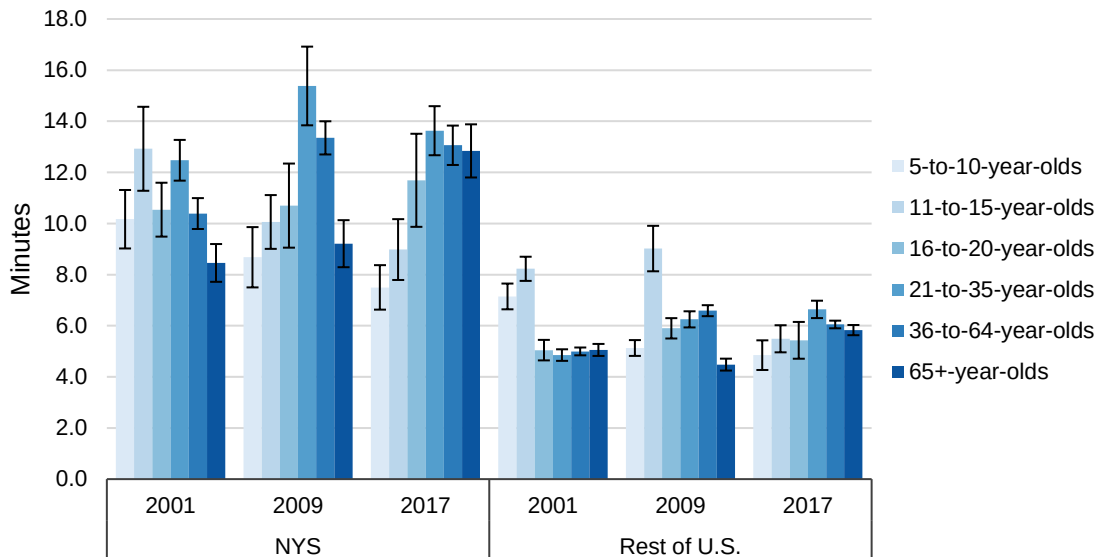


Figure 2-7. Average minutes spent walking and/or biking per day by age, year, and region.

The percentage of people in NYS who did not engage in biking was statistically significantly lower than that in the United States across all years and age groups (Figure 2-8). Overall, the percentage of people in NYS who did not walk or bike decreased statistically significantly from 2009 to 2017, except for the 16-to-20-year-old age group. This trend was also observed in the rest of the United States.

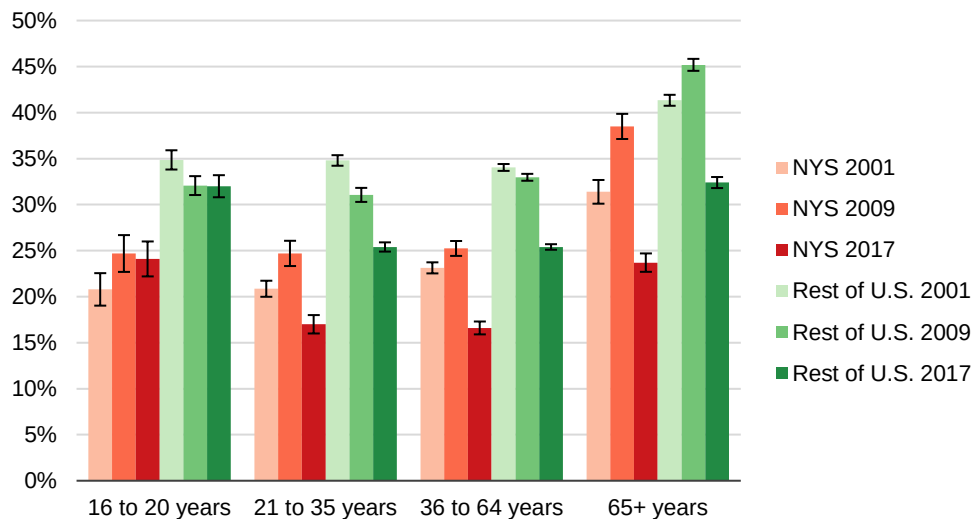


Figure 2-8. Percentages of people who reported no walk/bike trips in the last week by age, year, and region.

2.2.3 Purpose of Travel

Table 2-8 presents the distribution of person trips per person in five broad trip purpose categories: earning a living; family and personal business; civic, educational, and religious; social and recreational; and other. Trip purpose shares are essentially the same for NYS as they are in the rest of the nation. From 2009 to 2017, a 0.9% increase in work-related trips in NYS was observed. A decrease of 4.6% was observed in

family and personal business trips, whereas an increase was seen in social and recreational trips (1.7%) and other trips (2.8%).

Table 2-8. Distribution of daily person trips per person by trip purpose, year, and region.

Trip Purpose	NYS			Rest of US			% Changes in NYS	
	2001	2009	2017	2001	2009	2017	2001–2009	2009–2017
Person Trip per Person (%) ^a	3.73	3.64	3.29	4.11	3.81	3.38	–9.0%	–35.0%
Earning a Living	19.3%	18.3%	19.2%	18.8%	18.7%	19.0%	–1.0%	0.9%
Family & Personal	43.3%	43.2%	38.6%	43.9%	42.4%	38.5%	–0.1%	–4.6%
Civic, Educational, & Religious	9.7%	9.5%	9.7%	9.8%	9.6%	10.9%	–0.2%	0.2%
Social & Recreational	26.2%	27.0%	28.6%	26.6%	27.5%	27.5%	0.7%	1.7%
Other	1.2%	1.1%	3.9%	0.8%	0.8%	4.0%	–0.1%	2.8%
Unreported	0.3%	0.9%	0.0%	0.1%	1.0%	0.0%	0.6%	–0.9%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

2.2.4 Differences in Travel Pattern by Age

In 2017, the pattern of person trip rates by age group in NYS closely resembles that of the rest of the country (Table 2-9). As shown in Figure 2-9, there is no statistically significant difference in person trips between the two regions, except for the 20-to-29 and 50-to-59 age groups, where NYS had fewer trips than those in the rest of the United States. In 2017, the highest trip rates in NYS were observed in the 40-to-49 age group (3.8), with similarly high rates (3.9) in the rest of the United States. The high trip rates among individuals ages 40-to-49 may reflect this group’s increased employment stability, family responsibilities, or social activities, leading to a higher frequency of travel.

Table 2-9. Person trips per person by age, year, and region.

	NYS			Rest of U.S.		
	2001	2009	2017	2001	2009	2017
Person Trips per Person	3.73	3.64	3.29	4.11	3.81	3.38
Age Group						
5-to-15-year-olds	3.02	2.99	2.77	3.46	3.20	2.78
16-to-19-year-olds	3.53	3.32	2.71	4.19	3.54	2.79
20-to-29-year-olds	3.76	3.38	2.97	4.24	3.75	3.24
30-to-39-year-olds	4.16	4.12	3.65	4.57	4.21	3.73
40-to-49-year-olds	4.35	4.35	3.75	4.67	4.33	3.87
50-to-59-year-olds	3.97	3.91	3.44	4.30	4.13	3.67
60-to-64-year-olds	3.88	4.03	3.47	4.12	3.90	3.60
65+-year-olds	3.18	2.96	3.23	3.43	3.22	3.19
Unreported	3.33	0.00	0.00	3.71	0.00	0.00

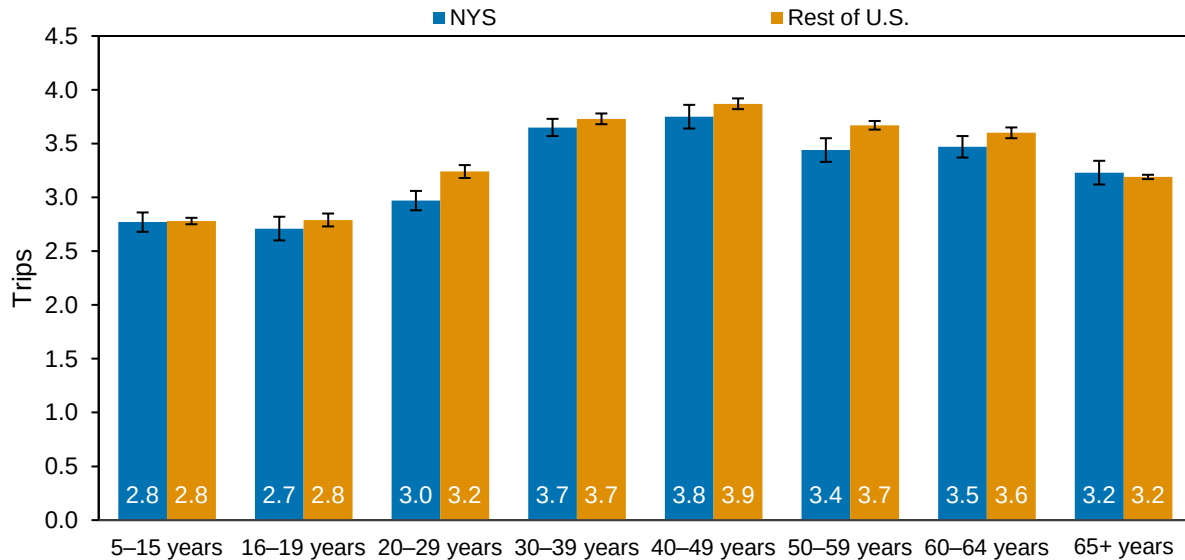


Figure 2-9. Person trips per person by age and region in 2017.

The average daily person trips per person in NYS for the 20-to-29 age group statistically significantly decreased from 2001 to 2009, and again from 2009 to 2017 (Figure 2-10). For age groups under 65, the reduction in trips from 2009 to 2017 was greater than the reduction from 2001 to 2009. However, the 65+ age group had a statistically significant increase in person trips, rising from 3.0 in 2009 to 3.2 in 2017.

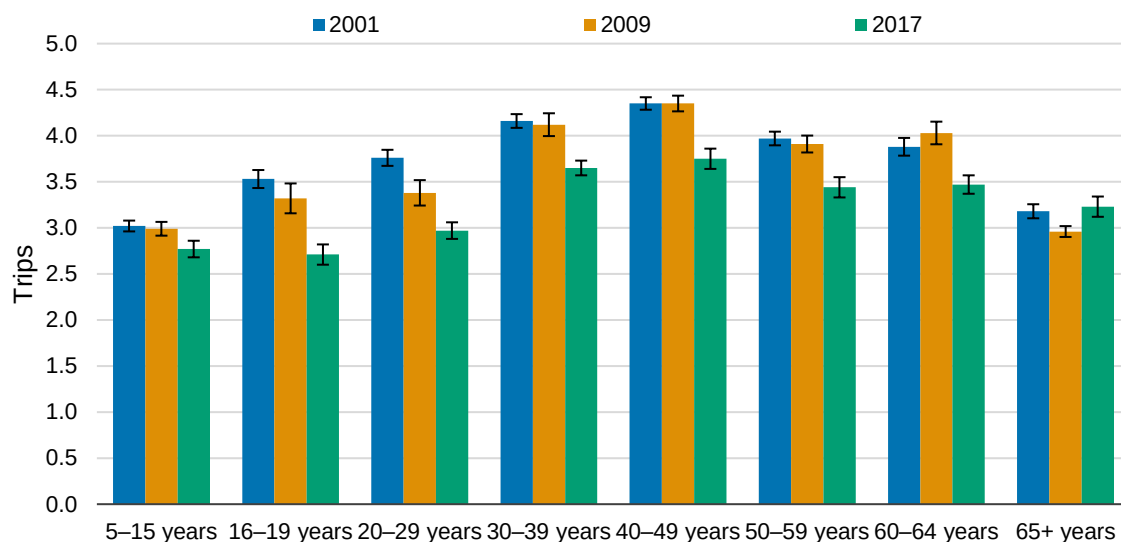


Figure 2-10. Person trips per person by age and year in NYS.

2.2.5 Travel Differences by Vehicle Ownership

Zero-vehicle households made statistically significantly fewer person trips than non-zero-vehicle households in both NYS and the rest of the United States across all years (Figure 2-11). For example, in 2017 in NYS, zero-vehicle households made an average of 3.1 trips, compared with 3.4 trips for non-zero-vehicle households. A declining trend in person trips was observed from 2009 to 2017. In the rest of

the United States, this reduction was statistically significant among non-zero-vehicle households, but not for zero-vehicle households. A similar trend was observed in NYS, as well.

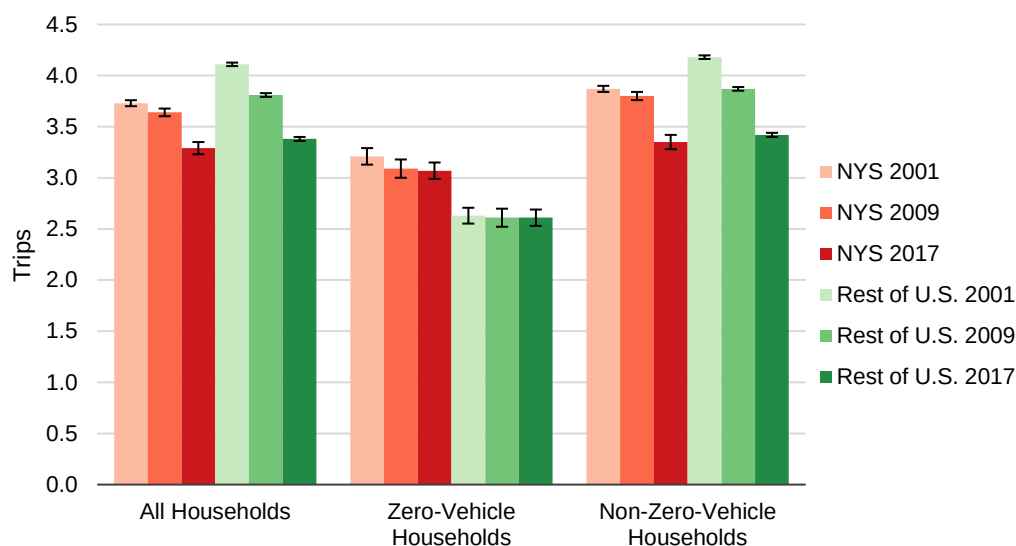


Figure 2-11. Person trips per person by household vehicle ownership, year, and region.

Figure 2-12 shows the mode share based on person trips for zero-vehicle and non-zero-vehicle households. For zero-vehicle households, most trips in NYS were made by walking, whereas in the rest of the United States, private vehicles still dominated. For example, in 2017, 52% of person trips in NYS were made by walking, compared with 39% in the rest of the United States. In contrast, only 6% of trips in NYS were made by private vehicles, whereas this figure was 27% in the rest of the United States. Overall, the share of private vehicle trips showed a declining trend from 2001 to 2009 and again from 2009 to 2017.

For non-zero-vehicle households, the share of public transportation and walking trips in NYS was statistically significantly higher than in the rest of the United States across all 3 years. For instance, in 2017, 7% of person trips in NYS were made using public transportation, compared with just 1% in the rest of the United States.

Figure 2-13 shows the comparison of population groups that did not walk or bike in the last week between 2017 and 2022. A notable trend is the significant increase in the percentage of people who did not engage in any walking or biking between 2017 and 2022, especially among younger age groups. In the Middle Atlantic region, the share of 16-to-20-year-olds reporting no walk or bike trips jumped statistically significantly from 30.2% in 2017 to 74.3% in 2022. A similar, though slightly less extreme, pattern is observed in the rest of the United States. This suggests a considerable decline in active travel participation among teenagers and young adults over this period. For the 21-to-35-year-old group, a significant increase in nonactive travel behavior is also evident, though not as drastic as in the youngest cohort. In the Middle Atlantic, this percentage rose from 20.9% to 36.8%, whereas in the rest of the United States, it increased more sharply from 25.4% to 55.1%. This might indicate shifting lifestyle patterns, including greater reliance on motorized transport, changes in work-from-home trends, or decreased engagement in recreational walking and biking. Among the 36-to-64-year-old population, the percentage of those reporting no walk or bike trips also increased but at a slightly lower rate compared with younger age groups. Older adults (65+ years) also showed a marked increase in nonwalking and biking behavior, though they already had relatively elevated levels in 2017. In the Middle Atlantic region,

the percentage rose from 27.4% to 52.8%, whereas in the rest of the United States, it increased from 32.5% to 62.4%. This could be due to aging-related mobility challenges, changing health conditions, or environmental factors that discourage active transportation.

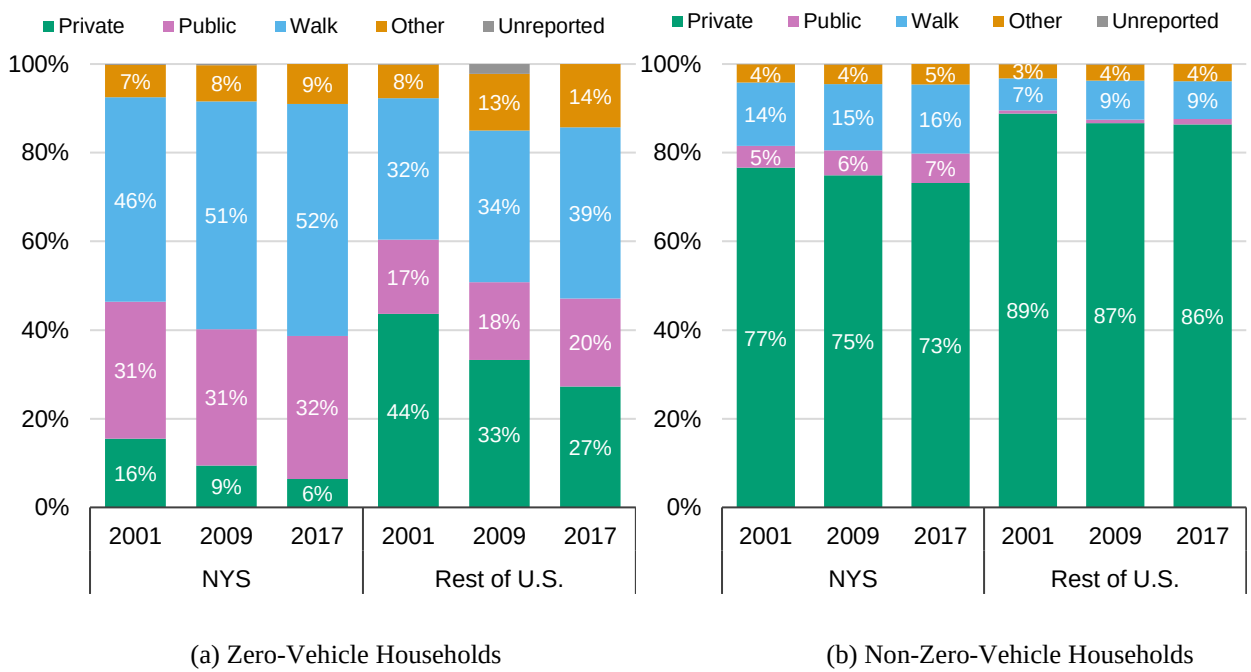


Figure 2-12. Mode shares by household vehicle ownership, year, and region. Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number. Unlabeled segments represent values lower than 3%.

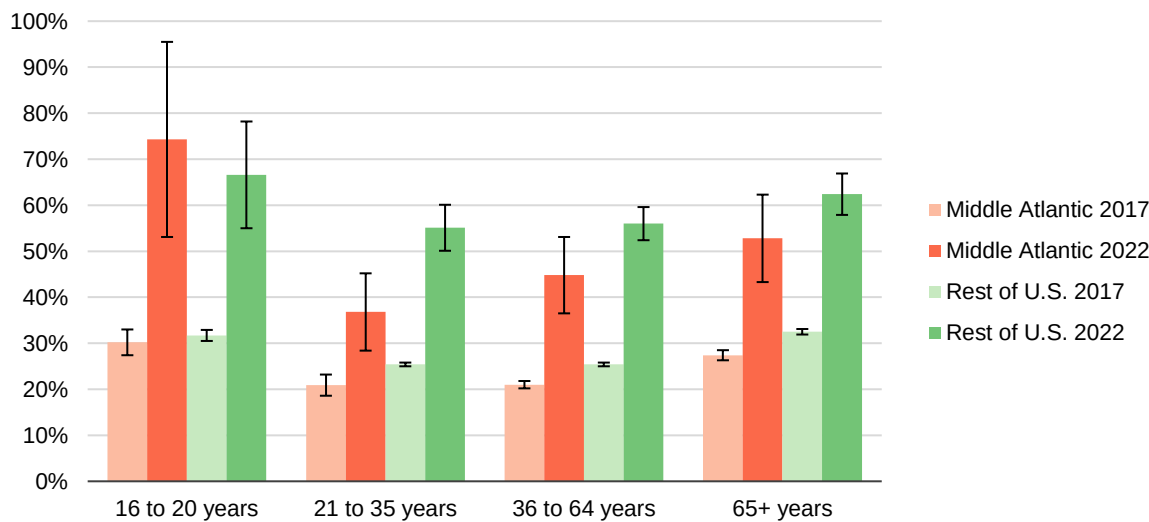


Figure 2-13. Percentages of people who reported no walk or bike trips in the last week by age, year, and region.

2.2.6 Characteristics of Zero-Vehicle Households

A higher share of zero-vehicle households in the rest of the United States had annual incomes below \$10,000 (Table 2-10). For example, in 2017, 25.2% of zero-vehicle households in the rest of the United States fell into this income category, which was statistically significantly higher than the share in NYS (11.5%). In NYS in 2017, the highest share of zero-vehicle households was in the \$10,000–\$14,999 income group, at 13.6%, with a relatively high share also in the less than \$10,000 group at 11.5%. From 2017 to 2022 in the Middle Atlantic Census Division (Figure 2-14), the share of zero-vehicle households slightly decreased, but this reduction was not statistically significant.

Table 2-10. Distribution of zero-vehicle households by household income in U.S.\$(2022), year, and region.

Household Income ^a	NYS			Rest of U.S.		
	2001	2009	2017	2001	2009	2017
Less than \$10,000	9.0%	11.8%	11.5%	21.8%	20.7%	25.2%
\$10,000 to \$14,999	10.8%	9.1%	13.6%	15.5%	19.0%	24.6%
\$15,000 to \$24,999	13.5%	14.3%	11.4%	18.5%	20.5%	17.1%
\$25,000 to \$34,999	10.5%	10.5%	7.5%	11.2%	10.2%	8.2%
\$35,000 to \$49,999	10.2%	8.8%	10.6%	8.4%	8.4%	6.1%
\$50,000 to \$74,999	10.7%	13.5%	9.7%	6.0%	5.7%	5.2%
\$75,000 to \$99,999	8.5%	6.7%	7.7%	2.0%	1.0%	3.1%
\$100,000 to \$124,999	2.8%	3.6%	7.2%	0.5%	0.8%	1.7%
\$125,000 to \$149,999	2.7%	3.6%	3.6%	0.0%	0.3%	1.1%
\$150,000 to \$199,999	2.9%	1.5%	4.2%	0.7%	1.3%	0.9%
\$200,000 or more	3.7%	6.1%	8.0%	1.0%	0.2%	1.5%
Unreported	14.6%	10.6%	4.9%	14.4%	11.9%	5.3%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 2-11 indicates that walking was the most common mode of transportation across all 3 years for New York residents living in zero-vehicle households. For instance, in 2017, 52.3% of trips among zero-vehicle households in NYS were walking, compared with 38.6% in the rest of the United States, a statistically significant difference. Regarding income, in 2017, households earning \$200,000 or more in the rest of the United States had the highest average person trips (4.5), whereas in NYS, the highest average (4.4) was among households earning \$125,000 to \$149,999.

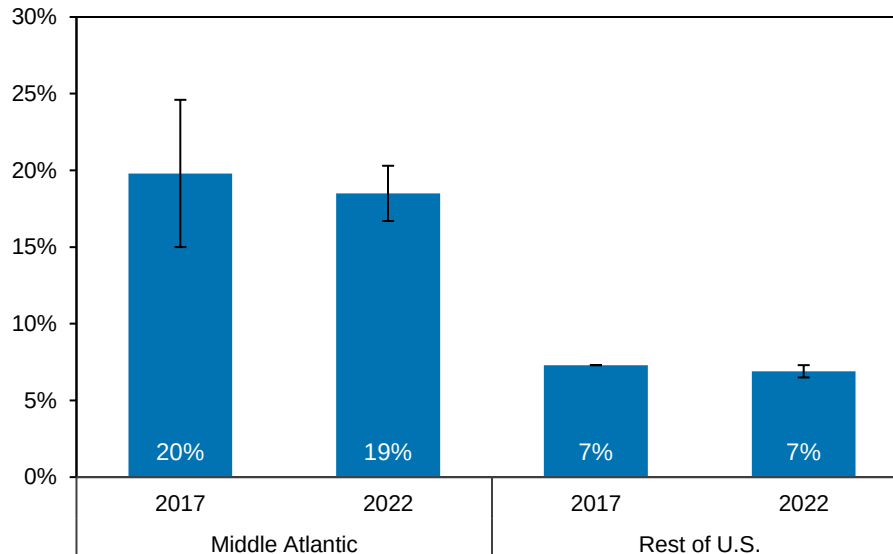


Figure 2-14. Share of zero-vehicle households by year and region.

Table 2-11. Distribution of person trips by travel mode and household income for zero-vehicle households by year and region.

	2001		2009		2017	
	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.
Person Trips per Person (%)	3.21	2.63	3.09	2.61	3.07	2.61
Private Vehicle Trips	15.5%	43.6%	9.5%	33.2%	6.4%	27.3%
Public Transit Trips	30.8%	16.7%	30.8%	17.6%	32.3%	19.8%
Walk Trips	46.1%	31.9%	51.4%	34.2%	52.3%	38.6%
Other Trips	7.3%	7.6%	8.2%	12.7%	9.0%	14.3%
Unreported	0.2%	0.1%	0.2%	2.2%	0.0%	0.0%
Household Income						
Less than \$10,000	2.42	2.28	2.31	2.49	2.52	2.47
\$10,000 to \$14,999	3.09	2.88	2.57	2.34	2.80	2.45
\$15,000 to \$24,999	3.02	2.27	3.11	2.60	2.91	2.65
\$25,000 to \$34,999	3.21	3.16	2.95	2.99	2.65	2.96
\$35,000 to \$49,999	3.10	2.69	3.17	3.20	3.23	2.81
\$50,000 to \$74,999	3.15	3.13	2.68	3.35	3.01	3.01
\$75,000 to \$99,999	3.98	3.73	4.24	3.31	3.35	2.53
\$100,000 to \$124,999	3.63	4.23	3.23	2.88	3.23	3.26
\$125,000 to \$149,999	3.95	1.93	4.33	3.85	4.42	2.34
\$150,000 to \$199,999	4.98	3.64	3.60	3.68	3.69	3.40
\$200,000 or more	4.29	4.34	5.07	3.88	4.05	4.50
Unreported	2.42	2.07	2.04	1.68	2.45	1.70

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

When comparing temporal differences, the relative change between 2009 and 2017 was smaller than the change from 2001 to 2009 in both NYS and the rest of the United States (Table 2-12). For example, person trips in NYS decreased by 0.6% from 2009 to 2017, compared with a larger decrease of 3.7% from 2001 to 2009. In the rest of the United States, the largest reduction in trips—39.2%—occurred among households earning \$125,000 to \$149,999, whereas the largest increase was among those earning \$200,000 or more, with a 16% rise. In contrast, NYS saw a 20.1% decrease in trips among households earning \$200,000 or more. Additionally, households earning \$50,000 to \$74,999 in NYS experienced a 12.3% increase in trips, whereas the same income group in the rest of the United States saw a 10.1% decrease.

Table 2-12. Temporal differences in person trips by household income for zero-vehicle households by period and region.

	2001 vs. 2009		2009 vs. 2017	
	% Difference in NYS	% Difference in Rest of U.S.	% Difference in NYS	% Difference in Rest of U.S.
Person Trips per Person	-3.7%	-0.8%	-0.6%	0.0%
Household Income				
Less than \$10,000	-4.5%	9.2%	9.1%	-0.8%
\$10,000 to \$14,999	-16.8%	-18.8%	8.9%	4.7%
\$15,000 to \$24,999	3.0%	14.5%	-6.4%	1.9%
\$25,000 to \$34,999	-8.1%	-5.4%	-10.2%	-1.0%
\$35,000 to \$49,999	2.3%	19.0%	1.9%	-12.2%
\$50,000 to \$74,999	-14.9%	7.0%	12.3%	-10.1%
\$75,000 to \$99,999	6.5%	-11.3%	-21.0%	-23.6%
\$100,000 to \$124,999	-11.0%	-31.9%	0.0%	13.2%
\$125,000 to \$149,999	9.6%	99.5%	2.1%	-39.2%
\$150,000 to \$199,999	-27.7%	1.1%	2.5%	-7.6%
\$200,000 or more	18.2%	-10.6%	-20.1%	16.0%

In terms of mode share (Figure 2-15), a significantly higher percentage of person trips were made by private vehicle in the rest of the United States (27%) compared with those in NYS (6%). This gap narrowed in 2017, with the rest of the United States at 21%, down from 24% in 2009. Although the rest of the United States consistently had a lower share of walking trips than that of NYS, the share of walking trips in the rest of the United States increased significantly over time, from 32% in 2001 to 34% in 2009 and reaching 39% in 2017.

2.3 VEHICLE TRAVEL COMPARISONS

In NYS, drivers had fewer average daily vehicle trips in 2017 than in either 2001 or 2009, as shown in Table 2-13. The share of social and recreational vehicle trips increased significantly from 20.8% in 2009 to 23.1% in 2017. Conversely, the share of family and personal business trips decreased significantly from 49% in 2009 to 44.7% in 2017.

In terms of age, the group with the highest number of vehicle trips in NYS in 2017 was those ages 55-to-64 (2.4 trips), whereas in the rest of the United States, it was the 35-to-54 age group (3.1 trips). Across both regions in 2001 and 2009, the 35-to-54 age group consistently logged the most vehicle trips. However, in NYS, vehicle trips among the 35-to-54 age group decreased significantly from 2.6 in 2009 to 2.3 in 2017. Regarding gender, males had statistically significantly higher vehicle trips than those of females in all 3 years across both regions.

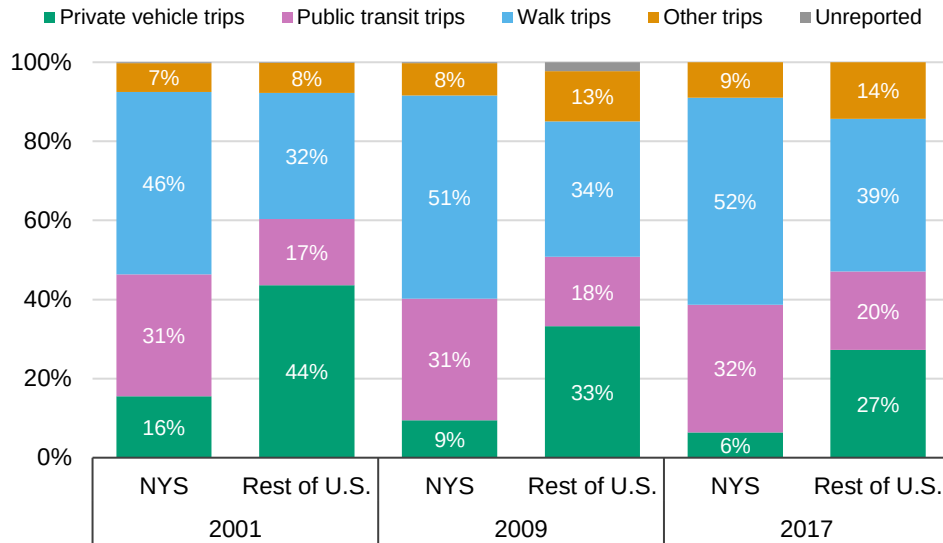


Figure 2-15. Distribution of modal shares for residents in zero-vehicle households by year and region.
Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number.
Unlabeled segments represent values lower than 6%.

Table 2-13. Distribution of vehicle trips per driver by trip purpose, age, gender, year, and region.

	2001		2009		2017	
	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.
Vehicle Trips per Driver	2.67	3.4	2.32	3.06	2.16	2.74
Trip Purpose (%) ^a						
Earning a Living	25.5%	26.2%	24.7%	26.1%	24.7%	26.2%
Family & Personal Business	49.0%	48.0%	49.0%	46.7%	44.7%	43.2%
Civic, Educational, & Religious	4.1%	4.9%	4.5%	4.9%	4.1%	5.3%
Social/Recreational	20.6%	20.3%	20.8%	21.1%	23.1%	21.9%
Other	0.6%	0.5%	0.3%	0.3%	3.4%	3.4%
Unreported	0.2%	0.1%	0.9%	0.9%	0.0%	0.0%
Age						
16-to-19 years	2.26	2.91	1.57	2.29	1.39	1.83
20-to-34 years	2.22	3.36	1.95	2.95	1.73	2.52
35-to-54 years	2.98	3.77	2.62	3.37	2.34	3.06
55-to-64 years	2.79	3.28	2.50	3.14	2.37	2.87
65+ years	2.65	2.85	2.25	2.70	2.32	2.57
Gender						
Male	2.79	3.61	2.45	3.14	2.26	2.81
Female	2.55	3.19	2.19	2.98	2.05	2.68

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

VMT in NYS slightly decreased from 19.7 in 2009 to 19.3 in 2017 (Table 2-14), although this change is not statistically significant. However, the share of commuting trip VMT decreased significantly from 34.1% in 2009 to 30.6% in 2017, with this reduction being greater than the change observed between 2001 and 2009.

The age group with the highest VMT across all 3 years in both regions is consistently those ages 35-to-54. In terms of gender, in 2017, males in NYS made an average of 2.3 vehicle trips, statistically significantly higher than that of females, who averaged 2.1 trips—a difference of 10.2%. Additionally, VMT for male drivers was 25.4, representing a 51.4% difference compared with female drivers, who had a VMT of 16.8. This indicates that not only do male drivers make more vehicle trips, but their trip lengths are also statistically significantly longer than those of female drivers.

Table 2-14. Distribution of vehicle miles of travel by trip purpose, age, gender, year, and region.

	2001		2009		2017	
	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.
Vehicle Miles of Travel per Driver	23.36	33.31	19.67	29.57	19.30	26.20
Trip Purpose (%) ^a						
Earning a Living	34.8%	35.5%	34.1%	34.8%	30.6%	33.5%
Family & Personal Business	34.8%	35.3%	34.0%	32.7%	34.4%	31.3%
Civic, Educational, & Religious	3.3%	3.8%	3.5%	4.5%	3.7%	5.5%
Social/Recreational	26.1%	24.4%	24.9%	24.4%	26.5%	24.2%
Other	0.8%	0.8%	0.4%	0.6%	4.8%	5.5%
Unreported	0.3%	0.2%	3.2%	3.2%	0.0%	0.1%
Age						
16-to-19 years	16.17	22.57	11.38	17.84	12.59	15.56
20-to-34 years	21.08	35.36	19.35	29.10	18.85	30.67
35-to-54 years	27.47	38.46	22.90	34.78	23.94	32.82
55-to-64 years	23.19	32.60	19.92	31.00	22.92	29.09
65+ years	17.82	21.34	15.05	19.97	19.27	22.68
Gender						
Male	28.58	41.63	23.44	35.76	25.42	33.21
Female	17.86	25.12	15.79	23.31	16.79	24.63

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 2-15 shows that weekend vehicle trips in NYS experienced a greater reduction (7.6%) compared with weekday trips (6.8%). VMT decreased by 2.3% in the rest of the United States but increased by 8.2% in NYS. The share of vehicle trips relative to person trips in NYS was 44.8% in 2009, declining by 6.5% to 41.9% in 2017. Similarly, the VMT/PMT share in NYS decreased from 53.6% in 2009 to 42.2% in 2017, representing a reduction of 21.3%.

Table 2-15. Temporal change from 2009 to 2017 in vehicle travel statistics by region.

	2009		2017		2009 vs. 2017 (% difference)	
	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.
Vehicle Trips per Driver	2.32	3.06	2.16	2.74	-6.9%	-10.5%
Weekday Vehicle Trips per Driver	2.36	3.15	2.20	2.82	-6.8%	-10.5%
Weekend Vehicle Trips per Driver	2.25	2.87	2.08	2.56	-7.6%	-10.8%
VMT per Driver	19.67	29.57	21.29	28.89	8.2%	-2.3%
Average Vehicle Trip Length	8.62	9.77	10.04	10.72	16.5%	9.7%
% Vehicle Trips/Person Trips	44.8%	60.6%	41.9%	60.5%	-6.5%	-0.2%
% VMT/PMT	53.6%	60.5%	42.2%	54.8%	-21.3%	-9.4%

Table 2-16 shows that the Middle Atlantic census division had statistically significantly lower VMT compared with the rest of the United States in both 2017 and 2022. The most substantial reduction in VMT in the Middle Atlantic census division was seen in the age group over 65, with a decrease of 44.7%. In contrast, the largest reduction in the rest of the United States was among the 20-to-34 age group, which experienced a decline of 32.1%. Additionally, male drivers in the Middle Atlantic census division exhibited a greater reduction in VMT compared with their counterparts in the rest of the United States, whereas this trend was not observed among female drivers.

2.3.1 Types and Ages of Vehicles in Use

The private vehicle market in the United States encompasses a diverse range of makes, models, and types. A comparison of major vehicle types in NYS and the rest of the country reveals notable differences (Table 2-16). For instance, the percentage of pickup trucks in the NYS fleet was 8% in 2017, compared with 16.3% in the rest of the nation. This gap narrowed slightly since 2009; between 2009 and 2017, the share of pickup trucks in the rest of the United States decreased significantly from 18.3% to 16.3%, whereas the share in NYS remained relatively stable during this period.

Table 2-16. Vehicle miles of travel per driver by age, gender, year, and region.

	2017		2022		2017 vs. 2022 (% difference)	
	Middle Atlantic	Rest of U.S.	Middle Atlantic	Rest of U.S.	Middle Atlantic	Rest of U.S.
VMT per Driver	24.18	29.08	16.67	22.52	-31.1%	-22.6%
Age						
16–19 years	13.12	15.71	9.68	11.61	-26.2%	-26.1%
20–34 years	24.22	30.82	16.26	20.94	-32.9%	-32.1%
35–54 years	26.00	33.18	18.79	26.90	-27.7%	-18.9%
55–64 years	26.57	29.09	21.11	24.48	-20.5%	-15.8%
65+ years	20.82	22.74	11.52	18.05	-44.7%	-20.6%
Gender						
Male	28.85	33.33	19.35	27.18	-32.9%	-18.5%
Female	19.35	24.88	13.86	17.91	-28.4%	-28.0%

Table 2-17. Distribution^a of household vehicle types by year and region.

	2001		2009		2017	
	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.
Auto	66.0%	56.4%	58.0%	49.7%	53.3%	49.4%
Van	9.9%	9.0%	9.1%	8.2%	6.5%	6.1%
Sport Utility	12.0%	12.1%	21.5%	19.4%	28.8%	23.4%
Pickup	9.3%	19.0%	8.6%	18.3%	8.0%	16.3%
Other Truck	0.3%	0.5%	0.2%	0.4%	0.4%	0.5%
RV	0.4%	0.7%	0.3%	0.5%	0.4%	0.6%
Motorcycle	2.1%	2.3%	2.4%	3.4%	2.4%	3.3%
Other POV	0.0%	0.0%	0.0%	0.0%	0.2%	0.4%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 2-18 reveals a statistically significant increase in the average vehicle age in NYS from 7.8 years in 2009 to 8.5 years in 2017. A similar trend occurred in the rest of the United States, where the average vehicle age rose significantly from 9.5 to 10.5 years during the same period. The median vehicle age followed this trend, as well. Additionally, the share of VMT by vehicles less than 10 years old in NYS decreased from 74.8%, whereas the share for vehicles older than 10 years increased from 25.2% to 31.6%.

The type of vehicle driven also indicates varying levels of utilization, as measured by average annual miles traveled. On average, automobiles are driven significantly fewer miles per vehicle compared with vans or sport utility vehicles (SUVs). In NYS, automobiles are also driven less than pickup trucks are driven (Figure 2-16).

Table 2-18. Characteristics of households' private vehicle fleets by year and region.

Statistics by Vehicle Age	NYS			Rest of U.S.		
	2001	2009	2017	2001	2009	2017
Statistics of Vehicle Age						
Average Vehicle Age	7.85	7.79	8.48	8.92	9.47	10.49
Median Vehicle Age	7.00	7.00	7.00	7.00	8.00	9.00
Percent VMT						
Less than 10 years	73.2%	74.8%	68.4%	71.0%	67.8%	58.6%
10 years or more	26.8%	25.2%	31.6%	29.0%	32.2%	41.4%
Percent Vehicle Fleet ^a						
Less than 10 years	66.5%	69.3%	61.9%	62.6%	60.7%	50.6%
10 years or more	33.5%	30.7%	38.1%	37.5%	39.3%	49.4%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

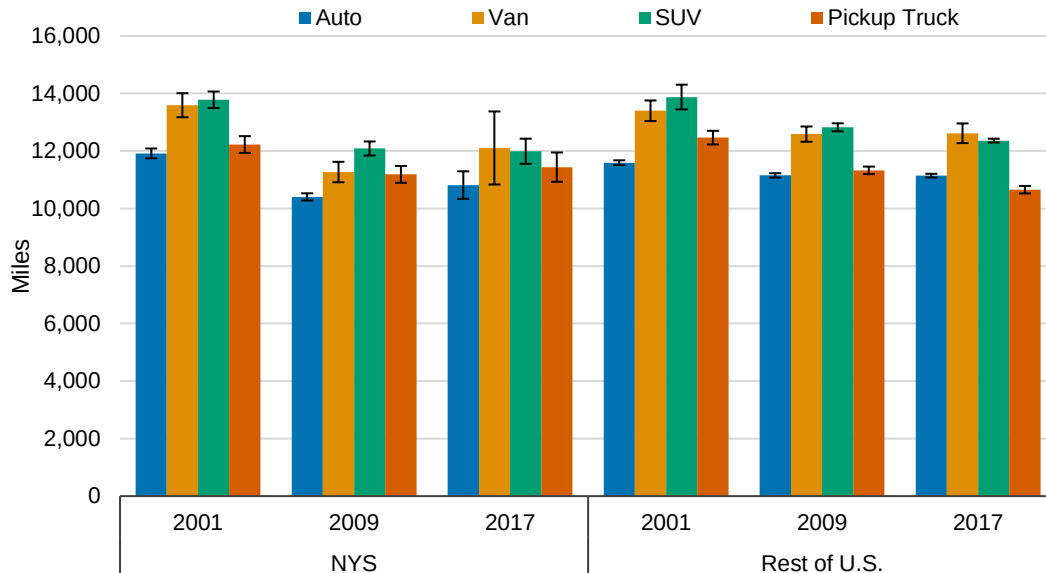


Figure 2-16. Average annual miles driven per household vehicle by vehicle type, year, and region.

Table 2-19 indicates significant reductions in annual vehicle utilization, measured in miles driven, for all four major types of private vehicles (automobiles, vans, SUVs, and pickup trucks) from 2001 to 2009. However, from 2009 to 2017, automobiles in NYS showed a statistically significant increase in utilization across all vehicle age groups. This trend was not observed in the rest of the United States.

Table 2-19. Average annual miles driven per household by vehicle type, vehicle age, year, and region.

Vehicle Type by Age	NYS			Rest of U.S.		
	2001	2009	2017	2001	2009	2017
Autos						
0 to 2 years	13,137	11,914	12,433	14,054	14,230	13,646
3 to 5 years	13,805	10,970	11,573	13,471	12,465	12,684
6 to 9 years	11,975	10,641	11,334	12,028	11,336	12,030
10 years or more	10,115	9,059	9,428	9,433	9,477	9,446
Unreported	12,081	10,503	11,477	11,292	11,425	10,406
SUVs						
0 to 2 years	13,400	13,230	12,680	14,968	15,038	14,238
3 to 5 years	14,336	12,609	13,244	15,235	13,672	13,205
6 to 9 years	15,471	11,741	12,820	14,308	12,860	12,647
10 years or more	11,496	10,286	9,596	10,687	10,184	10,827
Unreported	15,606	14,283	9,167	16,552	15,068	16,508
Vans						
0 to 2 years	15,273	13,652	12,600	15,862	15,495	18,935
3 to 5 years	14,597	13,154	13,701	15,526	14,105	14,066
6 to 9 years	13,559	12,202	12,793	13,597	13,389	13,413
10 years or more	10,759	8,599	11,124	10,274	10,119	10,941
Unreported	18,916	11,335	6,361	11,547	14,513	11,964
Pickup Trucks						

0 to 2 years	15,606	14,283	16,937	16,552	15,068	15,493
3 to 5 years	13,185	12,909	13,003	15,420	13,233	13,728
6 to 9 years	14,536	11,761	11,594	12,888	12,016	12,638
10 years or more	9,258	8,698	8,833	9,536	9,544	8,961
Unreported	9,057	10,959	14,150	11,879	11,759	11,435

2.4 COMPARISONS OF COMMUTING PATTERNS

Table 2-20 presents the distribution of commute person trips per worker and PMT per worker by transportation mode. In NYS, a statistically significantly higher share of commute trips was made by walking compared with the rest of the United States in all 3 years. A similar trend was observed for subway or elevated rail usage. In both regions, SOVs were the dominant mode for commute person trips. In 2017, 47% of commute person trips in NYS were by SOV, which was significantly higher than the second highest mode of subway or elevated rail (18.7%). In contrast, the second highest mode in the rest of the United States was MOV, which accounted for 17.9% of commute trips.

Regarding PMT, the average commute PMT per worker was statistically significantly lower in NYS than in the rest of the United States across all 3 years. In NYS, the average commute PMT in 2017 was 10.58 miles. The mode share in NYS remained dominated by SOV and subway or elevated rail, whereas in the rest of the United States, the mode share was dominated by SOV and MOV.

Table 2-20. Distribution of commute person trips and commuter PMT by transportation mode, year, and region.

Commute Trip Statistics	2001		2009		2017	
	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.
Person Trips per Worker (%) ^a	1.12	1.15	1.06	1.11	1.10	1.13
SOV	55.3%	77.7%	56.0%	78.4%	47.0%	72.1%
MOV	12.6%	16.4%	11.5%	14.6%	13.0%	17.9%
Amtrak/Commuter Rail	2.6%	0.4%	2.7%	0.4%	2.8%	0.5%
Subway/Elevated Rail	12.8%	0.5%	10.6%	0.3%	18.7%	1.4%
Other Public	5.5%	1.6%	6.6%	1.7%	5.3%	2.2%
Walk	8.7%	2.4%	8.5%	2.7%	9.7%	3.5%
Other	2.4%	0.9%	3.9%	1.8%	3.6%	2.3%
Unreported	0.1%	0.1%	0.2%	0.2%	0.0%	0.0%
PMT per Worker (%) ^a	10.79	13.78	10.15	12.99	10.58	13.06
SOV	64.7%	77.7%	67.5%	81.7%	54.1%	75.5%
MOV	12.8%	16.1%	13.3%	13.5%	13.5%	16.9%
Amtrak/Commuter Rail	5.5%	1.0%	3.7%	0.5%	7.5%	1.2%
Subway/Elevated Rail	10.9%	0.5%	7.3%	0.2%	18.4%	1.3%
Other Public	2.7%	1.0%	3.8%	1.1%	3.9%	2.0%
Walk	1.1%	0.2%	1.3%	0.2%	0.6%	0.4%
Other	2.1%	3.5%	2.8%	2.7%	2.0%	2.6%
Unreported	0.2%	0.1%	0.2%	0.2%	0.0%	0.1%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

The average commute trip speed (Table 2-21) for POVs is more than double that of public transit in all 3 years analyzed. For instance, in 2017, the average commute trip speed for POVs in NYS was 24.7 mph. In comparison, the rest of the United States recorded a statistically significantly higher POV speed of 28.5 mph in the same year.

In terms of trip distance (Table 2-21), POVs in NYS had a comparable distance to public transit across all 3 years, a trend mirrored in the rest of the United States. Regarding commute travel time, the average travel time in 2017 for POVs was statistically significantly lower at 25.2 min compared with travel times for public transit, which averaged 56.5 min. This pattern of lower POV travel times compared with those of public transit was also observed in 2001 and 2009. However, NYS exhibited average commute travel times for both POV and public transit that were slightly higher than those for the rest of the United States.

Comparing temporal trends, Figure 2-17 indicates that the rest of the United States had a statistically significant higher average commute trip distance than that of NYS. According to Table 2-22, trip distances in NYS in 2009 decreased by 1.3% compared with 2001, but the reduction was more pronounced from 2009 to 2017, at 14.6%. POV commute trip distance in NYS declined by 14.8% from 2009 to 2017, following a slight increase of 2.7% from 2001 to 2009.

In contrast, public transit in NYS experienced an opposite trend—a decrease of 20.1% in trip distance from 2001 to 2009, followed by a 10.6% increase from 2009 to 2017. Regarding commute travel speed, public transit speeds in NYS decreased by 35.6% from 2009 to 2017, whereas the reduction was smaller at 15.5% in the rest of the United States. Conversely, POV commute speed increased in both regions, by 8.8% in NYS and 2.2% in the rest of the United States.

Table 2-21. Commute travel distances, times, and average speeds by year and region.

Commute Trip Statistics	2001		2009		2017	
	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.
Trip Distance (miles)	10.39	12.22	10.25	11.89	8.75	10.66
POV	11.31	12.13	11.61	12.11	9.89	10.93
Public Transit	11.32	11.94	9.04	10.71	10.00	11.80
Travel Time (minutes)	30.40	23.11	29.89	23.45	31.79	24.64
POV	23.87	22.42	24.49	22.77	25.17	23.80
Public Transit	57.79	54.18	51.92	53.54	56.51	54.73
Trip Speed (mph)	26.40	32.82	26.08	31.45	16.80	26.59
POV	28.49	32.51	28.45	31.92	24.73	28.54
Public Transit	11.75	13.22	10.45	12.00	10.89	13.09

Table 2-22. Temporal difference in commute travel distances, times, and average speeds by year and region.

Commute Trip Statistics	2001–2009 % difference		2009–2017 % difference	
	NYS	Rest of U.S.	NYS	Rest of U.S.
Trip Distance	–1.3%	–2.7%	–14.6%	–10.3%
POV	2.7%	–0.2%	–14.8%	–9.7%
Public Transit	–20.1%	–10.3%	10.6%	10.2%
Travel Time	–1.7%	–0.2%	–14.8%	–9.7%
POV	2.6%	–10.3%	10.6%	10.2%
Public Transit	–10.2%	1.5%	6.4%	5.1%

Trip Speed	-1.2%	1.6%	2.8%	4.5%
POV	-0.1%	-1.2%	8.8%	2.2%
Public Transit	-11.1%	-4.2%	-35.6%	-15.5%

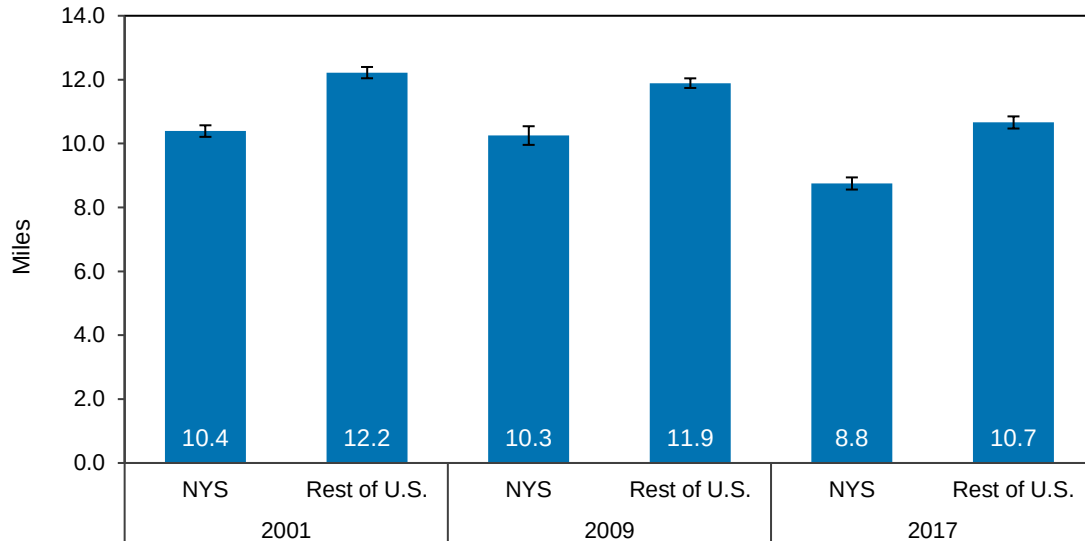
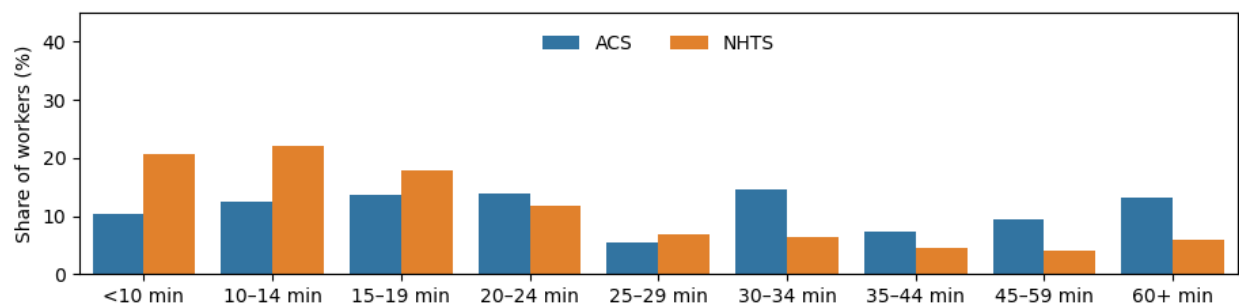


Figure 2-17. Temporal variations in average daily commute trip distances by year and region.

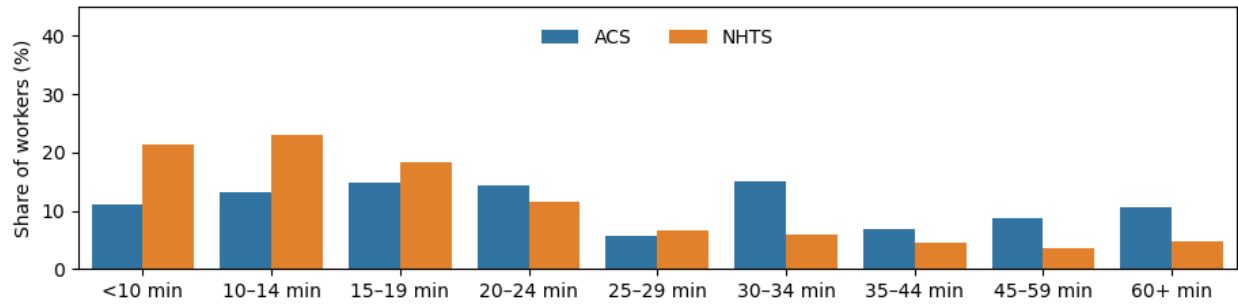
2.4.1 Comparison of ACS and NHTS

The American Community Survey (ACS) was compared with NHTS regarding commute travel time. Figure 2-18 illustrates the share of workers based on their commute travel time, using 2009 data for POV commute times. The ACS definition of public transport includes multiple modes of transportation, including buses, subways or elevated rail, long-distance trains, commuter rail, and ferryboats. The figure shows that a higher percentage of workers in NYS had a travel time of less than 20 min, according to the NHTS data.

Figure 2-19 compares public transport travel times, indicating that the NHTS reported a slightly lower share of workers in NYS than the ACS reported with travel times exceeding 60 min. A different trend was observed in the rest of the United States, where the NHTS showed a slightly higher share of workers experiencing over 60 min of public transit travel time compared with that reported by the ACS.

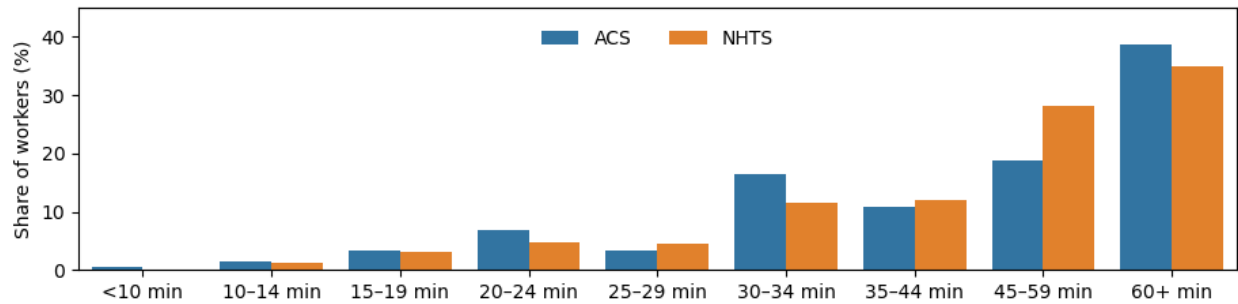


(a) POV in NYS

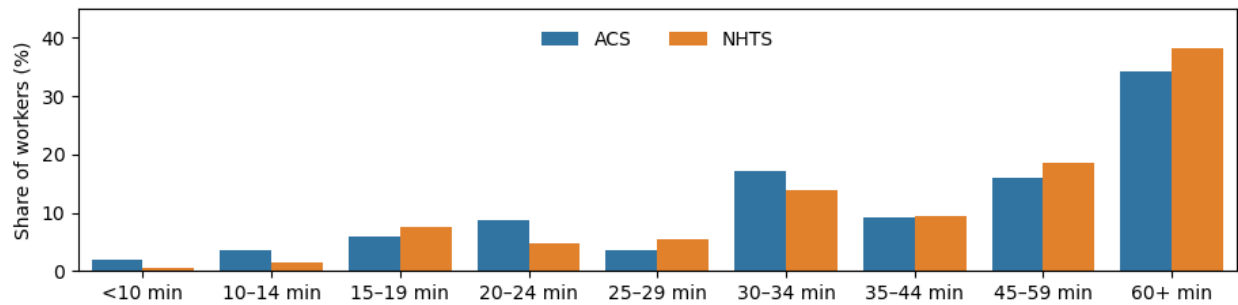


(b) POV in the rest of the United States

Figure 2-18. Commute travel time by POV in 2009, using data from ACS and NHTS.
 Note: The ACS data are based on a 5 year estimate of commute travel time from 2007 to 2011.



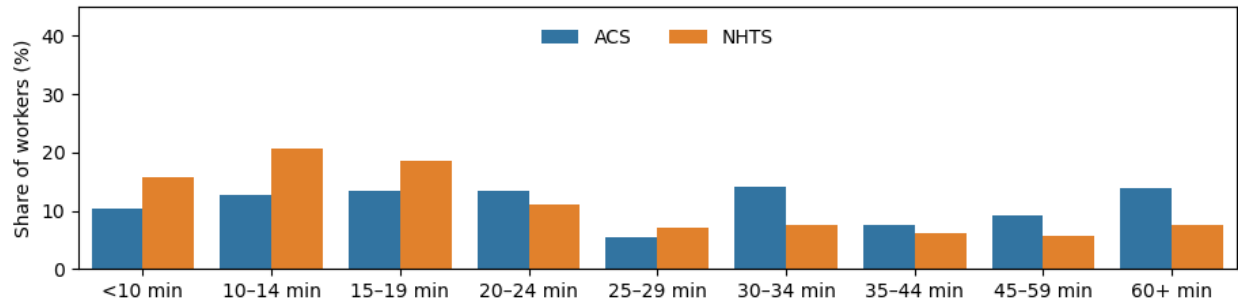
(a) Public transport in NYS



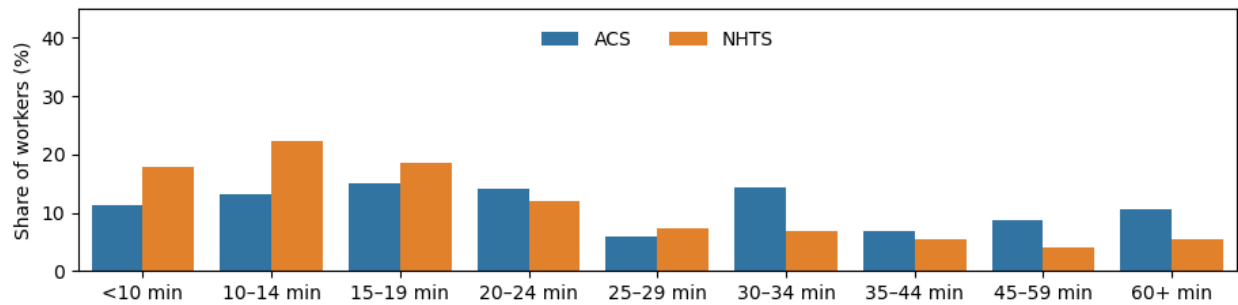
(b) Public transport in the rest of the United States

Figure 2-19. Commute travel time by public transport in 2009, using data from ACS and NHTS.
 Note: The ACS data are based on a 5 year estimate of commute travel time from 2007 to 2011.

The 5 year estimate of commute travel time from the 2017 ACS was compared with NHTS (Figure 2-20 and Figure 2-21). In NYS in 2017, both ACS and NHTS indicated a higher percentage of workers experiencing public transport travel times exceeding 60 min compared with reported times in 2009. This increase may reflect a range of factors, including heightened congestion in urban areas, reduced public transport efficiency, and potential service cuts or delays. Additionally, an increase in commuters relying on public transport owing to rising housing costs may have resulted in longer travel times as they traverse greater distances to reach transit hubs.

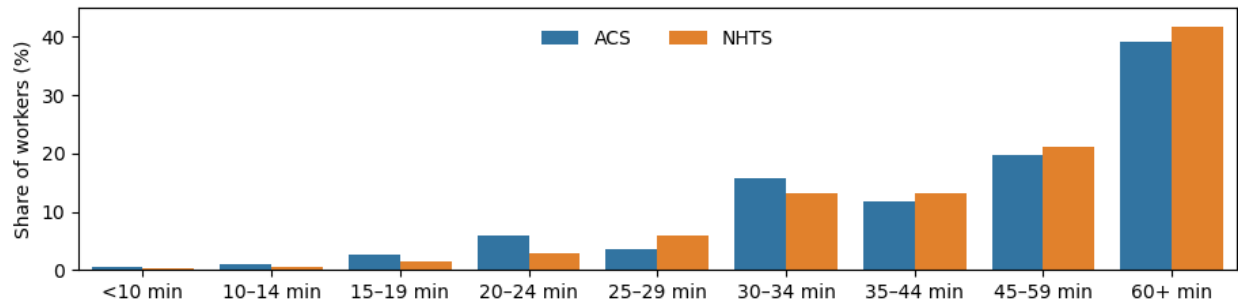


(a) POV in NYS

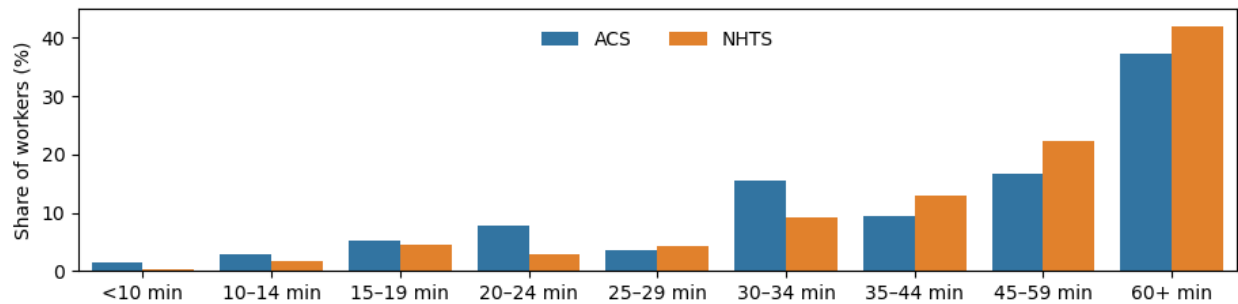


(b) POV in the rest of the United States

Figure 2-20. Commute travel time by POV in 2017, using data from ACS and NHTS.
 Note: The ACS data are based on a 5 year estimate of commute travel time from 2013 to 2017.



(a) Public transport in NYS



(b) Public transport in the rest of the United States

Figure 2-21. Commute travel time by public transport in 2017, using data from ACS and NHTS.
 Note: The ACS data are based on a 5 year estimate of commute travel time from 2013 to 2017.

3. INFLUENCE OF NEW YORK CITY ON NYS TRAVEL PATTERNS

This section presents an analysis that highlights significant differences in the travel behaviors of NYC residents compared with residents in the rest of NYS. Key aspects of this comparison include trip frequency, trip length, vehicle usage, and mode choice. The chapter specifically examines how travel habits in NYC contribute to these differences, emphasizing the unique characteristics of households within the city.

These temporal trends in household demographics and travel activities are explored using data from the 2001, 2009, and 2017 NHTS. The analysis focuses on three geographic regions:

- The NYC area (Figure 3-1), which includes the five boroughs: Bronx, Kings (Brooklyn), New York (Manhattan), Queens, and Richmond (Staten Island)
- The rest of NYS
- The remaining United States, excluding the NYS sample, with responses averaged across non-NYS households

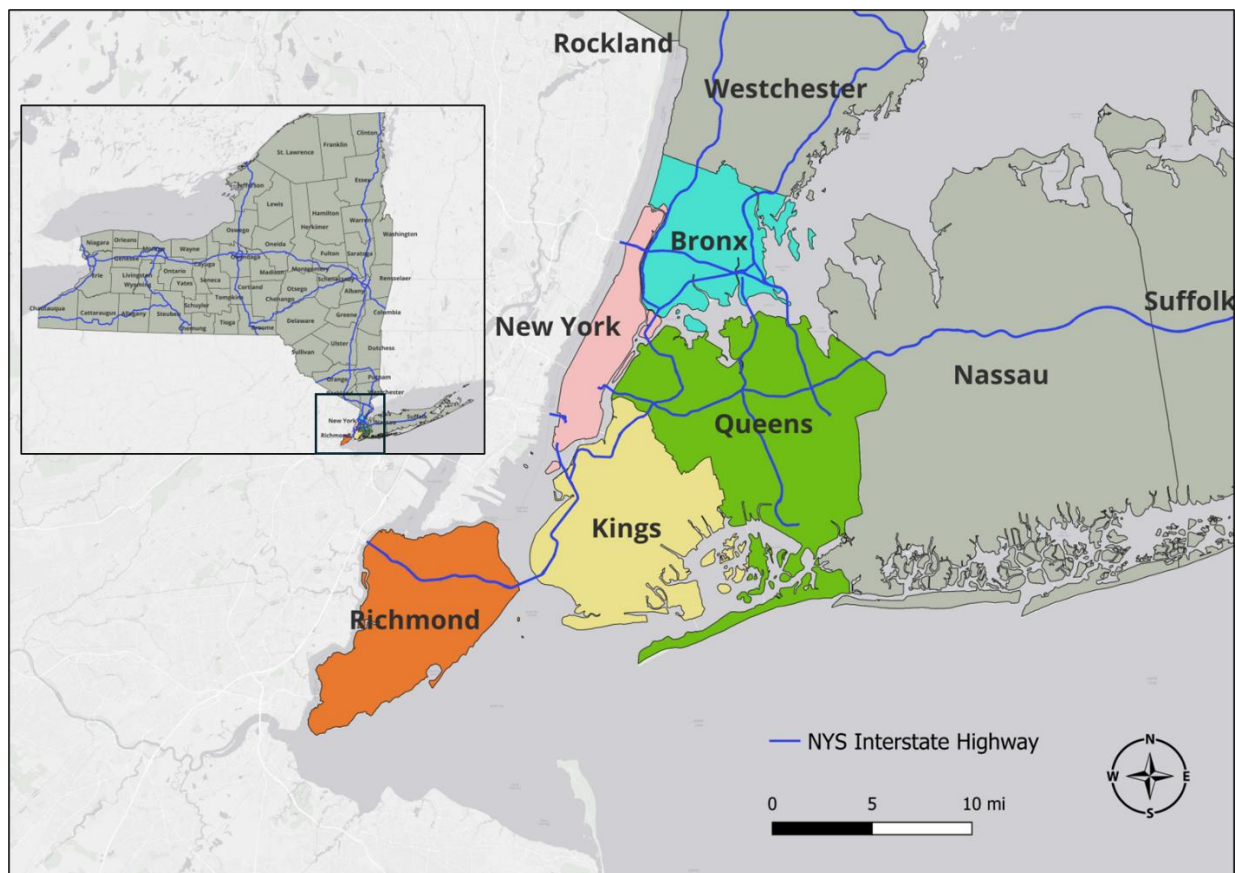


Figure 3-1. NYC's five boroughs, with an inset showing the larger map's location in NYS.

3.1 PROFILE OF NYC HOUSEHOLDS

According to Figure 3-2, NYC accounts for a percentage of total households and individuals over 5 years old comparable to that of the rest of NYS (43% vs. 57%). However, NYC has a statistically significantly lower share of vehicles (21%) compared with the rest of NYS (79%) and a significantly higher proportion of zero-vehicle households (80% in NYC vs. 20% in the rest of NYS). This disparity highlights NYC's reliance on public transit, walking, and cycling, supported by the city's dense infrastructure and extensive transit network that reduce the need for personal vehicles. When examining these figures over time in NYC and NYS (Figure 3-3), there are no notable changes.

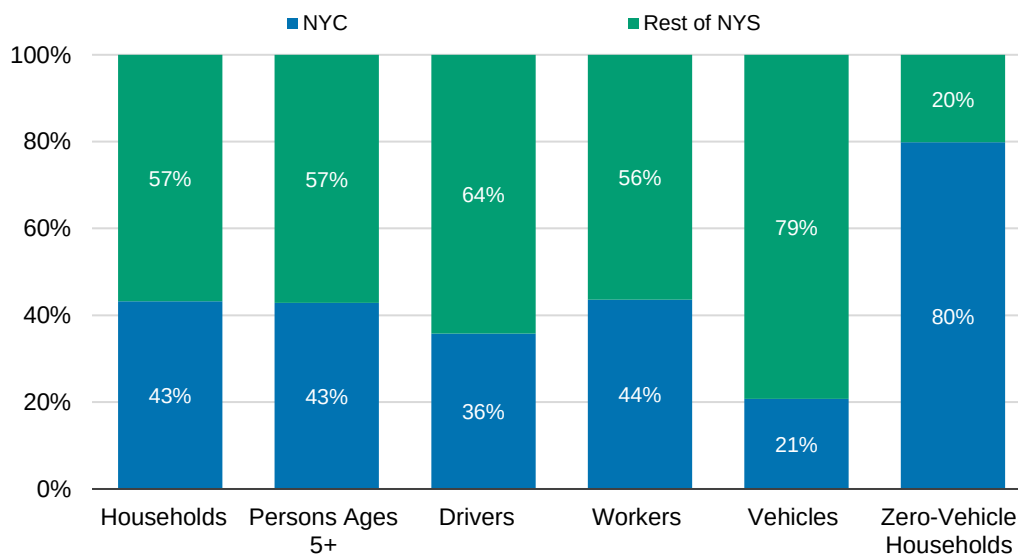
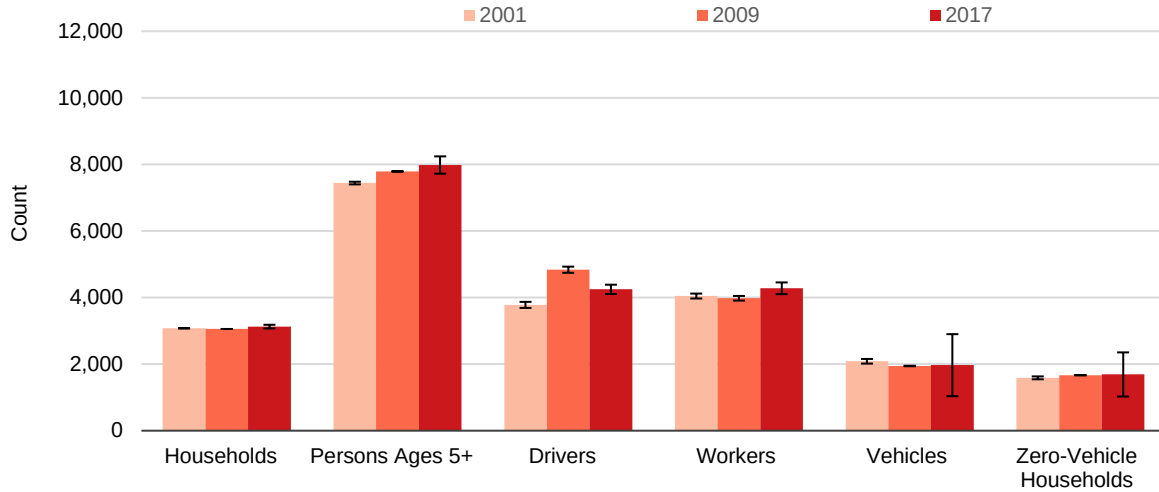


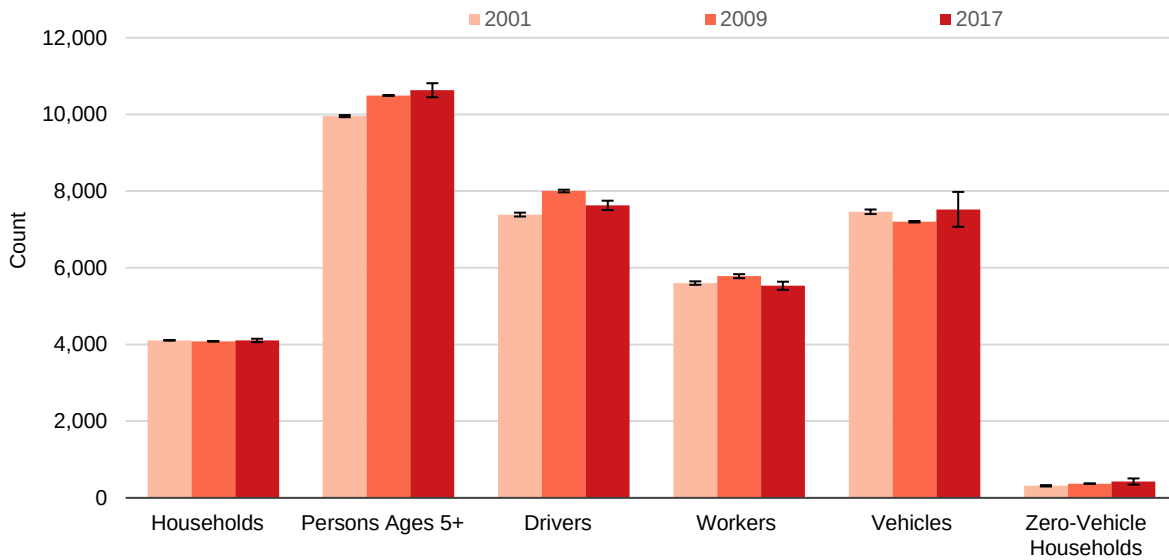
Figure 3-2. NYC and the rest of NYS traveler population statistics shown as percentages of NYS totals in 2017.

3.1.1 Household Incomes

A larger percentage of NYC households have income lower than \$10,000, compared with households in both the rest of NYS and the rest of the United States (Table 3-1). For instance, in 2017, 9.2% of NYC households have an annual income below \$10,000, which is statistically significantly higher than the 7.1% of households in the rest of NYS. The percentages for the two lowest income groups were slightly lower in the rest of NYS compared with the rest of the country, but by 2017, this gap narrowed. In 2017, 7.1% of households in the rest of NYS earned less than \$10,000, compared with 7.2% in the rest of the United States, with no statistically significant difference. At the higher end of the income scale, both NYC and the rest of the state have a greater percentage of households earning over \$200,000 annually compared with the national average.



(a) NYC



(b) Rest of NYS

Figure 3-3. Trends in traveler population (1,000) statistics by year.

Table 3-1. Distribution of households by income in U.S.\$(2016), year, and region.

Household Income ^a	NYC			Rest of NYS			Rest of U.S.		
	2001	2009	2017	2001	2009	2017	2001	2009	2017
Less than \$10,000	6.5%	8.1%	9.2%	3.5%	5.2%	7.1%	5.3%	6.6%	7.2%
\$10,000 to \$14,999	7.6%	5.3%	4.7%	4.3%	5.0%	6.0%	4.7%	5.5%	5.9%
\$15,000 to \$24,999	9.1%	11.1%	7.6%	8.4%	9.2%	8.4%	8.6%	10.4%	9.6%
\$25,000 to \$34,999	9.2%	8.4%	8.5%	8.1%	9.3%	8.2%	9.0%	10.6%	9.8%
\$35,000 to \$49,999	12.2%	12.8%	8.5%	13.3%	10.7%	9.6%	14.0%	11.6%	12.2%
\$50,000 to \$74,999	15.8%	14.4%	15.4%	17.5%	17.8%	14.7%	18.6%	17.4%	16.1%
\$75,000 to \$99,999	8.9%	9.3%	12.5%	12.1%	11.4%	12.9%	12.3%	11.8%	11.9%
\$100,000 to \$124,999	6.4%	7.2%	7.9%	9.3%	6.9%	9.8%	7.9%	7.1%	9.1%

\$125,000 to \$149,999	2.8%	2.3%	5.4%	5.0%	4.7%	5.9%	3.7%	3.9%	5.2%
\$150,000 to \$199,999	3.9%	5.8%	5.6%	4.0%	5.2%	6.1%	3.6%	4.0%	5.0%
\$200,000 or more	5.6%	5.8%	10.7%	5.5%	6.2%	7.5%	4.0%	4.3%	5.1%
Unreported	11.9%	9.4%	4.2%	9.1%	8.3%	3.8%	8.1%	6.8%	3.0%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

3.1.2 Vehicle Ownership

According to Table 3-2, the majority of individuals ages 16 and older held a driver's license across all years and regions, though the percentage was consistently lowest in NYC. In 2017, the licensing rate among men in NYC (70.3%) was statistically significantly higher than that among women (50%). This gender gap, though still statistically significant, was smaller in the rest of NYS and the rest of the United States. From 2001 to 2017, the licensing rate for both men and women decreased slightly in all regions.

Regarding vehicle ownership, the share of zero-vehicle households in NYC remained statistically significantly higher than that of the rest of NYS and the rest of the United States across all 3 years. For instance, in 2017, the share of households with no vehicles in NYC was 54%, compared with just 10.4% in the rest of NYS. As shown in Figure 3-4, the percentage of households with two or more vehicles in NYC was also statistically significantly lower than the percentages in both the rest of NYS and the rest of the United States throughout the 3 years.

Table 3-2. Vehicle ownership trends by year and region.

	NYC			Rest of NYS			Rest of U.S.		
	2001	2009	2017	2001	2009	2017	2001	2009	2017
Licensing Rate (%)									
Drivers/Persons 16+	61.9%	72.2%	59.5%	90.1%	89.1%	84.8%	90.5%	87.9%	87.9%
Female Drivers/Female 16+	49.7%	65.1%	50.0%	87.8%	86.5%	82.1%	87.7%	86.8%	86.4%
Male Drivers/Male 16+	71.5%	78.9%	70.3%	92.5%	91.7%	87.6%	93.3%	91.6%	89.4%
Vehicle Ownership (%) ^a									
0 Vehicle Households	51.6%	54.5%	54.0%	7.6%	9.1%	10.4%	6.8%	7.4%	7.6%
1 Vehicle Households	33.7%	31.4%	32.8%	32.8%	33.1%	33.4%	31.3%	32.3%	33.6%
2 Vehicle Households	11.3%	10.9%	10.6%	39.0%	39.2%	32.7%	37.9%	36.9%	33.7%
3 Vehicle Households	2.7%	2.2%	1.6%	14.3%	13.0%	14.2%	15.4%	14.8%	15.3%
4+ Vehicle Households	0.7%	0.9%	1.0%	6.3%	5.6%	9.3%	8.6%	8.6%	9.8%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

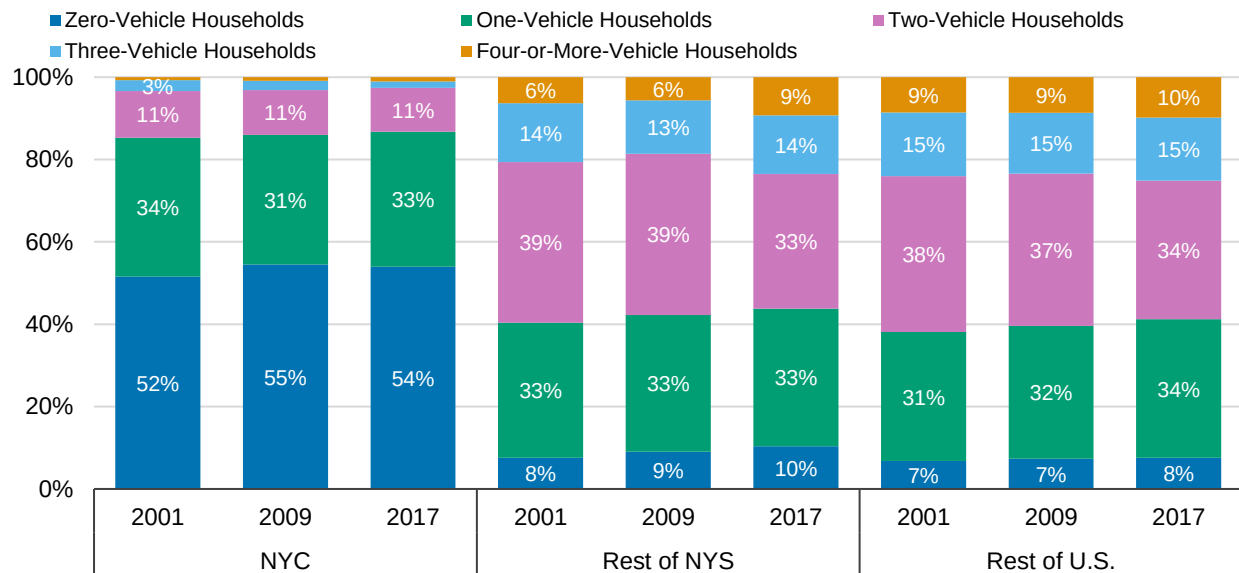


Figure 3-4. Distribution of households by number of vehicles by year and region. Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number. Unlabeled segments represent values lower than 3%.

3.2 PERSONAL TRAVEL COMPARISONS

3.2.1 Trip Frequencies and Trip Distances

Person trips decreased across all regions from 2001 to 2017 (Figure 3-5). In NYC, for example, overall person trips per person dropped statistically significantly from 3.46 in 2009 to 3.23 in 2017. In both 2001 and 2009, weekend trips were lower than weekday trips across all regions. However, by 2017, weekend trips slightly exceeded weekday trips, though this difference was not statistically significant.

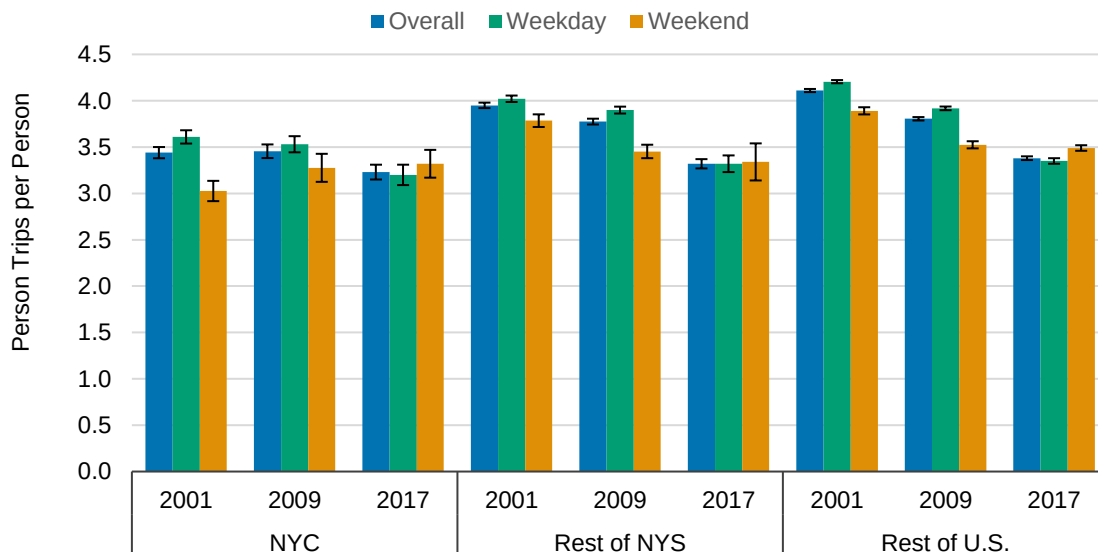


Figure 3-5. Person trip statistics by year and region.
Note: Weekend trips = Saturday and Sunday trips.

According to Figure 3-6, PMT in NYC increased statistically significantly from 16.1 in 2009 to 25.0 in 2017. A similar trend was observed in the rest of NYS and the rest of the United States. Regarding average person trip length, NYC experienced a statistically significant increase from 5.2 in 2009 to 7.8 in 2017, which was also reflected in the rest of NYS and the rest of the United States. However, in 2017, the average trip length in NYC (7.8) was statistically significantly lower than that in the rest of NYS (11.4).

3.2.2 Trip Purpose and Mode Choice

Regarding the share of person trips by mode of transportation, the percentage of trips made by SOVs declined across all regions over the years. For instance, in NYC, the share of trips by SOVs was 9.7% in 2017, which was statistically significantly lower than the share in 2009 (Table 3-3). As shown in Figure 3-7, walking accounted for the highest share of person trips in NYC in 2017, representing 40.3% of total trips. In contrast, most trips in the rest of NYS and the rest of the United States were made by SOVs and MOVs. When examining the share of person trips by trip purpose, the trip distribution was similar across regions over the 3 years.

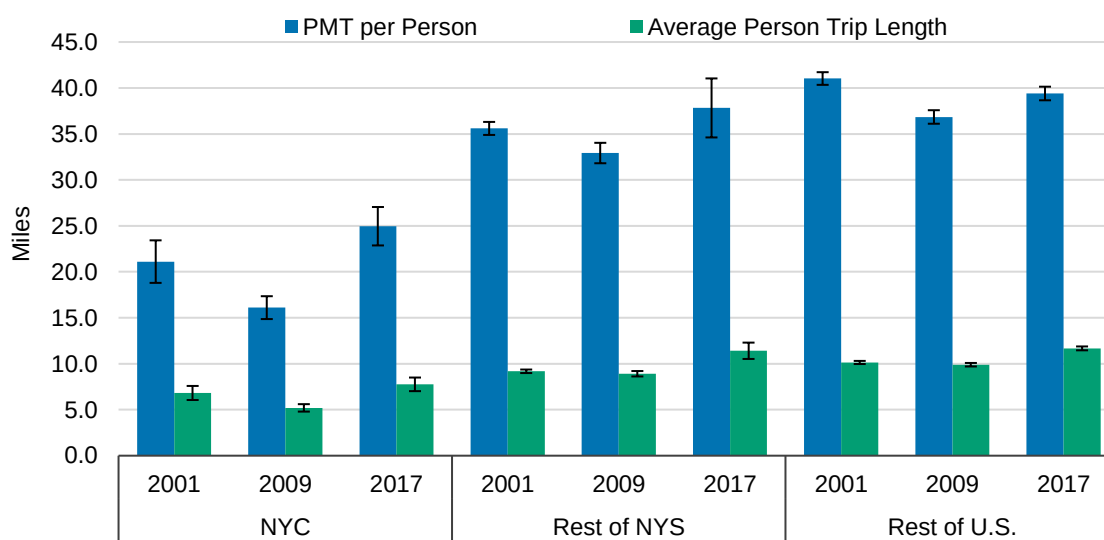


Figure 3-6. PMT statistics by year and region.

Table 3-3. Person trip by mode and trip purpose by year and region.

	NYC			Rest of NYS			Rest of U.S.		
	2001	2009	2017	2001	2009	2017	2001	2009	2017
Mode (%) ^a									
SOV	14.1%	12.0%	9.7%	40.8%	43.1%	37.9%	40.3%	40.5%	38.7%
MOV	21.9%	20.6%	19.0%	44.1%	39.2%	42.9%	47.3%	44.4%	45.4%
Amtrak	0.4%	0.4%	0.0%	0.1%	0.0%	0.0%	0.0%	0.0%	0.0%
Commuter Train	0.4%	1.6%	1.0%	0.4%	0.4%	0.9%	0.1%	0.1%	0.2%
Subway	13.0%	9.9%	17.3%	0.3%	0.2%	0.5%	0.2%	0.1%	0.5%
Other Public Transit	8.7%	10.5%	6.9%	0.6%	1.0%	1.3%	0.8%	1.0%	1.3%
Walk	36.8%	38.7%	40.3%	9.1%	10.6%	11.2%	7.9%	9.7%	9.7%
Other	4.6%	6.1%	5.9%	4.6%	5.2%	5.2%	3.3%	4.1%	4.3%
Unreported	0.1%	0.1%	0.0%	0.1%	0.2%	0.0%	0.1%	0.2%	0.0%
Purpose (%) ^a									

Earn a Living	20.1%	16.6%	20.1%	18.8%	19.5%	18.6%	18.8%	18.7%	19.0%
Family & Personal	42.7%	44.1%	37.7%	43.7%	42.7%	39.2%	43.9%	42.4%	38.5%
Civic, Educational, & Religious	11.1%	9.6%	9.0%	8.8%	9.4%	10.3%	9.8%	9.6%	10.9%
Social & Recreational	24.6%	27.3%	29.4%	27.3%	26.7%	27.9%	26.6%	27.5%	27.5%
Other	1.3%	1.3%	3.8%	1.1%	0.9%	4.0%	0.8%	0.8%	4.0%
Unreported	0.3%	1.0%	0.0%	0.3%	0.9%	0.0%	0.1%	1.0%	0.0%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

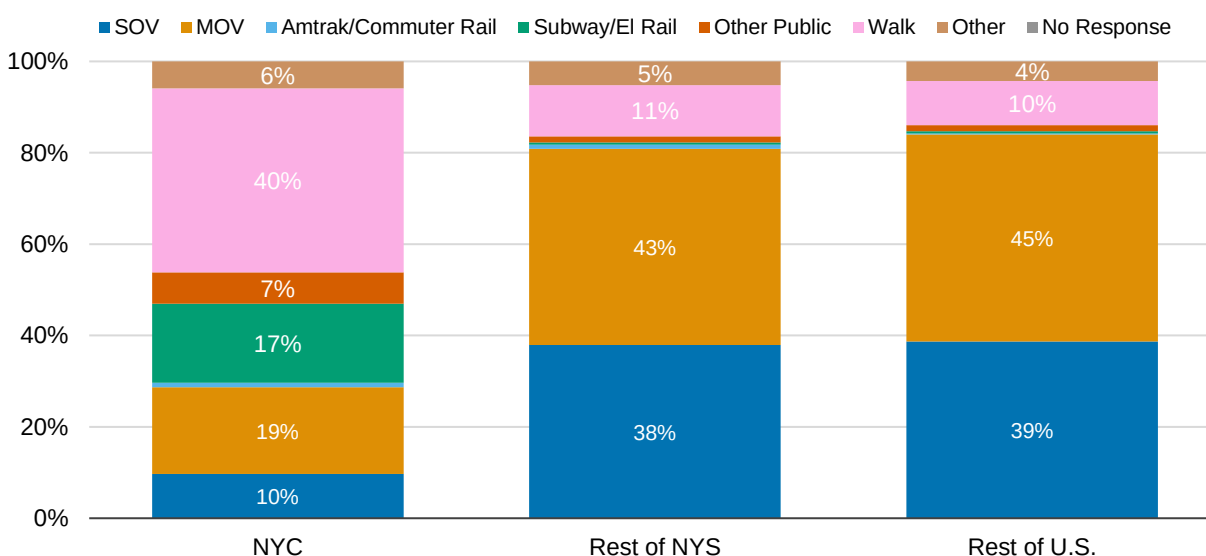


Figure 3-7. Person trip percentages by selected modes and by region in 2017. Notes: El Rail = elevated rail. Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number. Unlabeled segments represent values lower than 4%.

The pattern for PMT mirrored that of person trips, with the exception for walking modes (Table 3-4). In NYC, the share of PMT attributed to walking was 3.4% in 2017, whereas the majority of PMT was accounted for by SOV, MOV, and subway travel. A statistically significant higher share of PMT was observed for subway travel in NYC compared with the rest of NYS and the rest of the United States. Specifically, in 2017, the share of PMT by subway in NYC was 18.5%, whereas it was only 0.9% in the rest of NYS. Additionally, compared with 2009, the share of PMT by subway increased statistically significantly from 13.3% to 18.5% in 2017.

Regarding trip purposes, the share of PMT for family and personal business significantly decreased from 27.5% in 2009 to 19.9% in 2017 in NYC. However, a similar significant decrease was not observed in the rest of NYS and the rest of the United States. Furthermore, a statistically significant increase in PMT using other modes in 2017 was observed across all three regions when compared with 2001 and 2009.

3.2.3 Differences in Travel by Gender and Age

According to Table 3-5, the share of individuals who did not take walking trips in the week before to the survey was statistically significantly higher in the rest of NYS and the rest of the United States compared with NYC across all age groups. This contrast highlights NYC's unique urban environment, where walking is often a practical and necessary mode of transportation owing to the dense city layout, pedestrian infrastructure, and limited parking.

In NYC in 2017, the 65 and older group had the highest share of individuals who did not take walking trips (17%), whereas the lowest share was among those ages 21–35 (5.8%). Additionally, this younger age group experienced a statistically significant decrease from 12.5% in 2009. This shift might reflect a growing cultural emphasis on fitness and sustainability or increased awareness of the health benefits of walking among younger individuals.

Table 3-4. PMT by mode and trip purpose by year and region.

	NYC			Rest of NYS			Rest of U.S.		
	2001	2009	2017	2001	2009	2017	2001	2009	2017
Mode (%) ^a									
SOV	16.5%	16.5%	10.2%	38.6%	41.0%	32.5%	37.6%	38.9%	32.8%
MOV	28.2%	43.3%	27.1%	49.5%	44.9%	42.5%	51.3%	49.9%	46.2%
Amtrak	1.0%	0.1%	0.0%	0.4%	0.2%	0.0%	0.1%	0.1%	0.0%
Commuter Train	0.6%	2.4%	3.7%	1.3%	1.3%	2.5%	0.2%	0.3%	0.7%
Subway	14.2%	13.3%	18.5%	0.4%	0.2%	0.9%	0.1%	0.1%	0.4%
Other Public Transit	5.4%	8.3%	3.9%	0.4%	0.6%	0.6%	0.5%	0.6%	0.8%
Walk	3.9%	6.5%	3.4%	0.8%	0.8%	2.0%	0.6%	0.7%	0.7%
Other	29.1%	9.6%	33.2%	8.1%	10.8%	18.9%	9.3%	9.4%	18.4%
Unreported	0.9%	0.1%	0.0%	0.4%	0.2%	0.0%	0.5%	0.1%	0.0%
Purpose (%) ^a									
Earn a Living	34.2%	29.3%	20.0%	25.9%	26.5%	23.6%	27.5%	25.2%	22.2%
Family & Personal	23.6%	27.5%	19.9%	32.9%	30.0%	27.9%	32.9%	29.6%	26.3%
Civic, Educational, & Religious	5.3%	5.5%	4.5%	5.1%	5.7%	5.7%	5.9%	6.2%	6.7%
Social & Recreational	24.3%	34.2%	23.5%	33.3%	28.0%	28.9%	30.0%	30.3%	27.2%
Other	9.1%	1.3%	32.0%	2.4%	5.5%	13.9%	3.4%	5.0%	17.5%
Unreported	3.6%	2.2%	0.0%	0.5%	4.4%	0.0%	0.3%	3.7%	0.0%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 3-5. Statistics on walking by age by year and region.

Age Group	NYC			Rest of NYS			Rest of U.S.		
	2001	2009	2017	2001	2009	2017	2001	2009	2017
Percent Not Taking Walk Trips in the Last Week	15.5%	16.8%	8.7%	29.9%	33.4%	27.3%	35.5%	32.8%	27.4%
5-to-10-year-olds ^a	—	18.6%	9.9%	—	27.2%	28.3%	—	24.3%	25.4%
11-to-15-year-olds ^a	—	11.5%	13.0%	—	24.6%	22.0%	—	24.0%	23.4%
16-to-20-year-olds	13.2%	21.2%	8.9%	28.7%	27.3%	32.6%	34.9%	32.1%	33.1%
21-to-35-year-olds	11.6%	12.5%	5.8%	29.8%	36.6%	27.8%	34.8%	31.1%	25.9%
36-to-64-year-olds	16.1%	15.9%	6.1%	27.9%	31.4%	25.5%	34.0%	33.0%	26.1%
65+ year-olds	23.8%	26.2%	17.0%	36.1%	46.6%	30.5%	41.3%	45.2%	32.9%
Average Minutes Spent Walking per Day	16.11	20.35	18.04	5.61	5.51	6.27	4.71	5.47	5.28
5-to-10-year-olds	14.79	14.49	12.71	4.13	3.71	4.22	4.89	3.89	4.20
11-to-15-year-olds	13.18	14.10	13.66	8.58	5.33	5.01	6.18	7.43	4.57
16-to-20-year-olds	14.14	17.68	19.46	6.47	4.72	7.27	4.35	5.19	4.82

21-to-35-year-olds	19.24	23.79	17.94	5.02	5.85	7.44	4.42	5.66	5.80
36-to-64-year-olds	15.91	22.38	18.76	6.01	6.35	6.02	4.44	5.82	5.33
65+ year-olds	14.17	16.62	18.97	4.44	4.03	6.58	4.79	4.10	5.51

In contrast, the age group with the lowest percentage of not taking walk trips before to the survey day in the rest of NYS and the rest of the United States was those ages 11–15. This finding suggests that younger individuals in these regions may rely on walking for school commutes or leisure activities in local neighborhoods, despite the overall lower rates of walking in these two regions (McDonald, 2008).

Regarding the average minutes spent walking, most age groups in NYC saw a decrease from 2009 to 2017, except for those ages 16–20 and those over 65, although the changes in these two groups were not statistically significant. Across all years, NYC had a statistically significantly higher average walking time per day compared with averages in the rest of NYS and the rest of the United States.

Table 3-6 presents statistics on biking trips, indicating that the percentage of individuals who did not take bike trips was higher than that for walking trips (Table 3-5). The lowest share of those not taking bike trips before the survey day in 2017 was among those ages 11–15 in all regions, suggesting that this age group was the most active when it comes to biking. In NYC, the share of people who did not bike was 80.9%, which was statistically significantly higher than the 73.2% in the rest of NYS and 72.7% in the rest of the United States.

In terms of time spent biking, in 2017 individuals ages 36–64 had the highest average biking time per day in NYC at 3.26 min, which was statistically significantly higher than that of other age groups in the same region. In the rest of NYS, the age group with the highest biking time was those ages 11–15, consistent with the finding that they had the lowest share of individuals who did not bike in the week prior to the survey.

When combining biking and walking, the average time spent on both activities in NYC in 2017 was approximately 20 min, which was statistically significantly higher than the 6.73 min in the rest of NYS and 5.96 min in the rest of the United States.

Males had daily person trips comparable to those of females across all 3 years and regions (Table 3-7). Among age groups, those ages 35–54 exhibited the highest daily person trips in all regions. In NYC, the average daily person trips for this age group in 2017 was 3.59, which was statistically significantly higher than that of the second highest group (ages 20–34) at 3.36. In the rest of NYS, the average daily person trips for those ages 35–54 followed a decreasing trend, dropping from 4.32 in 2009 to 3.73 in 2017, a statistically significant decrease. A similar pattern was observed in the rest of the United States and NYC, although the decline in NYC was not statistically significant. For all 3 years, individuals over 65 in NYS slightly increased their average daily person trips, although this change was not statistically significant.

Table 3-6. Statistics on biking by age by year and region.

Age Group	NYC			Rest of NYS			Rest of U.S.		
	2001	2009	2017	2001	2009	2017	2001	2009	2017
Percent Not Taking Bike Trips in the Last Week	94.4%	93.4%	88.4%	92.2%	88.9%	89.4%	92.6%	87.2%	87.8%
5-to-10-year-olds ^a	—	79.7%	72.0%	—	58.6%	65.0%	—	50.3%	57.4%
11-to-15-year-olds ^a	—	92.7%	80.9%	—	70.0%	73.2%	—	68.6%	72.7%
16-to-20-year-olds	88.5%	95.3%	93.0%	84.8%	88.0%	88.6%	87.4%	88.9%	91.1%
21-to-35-year-olds	92.8%	89.7%	84.8%	91.5%	92.7%	92.6%	91.2%	91.7%	90.6%
36-to-64-year-olds	95.3%	95.5%	89.2%	92.1%	92.6%	92.7%	93.2%	92.1%	91.2%
65+ year-olds	99.0%	98.2%	97.1%	96.4%	98.0%	97.3%	96.3%	97.0%	96.0%
Average Minutes Spent Biking per Day	0.65	0.38	2.00	0.64	0.61	0.46	0.79	0.78	0.68
5-to-10-year-olds	2.04	0.19	0.93	1.44	0.96	0.43	2.25	1.24	0.65
11-to-15-year-olds	2.92	0.02	0.23	2.60	1.47	0.87	2.05	1.59	0.94
16-to-20-year-olds	0.08	0.14	—	0.26	0.74	0.49	0.70	0.71	0.61
21-to-35-year-olds	0.54	0.42	1.97	0.37	0.73	0.35	0.44	0.59	0.84
36-to-64-year-olds	0.40	0.57	3.26	0.34	0.49	0.44	0.56	0.77	0.72
65+ year-olds	0.05	0.15	0.58	0.47	0.18	0.50	0.26	0.38	0.31
Average Minutes Spent Biking and/or Walking per Day	16.76	20.73	20.04	6.25	6.12	6.73	5.50	6.24	5.96
5-to-10-year-olds	16.83	14.68	13.64	5.57	4.67	4.64	7.15	5.13	4.85
11-to-15-year-olds	16.09	14.12	13.90	11.18	6.80	5.87	8.23	9.02	5.51
16-to-20-year-olds	14.22	17.82	19.46	6.72	5.46	7.76	5.05	5.90	5.43
21-to-35-year-olds	19.78	24.22	19.91	5.39	6.58	7.79	4.85	6.25	6.64
36-to-64-year-olds	16.31	22.95	22.02	6.35	6.84	6.46	5.00	6.59	6.05
65+ year-olds	14.22	16.77	19.56	4.90	4.21	7.09	5.05	4.48	5.82

Table 3-7. Person trips by demographics, year, and region.

	NYC			Rest of NYS			Rest of U.S.		
	2001	2009	2017	2001	2009	2017	2001	2009	2017
Person Trips per Person	3.44	3.46	3.23	3.95	3.78	3.32	4.11	3.81	3.38
By Gender									
Male	3.54	3.46	3.27	3.93	3.76	3.29	4.11	3.76	3.32
Female	3.35	3.45	3.19	3.97	3.79	3.35	4.12	3.85	3.44
By Age									
5-to-10-year-olds	2.71	2.76	2.78	3.21	3.17	2.74	3.46	3.20	2.78
11-to-15-year-olds	3.08	2.95	2.57	3.97	3.61	2.79	4.19	3.54	2.79
16-to-20-year-olds	3.77	3.52	3.36	3.98	3.64	2.99	4.30	3.91	3.39
21-to-35-year-olds	3.89	4.04	3.59	4.47	4.32	3.73	4.64	4.27	3.81
36-to-64-year-olds	3.66	3.96	3.15	4.18	3.98	3.62	4.13	3.99	3.61
65+ year-olds	2.57	2.55	2.94	3.56	3.24	3.50	3.43	3.22	3.18

Table 3-8 presents daily PMT by gender and age. From 2009 to 2017, PMT increased significantly across all regions. In NYC, PMT for females rose by 78%, from 13 to 23 miles, whereas PMT for males rose by

only 38%. Among age groups, 20-to-34-year-olds had the highest PMT in NYC (37 miles in 2017), which could reflect this group's high mobility owing to work, social engagements, and urban commuting patterns, whereas 35-to-54-year-olds led in the rest of NYS (47 miles in 2017). PMT among those over 20 years old increased significantly in NYC during this period, with only minor changes in the rest of NYS and the United States.

Table 3-8. PMT by demographics, year, and region.

	NYC			Rest of NYS			Rest of U.S.		
	2001	2009	2017	2001	2009	2017	2001	2009	2017
PMT per Person	21.11	16.10	24.96	35.60	32.92	37.83	41.03	36.85	39.40
By Gender									
Male	25.92	19.37	26.70	39.66	36.94	43.11	45.84	41.66	42.89
Female	16.88	13.14	23.39	31.82	29.06	32.73	36.47	32.17	36.03
By Age									
5-to-10-year-olds	5.61	8.57	11.86	20.19	22.64	22.78	25.18	25.79	25.29
11-to-15-year-olds	10.87	13.24	28.06	30.99	27.26	30.08	37.58	29.89	28.50
16-to-20-year-olds	31.58	22.45	36.96	36.91	36.16	38.33	46.31	37.67	44.09
21-to-35-year-olds	27.24	17.53	30.91	45.73	40.57	46.66	52.02	46.16	46.45
36-to-64-year-olds	19.71	19.78	17.21	41.15	33.32	39.71	42.62	41.58	43.41
65+ year-olds	8.74	8.78	12.54	27.49	25.11	37.49	28.62	25.42	34.85

In 2017, individuals ages 35–54 exhibited the highest daily PMT in the rest of NYS and the rest of the United States compared with other age groups (Figure 3-8). In contrast, the 20-to-34-year-old age group had the highest PMT in NYC. For this group, PMT in NYC was comparable to that in the rest of NYS and the rest of the United States. However, for those ages 35 and older, PMT in NYC was statistically significantly lower than in both the rest of NYS and the rest of the United States.

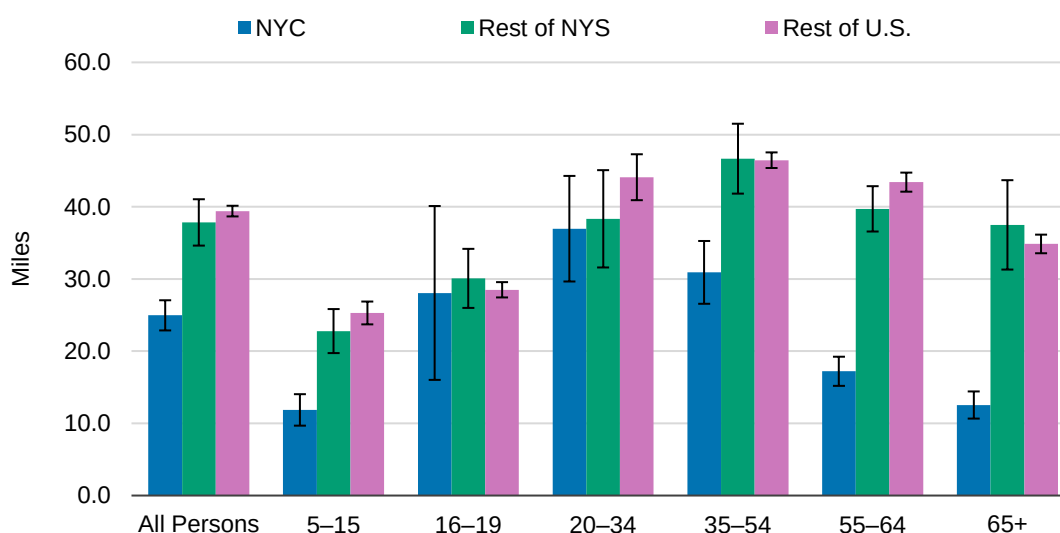


Figure 3-8. PMT by age and region in 2017.

3.2.4 Travel Differences and Vehicle Ownership

The income distribution among zero-vehicle households in the rest of NYS and the rest of the United States was relatively similar across all 3 years (Table 3-9). However, NYC exhibited a statistically significant lower percentage of zero-vehicle households with annual incomes below \$10,000 compared with the rest of NYS and the rest of the United States. In NYC, the \$50,000–\$99,999 income group had the second highest proportion of zero-vehicle households. Additionally, NYC had a statistically significant higher percentage of zero-vehicle households earning over \$200,000 compared with both the rest of NYS and the rest of the United States in each of the 3 years analyzed. For income brackets exceeding \$75,000 in the rest of NYS and the rest of the United States, the share of zero-vehicle households remained below 3% across all 3 years.

Table 3-9. Percentage of zero-vehicle households by annual income classes in U.S.\$(2016), year, and region.

	NYC			Rest of NYS			Rest of U.S.		
	2001	2009	2017	2001	2009	2017	2001	2009	2017
Zero-Vehicle Households (%)	51.6%	54.5%	54.0%	7.6%	9.1%	10.4%	6.8%	7.4%	7.6%
Household Income (%) ^a									
Less than \$10,000	10.7%	13.6%	15.8%	17.9%	27.6%	34.7%	27.7%	31.6%	39.4%
\$10,000 to \$14,999	11.4%	7.1%	6.7%	18.5%	18.1%	25.0%	18.5%	17.0%	19.3%
\$15,000 to \$24,999	13.3%	16.1%	9.0%	22.4%	21.6%	16.1%	17.1%	18.6%	14.1%
\$25,000 to \$34,999	9.5%	8.5%	9.3%	13.7%	9.8%	11.0%	8.1%	8.8%	6.9%
\$35,000 to \$49,999	11.0%	11.7%	9.7%	7.7%	5.6%	4.3%	7.1%	5.7%	4.2%
\$50,000 to \$74,999	14.1%	13.6%	12.3%	2.5%	3.3%	2.3%	4.2%	3.5%	4.9%
\$75,000 to \$99,999	4.1%	5.3%	11.0%	0.9%	0.7%	1.8%	1.1%	0.9%	2.2%
\$100,000 to \$124,999	4.0%	4.8%	5.9%	0.2%	0.2%	0.3%	0.1%	0.4%	1.3%
\$125,000 to \$149,999	1.5%	0.5%	3.6%	0.0%	0.1%	0.0%	0.4%	0.6%	0.8%
\$150,000 to \$199,999	2.4%	4.1%	4.1%	0.6%	0.9%	0.2%	0.8%	0.8%	0.9%
\$200,000 or more	3.5%	4.4%	7.4%	0.4%	0.1%	0.2%	0.5%	0.1%	0.7%
Unreported	14.5%	10.3%	5.1%	15.2%	12.1%	4.1%	14.4%	11.9%	5.3%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

The temporal trends regarding the share of zero-vehicle households by income are illustrated in Figure 3-9 (NYC), Figure 3-10 (rest of NYS), and Figure 3-11 (rest of the United States). In NYC, the percentage of zero-vehicle households earning less than \$10,000 experienced a slight increase, though this change was not statistically significant. A similar pattern can be observed in the rest of NYS and the rest of the United States, where the share of households in the lowest income bracket in 2017 was statistically significantly higher than the share in 2009. Conversely, the proportion of households with an annual income between \$50,000 and \$74,000 showed a decreasing trend across all three regions. This observation may suggest that middle-income households may be increasingly able to afford vehicle ownership or view it as a necessary investment for mobility. This could reflect suburbanization trends, changes in job accessibility, or evolving transportation infrastructure that favors car use for this demographic (Uddin et al., 2024). Additionally, the share of households with unreported incomes declined over the years, with the percentage of unreported households in 2017 being statistically significantly lower than the percentages in 2001 and 2009 in all three regions.

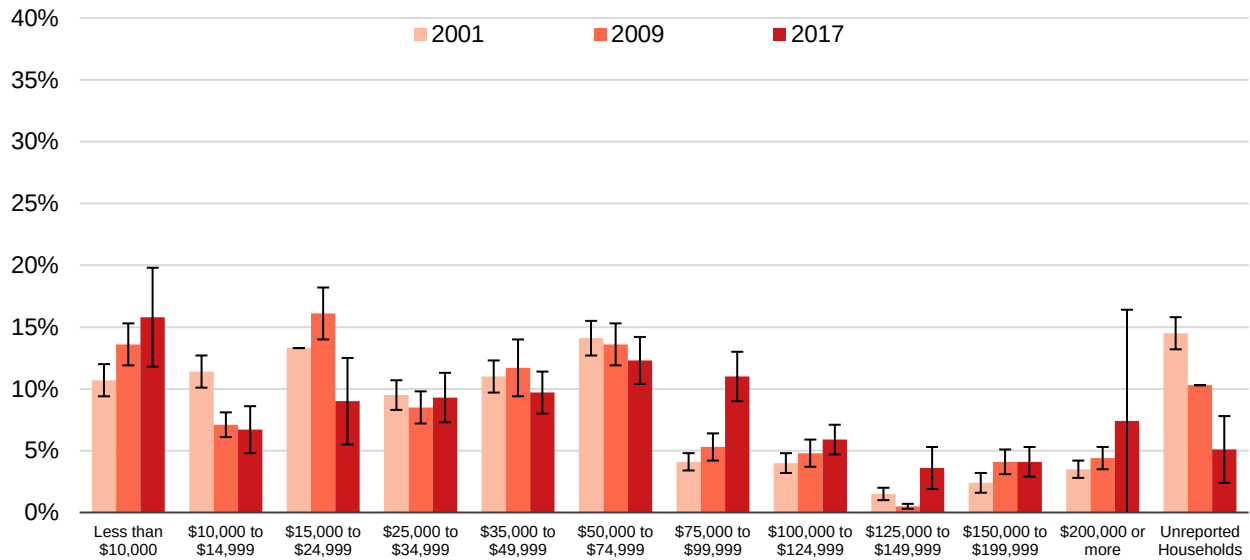


Figure 3-9. Distribution of zero-vehicle households by annual income in U.S.\$(2016) and year in NYC.

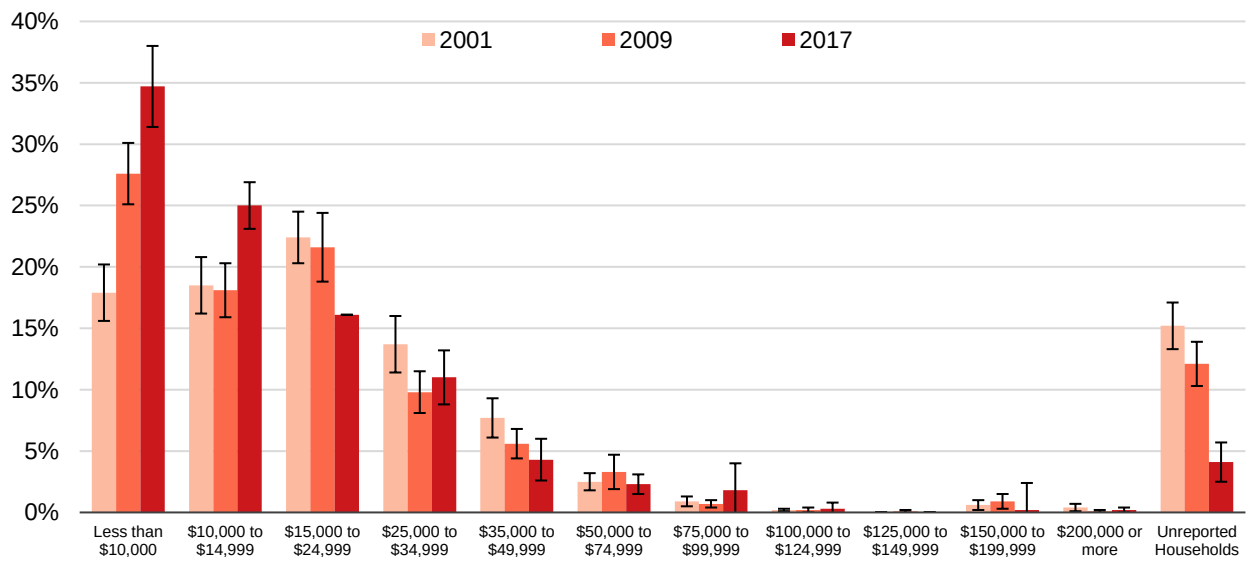


Figure 3-10. Distribution of zero-vehicle households by annual income in U.S.\$(2016) and year in the rest of NYS.

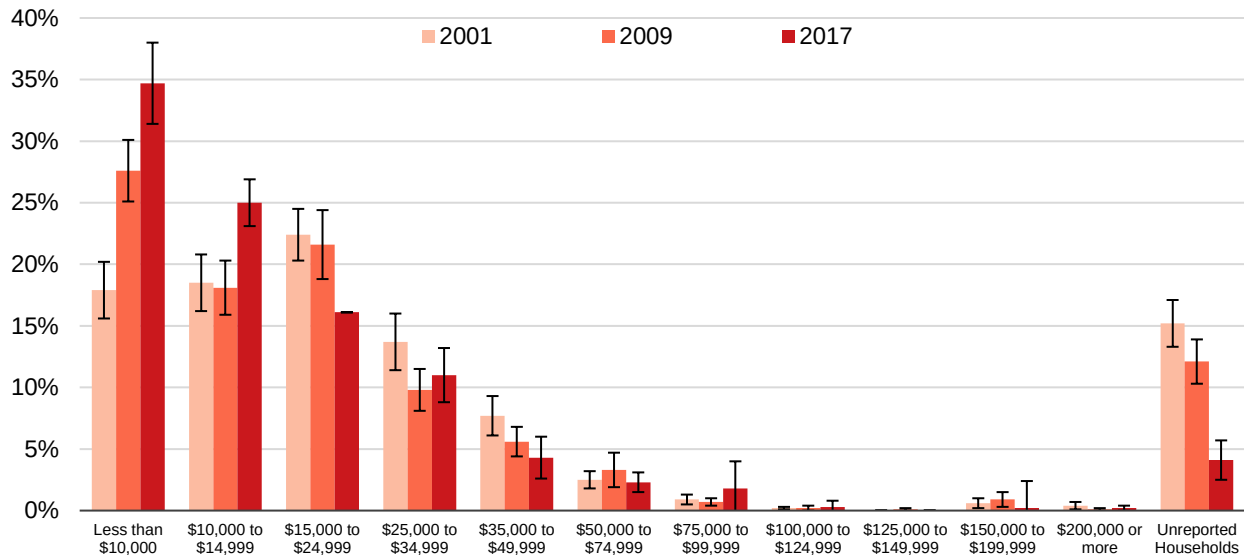


Figure 3-11. Distribution of zero-vehicle households by annual income in U.S. \$(2016) and year in the rest of the United States.

As shown in Table 3-10, zero-vehicle households in NYC averaged significantly more trips compared with those of their counterparts in other areas. In 2017, zero-vehicle households in NYC recorded a statistically significant increase in trip frequency compared with those in the rest of NYS and outside of NYS. Additionally, the proportion of trips made via public transit was notably higher in NYC, at 35.2%, compared with 18.0% in the rest of NYS and 19.8% in the rest of the United States.

Table 3-10. Average person trips by mode for zero-vehicle households by year and region.

	NYC			Rest of NYS			Rest of U.S.		
	2001	2009	2017	2001	2009	2017	2001	2009	2017
Person Trips per Person	3.30	3.20	3.19	2.53	2.50	2.61	2.63	2.61	2.61
Mode (%) ^a									
Private Vehicle	13.3%	7.4%	4.3%	36.8%	24.3%	16.3%	43.6%	33.2%	27.3%
Public Transit	32.7%	32.5%	35.2%	13.5%	18.5%	18.0%	16.7%	17.6%	19.8%
Walk	47.2%	52.7%	52.6%	35.8%	41.8%	50.9%	31.9%	34.2%	38.6%
Other	6.6%	7.3%	7.8%	13.8%	14.3%	14.8%	7.6%	12.7%	14.3%
Unreported	0.2%	0.1%	0.0%	0.2%	1.2%	0.0%	0.1%	2.2%	0.0%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Figure 3-12 demonstrates that, in both the rest of NYS and the rest of the United States, zero-vehicle households recorded a statistically significant decrease in the share of person trips compared with all households across all 3 years. This finding suggests that zero-vehicle households may face greater challenges in accessing transportation options, which limits their mobility and ability to engage in daily activities. Notably, this gap was narrower in NYC, indicating the presence of a potentially more robust public transit infrastructure that may offer greater mobility for residents without vehicles.

Notably, this gap was narrower in NYC. Regarding PMT (Figure 3-13), the disparity between zero-vehicle households and all households was more pronounced. In 2017, zero-vehicle households in the rest of NYS accounted for less than one-fourth of the PMT compared with all households. This significant

difference underscores the reliance of vehicle-owning households on personal transportation for longer trips, as opposed to public transit or alternative modes. Access to a vehicle significantly influences travel behavior, highlighting the mobility constraints faced by zero-vehicle households (Blumenberg et al., 2020). Therefore, addressing these disparities through improved public transit options and support services could enhance the travel experience and accessibility for zero-vehicle households in both urban and rural contexts.

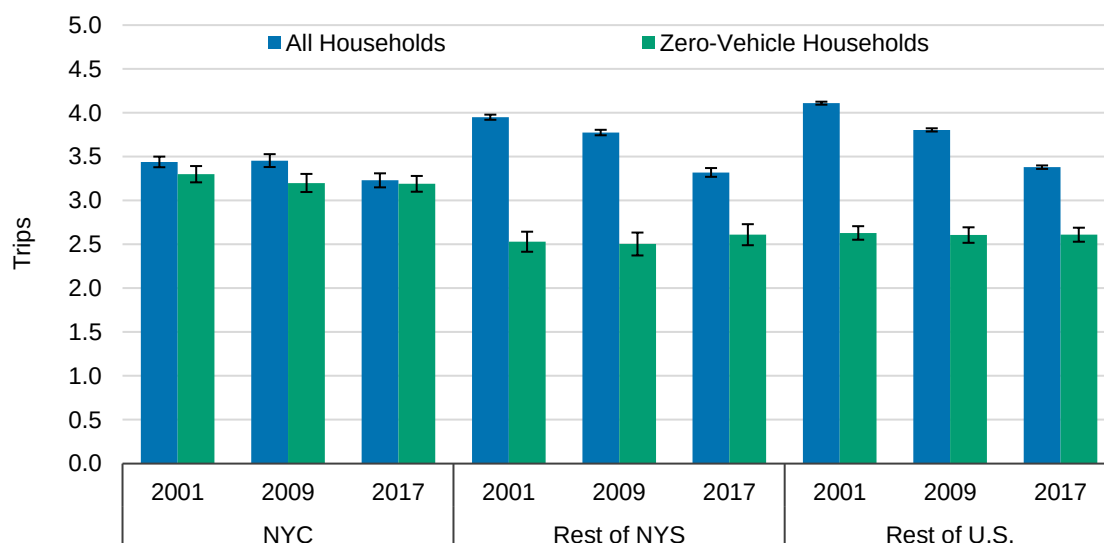


Figure 3-12. Effects of not owning a vehicle are shown on the number of person trips by household, year, and region.

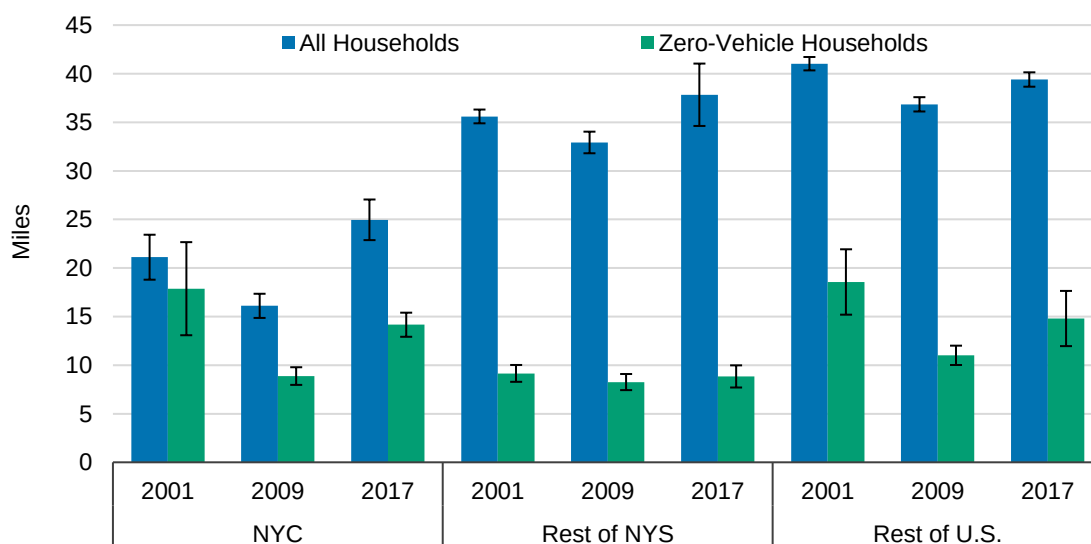


Figure 3-13. Effects of not owning a vehicle are shown on PMT by household, year, and region.

3.3 VEHICLE TRAVEL COMPARISONS

3.3.1 Vehicle Trip Frequencies and Trip Distances

The number of vehicle trips per day decreased from 2001 to 2017 in the rest of NYS and the rest of the United States (Figure 3-14). In 2017, the daily vehicle trip rate in NYC was 1.1, whereas the daily rate was 2.7 in the rest of NYS and the United States. Although vehicle trip rates varied by region, the purpose of these trips was comparable across regions in 2017 (Figure 3-15).

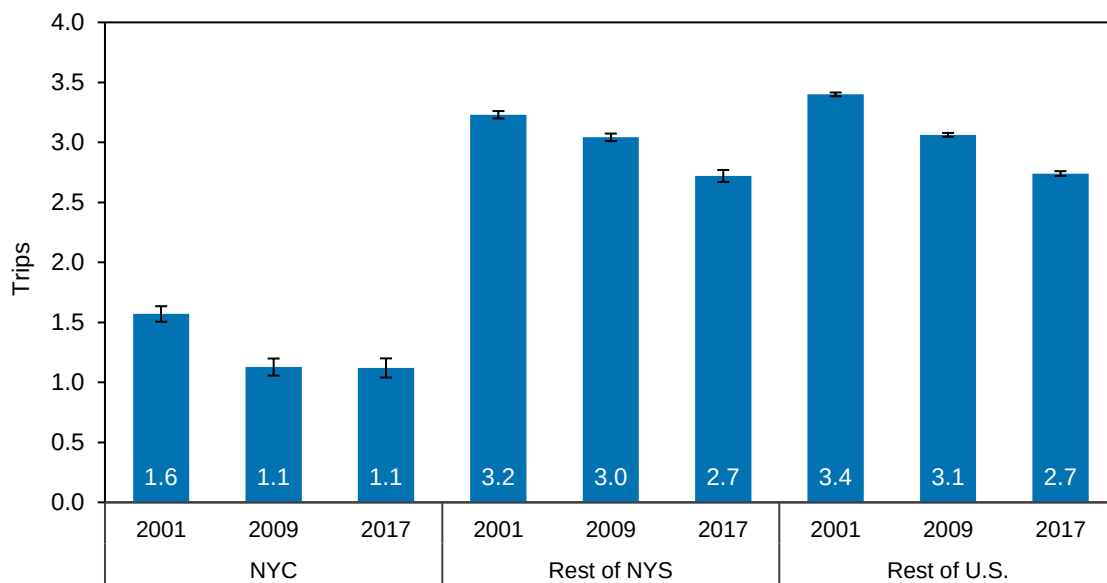


Figure 3-14. Vehicle trips per driver per day by year and region.

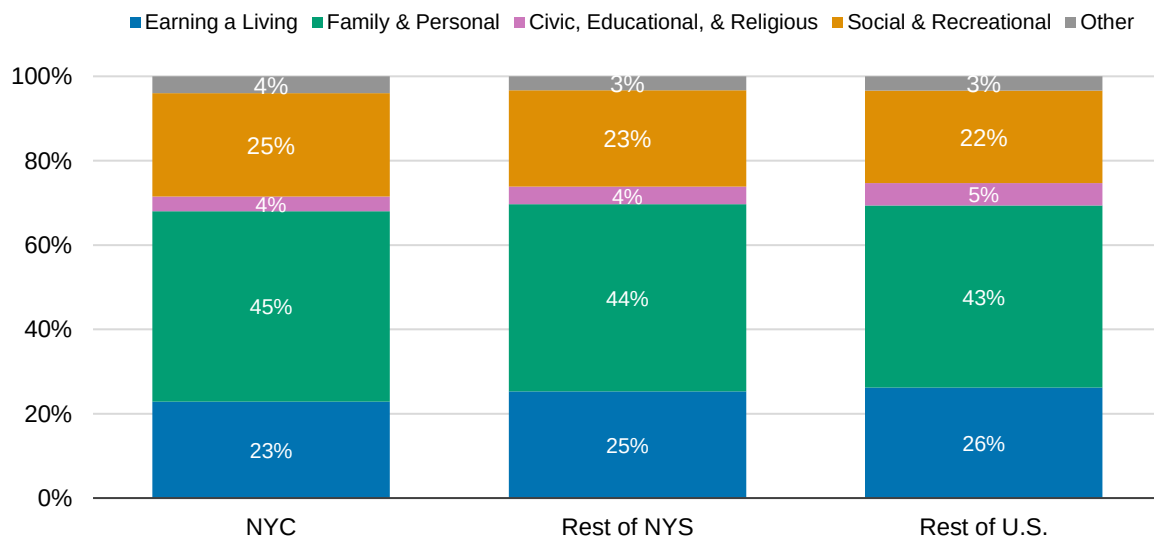


Figure 3-15. Shares of driver trips by purpose and region in 2017. Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number.

Vehicle trips decreased slightly across all regions over time, with NYC showing statistically significantly fewer vehicle trips than both the rest of NYS and the rest of the United States in each year analyzed

(Table 3-11). This may reflect NYC's greater availability of alternative transportation options, such as public transportation, biking, and walking, thus reducing residents' reliance on personal vehicles. The distribution of vehicle trips by purpose was similar across regions. However, NYC saw a significant decline in family and personal business trips from 55.5% in 2009 to 45.2% in 2017. This decline was less pronounced in the rest of NYS and the United States. In terms of age groups, people ages 65 and over in NYC had the highest average vehicle trips (1.35), whereas those under 34 had the lowest. In the rest of NYS and the United States, the 35-to-54-year-old age group showed the highest average vehicle trips. Gender differences were also notable in NYC, where men had significantly more vehicle trips (1.33) than women (0.86). Additionally, NYC saw significantly higher vehicle trips on weekends compared with those on weekdays, possibly owing to more leisure and family-related activities. This pattern was not observed in the rest of NYS or the rest of the United States.

Table 3-11. Vehicle trips statistics by year and region.

	NYC			Rest of NYS			Rest of U.S.		
	2001	2009	2017	2001	2009	2017	2001	2009	2017
Vehicle Trips per Driver	1.57	1.13	1.12	3.23	3.04	2.72	3.40	3.06	2.74
Purpose (%) ^a									
Earning a Living	25.6%	18.2%	22.9%	25.5%	26.1%	25.3%	26.2%	26.1%	26.2%
Family & Personal	50.4%	55.5%	45.2%	48.7%	47.5%	44.3%	48.0%	46.7%	43.2%
Civic, Educational, & Religious	3.8%	3.4%	3.5%	4.2%	4.7%	4.2%	4.9%	4.9%	5.3%
Social & Recreational	19.0%	21.3%	24.5%	21.0%	20.7%	22.8%	20.3%	21.1%	21.9%
Other	1.0%	0.2%	4.0%	0.5%	0.3%	3.3%	0.5%	0.3%	3.4%
Unreported	0.1%	1.5%	0.0%	0.2%	0.7%	0.0%	0.1%	0.9%	0.0%
Age									
16-to-19-year-olds	1.34	0.41	0.21	2.60	2.15	1.50	2.91	2.29	1.83
20-to-34-year-olds	0.98	0.83	0.84	3.05	2.78	2.30	3.36	2.95	2.52
35-to-54-year-olds	1.87	1.30	1.21	3.58	3.42	2.99	3.77	3.37	3.06
55-to-64-year-olds	2.06	1.50	1.19	3.12	3.09	2.93	3.28	3.14	2.87
65+ year-olds	1.67	1.07	1.35	2.96	2.76	2.88	2.85	2.70	2.57
Gender									
Male	1.75	1.34	1.33	3.39	3.14	2.80	3.61	3.14	2.80
Female	1.33	0.91	0.86	3.08	2.95	2.64	3.19	2.98	2.67
Weekday Vehicle Trips per Driver	1.61	1.08	1.04	3.39	3.22	2.82	3.60	3.26	2.82
Weekend Vehicle Trips per Driver	1.47	1.26	1.32	2.85	2.58	2.51	2.88	2.56	2.55

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 3-12 shows daily VMT per driver, highlighting notable regional and demographic differences. In NYC, the share of VMT per driver of social and recreational trips significantly decreased from 37% to 30% between 2009 and 2017. Conversely, the share of VMT in other types saw increases, ranging from 0.4% to 5.7%, possibly owing to the trip type not being able to be categorized in the current survey. In the rest of NYS, the share of work-related trips statistically significantly decreased from 36% to 31%, which may indicate changes in commuting patterns, such as increased remote work or alternative work schedules outside of NYC. By age, the 35-to-54-year-old age group had the highest VMT across all regions, likely reflecting the increased travel needs for work and family responsibilities. However, only NYC saw a statistically significant increase in VMT for this age group, whereas the rest of NYS and the United States showed declines. This trend may suggest that NYC's 35-to-54-year-old age group

experienced a shift in travel needs, potentially tied to longer travel distances or more suburban commuting or other personal travel patterns.

Figure 3-16 shows the average length of vehicle trips across years, comparing NYC with the rest of NYS and the rest of the United States.

Table 3-12. VMT statistics by year and region.

	NYC			Rest of NYS			Rest of U.S.		
	2001	2009	2017	2001	2009	2017	2001	2009	2017
VMT per Driver	11.86	8.13	10.11	29.24	26.64	27.59	33.31	29.57	28.88
Purpose (%) ^a									
Earning a Living	34.6%	24.1%	26.5%	34.8%	35.9%	30.7%	35.5%	34.7%	32.7%
Family & Personal	35.7%	34.5%	36.3%	34.7%	33.9%	34.2%	35.3%	32.7%	31.7%
Civic, Educational, & Religious	3.9%	2.3%	1.8%	3.2%	3.7%	4.2%	3.8%	4.4%	5.5%
Social & Recreational	25.1%	36.8%	29.6%	26.3%	22.7%	26.3%	24.4%	24.4%	24.5%
Other	0.6%	0.4%	5.7%	0.9%	0.4%	4.7%	0.8%	0.6%	5.5%
Unreported	0.2%	1.9%	0.0%	0.3%	3.5%	0.0%	0.2%	3.2%	0.1%
Age									
16-to-19-year-olds	5.57	2.30	0.89	20.05	15.88	27.59	22.57	17.84	15.56
20-to-34-year-olds	9.28	9.32	9.22	29.02	26.78	13.69	35.36	29.10	30.67
35-to-54-year-olds	14.52	8.39	12.05	34.35	31.68	24.93	38.46	34.78	32.81
55-to-64-year-olds	12.54	9.36	9.05	27.83	26.24	31.31	32.60	31.00	29.09
65+ year-olds	8.98	6.45	8.63	20.63	18.78	29.56	21.34	19.97	22.67
Gender									
Male	15.02	10.56	13.40	36.47	31.46	32.75	41.63	35.76	33.21
Female	7.88	5.54	6.10	22.28	21.80	22.28	25.12	23.31	24.62
Average Vehicle Trip Length	7.91	7.55	9.34	9.16	8.85	10.32	9.91	9.77	10.72
% Vehicle Trips/Person Trips	23.2%	20.3%	18.5%	60.7%	61.5%	58.8%	61.6%	60.6%	60.5%
% VMT/PMT	28.6%	31.4%	21.6%	61.0%	61.7%	52.3%	60.6%	60.5%	54.8%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

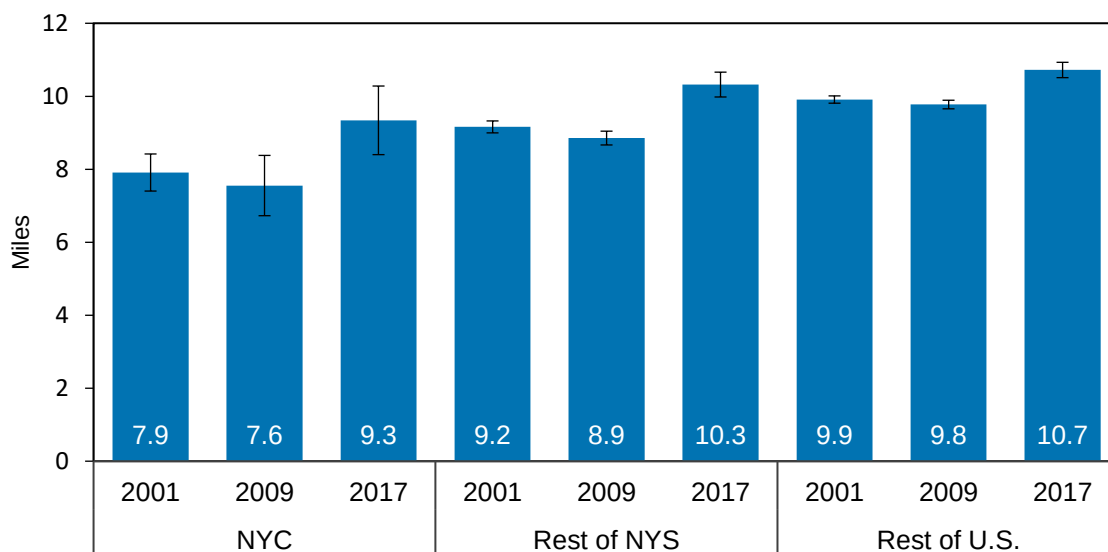


Figure 3-16. Average vehicle trip length by year and region.

3.3.2 Differences in Driving by Gender and Age

Males drove statistically significantly higher VMT than females across all regions. In NYC in 2017, men drove nearly twice as many miles as women drove (Figure 3-17). In the rest of NYS and the United States, this gender gap in VMT was also present, though relatively smaller, suggesting that outside of NYC, the differences in travel behavior or access to alternative transportation by gender were less pronounced.

Across all 3 years, the 35-to-54-year-old age group consistently had the highest daily VMT regardless of the region (Figure 3-18). However, in the rest of the United States, this age group experienced a statistically significant decrease in VMT over time. Meanwhile, in both the rest of NYS and the rest of the United States, VMT significantly increased for those ages 65 and older from 2009 to 2017, possibly reflecting extended workforce participation or increased mobility among older adults.

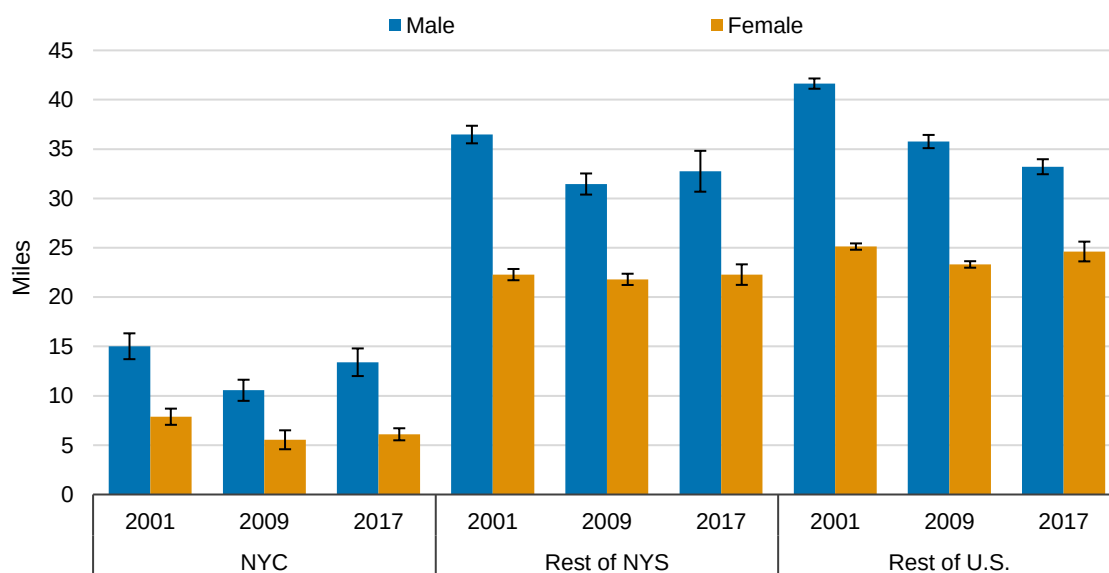


Figure 3-17. VMT by gender, year, and region.

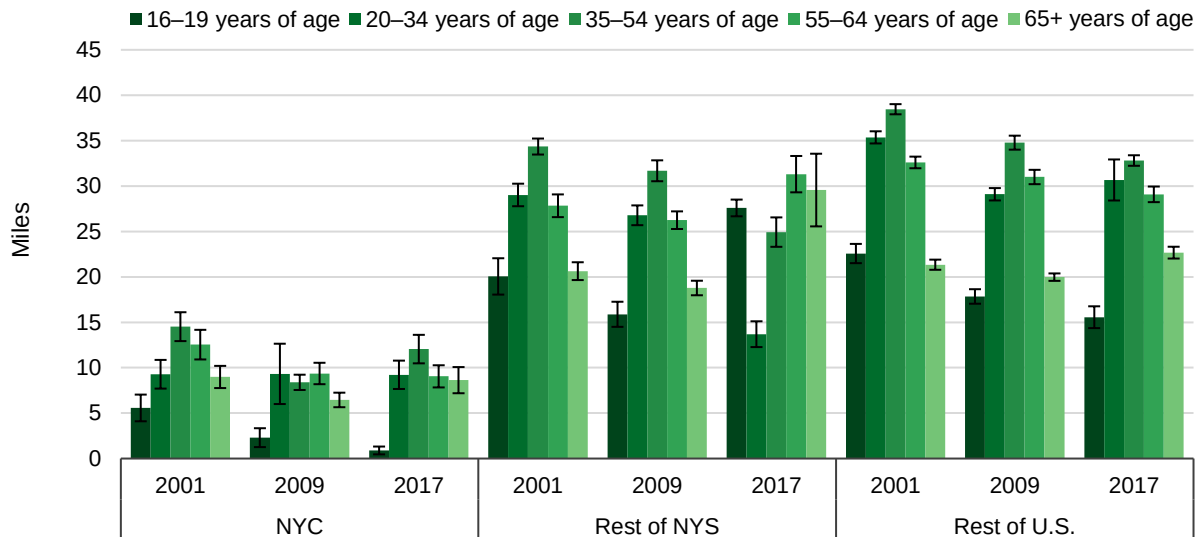


Figure 3-18. VMT by age, year, and region.

3.3.3 Types and Ages of Vehicles in Use

Table 3-13 indicates that the ownership share of automobiles and SUVs has been the highest among vehicle types across all years and regions. In NYC, the share of automobiles among all vehicle types in 2017 was 59.8%, significantly higher than the 51.7% observed in the rest of NYS. This share has shown a decreasing trend, dropping significantly from 64.7% in 2009. Additionally, a statistically significant increase in SUV ownership was noted across all regions from 2001 to 2017.

Table 3-14 reveals that the average vehicle age in NYC in 2017 was 8.24 years, comparable to the average of 8.56 years in the rest of NYS. However, this average was significantly lower than the national average of 10.49 years, indicating a trend toward newer vehicles in urban areas like NYC. In terms of VMT, 62.4% of the VMT in NYC came from vehicles less than 10 years old, which was significantly lower than the 69.5% observed in the rest of NYS. This suggests that newer vehicles in NYC are driven less than those in the rest of NYS. Similarly, 59% of fuel consumption in NYC was attributed to vehicles less than 10 years old in 2017, a rate that was significantly lower than the 66.2% in the rest of NYS. This could reflect higher adoption of hybrid and electric vehicles, which consume less fuel per mile (Pan et al., 2023).

Table 3-13. Share^a of fleet by vehicle types, year, and region.

Vehicle Type	NYC			Rest of NYS			Rest of U.S.		
	2001	2009	2017	2001	2009	2017	2001	2009	2017
Auto	75.7%	64.7%	59.8%	63.2%	56.2%	51.7%	56.4%	49.7%	49.4%
Van	9.3%	9.9%	6.3%	10.1%	8.9%	6.6%	9.0%	8.2%	6.0%
SUV	12.4%	22.6%	30.5%	11.9%	21.1%	28.1%	12.1%	19.4%	23.4%
Pickup	1.0%	1.2%	1.4%	11.6%	10.6%	9.9%	19.0%	18.3%	16.3%
Other Truck	0.4%	0.0%	0.2%	0.3%	0.3%	0.5%	0.5%	0.4%	0.5%
RV	0.0%	0.0%	0.2%	0.5%	0.4%	0.4%	0.7%	0.5%	0.6%
Motorcycle	1.3%	1.6%	1.6%	2.3%	2.6%	2.6%	2.3%	3.4%	3.3%
Unreported	0.0%	0.0%	0.0%	0.0%	0.0%	0.3%	0.0%	0.0%	0.4%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 3-14. Share of households' private vehicle fleets by age, year, and region.

Vehicle Age	NYC			Rest of NYS			Rest of U.S.		
	2001	2009	2017	2001	2009	2017	2001	2009	2017
Age Statistics									
Average Vehicle Age	8.18	7.98	8.24	7.76	7.74	8.56	8.92	9.47	10.49
Median Vehicle Age	7	7	7	7	7	7	7	8	9
Percent VMT									
Less than 10 years	69.3%	71.1%	62.4%	74.2%	75.6%	69.6%	71.0%	67.8%	58.6%
10 years or more	30.7%	28.9%	37.6%	25.8%	24.4%	30.4%	29.0%	32.2%	41.4%
Percent Vehicle Fleet ^a									
Less than 10 years	61.3%	64.9%	61.3%	68.0%	70.5%	62.0%	62.6%	60.7%	50.6%
10 years or more	38.7%	35.1%	38.7%	32.0%	29.5%	38.0%	37.5%	39.3%	49.4%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 3-15 illustrates how annual miles driven vary according to vehicle age and type, including autos, vans, SUVs, pickup trucks, and other POVs. This variation reflects factors like fuel efficiency, vehicle reliability, and the driving purposes associated with each vehicle type. For instance, SUVs and pickup trucks, often used for long-distance travel or work-related purposes, may show higher annual miles compared with those of autos and vans, which are frequently used for shorter, everyday trips.

Table 3-16 shows the percentage of vehicles in the fleets of NYC, the rest of NYS, and the rest of the nation, cross-classified by vehicle types and age groups. For example, NYC's dense infrastructure might favor smaller vehicles, whereas rural and suburban areas in NYS and the rest of the United States had a higher proportion of larger vehicles like pickup trucks and SUVs. The fleet sizes are noted in the top data line, enabling readers to compute the average annual vehicle miles for each vehicle type and age group cell.

Table 3-15. Average vehicle miles driven per household vehicle, by vehicle type, vehicle age group, year, and region.

Vehicle Age	NYC			Rest of NYS			Rest of U.S.		
	2001	2009	2017	2001	2009	2017	2001	2009	2017
All Vehicles*	11,446	9,460	9,933	12,338	11,033	11,387	12,019	11,314	11,133
0 to 2 years	12,583	10,500	10,765	13,712	12,977	13,449	14,701	14,216	14,117
3 to 5 years	13,799	9,871	10,717	13,796	11,981	12,521	14,191	12,735	12,790
6 to 9 years	12,177	10,348	8,534	12,722	11,040	12,326	12,512	11,754	12,066
10 years +	9,178	7,876	9,630	10,080	9,173	9,190	9,360	9,319	9,371
Unreported	11,854	10,228	10,728	12,443	10,423	10,480	10,925	11,143	10,106
Auto	11,003	9,529	9,417	12,220	10,665	11,223	11,596	11,154	11,144
0 to 2 years	11,992	10,065	10,164	13,471	12,522	13,436	14,054	14,230	13,644
3 to 5 years	14,235	8,898	9,526	13,681	11,471	12,020	13,471	12,465	12,685
6 to 9 years	11,264	10,836	7,891	12,183	10,588	12,167	12,028	11,336	12,030
10 years+	8,861	8,383	9,588	10,610	9,282	9,374	9,433	9,477	9,446
Unreported	11,407	11,254	17,419	12,448	10,153	9,781	11,292	11,425	10,406
Van	14,728	7,991	11,164	13,307	12,237	12,342	13,400	12,587	12,618
0 to 2 years	19,632	8,130	10,522	14,270	15,848	13,260	15,862	15,495	18,935
3 to 5 years	12,969	11,890	13,620	14,863	13,365	13,719	15,526	14,105	14,066
6 to 9 years	14,718	10,315	9,804	13,292	12,389	13,194	13,597	13,389	13,413
10 years+	12,594	6,385	11,317	10,086	9,737	11,072	10,274	10,119	10,941
Unreported	18,233	8,453	2,117	19,268	13,635	11,174	11,547	14,513	11,964
SUV	13,108	10,475	10,840	13,972	12,552	12,354	13,877	12,825	12,360
0 to 2 years	11,628	12,190	11,405	13,982	13,505	13,068	14,968	15,038	14,238
3 to 5 years	12,296	11,150	11,930	14,753	12,992	13,464	15,235	13,672	13,212
6 to 9 years	17,161	9,786	9,863	15,062	12,329	13,455	14,308	12,860	12,647
10 years+	12,325	8,577	9,977	11,217	10,829	9,718	10,687	10,184	10,819
Unreported	15,521	9,457	6,195	11,579	11,770	9,425	11,117	13,167	16,508
Pickup Truck	9,597	9,467	14,835	12,291	11,242	11,310	12,465	11,329	10,656
0 to 2 years	16,542	22,178	21,356	15,564	14,145	16,637	16,552	15,068	15,493
3 to 5 years	6,833	12,504	17,298	13,240	12,917	12,595	15,420	13,233	13,730
6 to 9 years	6,792	8,505	9,522	14,660	11,884	11,602	12,888	12,016	12,638
10 years+	7,071	7,422	5,130	9,330	8,742	8,913	9,536	9,544	8,961
Unreported	11,516	6,107	4,642	9,044	11,121	14,439	11,879	11,759	11,435
Other POV	1,182	1,270	1,874	3,626	2,885	3,586	5,094	3,459	3,130
0 to 2 years	1,944	3,665	1,027	5,379	4,765	6,829	7,612	5,078	6,341
3 to 5 years	2,500	1,777	1,241	4,238	3,144	10,579	8,478	4,077	4,557
6 to 9 years	6,000	1,386	365	3,717	2,809	2,716	7,350	4,198	2,917
10 years+	1,042	479	2,416	2,406	2,060	2,827	3,101	2,470	2,475
Unreported	632	171	**	9,504	3,613	2,439	2,189	2,666	3,892

* Includes all autos, SUVs, vans, pickup trucks, and “other POVs” not otherwise specified.

** Data not reliable.

Table 3-16. Share^a of households by vehicle type, vehicle age, year, and region.

Vehicle Age	NYC			Rest of NYS			Rest of U.S.		
	2001	2009	2017	2001	2009	2017	2001	2009	2017
# Household Vehicles	2,085	1,943	1,968	7,461	7,201	7,524	193,041	202,357	213,086
0 to 2 years	17.7%	17.9%	23.4%	18.3%	16.5%	18.0%	15.6%	12.0%	12.6%
3 to 5 years	18.5%	19.3%	20.4%	23.2%	23.8%	21.8%	21.3%	20.6%	18.5%
6 to 9 years	21.8%	24.2%	17.2%	24.6%	28.0%	21.7%	23.5%	25.8%	19.1%
10 years +	36.6%	33.2%	38.5%	31.2%	28.6%	37.8%	36.2%	37.9%	49.0%
Unreported	5.4%	5.5%	0.5%	2.7%	3.2%	0.7%	3.4%	3.7%	0.8%
Auto	75.7%	64.7%	59.8%	63.2%	56.2%	51.7%	56.4%	49.7%	49.4%
0 to 2 years	11.4%	11.0%	12.9%	10.8%	8.6%	8.0%	7.7%	5.8%	5.9%
3 to 5 years	13.9%	11.2%	10.8%	13.6%	12.2%	11.2%	11.5%	9.4%	10.0%
6 to 9 years	17.1%	16.5%	11.3%	16.4%	16.2%	11.9%	14.1%	12.9%	10.2%
10 years +	29.4%	22.6%	24.6%	20.8%	17.4%	20.3%	21.4%	19.8%	23.0%
Unreported	3.8%	3.4%	0.3%	1.6%	1.7%	0.3%	1.7%	1.7%	0.2%
Van	9.3%	9.9%	6.3%	10.1%	8.9%	6.6%	9.0%	8.2%	6.0%
0 to 2 years	1.5%	1.6%	0.8%	1.9%	1.0%	0.7%	1.5%	0.7%	0.6%
3 to 5 years	1.6%	1.3%	1.2%	2.7%	2.1%	1.4%	2.1%	1.8%	0.9%
6 to 9 years	2.4%	1.1%	0.8%	2.9%	2.8%	1.5%	2.5%	2.5%	1.2%
10 years +	3.2%	5.1%	3.3%	2.4%	2.7%	3.0%	2.6%	3.0%	3.4%
Unreported	0.6%	0.9%	0.2%	0.3%	0.3%	0.0%	0.4%	0.3%	0.0%
SUV	12.4%	22.6%	30.5%	11.9%	21.1%	28.1%	12.1%	19.4%	23.4%
0 to 2 years	4.3%	5.0%	9.0%	3.5%	5.3%	7.5%	3.1%	3.3%	4.4%
3 to 5 years	2.8%	6.5%	7.3%	3.8%	6.5%	7.1%	3.5%	5.5%	5.1%
6 to 9 years	2.1%	5.7%	5.0%	2.5%	5.2%	5.9%	2.6%	5.3%	4.7%
10 years +	2.5%	4.2%	9.3%	1.9%	3.6%	7.5%	2.6%	4.7%	9.2%
Unreported	0.6%	1.1%	0.0%	0.3%	0.5%	0.1%	0.3%	0.5%	0.1%
Pickup Truck	1.0%	1.2%	1.4%	11.6%	10.6%	9.9%	19.0%	18.3%	16.3%
0 to 2 years	0.3%	0.1%	0.4%	1.7%	1.2%	1.5%	2.9%	1.6%	1.5%
3 to 5 years	0.1%	0.2%	0.6%	2.7%	2.3%	1.8%	3.6%	3.1%	1.9%
6 to 9 years	0.1%	0.4%	0.0%	2.5%	3.0%	1.7%	3.9%	4.2%	2.3%
10 years +	0.5%	0.5%	0.4%	4.4%	3.7%	4.7%	8.0%	8.6%	10.5%
Unreported	0.0%	0.0%	0.0%	0.3%	0.4%	0.1%	0.7%	0.8%	0.1%
Other POV	1.7%	1.7%	2.0%	3.1%	3.2%	3.7%	3.5%	4.3%	4.7%
0 to 2 years	0.2%	0.2%	0.3%	0.4%	0.4%	0.3%	0.5%	0.6%	0.3%
3 to 5 years	0.1%	0.1%	0.7%	0.4%	0.7%	0.4%	0.5%	0.8%	0.6%
6 to 9 years	0.1%	0.5%	0.2%	0.4%	0.7%	0.6%	0.5%	0.8%	0.7%
10 years +	0.9%	0.8%	0.9%	1.6%	1.2%	2.3%	1.7%	1.8%	2.9%
Unreported	0.4%	0.0%	—	0.2%	0.3%	0.1%	0.3%	0.4%	0.2%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 3-17. Daily commute statistics by mode^a, year, and region.

	NYC			Rest of NYS			Rest of U.S.		
	2001	2009	2017	2001	2009	2017	2001	2009	2017
Commute Person Trips per Worker	1.08	0.95	1.11	1.14	1.14	1.10	1.15	1.11	1.13
SOV	25.0%	19.9%	16.9%	76.1%	76.7%	70.6%	77.7%	78.4%	72.1%
MOV	9.7%	7.7%	8.0%	14.7%	13.7%	16.4%	16.4%	14.6%	18.0%
Amtrak/Commuter Rail	2.7%	4.8%	2.0%	2.5%	1.4%	3.5%	0.4%	0.0%	0.5%
Subway/Elevated Rail	30.3%	28.7%	41.2%	0.7%	0.3%	1.3%	0.5%	0.3%	1.4%
Other Public ^b	12.2%	14.9%	9.3%	1.0%	1.9%	2.2%	1.6%	0.3%	2.2%
Walk	16.0%	17.5%	16.6%	3.8%	3.3%	4.3%	2.4%	1.7%	3.5%
Other	4.2%	6.5%	6.0%	1.2%	2.3%	1.7%	0.9%	2.7%	2.3%
No Response	0.0%	0.0%	0.0%	0.2%	0.2%	0.0%	0.1%	1.8%	0.0%
Commute PMT per Worker	7.48	6.17	8.54	13.18	12.89	13.12	13.78	12.99	13.95
SOV	33.5%	30.7%	24.0%	77.4%	79.7%	71.1%	77.7%	81.7%	75.9%
MOV	13.0%	15.5%	10.8%	12.8%	12.6%	15.2%	16.1%	13.5%	17.0%
Amtrak/Commuter Rail	3.7%	3.6%	4.2%	6.2%	3.7%	8.7%	1.0%	0.5%	1.1%
Subway/Elevated Rail	35.1%	28.1%	47.4%	1.0%	0.5%	2.5%	0.5%	0.2%	1.2%
Other Public ^b	7.7%	11.9%	8.3%	0.6%	1.1%	1.5%	1.0%	1.1%	1.9%
Walk	2.8%	4.7%	1.2%	0.4%	0.2%	0.2%	0.2%	0.2%	0.4%
Other	4.1%	5.5%	3.9%	1.3%	1.9%	0.8%	3.5%	2.7%	2.4%
No Response	0.0%	0.0%	0.0%	0.3%	0.3%	0.0%	0.1%	0.2%	0.1%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

^b Other Public is mostly bus transit.

3.4 COMMUTING COMPARISONS

Providing efficient transportation options for commuting is vital to both state and national economies. Reducing daily commute times also contributes significantly to quality of life because many people consider shorter commutes an essential factor in overall well-being.

3.4.1 Commuting Trip Frequencies and Trip Lengths

Commute person trips and PMT are shown in Table 3-17. In NYC, both commute person trips and PMT saw a statistically significant increase from 2009 to 2017. In the rest of NYS, whereas commute person trips slightly declined, PMT increased, possibly reflecting longer average commute distances due to urban sprawl or limited public transit options; however, these changes were not statistically significant. In 2017, subway or elevated rail held the largest share of commute person trips in NYC, accounting for 41.2%. Meanwhile, in both the rest of NYS and the rest of the United States, SOVs dominated, comprising over 70% of commute person trips—a pattern also reflected in PMT. This pattern indicates a reliance on private vehicles in areas where public transit is less accessible or convenient, likely contributing to longer commute times and higher PMT in suburban and rural regions.

Figure 3-19 to Figure 3-21 illustrate the temporal trends in commute mode share across NYC, the rest of NYS, and the rest of the United States. Figure 3-19 shows a statistically significant increase in the share of subway or elevated rail trips in NYC from 2009 to 2017, highlighting a shift toward greater reliance on public transit within the city. However, the share of Amtrak or commuter rail trips and other public modes decreased significantly over the same period, possibly indicating that subway services are more effective than longer-distance rail options are for meeting commuting needs.

Although SOVs remained the dominant mode for commuting in the rest of NYS (Figure 3-10), the share of SOV trips significantly declined from 76.7% in 2009 to 70.6% in 2017. This shift may suggest a gradual move toward alternative commuting options in suburban and rural parts of NYS, potentially influenced by rising fuel costs or the availability of carpooling programs. Similar patterns can be observed in the rest of the United States (Figure 3-21), where reliance on SOVs also decreased, hinting at a broader national shift in commuting preferences that could be driven by factors such as urbanization, environmental awareness, or investment in local transit infrastructure.

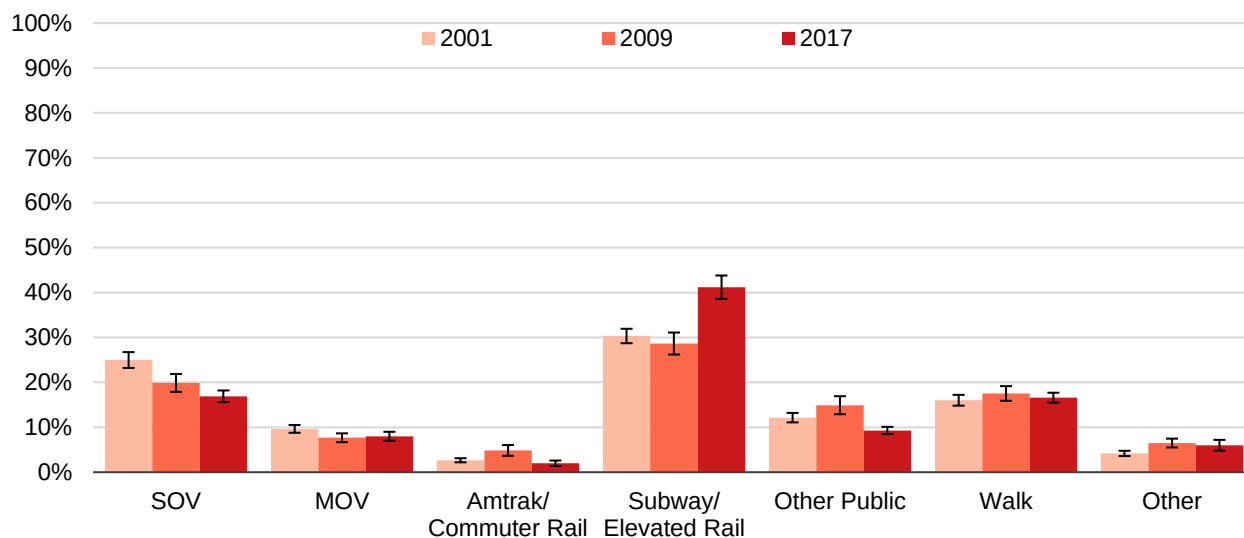


Figure 3-19. Trends in daily commute trip mode share by year for NYC.

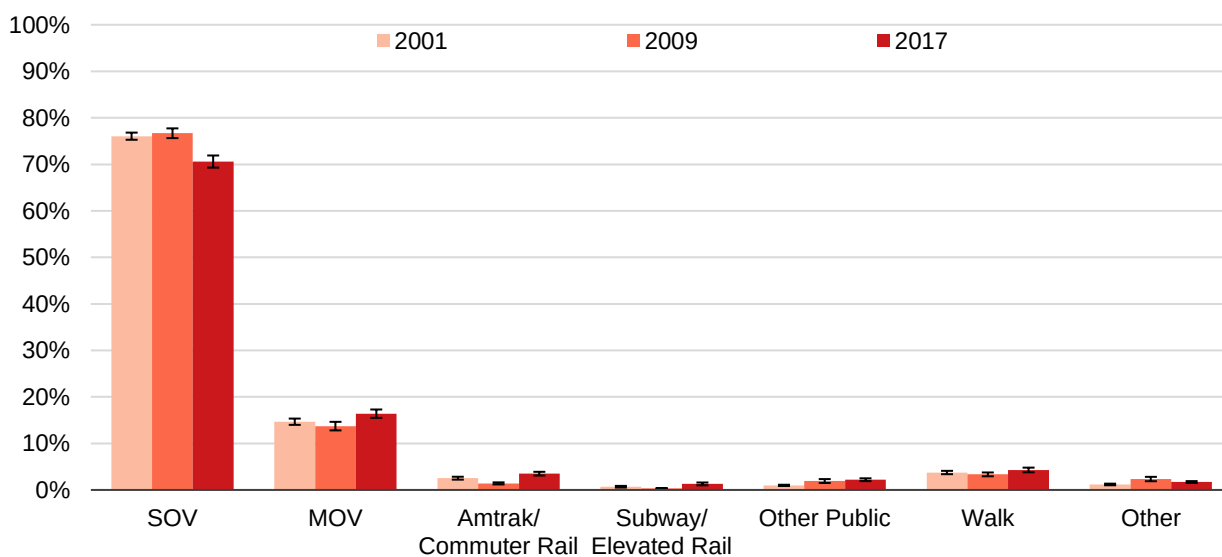


Figure 3-20. Trends in daily commute trip mode share by year for the rest of NYS.

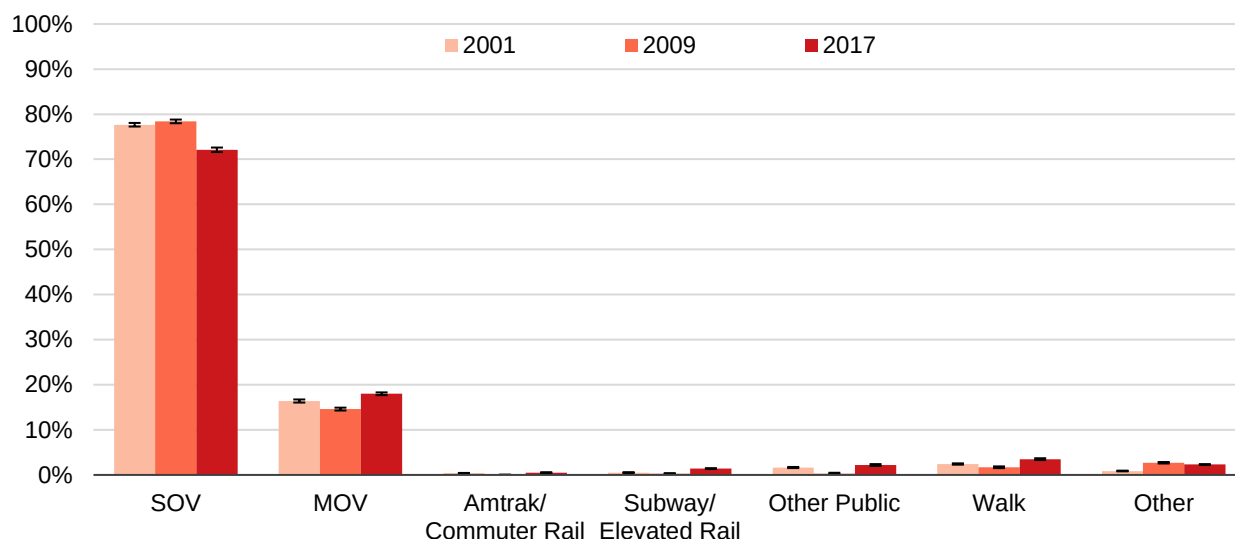


Figure 3-21. Trends in daily commute trip mode share by year for the rest of the United States.

3.4.2 Commuting Times and Commuting Speeds

Table 3-18 presents data on commute distance and travel time by public and private modes of transportation. Total commute distance decreased across all three regions from 2009 to 2017, though these reductions were not statistically significant. Notably, public transportation modes in the rest of NYS saw a statistically significant increase in average travel distance, rising from 16 to 21 miles. This suggests a possible trend toward longer public transit commutes, potentially driven by expanded service areas or increased reliance on public transit for longer trips. In contrast, no statistically significant change in commute distance was observed in NYC or the rest of the United States. When looking at travel time, private mode commutes in NYC increased significantly from 33 min in 2009 to 41 min in 2017, possibly reflecting the effects of congestion or slower traffic speeds because of rising vehicle volumes.

Table 3-18. Commuting statistics for private and public modes of transportation by year and region.

Trip Statistics	NYC			Rest of NYS			Rest of U.S.		
	2001	2009	2017	2001	2009	2017	2001	2009	2017
Length (miles)									
All ^a	8.02	7.78	7.04	11.82	11.45	10.85	12.22	11.89	11.38
Private	9.87	11.52	10.09	11.67	11.63	10.77	12.13	12.11	11.73
Public	9.33	7.90	8.40	24.43	15.91	20.75	11.94	10.71	11.80
Time (minutes)									
All ^a	40.21	40.20	41.14	24.00	24.01	24.82	23.11	23.45	24.64
Private	31.36	33.46	40.51	21.90	22.93	21.90	22.42	22.77	23.80
Public	55.78	51.02	53.98	72.48	58.93	73.54	54.18	53.54	54.73

^a “All” trips include walking and other modal trips.

Figure 3-22 compares commute distances for private and public transportation modes across NYC, the rest of NYS, and the rest of the United States. In the rest of NYS, public modes had a statistically significantly greater commute distance than that of private modes across all 3 years, suggesting that public transportation modes serve more distant commuters in this region. This pattern was not seen in

NYC or the rest of the United States, where commute distances for public and private modes were comparable, possibly reflecting the shorter average travel distances within dense urban areas or more evenly distributed commute patterns in other parts of the country.

Figure 3-23 shows that public modes had significantly higher travel times than private modes across all regions. This discrepancy may be due to the additional stops and transfers required in public transit systems, as well as lower speeds in urban settings. The longer travel times associated with public transportation modes emphasize the potential trade-off for commuters who prioritize cost savings and reduced environmental impact over shorter, more direct travel times offered by private vehicles.

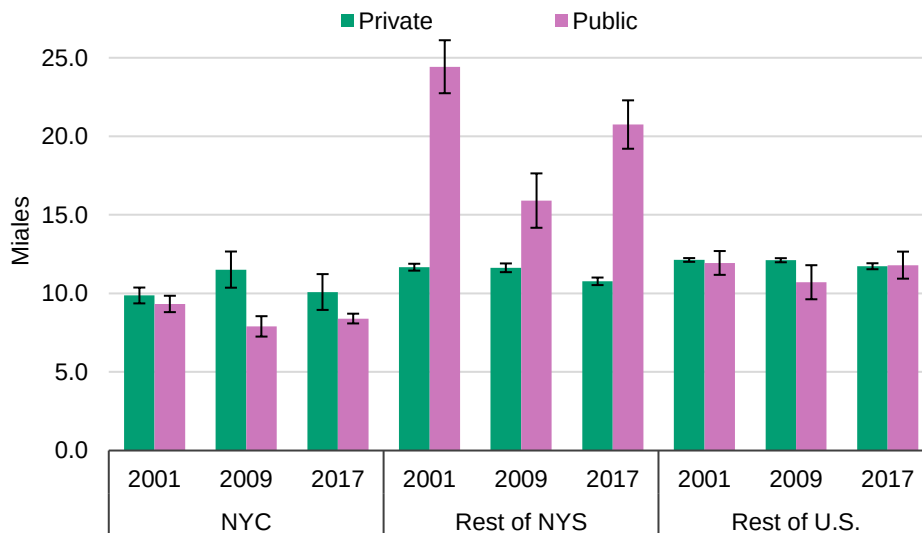


Figure 3-22. Commute distance by year and region.

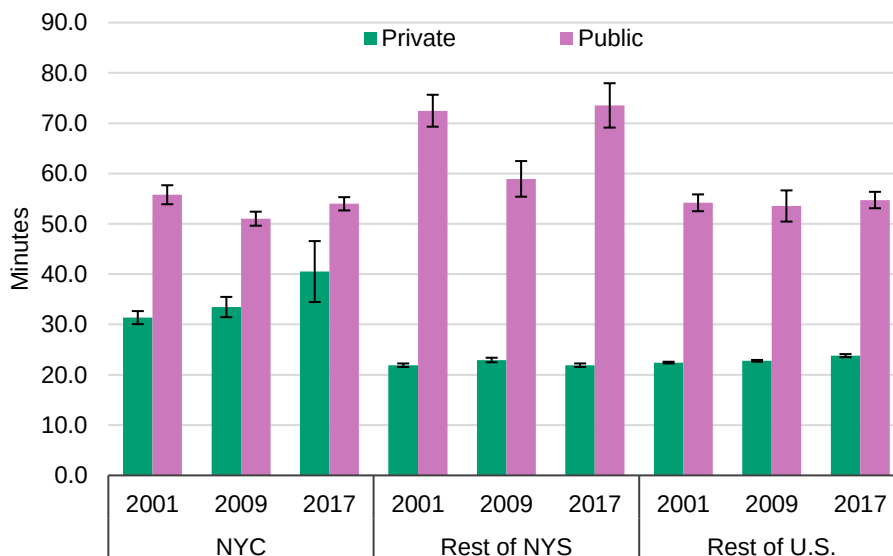


Figure 3-23. Commute travel time by year and region.

4. URBAN TRAVEL AND POPULATION DENSITY

In 2017, the population distribution in NYS reflects variation in density across metropolitan areas and regions. As shown in Table 4-1, a sizable portion of the population is concentrated in NYC and surrounding areas, particularly in regions with more than 50,000 people per square mile. For example, nearly 1.9 million people in NYC reside in such densely populated areas, making up a substantial share of the state's total population. By contrast, areas like Watertown, Elmira, and Glens Falls have most of their populations residing in less dense regions, with fewer than 2,000 people per square mile. Meanwhile, midsized metropolitan areas, including Syracuse, Rochester, and Buffalo, exhibit more balanced distributions, with populations spread across a range of density categories.

Table 4-1. 2017 NYS household distribution by population density.

	Total	Population Density (population per square mile)						
		[0, 500)	[500, 2000)	[2000, 4000)	[4000, 10000)	[10000, 20000)	[20000, 50000)	[50000, ∞)
NYS	7,261,081	1,103,886	817,473	831,562	1,017,849	584,044	975,767	1,892,721
Capital District	322,961	69,631	106,375	64,184	60,621	17,302	4,846	—
Glens Falls, Moreau	40,991	23,588	7,877	4,044	5,482	—	—	—
Utica–Rome	102,947	32,523	25,627	17,357	18,053	5,255	—	—
Syracuse	219,738	39,506	56,902	47,683	49,742	18,030	—	—
Ithaca	55,391	31,431	10,916	4,510	7,815	—	718	—
Rochester	360,008	53,712	102,117	85,827	82,457	35,895	—	—
Buffalo	462,364	54,754	84,546	101,363	158,313	63,388	—	—
Elmira	37,961	11,371	9,710	7,569	9,311	—	—	—
Poughkeepsie	127,933	52,365	47,309	11,851	12,136	4,271	—	—
Binghamton	85,832	29,284	15,982	16,142	21,508	2,916	—	—
Newburgh–Middletown	148,914	52,387	34,006	20,529	22,780	13,345	5,867	—
Kingston	60,190	36,292	13,685	8,250	1,963	—	—	—
Watertown	31,653	11,477	8,937	8,661	2,578	—	—	—
NYMTC								
New York City	3,127,319	2,127	3,130	6,007	84,669	254,278	884,388	1,892,721
Bronx	557,615	2,127	—	—	21,254	19,183	208,837	306,214
Kings	774,376	—	—	—	2,130	23,693	300,243	448,309
New York	974,961	—	—	—	1,362	13,835	60,241	899,522
Queens	646,794	—	—	—	20,530	109,681	277,907	238,675
Richmond	173,574	—	3,130	6,007	39,393	87,886	37,159	—
Nassau, Suffolk	787,548	38,554	103,651	225,372	321,529	84,993	13,449	—
Putnam, Rockland, Westchester	544,107	40,861	119,924	106,021	126,722	84,371	66,208	—
Rest of State	745,226	524,024	66,778	96,192	32,168	—	292	—

NYMTC = New York Metropolitan Transportation Council.

In 2009, nearly 90% of households in NYS were concentrated within the 13 metropolitan strata (Table 4-2). Approximately 43% of these households were in the five boroughs of NYC, underscoring the significant population density in this urban core compared with other parts of the state and the nation. Notably, 38% of NYS households in 2009 lived in regions with densities exceeding 20,000 people per square mile—much higher than the 3% of households observed nationwide. This distribution has remained largely consistent since 2001 (Table 4-3)

Table 4-2. 2009 NYS household distribution by population density.

	Total	Population Density (population per square mile)						
		[0, 500)	[500, 2000)	[2000, 4000)	[4000, 10000)	[10000, 20000)	[20000, 50000)	[50000, ∞)
NYS	7,137,482	1,178,437	873,916	780,844	1,065,752	522,296	926,455	1,789,781
Albany	320,181	77,607	117,979	63,311	53,997	7,287	—	—
Glens Falls	58,260	36,617	10,173	3,465	8,005	—	—	—
Utica–Rome	116,223	47,118	34,714	20,101	10,875	3,415	—	—
Syracuse	184,815	30,393	64,786	34,400	49,540	5,696	—	—
Ithaca	37,443	20,966	6,380	5,489	4,508	—	100	—
Rochester	383,077	102,330	107,713	84,962	64,010	24,061	—	—
Buffalo	471,544	61,838	109,723	99,956	160,510	39,517	—	—
Elmira	34,853	12,719	10,944	5,261	5,930	—	—	—
Poughkeepsie	103,204	37,783	43,639	13,051	6,481	2,250	—	—
Binghamton	97,837	36,063	26,441	12,026	21,512	1,795	—	—
Newburgh	122,233	50,182	31,491	20,620	12,459	7,481	—	—
NYMTC								
New York City	3,055,273	4,303	4,262	7,988	80,107	287,547	882,485	1,788,582
Bronx	473,884	—	—	—	3,120	16,370	119,541	334,853
Kings	888,404	1,396	—	—	4,118	31,565	341,876	509,449
New York	749,335	—	—	—	4,701	4,117	75,197	665,320
Queens	777,224	1,612	—	3,412	34,042	138,638	320,560	278,959
Richmond	166,425	1,295	4,262	4,575	34,126	96,857	25,310	—
Nassau, Suffolk	911,037	38,832	129,158	232,549	433,867	63,356	13,275	—
Putnam, Rockland, Westchester	469,399	34,618	109,245	97,446	116,404	79,891	30,595	1,200
Rest of State	702,055	545,452	51,694	70,990	33,918	—	—	—

NYMTC = New York Metropolitan Transportation Council.

Table 4-3. 2001 NYS household distribution by population density.

	Total	Population Density (population per square mile)						
		[0, 500)	[500, 2000)	[2000, 4000)	[4000, 10000)	[10000, 20000)	[20000, 50000)	[50000, ∞)
NYS	7,183,208	1,186,731	813,608	752,260	1,073,037	635,381	1,001,682	1,719,602
Albany	323,953	76,618	113,160	56,277	59,204	18,178	96	—
Glens Falls	49,047	33,958	6,500	4,608	3,980	—	—	—
Utica–Rome	118,311	50,849	32,415	20,238	12,794	2,014	—	—
Syracuse	184,396	27,098	49,006	38,194	51,809	18,290	—	—
Ithaca	37,072	21,299	4,853	5,906	3,876	—	1,138	—
Rochester	291,642	23,949	89,634	70,432	66,215	41,412	—	—
Buffalo	477,111	65,689	94,217	88,470	174,441	54,294	—	—
Elmira	35,677	13,416	10,872	3,726	7,664	—	—	—
Poughkeepsie	101,318	33,897	40,667	13,606	10,736	2,412	—	—
Binghamton	102,273	40,523	22,557	11,024	26,518	1,651	—	—
Newburgh	116,843	53,270	34,915	15,139	7,757	5,393	—	—
NYMTC								
New York City	3,075,688	15,957	5,155	11,285	59,211	304,948	961,727	1,717,404
Bronx	471,506	1,256	—	370	6,983	25,615	151,096	286,185
Kings	896,496	780	3,335	—	1,453	31,479	337,402	522,047
New York	751,869	9,521	265	1,704	1,329	11,035	54,347	673,667
Queens	796,677	2,485	385	4,243	17,462	146,039	391,161	234,902
Richmond	159,140	1,914	1,169	4,968	31,984	90,780	27,722	604
Nassau, Suffolk	933,098	37,360	117,460	231,227	438,220	102,188	6,644	—
Putnam, Rockland, Westchester	470,801	36,488	116,293	93,432	105,594	84,601	32,078	2,198
Rest of State	865,978	656,360	75,905	88,696	45,017	—	—	—

NYMTC = New York Metropolitan Transportation Council.

Comparing the 2017 and 2009 data, the state experienced population growth of 123,599 individuals, increasing from 7,137,482 to 7,261,081—a relatively modest 1.7% expansion over 8 years. Rural and least-dense areas in NYS experienced a notable decline, with the lowest population density category (fewer than 500 people per square mile) shrinking by 74,551 households. This represents a 6.3% reduction, signaling a continued trend of rural out-migration. Conversely, high-density areas saw substantial growth, with population centers of 10,000 or more people per square mile expanding by 229,398 households, a robust 7.1% increase.

Suburban regions demonstrated the most dynamic population movements. Nassau and Suffolk counties, for instance, saw substantial redistribution across population density categories, reflecting changing residential preferences and economic dynamics. For example, in 2009, 62% of the population in Nassau and Suffolk lived in areas with 4,000–10,000 people per square mile. In 2017, the proportion reduced to 41%, with more people moved to sparser areas with population density lower than 4,000 people per square mile. The rest of the United States represented more rural areas, which continued to experience a gradual depopulation, with residents seemingly gravitating towards more densely populated regions.

Comparing 2001, 2009, and 2017, the share of NYS households by population density is similar (Figure 4-1). In 2017, the highest population density category is more than 50,000 people per square mile, with 26% of the households. The lowest category is between 10,000 and 20,000, with 8% in 2017. For the rest

of the population density categories, the share of households was relatively evenly distributed, with around 10%–15%.

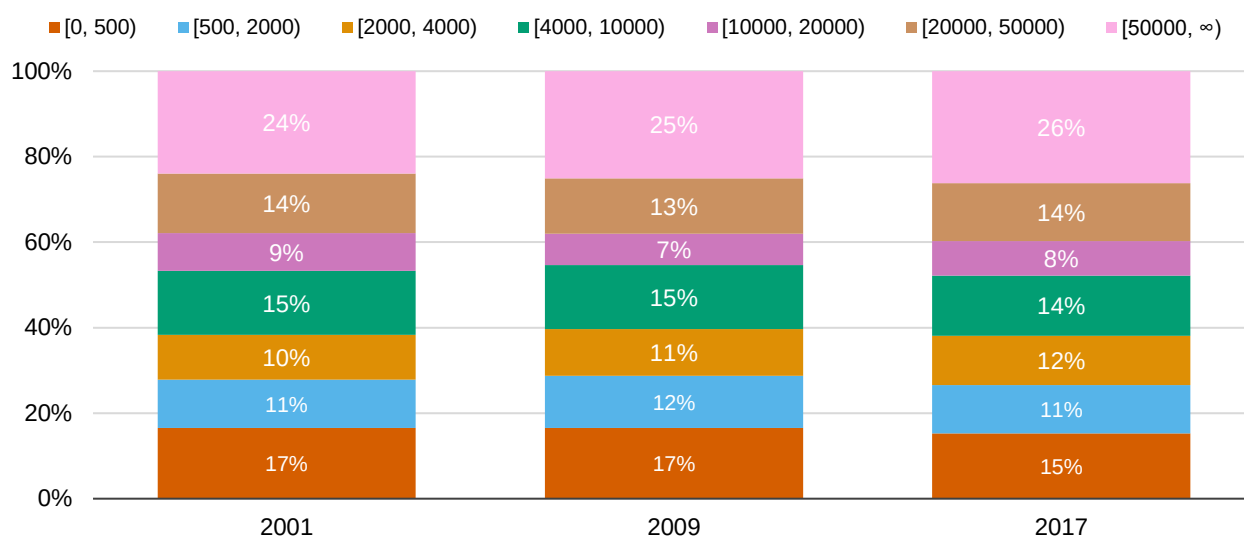


Figure 4-1. Distribution of NYS households by population density (persons per square mile) and year.

4.1 PROFILE OF NYS ADD-ON STRATA HOUSEHOLDS BY POPULATION DENSITY

Figure 4-2 presents the distribution of households in NYS and the rest of the United States across population density categories. In census tracts with fewer than 50,000 people per square mile, the rest of the United States exhibited a statistically significantly higher number of households compared with NYS. Conversely, in areas with a population density exceeding 50,000 people per square mile, NYS had a statistically significantly higher number of households. This stark contrast highlights the unique urban concentration within NYS, particularly in high-density areas like New York City, compared with the more dispersed population distribution across the rest of the country.

When examining the share of households (Figure 4-3), 29% of households in NYS were located in areas with more than 50,000 people per square mile, whereas the corresponding proportion in the rest of the United States was less than 1%. This striking difference underscores the dominant role of urban centers in NYS and their capacity to accommodate a sizable portion of the population in high-density living conditions. Meanwhile, in the rest of the United States, the largest share of households—28%—was found in areas with a population density between 4,000 and 10,000 people per square mile, reflecting a more suburban or moderately urban settlement pattern.

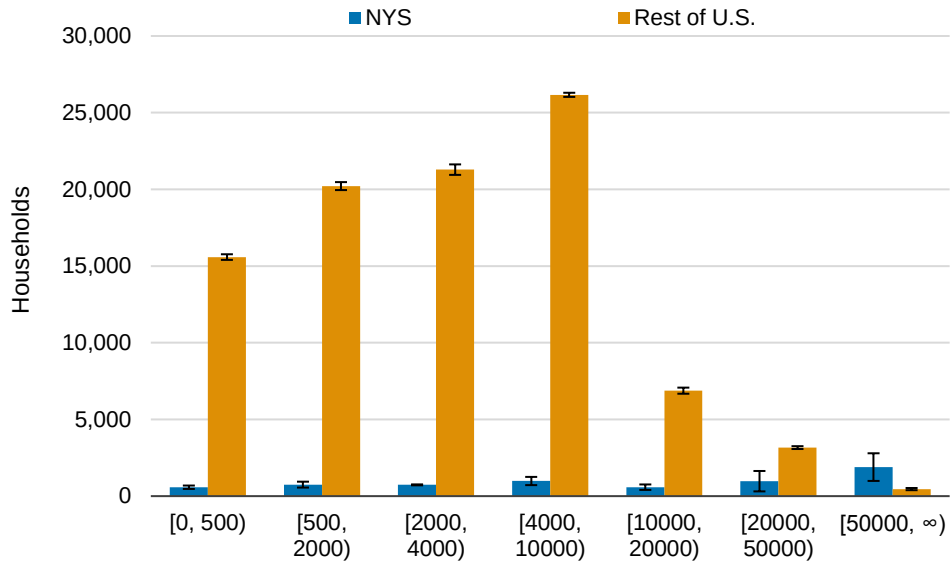


Figure 4-2. Number of households by population density and region in 2017.

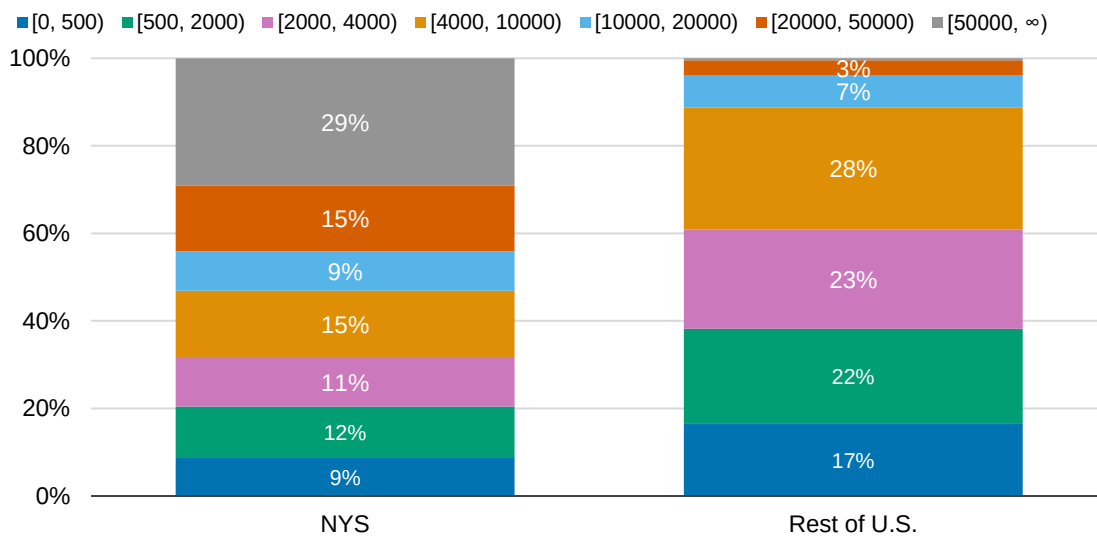


Figure 4-3. Share of households by population density and region in 2017.

4.1.1 Vehicle Ownership and Rail Access

Table 4-4, Table 4-5, and Table 4-6 show the household characteristics and vehicle ownership patterns across NYS metro areas compared with metro areas in the rest of United States, examining trends from 2001 through 2017. Access to rail transit shows a steady pattern across these years, with NYS consistently maintaining higher percentages of households with rail access compared with the rest of the country. Vehicle ownership patterns reveal that in 2001, even the lowest density areas of NYS (<500 people per square mile) showed vehicle ownership rates slightly lower than those of their counterparts in the rest of the country—2.08 versus 2.28 vehicles per household. This gap persisted in 2017. The most dramatic differences emerge in high-density areas, particularly those with populations exceeding 50,000 per square mile. In 2017, NYS households in these areas averaged just 0.40 vehicles per household, whereas their counterparts in other states maintained 0.92 vehicles per household. This difference reflects

NYS's unique urban environment, particularly in NYC, where robust public transportation infrastructure and dense urban development make car ownership less necessary.

Whereas vehicle ownership consistently decreased with density, the number of workers per household showed more complex patterns. In 2017, NYS maintained relatively stable worker-per-household ratios across density categories until the highest density areas, where the ratio dropped to 1.25 workers per household. Interestingly, the rest of the United States showed a spike to 1.64 workers per household in the highest density category, suggesting different patterns of urban employment and household composition.

In 2001, the difference between NYS and the rest of the United States in high-density areas was notable but not extreme—62% versus 49% of households owned no vehicles in areas with 50,000+ people per square mile. By 2017, this gap had widened, with 66% of NYS households in the highest density areas owning no vehicles, compared with just 31% in similar density areas elsewhere in the country.

Table 4-4. 2017 household characteristics and vehicle ownership patterns by population density and region.

	Population Density (population per square mile)							
	[0, 500)		[500, 2000)		[2000, 4000)		[4000, 10000)	
	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.
Households	580	15,586	751	20,214	735	21,287	986	26,164
% Households in an MSA with Rail	32.2%	10.5%	41.0%	21.4%	50.3%	23.5%	57.6%	32.0%
Workers per Household	1.34	1.32	1.4	1.38	1.27	1.33	1.48	1.38
Vehicles per Household	2.29	2.36	2.09	2.05	1.81	1.86	1.59	1.72
Vehicles per Driver	1.21	1.16	1.07	1.02	0.95	0.97	0.83	0.93
% Households by Vehicle Ownership ^a								
0	3.3%	2.7%	3.7%	4.1%	8.1%	5.7%	15.5%	9.1%
1	26.4%	23.5%	28.9%	29.9%	34.7%	35.9%	38.5%	38.9%
2	36.3%	36.0%	38.4%	38.2%	36.6%	36.3%	27.8%	32.9%
3	17.4%	22.0%	17.3%	17.7%	12.7%	14.2%	11.1%	12.2%
4+	16.6%	15.8%	11.6%	10.2%	7.9%	7.9%	7.2%	6.9%

	Population Density (population per square mile)					
	[10000, 20000)		[20000, 50000)		[50000, ∞)	
	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.
Households	584	6,875	975	3,164	1,893	459
% Households in an MSA with Rail	75.6%	67.0%	99.4%	88.2%	100.0%	97.9%
Workers per Household	1.45	1.36	1.55	1.4	1.25	1.64
Vehicles per Household	1.29	1.39	0.87	0.99	0.4	0.92
Vehicles per Driver	0.76	0.82	0.52	0.65	0.36	0.56
% Households by Vehicle Ownership ^a						
0	23.6%	19.7%	38.9%	30.2%	66.3%	30.7%
1	36.9%	41.2%	40.9%	48.0%	28.3%	47.3%
2	28.8%	26.1%	16.3%	16.2%	4.7%	21.6%
3	8.5%	8.5%	2.5%	3.8%	0.5%	—
4+	2.1%	4.5%	1.4%	1.7%	0.1%	0.4%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 4-5. 2009 household characteristics and vehicle ownership patterns by population density and region.

	Population Density (population per square mile)							
	[0, 500)		[500, 2000)		[2000, 4000)		[4000, 10000)	
	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.
Households	633	14,480	822	19,448	710	19,717	1,032	22,496
% Households in an MSA with Rail	26.2%	15.8%	38.7%	25.4%	52.4%	26.3%	62.9%	36.0%
Workers per Household	1.53	1.43	1.45	1.38	1.44	1.33	1.40	1.35
Vehicles per Household	2.07	2.28	1.86	1.96	1.83	1.83	1.55	1.70
Vehicles per Driver	1.00	1.12	0.93	1.01	0.90	0.97	0.80	0.94
% Households by Vehicle Ownership ^a								
0	4.0%	2.8%	6.1%	5.1%	6.7%	6.2%	12.7%	9.4%
1	25.5%	24.4%	32.0%	28.9%	30.8%	35.2%	39.5%	37.3%
2	44.6%	39.1%	41.8%	42.8%	43.5%	38.6%	32.8%	36.1%
3	16.9%	19.6%	13.7%	15.7%	13.8%	13.6%	11.7%	11.4%
4+	9.0%	14.1%	6.4%	7.6%	5.2%	6.4%	3.3%	5.8%

	Population Density (population per square mile)					
	[10000, 20000)		[20000, 50000)		[50000, ∞)	
	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.
Households	522	5,711	926	2,430	1,790	133
% Households in an MSA with Rail	84.3%	74.1%	100.0%	92.7%	100.0%	90.1%
Workers per Household	1.36	1.21	1.35	1.32	1.26	1.75
Vehicles per Household	1.30	1.35	0.80	1.11	0.40	0.82
Vehicles per Driver	0.71	0.81	0.48	0.67	0.27	0.54
% Households by Vehicle Ownership ^a						
0	21.9%	18.8%	43.1%	30.1%	67.9%	27.0%
1	40.1%	43.6%	38.4%	40.1%	25.6%	65.5%
2	28.0%	26.9%	15.0%	23.0%	5.4%	6.2%
3	6.9%	7.2%	2.3%	5.1%	1.0%	1.4%
4+	3.1%	3.5%	1.3%	1.8%	0.1%	0.0%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 4-6. 2001 household characteristics and vehicle ownership patterns by population density and region.

	Population Density (population per square mile)							
	[0, 500)		[500, 2000)		[2000, 4000)		[4000, 10000)	
	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.
Households	530	14,822	738	16,982	664	17,629	1,028	21,548
% Households in an MSA with Rail	33.4%	15.3%	42.6%	24.5%	55.0%	25.5%	60.5%	35.1%
Workers per Household	1.42	1.46	1.42	1.41	1.33	1.32	1.39	1.35
Vehicles per Household	2.08	2.28	1.96	2.02	1.77	1.85	1.66	1.71
Vehicles per Driver	1.05	1.15	1.03	1.06	0.98	1.05	0.96	1.00
% Households by Vehicle Ownership ^a								
0	4.0%	3.3%	3.2%	3.5%	7.5%	5.6%	10.2%	8.7%
1	24.0%	22.9%	30.2%	28.4%	33.7%	33.9%	35.6%	37.1%
2	45.0%	39.9%	44.1%	43.1%	40.7%	39.2%	37.4%	36.6%
3	18.0%	20.1%	15.9%	16.4%	12.9%	15.3%	12.6%	12.4%
4+	9.0%	13.8%	6.7%	8.6%	5.3%	6.0%	4.3%	5.2%

	Population Density (population per square mile)					
	[10000, 20000)		[20000, 50000)		[50000, ∞)	
	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.
Households	635	5,893	1,002	2,380	1,720	197
% Households in an MSA with Rail	78.6%	68.3%	99.9%	92.1%	100.0%	95.6%
Workers per Household	1.50	1.40	1.28	1.27	1.26	1.11
Vehicles per Household	1.37	1.45	0.78	1.11	0.47	0.58
Vehicles per Driver	0.80	0.93	0.61	0.84	0.44	0.62
% Households by Vehicle Ownership ^a						
0	21.8%	15.6%	45.0%	28.0%	61.5%	49.1%
1	38.2%	41.7%	36.6%	42.2%	32.5%	43.7%
2	26.5%	30.1%	15.2%	22.6%	5.2%	6.9%
3	10.2%	8.4%	2.5%	5.5%	0.6%	0.3%
4+	3.3%	4.2%	0.8%	1.8%	0.3%	—

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Figure 4-4 illustrates an increasing share of households as population density rises, a pattern consistent across all 3 years analyzed. However, there is no statistically significant difference in household share across most density categories, except for the highest density category. In areas in NYS with population density exceeding 50,000 people per square mile, the share of households in 2009 was statistically significantly higher than the share in 2001. Conversely, in the rest of the United States, 2001 showed a statistically significantly higher share than that in 2009. Notably, 2017 exhibited a relatively higher margin of error compared with other years, particularly in the highest-density areas of NYS.

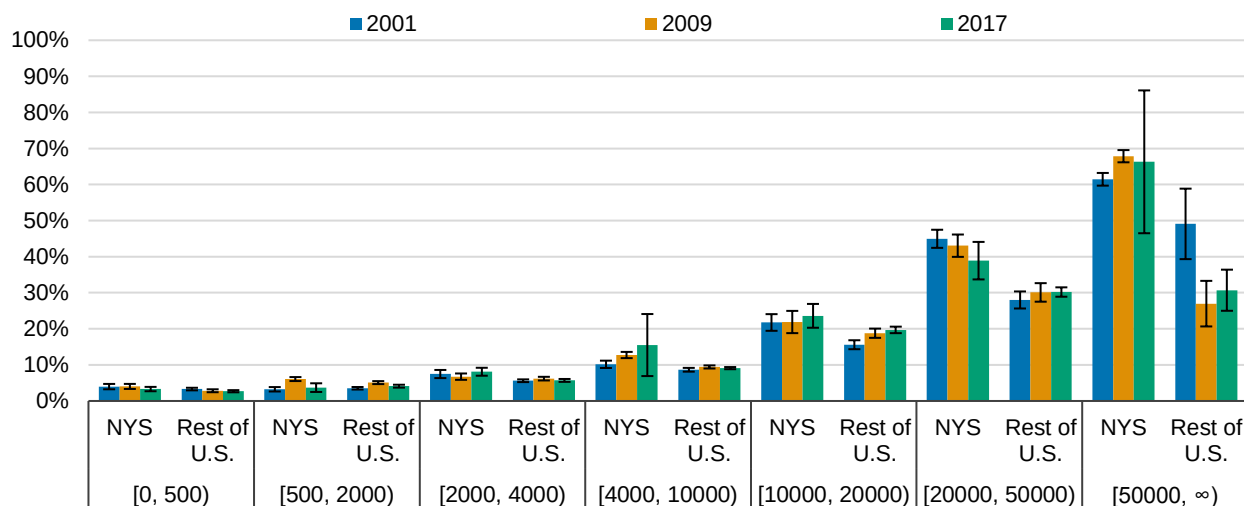


Figure 4-4. Share of households without a vehicle by year, region, and population density.

4.1.2 Driver Rate

NHTS respondents were asked for their driver status (i.e., “Are you a driver?”). Based on responses to that question, Figure 4-5 shows a decrease in the share of drivers among females and males over 16 years old as population density increased in 2017. Furthermore, the gap between NYS and the rest of the United States metros widened with increasing population density. For example, in sparsely populated areas with fewer than 500 people per square mile, the male driver percentage in NYS was 92%, compared with 93% in the rest of the United States. In contrast, in the densest areas with more than 50,000 people per square mile, the male driver percentage in NYS dropped to 67%, whereas it was 82% in the rest of the United States. This trend likely reflects the differences in transportation infrastructure and urban living patterns. In high-density areas such as those in NYS, public transit options are more prevalent and convenient, leading to reduced reliance on driving. Similar patterns can also be observed in 2009 (Figure 4-6) and 2001 (Figure 4-7).

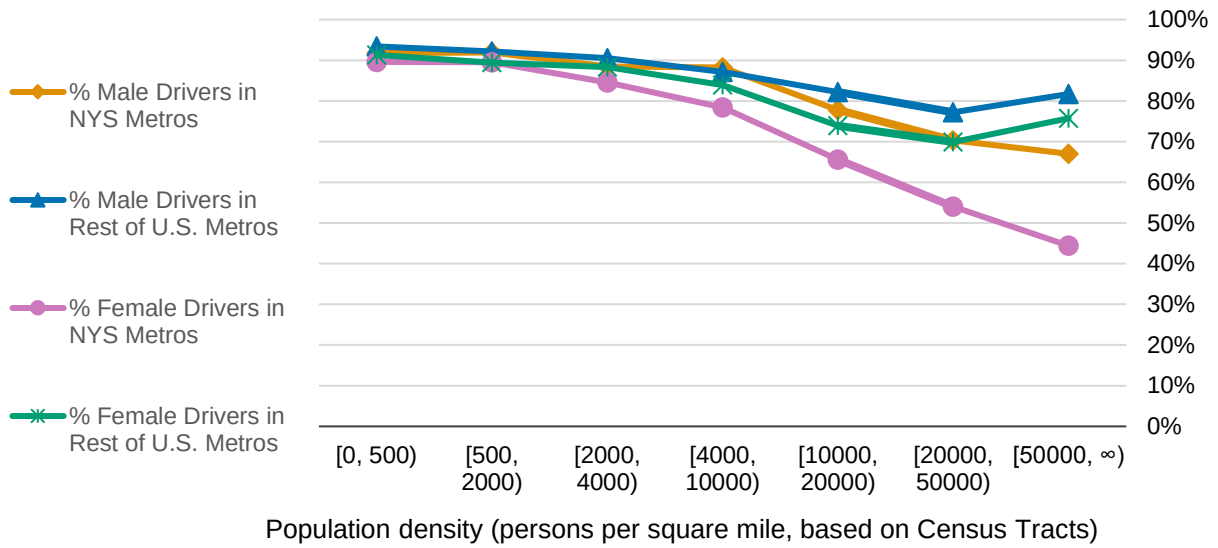


Figure 4-5. Driver rates by gender, region, and population density in 2017.

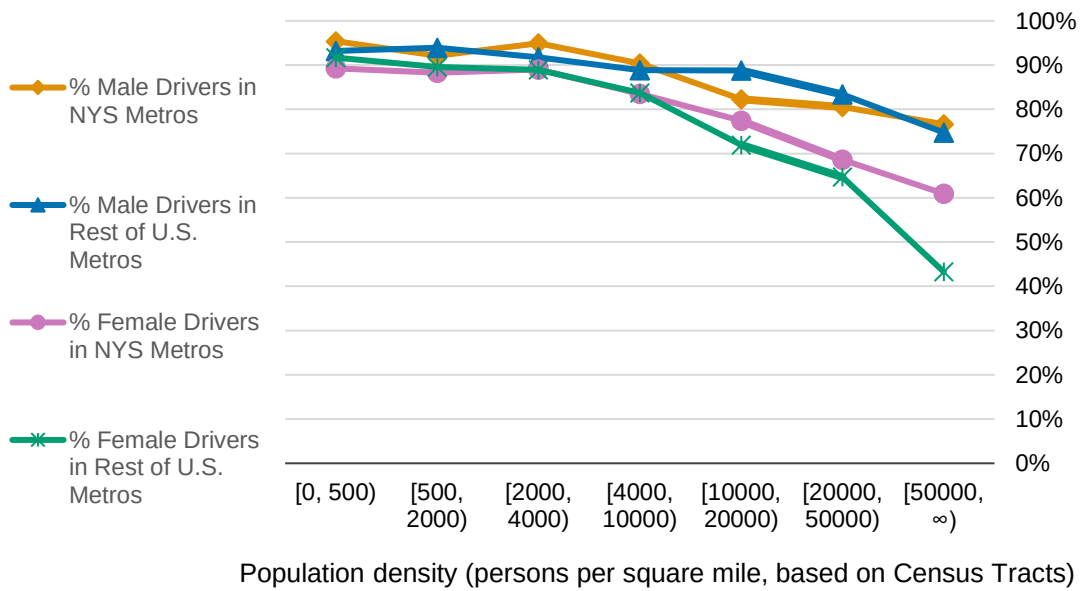


Figure 4-6. Driver rates by gender, region, and population density in 2009.

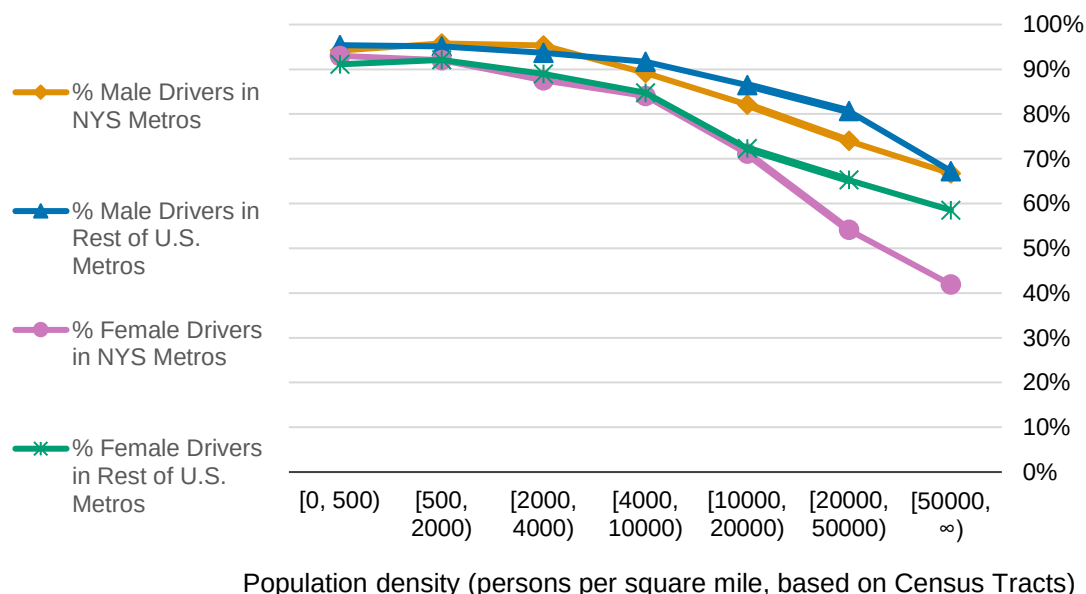


Figure 4-7. Driver rates by gender, region, and population density in 2001.

4.2 PERSONAL TRAVEL COMPARISONS

The daily person trip data reveal distinct patterns across population density, geography, and time, as shown in Figure 4-8, Figure 4-9, and Figure 4-10. When examining the effect of population density, both NYS and the rest of the United States metros show relatively consistent trip rates in low- to medium-density areas. However, in 2017, in very high-density areas (20,000+ people per square mile), the rest of the United States demonstrated statistically significantly higher trip rates than those in NYS. For example, in areas with 50,000+ people per square mile, the rest of the United States reported 4.28 trips per day, compared with NYS's 3.35 trips. This pattern, however, was not observed in 2009 or 2001.

Geographically, NYS exhibits more stable trip patterns across different density levels, whereas the rest of the United States shows greater variation, as reflected by the larger margin of error in very high-density areas (50,000+ people per square mile) across all 3 years. This suggests differing urban transportation dynamics, where NYS's dense urban infrastructure, such as in NYC, provides more consistent transportation options, whereas other US metropolitan areas exhibit greater variability that is likely due to regional differences in transit availability and urban design.

The temporal analysis reveals a downward trend in trip rates from 2001 to 2017 across both regions. This decline may reflect broader societal changes, such as increased telecommuting, shifts in lifestyle, or improved online services reducing the need for travel. It could also indicate rising transportation costs, congestion in urban areas, or policy interventions aimed at reducing travel demand. In 2001, trip rates peaked at over four daily trips in low- to medium-density areas in both regions. By 2009, these rates had moderated, maintaining similar patterns across density levels. By 2017, the data reveal the lowest trip rates in most density categories, except for the rest of the United States, which maintained higher trip rates in very dense areas—a shift from the more uniform patterns observed in earlier years.

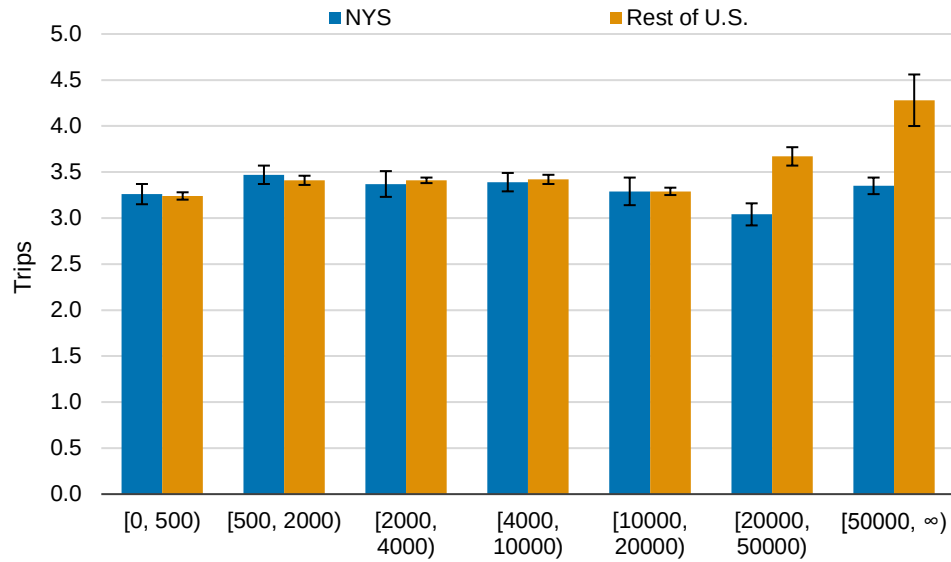


Figure 4-8. Person trips by region and population density in 2017.

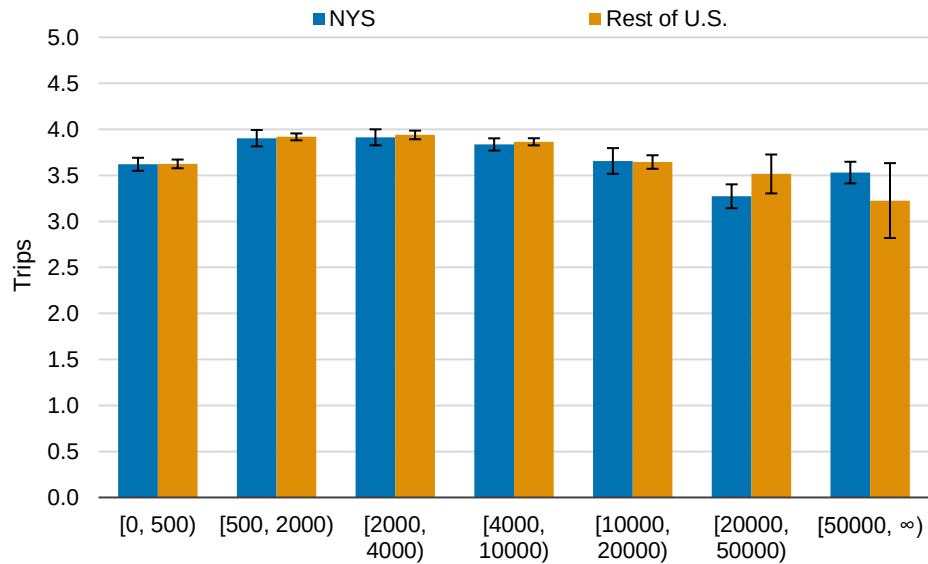


Figure 4-9. Person trips by region and population density in 2009.

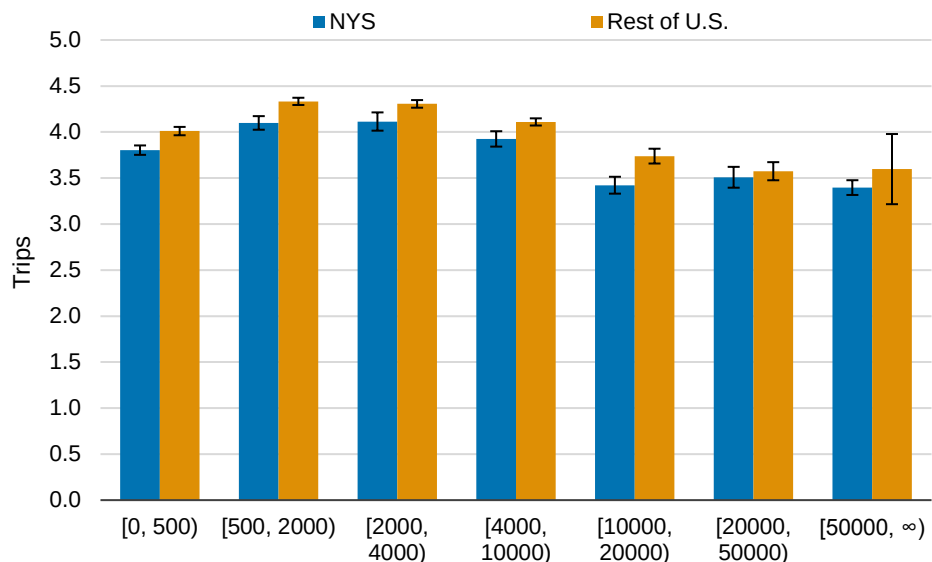


Figure 4-10. Person trips by region and population density in 2001.

NHTS reveals distinct temporal patterns in trip rates across gender and population density between two periods: 2001–2009 (Figure 4-11) and 2009–2017 (Figure 4-12). During 2001–2009, NYS showed relatively modest declines (–5%) in trip rates in low-density areas while experiencing some growth in higher-density areas, particularly in the 10,000–20,000 people per square mile density range (+7%). In contrast, the rest of the United States experienced more substantial decreases in trip rates across most density levels, with significant declines in very low-density (–10%) and very high-density (–10%) areas.

In the 2009–2017 period, more pronounced changes occurred, especially in high-density areas. Both regions experienced consistent decreases (–10% to –14%) in trip rates in lower-density areas. However, their patterns diverged in areas with population densities above 20,000 people per square mile. The rest of the United States saw increases in trip rates in these high-density areas (+33% overall, +49% for males, and +15% for females), whereas NYS continued to experience modest declines (–5% to –7%).

Gender differences also became more pronounced during this period. In the rest of the United States, males exhibited a higher rate of change in trip rates in the highest-density areas (50,000+ people per square mile) compared with females, highlighting a potential gender-based variation in travel behavior that warrants further investigation.

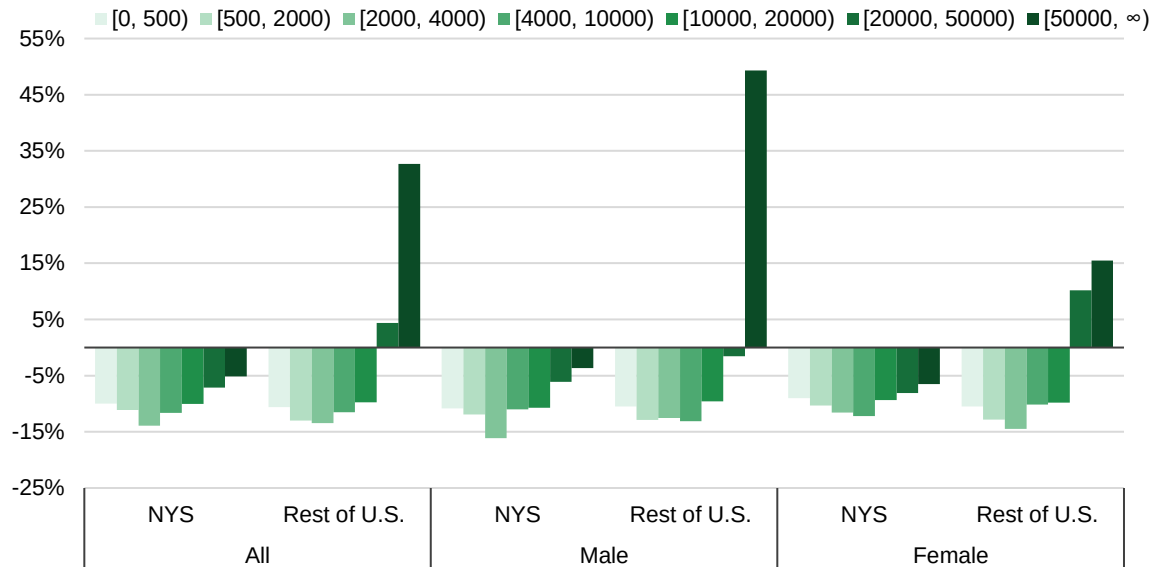


Figure 4-11. Percentage changes in person trip by gender, region, and population density from 2009 to 2017.

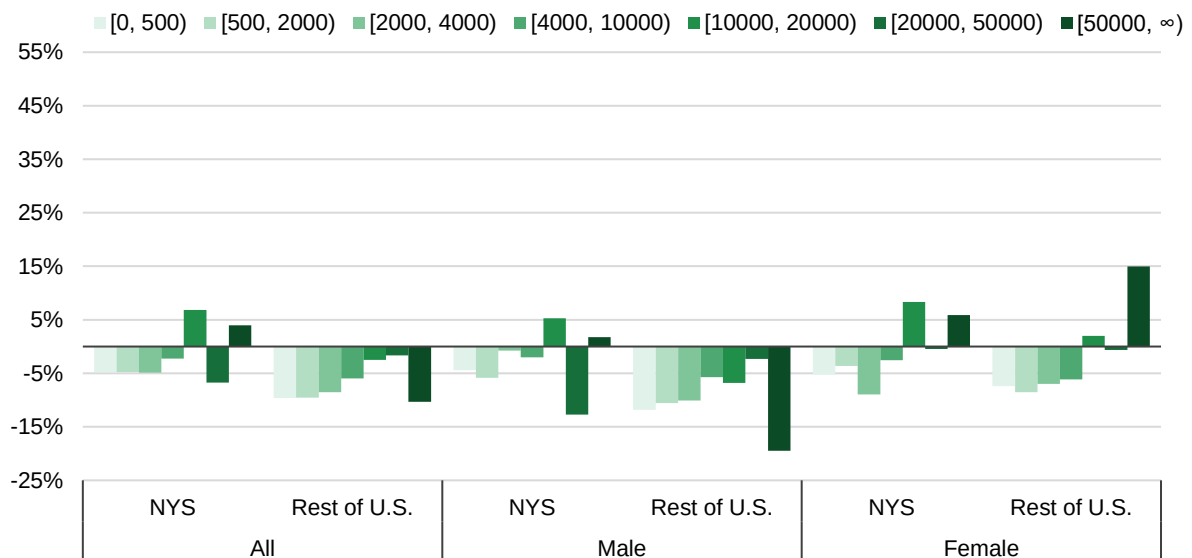


Figure 4-12. Percentage changes in person trip by gender, region, and population density from 2001 to 2009.

4.2.1 Trip Length

Looking at trip lengths across density levels and time, several key patterns emerge (Figure 4-13). Trip lengths consistently decreased as population density increased across all years, with the lowest density areas (fewer than 500 people per square mile) showing the longest trips (10–13 miles) and high-density areas generally having shorter trips (4–8 miles). This pattern holds true for both NYS and the rest of the United States. In particular, the trip length of the lowest-density area was statistically significantly higher than that of the second lowest density area across all years and regions, apart from 2017 NYS.

Comparing regions, the rest of the United States generally showed slightly longer trip lengths than those of NYS (though not always statistically significant) across most density categories. In 2017, NYS showed some unusual spikes in trip lengths, especially in the 2,000–4,000 density range (14.51 miles), although this was with a high margin of error.

From 2009 to 2017, both regions experienced increases in trip lengths in lower-density areas. The most changes occurred in the highest-density areas (50,000+), where trip lengths in NYS increased statistically significantly from 4.93 to 8.02 miles and in the rest of the United States showed a statistically significant increase from 2.59 to 7.55 miles.

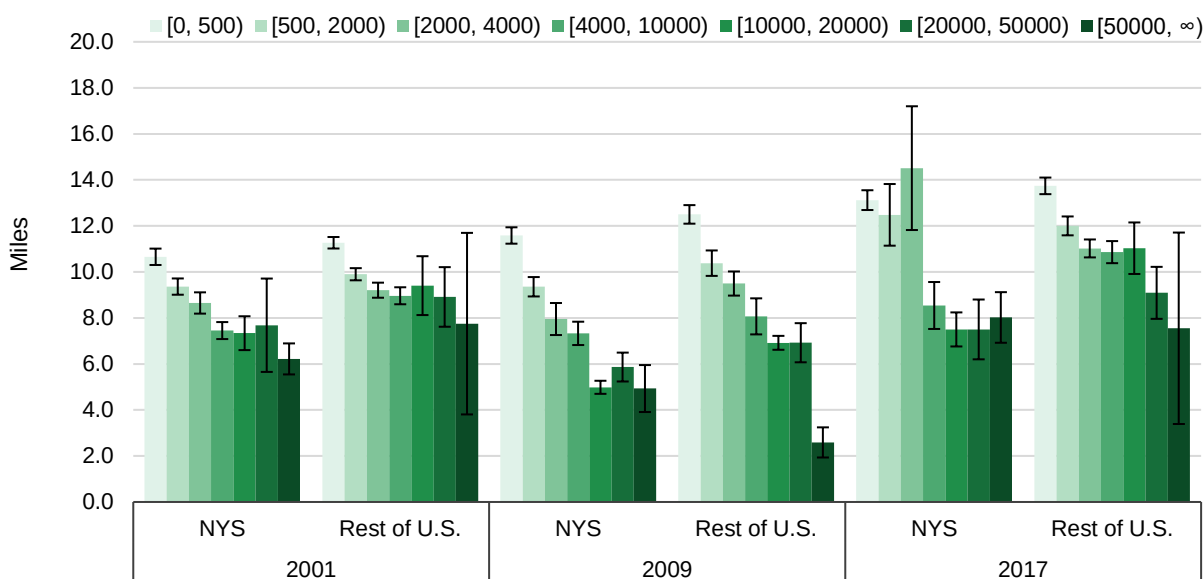


Figure 4-13. Average person trip lengths by year, region, and population density.

The trip length changes between 2001–2009 (Figure 4-14) and 2009–2017 (Figure 4-15) show some different patterns. From 2001 to 2009, both regions experienced decreases in trip lengths in higher-density areas, with particularly sharp declines in areas of more than 10,000 people per square mile. The rest of the United States exhibited the most significant decrease (–67%) in the highest density areas (50,000+ people per square mile) while maintaining slight increases in lower-density areas.

In contrast, the 2009–2017 period showed a reversal, with substantial increases in trip lengths across all density levels. NYS recorded an 82% increase in trip lengths in the 2,000–4,000-density range, whereas the rest of the United States showed a 192% increase in the highest density areas (50,000+ people per square mile). Both regions also experienced moderate growth in medium-density areas, with increases ranging from 30% to 60%. These shifts could reflect changes in urban transportation dynamics and land use patterns. For instance, the decrease in trip lengths from 2001 to 2009 in higher-density areas may align with increased accessibility to services, employment, or amenities within compact urban centers. The reversal from 2009 to 2017 may indicate evolving travel behaviors, possibly because of urban sprawl, increased commuting distances, or shifts in economic activity to peripheral areas.

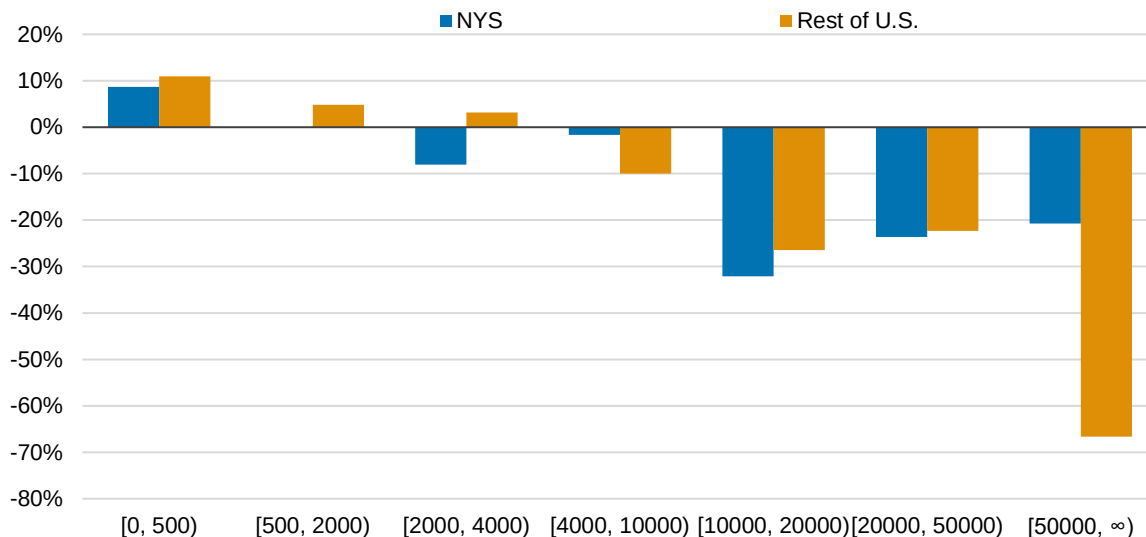


Figure 4-14. Percentage differences in trip length from 2001 to 2009 by region and population density.

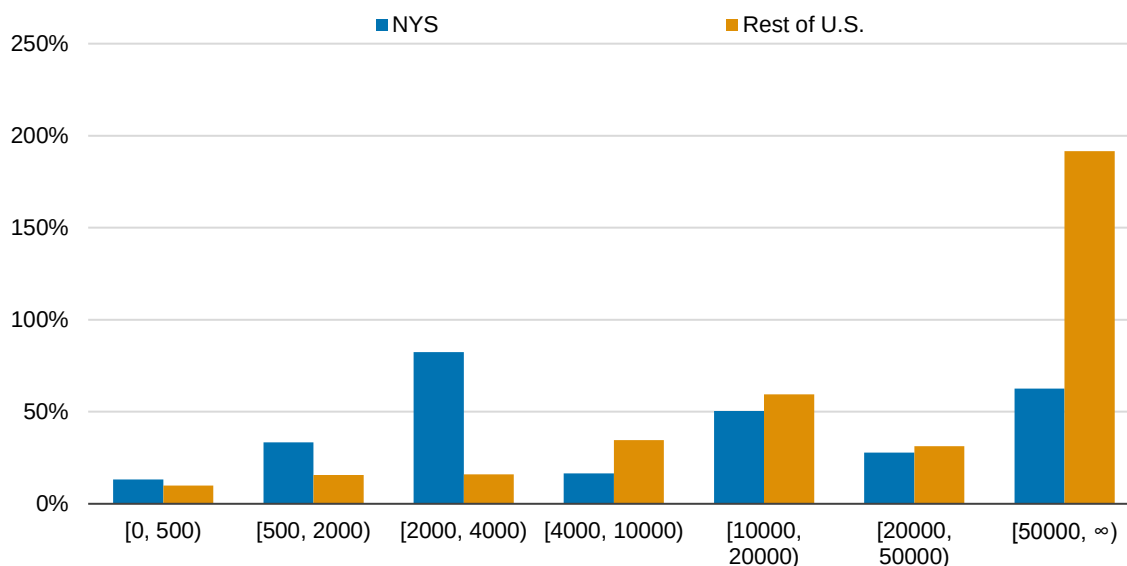


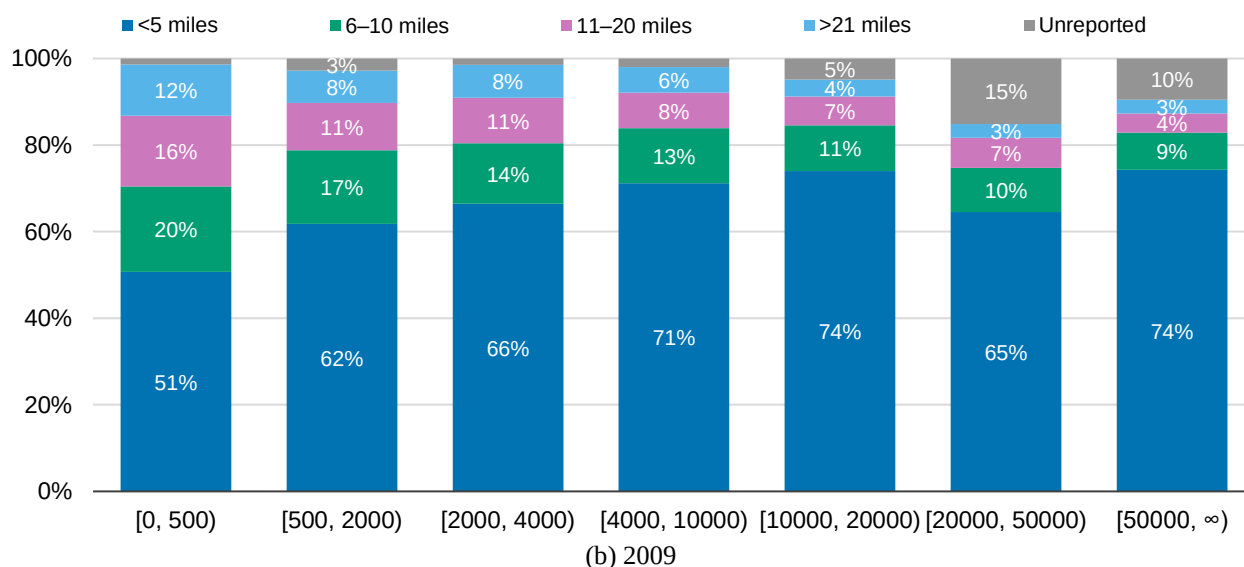
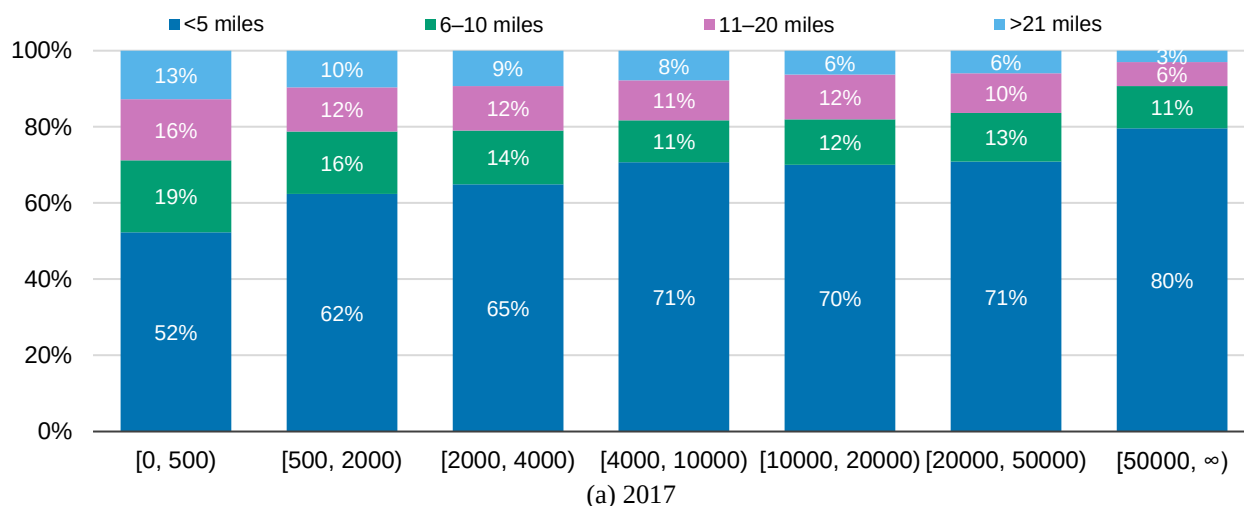
Figure 4-15. Percentage differences in trip length from 2009 to 2017 by region and population density.

Figure 4-16 shows the trip length distribution data for NYS in 2001, 2009, and 2017. The data show consistent patterns across all density levels, with short trips dominating but significant changes occurring over time. In 2017, the proportion of trips under 5 miles increased steadily with density, from 52% in low-density areas to 80% in the highest density areas. This pattern remains stable across all three time periods.

Comparing 2001, 2009, and 2017 reveals the increasing concentration of shorter trips in higher-density areas. In the highest-density area, the share of trips under 5 miles increased from 71% in 2001 to 80% in 2017. Long-distance trips (21+ miles) maintain their highest share in low-density areas (11%–13%), gradually decreasing as density increases. One notable observation occurs in the share of unreported

trips—whereas 2017 shows virtually no unreported trips, 2009 and 2001 had significant percentages, particularly in high-density areas (reaching 10%–15% in 2009 for the highest density categories).

Medium-length trips (6–10 and 11–20 miles) show gradual declines as density increases across all years, though the pattern is most pronounced in 2017. The improved reporting quality in 2017 (0% unreported across all densities) suggests more reliable data compared with that of previous years.



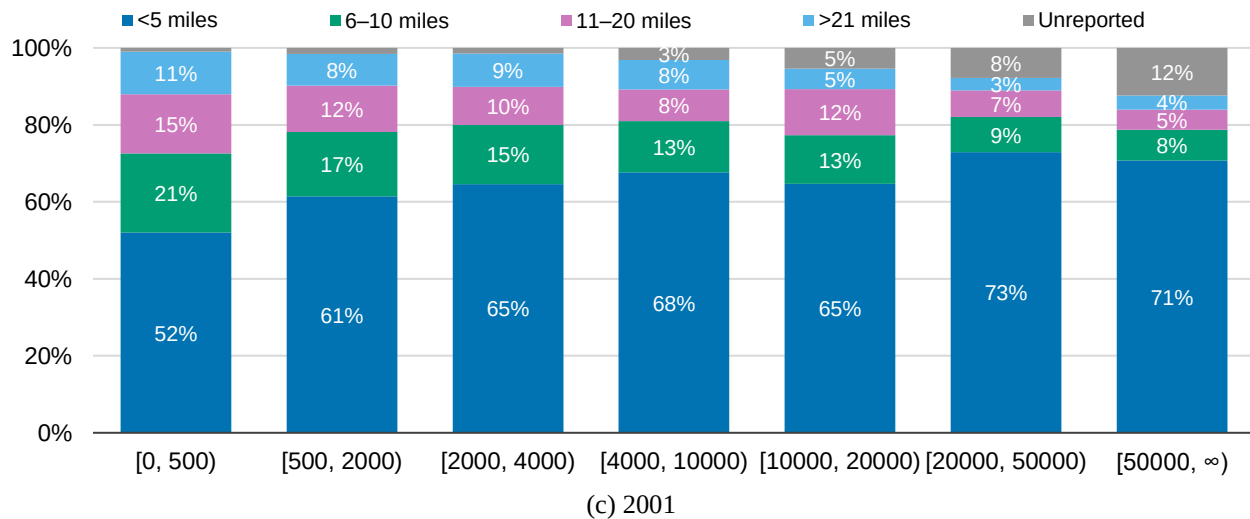


Figure 4-16. Distribution of person trips by trip length, year, and population density. Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number. Unlabeled segments represent values lower than 3%.

4.2.2 PMT Pattern

The PMT pattern (Figure 4-17) shows that in low-density areas (<500 people per square mile), both NYS and the rest of the United States maintain relatively stable PMT (40–45 miles) across all years. However, medium-density areas show diverging trends. For example, NYS’s PMT increased statistically significantly from 30 miles in 2009 to almost 50 miles in 2017 in 2,000–4,000 density areas, whereas the rest of the United States remained stable.

High-density areas (>20,000 people per square mile) show the most volatility. NYS experienced a statistically significant decline from 2001 to 2009 (24.8 to 16.3 miles in 20,000–50,000 density areas) before rebounding in 2017 (22.7 miles). The rest of the United States showed similar patterns but with even greater fluctuations, particularly in the highest density areas where PMT dropped in 2009 (7.1 miles) before recovering strongly in 2017 (32.3 miles).

Table 4-7 and Table 4-8 show person trip statistics in 2017. The mode referred to as “Other” includes commercial and private air, bus (e.g., commuter, school, charter, city to city), water, taxi, limousine, airport shuttle, bicycle, and other nonspecified vehicles. Analysis of Table 4-7’s mode share data in 2017 showed that SOV and MOV trips dominated in low-density areas (combined 84% in NYS and 91% in the rest of the United States) but decreased dramatically in high-density areas (combined 18% in NYS and 35% in the rest of the United States). Walking increased significantly with density (ranging from ~5%–8% to 44%–48% in the highest density areas). Subway use shows marked differences between regions: NYS reaches 19.2% in highest density areas versus 6.9% for the rest of the United States.

Table 4-8

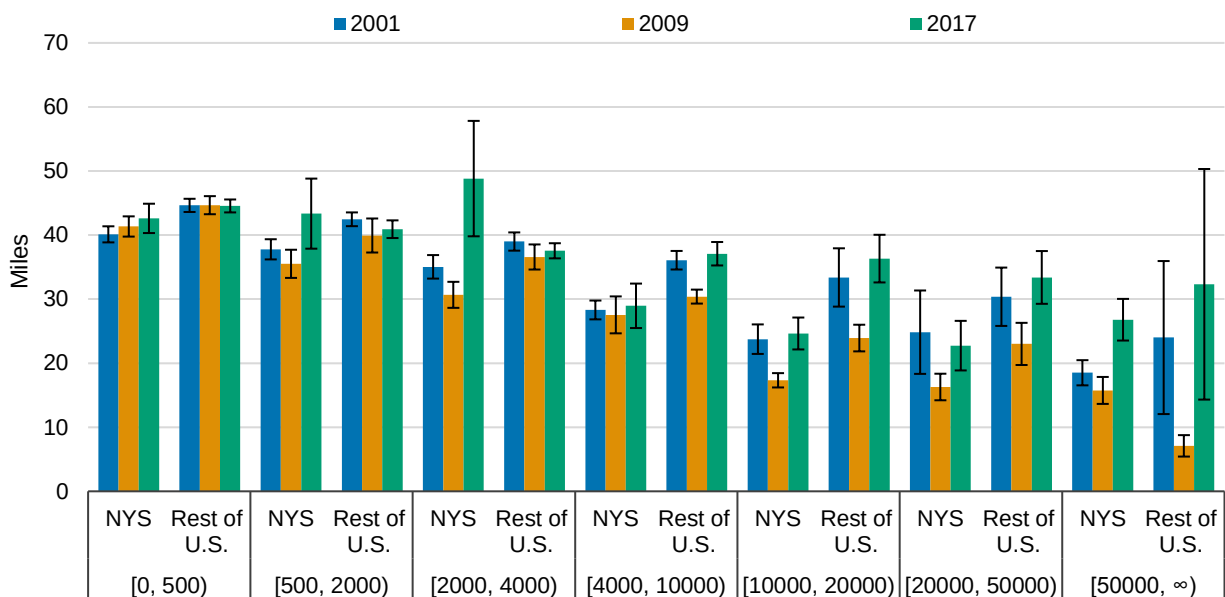
**Figure 4-17. PMT statistics by year, region, and population density.**

Table 4-8's PMT data show that personal vehicle modes including SOV and MOV accounted for most miles in low-density areas (combined 83% in NYS and 89% in the rest of the United States) in 2017. In highest density areas, the "Other" mode became dominant (44% in NYS and 48% in the rest of the United States). Rail modes showed some regional differences: NYS had a higher subway/rail PMT share in the densest areas (25% combining Amtrak, subway, and other public) than that of the rest of the United States (12%). The PMT share for walking remained relatively low (<3% in both NYS and the rest of the United States) even in the highest density areas despite high trip share, reflecting shorter trip lengths.

Table 4-9 to Table 4-12 show the person trip and PMT statistics in 2009 and 2001. Comparing mode share patterns between 2009 and 2017 reveals shifts in travel behavior across density levels. Private vehicle usage (SOV and MOV combined) declined in several density categories in NYS (2009: 87% in lowest density to 21% in highest; 2017: 84% to 18%). Walking maintained its dominant position in the highest density areas at around 47% of person trips in NYS. Public transit patterns diverged between regions. NYS saw substantial growth in subway utilization, particularly in the highest density areas, where it increased statistically significantly from 11% to 19%, while the rest of the United States showed more modest gains from 4.7% to 6.9% that were not statistically significant. PMT patterns also underwent some changes, especially in high-density areas. Total PMT in the highest density areas increased substantially, with NYS rising statistically significantly from 15.75 to 26.8 miles and the rest of the United States showing a statistically significant increase from 7.10 to 32.3 miles. The composition of these miles also shifted, with private vehicle modes (SOV and MOV) accounting for a smaller share of PMT in high-density areas in 2017. Transit PMT share evolved differently between regions, increasing slightly in NYS while declining slightly in the rest of the United States. Average trip lengths increased across most density levels between 2009 and 2017, with particularly notable increases in NYS's mid-density areas. This trend, combined with the shifts in mode share, suggests a complex evolution in travel patterns that varies significantly by both density level and region.

Table 4-7. Person trip statistics in 2017 by region and population density.

	Population Density (population per square mile)							
	[0, 500)		[500, 2000)		[2000, 4000)		[4000, 10000)	
	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.
Person Trips	3.26	3.24	3.47	3.41	3.37	3.41	3.39	3.42
Mode (%) ^a								
SOV	40.1%	41.8%	41.3%	41.3%	40.8%	40.7%	35.1%	37.9%
MOV	44.3%	49.0%	44.6%	46.2%	43.3%	45.4%	40.0%	43.9%
Amtrak/Commuter Rail	0.3%	0.0%	0.7%	0.1%	1.1%	0.2%	1.4%	0.2%
Subway/Elevated Rail	0.1%	0.0%	0.4%	0.1%	0.2%	0.2%	0.9%	0.5%
Other Public	0.4%	0.2%	0.6%	0.7%	1.0%	1.0%	2.3%	1.8%
Walk	8.2%	5.3%	7.1%	7.0%	9.9%	8.4%	15.0%	11.7%
Other	6.5%	3.7%	5.2%	4.6%	3.8%	4.0%	5.3%	3.9%
Day of Week								
Weekday	3.31	3.17	3.53	3.43	3.29	3.29	3.43	3.44
Weekend	3.15	3.47	3.36	3.40	3.58	3.72	3.32	3.39
Average Person Trip Length	13.12	13.74	12.48	12.00	14.51	11.02	8.54	10.86

	Population Density (population per square mile)					
	[10000, 20000)		[20000, 50000)		[50000, ∞)	
	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.
Person Trips	3.29	3.29	3.04	3.67	3.35	4.28
Mode (%) ^a						
SOV	25.3%	31.5%	12.6%	20.5%	6.0%	13.1%
MOV	36.0%	37.7%	26.7%	27.3%	12.2%	22.0%
Amtrak/Commuter Rail	1.1%	0.2%	0.9%	1.1%	1.1%	0.7%
Subway/Elevated Rail	6.9%	2.3%	15.3%	5.0%	19.2%	6.9%
Other Public	4.6%	4.9%	6.9%	5.8%	6.9%	7.8%
Walk	20.7%	18.3%	32.5%	31.6%	47.8%	44.2%
Other	5.3%	5.1%	5.1%	8.6%	6.9%	5.3%
Day of Week						
Weekday	3.30	3.24	2.88	3.61	3.35	4.45
Weekend	3.29	3.45	3.45	3.87	3.37	3.92
Average Person Trip Length	7.50	11.03	7.50	9.09	8.02	7.55

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 4-8. PMT by mode in 2017 by region and population density.

Mode ^a	Population Density (population per square mile)							
	[0, 500)		[500, 2000)		[2000, 4000)		[4000, 10000)	
	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.
PMT	42.6	44.6	43.4	40.9	48.8	37.6	29.0	37.1
SOV	36.4%	36.9%	36.8%	35.2%	23.7%	33.6%	33.3%	29.2%
MOV	47.1%	52.2%	38.6%	45.6%	36.5%	41.7%	42.4%	44.5%
Amtrak/Commuter Rail	1.7%	0.1%	2.2%	0.5%	2.6%	0.8%	4.7%	0.8%
Subway/Elevated Rail	0.1%	0.1%	0.8%	0.2%	0.2%	0.3%	1.8%	0.7%
Other Public	0.2%	0.2%	0.5%	0.7%	0.3%	1.0%	1.7%	1.1%
Walk	0.7%	0.3%	0.5%	0.8%	0.4%	0.7%	8.0%	0.7%
Other	13.9%	10.2%	20.8%	17.1%	36.3%	22.1%	8.2%	22.8%
Unreported	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.1%

Mode ^a	Population Density (population per square mile)					
	[10000, 20000)		[20000, 50000)		[50000, ∞)	
	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.
PMT	24.6	36.3	22.7	33.4	26.8	32.3
SOV	27.3%	23.2%	12.2%	19.3%	7.2%	12.8%
MOV	39.9%	35.7%	37.2%	41.0%	20.4%	25.0%
Amtrak/Commuter Rail	2.8%	1.0%	2.2%	7.5%	4.9%	1.5%
Subway/Elevated Rail	11.6%	1.8%	20.8%	3.2%	17.2%	5.3%
Other Public	4.3%	2.6%	4.2%	3.1%	3.0%	5.6%
Walk	5.9%	1.0%	2.0%	1.8%	2.9%	2.2%
Other	8.2%	34.8%	21.4%	24.2%	44.4%	47.5%
Unreported	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 4-9. Person trip statistics in 2009 by region and population density.

	Population Density (population per square mile)							
	[0, 500)		[500, 2000)		[2000, 4000)		[4000, 10000)	
	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.
Person Trips	3.62	3.62	3.90	3.92	3.91	3.94	3.84	3.86
Mode (%) ^a								
SOV	45.3%	41.9%	45.9%	41.0%	44.2%	41.2%	40.9%	39.9%
MOV	41.8%	46.9%	38.4%	46.3%	38.8%	43.9%	37.8%	43.4%
Amtrak	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Commuter Train	0.1%	0.0%	0.4%	0.1%	0.6%	0.1%	0.6%	0.2%
Subway/Elevated Rail	0.1%	0.0%	0.2%	0.0%	0.3%	0.1%	0.6%	0.1%
Other Public	0.2%	0.1%	0.6%	0.3%	0.4%	0.7%	1.5%	1.4%
Walk	6.9%	6.8%	8.5%	7.7%	10.4%	9.8%	14.0%	11.3%
Other	5.5%	4.1%	5.9%	4.5%	5.1%	4.1%	4.6%	3.4%
Unreported	0.2%	0.2%	0.1%	0.2%	0.3%	0.1%	0.1%	0.3%
Day of Week								
Weekday	3.42	3.19	3.50	3.68	3.73	3.75	3.41	3.60
Weekend	3.69	3.81	4.07	4.01	4.00	4.01	3.96	3.97
Average Person Trip Length	11.58	12.50	9.35	10.38	7.95	9.49	7.33	8.07

	Population Density (population per square mile)					
	[10000, 20000)		[20000, 50000)		[50000, ∞)	
	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.
Person Trips	3.66	3.65	3.27	3.52	3.53	3.23
Mode (%) ^a						
SOV	33.3%	32.4%	15.1%	24.2%	6.0%	19.3%
MOV	33.3%	38.3%	24.8%	30.6%	15.2%	5.6%
Amtrak	0.1%	0.1%	1.1%	0.1%	0.2%	0.4%
Commuter Train	0.6%	0.1%	1.9%	1.1%	1.7%	0.6%
Subway/Elevated Rail	2.0%	0.4%	10.5%	2.5%	11.0%	4.7%
Other Public	7.4%	4.4%	11.0%	9.4%	10.9%	16.9%
Walk	18.4%	19.0%	30.5%	26.6%	47.6%	47.9%
Other	4.4%	5.1%	4.9%	5.4%	7.3%	4.7%
Unreported	0.4%	0.3%	0.2%	0.1%	0.0%	0.0%
Day of Week						
Weekday	3.62	3.45	3.14	3.07	3.30	4.16
Weekend	3.67	3.72	3.33	3.71	3.62	3.10
Average Person Trip Length	4.98	6.92	5.86	6.92	4.93	2.59

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent

Table 4-10. PMT by mode in 2009 by region and population density.

Mode ^a	Population Density (population per square mile)							
	[0, 500)		[500, 2000)		[2000, 4000)		[4000, 10000)	
	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.
PMT	41.34	44.67	35.52	39.94	30.67	36.58	27.54	30.40
SOV	43.4%	41.5%	41.8%	39.3%	43.7%	36.2%	37.1%	38.1%
MOV	47.9%	51.5%	45.1%	45.5%	40.2%	52.2%	37.4%	47.5%
Amtrak	0.0%	0.0%	0.1%	0.1%	0.1%	0.1%	0.0%	0.1%
Commuter Train	0.4%	0.0%	1.4%	0.1%	2.6%	0.3%	1.8%	0.2%
Subway/Elevated Rail	0.2%	0.0%	0.3%	0.0%	0.3%	0.1%	0.7%	0.2%
Other Public	0.1%	0.1%	0.4%	0.2%	0.3%	0.5%	1.1%	1.3%
Walk	0.4%	0.4%	0.6%	0.5%	1.0%	0.7%	1.1%	1.0%
Other	7.5%	6.5%	10.2%	14.2%	11.4%	9.8%	20.9%	11.5%
Unreported	0.1%	0.1%	0.1%	0.1%	0.4%	0.1%	0.0%	0.2%

Mode ^a	Population Density (population per square mile)					
	[10000, 20000)		[20000, 50000)		[50000, ∞)	
	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.
PMT	17.33	23.93	16.29	23.01	15.75	7.10
SOV	37.8%	38.2%	16.2%	34.4%	11.6%	38.7%
MOV	38.5%	46.6%	43.4%	38.3%	44.5%	7.3%
Amtrak	2.5%	0.3%	0.3%	0.2%	0.1%	0.2%
Commuter Train	2.0%	0.2%	1.6%	11.4%	3.3%	4.5%
Subway/Elevated Rail	2.3%	0.3%	13.4%	2.7%	15.1%	4.8%
Other Public	7.5%	2.7%	6.9%	5.9%	9.5%	17.7%
Walk	2.7%	2.0%	4.6%	2.7%	8.6%	14.8%
Other	6.7%	9.5%	13.2%	4.3%	7.3%	12.0%
Unreported	0.1%	0.1%	0.5%	0.0%	0.0%	0.0%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 4-11. Person trip statistics in 2001 by region and population density.

	Population Density (population per square mile)							
	[0, 500)		[500, 2000)		[2000, 4000)		[4000, 10000)	
	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.
Person Trips	3.80	4.01	4.10	4.33	4.11	4.31	3.93	4.11
Mode (%) ^a								
SOV	42.1%	41.8%	44.2%	40.8%	39.9%	42.3%	39.0%	39.0%
MOV	44.2%	48.7%	43.0%	49.0%	46.3%	46.7%	41.6%	47.0%
Amtrak	0.0%	0%	0.1%	0%	0.1%	0.1%	0.2%	0.0%
Commuter Train	0.2%	0.0%	0.2%	0.0%	0.6%	0.1%	0.9%	0.2%
Subway/Elevated Rail	0.4%	0.0%	0.3%	0.1%	0.5%	0.1%	0.8%	0.2%
Other Public	0.7%	0.1%	0.1%	0.2%	0.5%	0.5%	1.2%	1.3%
Walk	7.2%	5.5%	7.3%	5.9%	7.7%	7.5%	11.6%	9.7%
Other	5.1%	3.8%	4.6%	3.8%	4.4%	2.6%	4.7%	2.6%
Unreported	0.1%	0.1%	0.1%	0.1%	0.1%	0.1%	0.1%	0.1%
Day of Week								
Weekday	3.59	3.69	3.96	4.20	3.98	4.01	3.78	3.97
Weekend	3.89	4.14	4.15	4.38	4.17	4.43	3.98	4.17
Average Person Trip Length	10.66	11.27	9.36	9.90	8.65	9.21	7.45	8.96

	Population Density (population per square mile)					
	[10000, 20000)		[20000, 50000)		[50000, ∞)	
	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.
Person Trips	3.42	3.74	3.51	3.57	3.40	3.60
Mode (%) ^a						
SOV	32.5%	33.8%	18.2%	27.6%	8.0%	22.4%
MOV	34.6%	41.9%	26.8%	38.0%	15.6%	15.3%
Amtrak	0.5%	0.1%	0.3%	0.0%	0.4%	0.0%
Commuter Train	0.4%	0.1%	0.5%	0.2%	0.4%	0.0%
Subway/Elevated Rail	4.8%	0.8%	10.0%	1.7%	16.1%	5.0%
Other Public	4.0%	3.9%	7.4%	7.7%	10.3%	19.5%
Walk	19.3%	16.2%	32.4%	20.4%	43.9%	34.0%
Other	3.7%	3.4%	4.3%	4.3%	5.2%	3.8%
Unreported	0.2%	0.0%	0.1%	0.1%	0.1%	0.1%
Day of Week						
Weekday	3.17	3.84	3.03	3.40	3.00	2.63
Weekend	3.52	3.70	3.70	3.65	3.58	3.93
Average Person Trip Length	7.34	9.40	7.68	8.91	6.22	7.75

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 4-12. PMT by mode in 2001 by region and population density.

Mode ^a	Population Density (population per square mile)							
	[0, 500)		[500, 2000)		[2000, 4000)		[4000, 10000)	
	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.
PMT	40.12	44.64	37.78	42.47	35.05	39.00	28.31	36.08
SOV	42.5%	40.3%	41.5%	38.5%	39.7%	39.6%	36.8%	36.6%
MOV	49.3%	53.7%	44.8%	49.6%	49.0%	50.7%	47.4%	49.2%
Amtrak	0.6%	0.0%	0.4%	0.0%	0.2%	0.0%	0.6%	0.2%
Commuter Train	0.7%	0.1%	0.7%	0.2%	2.2%	0.1%	2.9%	0.5%
Subway/Elevated Rail	0.5%	0.0%	0.3%	0.2%	0.5%	0.1%	1.5%	0.2%
Other Public	0.6%	0.1%	0.2%	0.2%	0.4%	0.4%	1.0%	1.0%
Walk	0.6%	0.4%	0.7%	0.5%	0.7%	0.6%	1.3%	0.8%
Other	5.2%	5.4%	11.4%	10.7%	7.1%	6.3%	6.8%	11.6%
Unreported	0.0%	0.2%	0.1%	0.1%	0.2%	2.2%	1.8%	0.1%

Mode ^a	Population Density (population per square mile)					
	[10000, 20000)		[20000, 50000)		[50000, ∞)	
	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.
PMT	23.75	33.39	24.84	30.37	18.51	24.00
SOV	31.4%	27.9%	16.3%	24.5%	13.0%	22.8%
MOV	47.9%	41.8%	20.9%	42.7%	29.4%	58.0%
Amtrak	0.9%	0.4%	0.2%	0.0%	1.7%	0.0%
Commuter Train	1.4%	0.3%	0.9%	0.2%	0.5%	0.0%
Subway/Elevated Rail	6.9%	0.7%	9.8%	1.1%	18.2%	3.6%
Other Public	3.5%	2.3%	4.1%	3.9%	6.5%	12.0%
Walk	2.2%	1.2%	3.4%	1.5%	4.9%	2.9%
Other	5.7%	25.4%	44.5%	26.2%	24.1%	0.7%
Unreported	0.2%	0.0%	0.0%	0.0%	1.9%	0.0%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

4.2.3 Public Transit, Walk, and Cycle Trips

Across all years, public transit use is consistently higher in NYS than in the rest of the United States, with the most notable differences occurring in higher-density areas (Figure 4-18). In low-density regions (<500 people per square mile), transit use is marginal in both NYS and the rest of the United States, with values below 1.5%. However, in the highest-density areas (50,000+ people per square mile), public transit accounted for a substantial share of daily trips, reaching 27.2% in NYS in 2017. This is significantly higher than the 15.4% observed in the rest of the United States for the same density category. The difference is statistically significant. For areas with moderate density (10,000–20,000 people per square mile), transit use in NYS was 12.7% in 2017, notably higher than the 7.4% in the rest of the US. The difference exceeds the combined margin of error (2.3%), indicating a statistically significant gap. Similarly, for 20,000–50,000 density areas, transit use in NYS remained consistently higher than transit use in the rest of the country throughout the years.

From 2001 to 2017, public transit use showed mixed trends across different density levels. In high-density areas (50,000+ people per square mile), NYS saw a relatively stable trend, with transit use rising slightly from 26.8% in 2001 to 27.2% in 2017. However, in the rest of the United States, transit use in these high-density areas declined significantly from 24.4% in 2001 to 15.4% in 2017. For medium-density areas (10,000–20,000 people per square mile), public transit use in NYS increased from 9.2% in 2001 to 12.7% in 2017, with a statistically significant jump after 2009. In contrast, the rest of the United States saw a much more modest increase from 4.7% to 7.4% over the same period. In low-density areas (<500 people per square mile), transit use in NYS fluctuated, dropping from 1.3% in 2001 to 0.4% in 2009, before recovering to 0.9% in 2017. A similar but even lower trend was observed in the rest of the United States, where transit use stayed below 0.3% across all years.

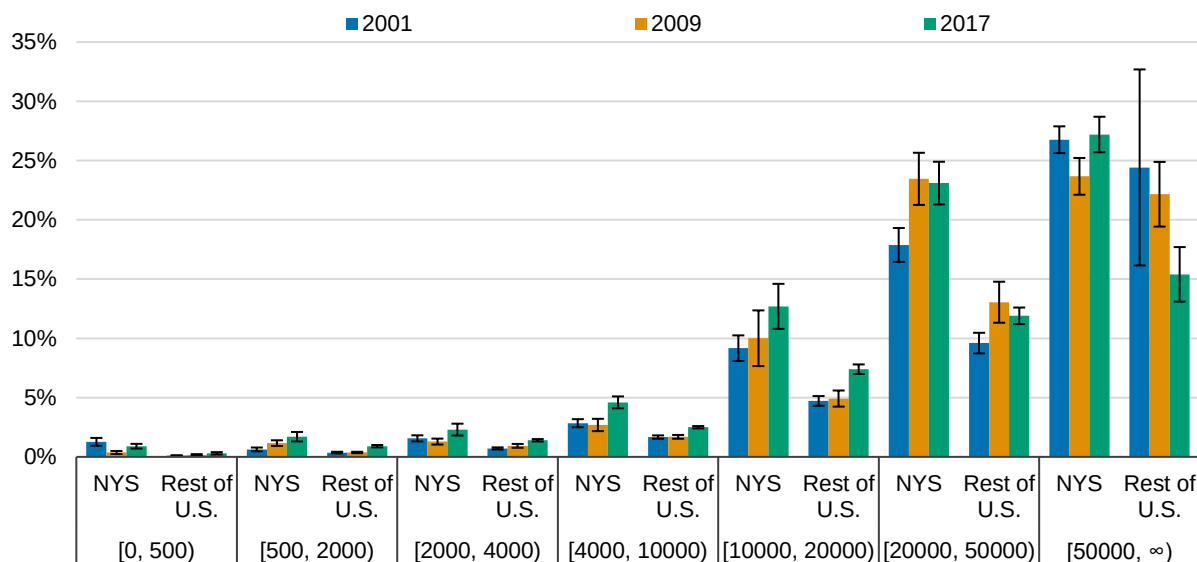


Figure 4-18. Share of person trips via public transit by year, region, and population density.

The walking trip percentages of daily person trips by population density, as presented in Figure 4-19, shows the trends across NYS and the rest of the United States from 2001 to 2017. In 2001, NYS exhibited a higher percentage of daily person trips taken by walking compared with the rest of the United States across most population density categories. In the highest density areas (50,000+ persons per square mile), 43.9% of trips in NYS were walking trips, compared with 34.0% in the rest of the United States. NYS's higher percentage in this category is statistically significant. In the 20,000–50,000 persons per square mile category, NYS recorded 32.4%, statistically significantly higher than the 20.4% in the rest of the United States. In the 10,000–20,000 persons per square mile category, NYS had 19.3%, compared with 16.2% in the rest of the United States. This difference is not statistically significant. A similar trend appears in the 4,000–10,000 persons per square mile category, where NYS recorded 11.6%, whereas the rest of the United States recorded 9.7%, though the difference is not statistically significant. At lower population densities, NYS had 7.7% walking trips in the 2,000–4,000 persons per square mile category, compared with 7.5% in the rest of the United States. This difference is not statistically significant. Similarly, in the 500–2,000 persons per square mile category, NYS had 7.3%, compared with 5.9% in the rest of the United States, though the difference is not statistically significant. In the lowest density category (fewer than 500 persons per square mile), NYS recorded 7.2%, slightly higher than 5.5% in the rest of the United States. However, this difference is not statistically significant.

In 2009, walking trips increased in some categories while declining in others. In the highest density areas (50,000+ persons per square mile), NYS saw an increase to 47.6%, whereas the rest of the United States also rose to 47.9%. The change is not statistically significant for either region. In 20,000–50,000 persons per square mile areas, NYS declined slightly to 30.5%, whereas the rest of the United States increased slightly to 26.6%. In 10,000–20,000 persons per square mile areas, NYS dropped slightly to 18.4%, whereas the rest of the United States increased slightly to 19.0%. In the 4,000–10,000 persons per square mile category, NYS increased slightly to 14.0%. In the 2,000–4,000 persons per square mile category, NYS saw a statistically significant increase to 10.4%, slightly higher than the 9.8% in the rest of the United States. In the 500–2,000 persons per square mile category, NYS increased to 8.5%, compared with 7.7% in the rest of the United States. This increase in NYS is statistically significant.

In 2017, walking trips increased in NYS across most density categories. In the highest density areas (50,000+ persons per square mile), NYS rose slightly to 47.8%, whereas the rest of the United States declined to 44.2%. In 20,000–50,000 persons per square mile areas, NYS remained stable at 32.5%, whereas the rest of the United States increased slightly to 31.6%. In 10,000–20,000 persons per square mile areas, NYS increased to 20.7%, whereas the rest of the United States decreased to 18.3%. In the 4,000–10,000 persons per square mile category, NYS increased statistically significantly to 15.0%, compared with 11.7% in the rest of the United States. In the 2,000–4,000 persons per square mile category, NYS slightly declined to 9.9%, whereas the rest of the United States remained at 8.4%. In the 500–2,000 persons per square mile category, NYS decreased to 7.1%, whereas the rest of the United States stayed at 7.0%. However, in the fewer than 500 persons per square mile category, NYS increased to 8.2%, whereas the rest of the United States declined to 5.3%. The increase in NYS is statistically significant.

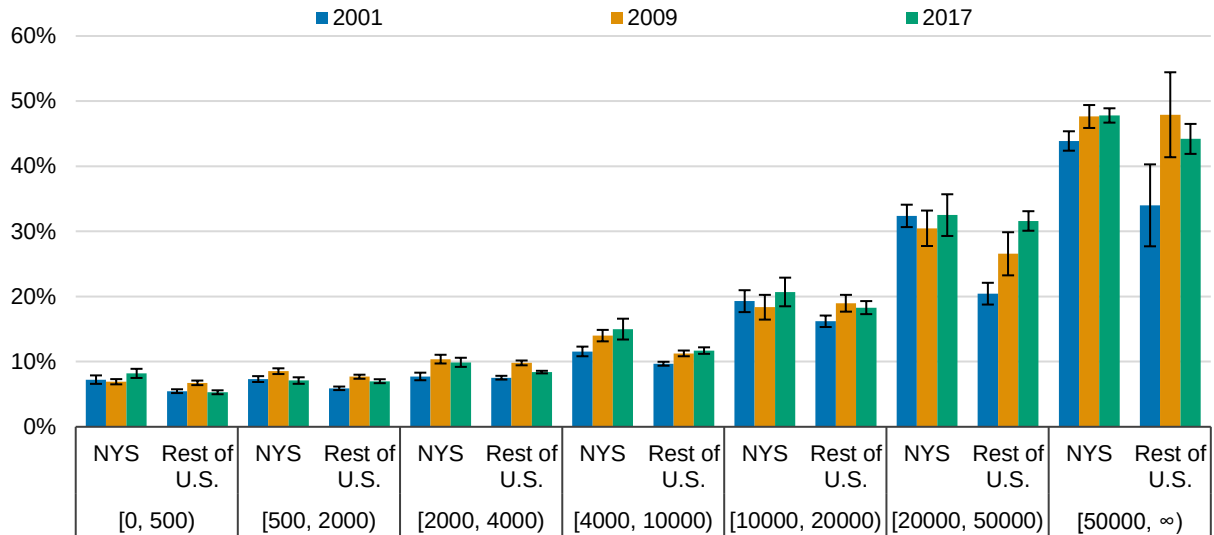


Figure 4-19. Share of person trips via walking by year, region, and population density.

The average minutes spent on walking trips per day by individuals, as presented in Figure 4-20, vary from 2001 to 2017 across population density categories and between NYS and the rest of the United States. In 2001, NYS showed a higher average walking time per day than the rest of the United States in most population density categories. In the highest density areas (50,000+ persons per square mile), individuals in NYS spent 17.76 min per day walking, compared with 21.58 min in the rest of the United States. However, this difference is not statistically significant. In the 20,000–50,000 persons per square mile category, NYS recorded 16.31 min, statistically higher than the 8.59 min observed in the rest of the United States. This observation shows that individuals without a vehicle in NYS walked significantly more in this density category than their counterparts walked in the rest of the United States. In the 10,000–20,000 persons per square mile category, NYS recorded 8.14 min, slightly below the 8.93 min in the rest of the United States. This difference is not statistically significant. A similar pattern is observed in the 4,000–10,000 persons per square mile category, where NYS reported 8.34 min, statistically significantly higher than the 5.83 min in the rest of the United States. For lower-density areas, in the 2,000–4,000 persons per square mile category, NYS reported 4.56 min, whereas the rest of the United States recorded 4.55 min. This difference is not statistically significant. Similarly, in the 500–2,000 persons per square mile category, NYS reported 5.19 min, which is statistically higher than the 4.05 min in the rest of the United States. Finally, in the lowest-density category (fewer than 500 persons per square mile), NYS recorded 3.76 min per day, slightly higher than the 3.05 min recorded in the rest of the United States, but this difference is not statistically significant.

By 2009, average walking time per day changed across density categories. In areas with 50,000+ persons per square mile, NYS saw a substantial increase to 25.58 min, whereas the rest of the United States increased to 28.10 min. However, this increase is not statistically significant for either NYS or the rest of the United States. In areas of 20,000–50,000 persons per square mile, NYS slightly decreased to 15.43 min, whereas the rest of the United States increased to 12.78 min. This change for both regions is not statistically significant. In areas of 10,000–20,000 persons per square mile, NYS walking time increased to 9.58 min, whereas walking time in the rest of the United States increased to 10.72 min. Both increases are not statistically significant. However, in areas with 4,000–10,000 persons per square mile, NYS saw a statistically significant decrease to 6.98 min, whereas the rest of the United States slightly increased to 6.95 min. In areas with 2,000–4,000 persons per square mile, NYS saw an increase to 6 min, whereas the rest of the United States rose to 5.58 min. Both regions' changes are not statistically significant. In areas

of 500–2,000 persons per square mile, NYS saw a slight decline to 4.66 min, whereas the rest of the United States increased slightly to 4.36 min. In the lowest density category (fewer than 500 persons per square mile), NYS walking time decreased slightly to 3.39 min, whereas walking time in the rest of the United States increased slightly to 3.68 min.

By 2017, NYS walking times rebounded across most density categories. In areas with 50,000+ persons per square mile, NYS declined slightly to 21.65 min, whereas the rest of the United States dropped to 22.60 min. The change is not statistically significant. In areas with 20,000–50,000 persons per square mile, NYS further declined to 14.00 min, whereas the rest of the United States increased to 15.89 min. This difference is not statistically significant. In areas with 10,000–20,000 persons per square mile, NYS saw a slight increase to 10.02 min, whereas the rest of the United States declined to 9.17 min. This difference is not statistically significant. In areas with 4,000–10,000 persons per square mile, NYS increased statistically significantly to 9.41 min, whereas the rest of the United States decreased statistically significantly to 6.54 min. In areas with 2,000–4,000 persons per square mile, NYS declined slightly to 5.20 min, whereas the rest of the United States declined slightly to 4.96 min. In areas with 500–2,000 persons per square mile, NYS declined slightly further to 4.14 min, whereas the rest of the United States declined slightly to 3.77 min. In areas with fewer than 500 persons per square mile, NYS increased slightly to 4.59 min, whereas the rest of the United States declined slightly to 2.97 min.

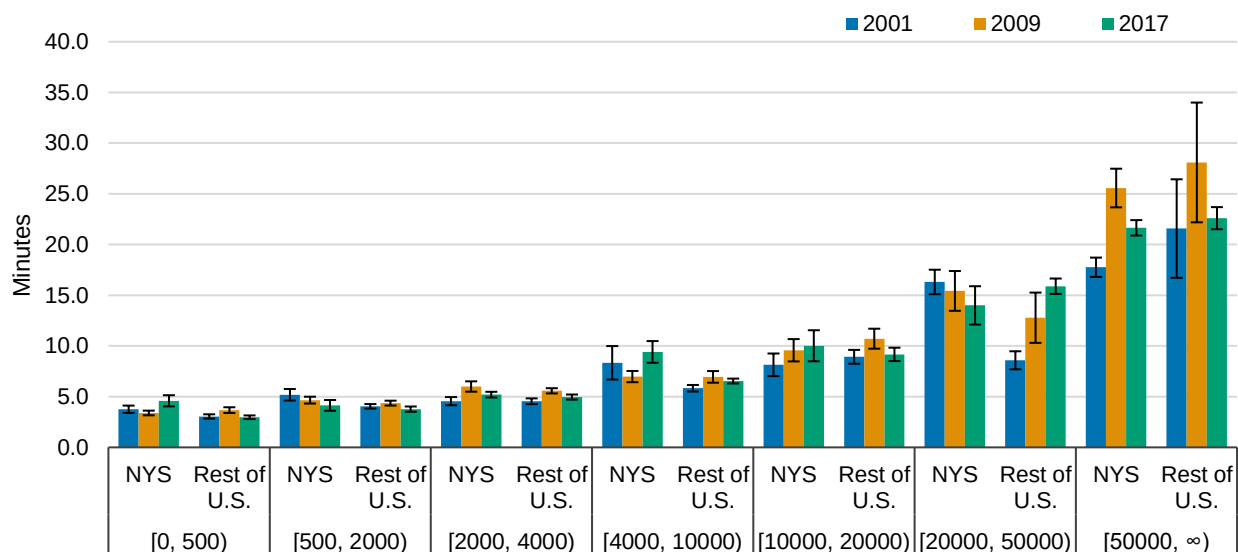


Figure 4-20. Average minutes spent on walking by year, region, and population density.

4.2.4 Travel by Zero-Vehicle Households

Figure 4-21 shows that in all 3 years, the number of daily trips per person without a vehicle tends to be higher in NYS metros compared with metros in the rest of United States, particularly in higher-density areas. In low-density areas (fewer than 500 people per square mile), however, NYS had more trips per person than those in the rest of the United States in 2001 (2.97 vs. 2.32) and 2017 (2.94 vs. 1.87) but recorded a sharp drop in 2009 (1.90 trips), falling below the national average (2.21 trips). These regional differences are statistically significant. In moderate-density areas (2,000–10,000 people per square mile), individuals without a vehicle in NYS metros consistently took more trips per day than their counterparts in the rest of the United States. For instance, in 2017, individuals in NYS metros in the 10,000–20,000-density range took an average of 3.32 trips per day, compared with 2.88 trips in the rest of the US, a statistically significant difference. Similarly, in higher-density areas (50,000+ people per square mile),

NYS metros consistently reported higher or comparable trip rates across all 3 years, with 3.24 trips per day in 2017, compared with 3.42 in the rest of the United States.

Between 2001 and 2017, the number of daily trips taken by individuals without a vehicle in NYS metros fluctuated, with a sharp decline in 2009 followed by a recovery in 2017. This pattern is particularly noticeable in low-density areas (fewer than 500 people per square mile), where the average trips per day dropped from 2.97 in 2001 to 1.90 in 2009 before rebounding to 2.94 in 2017. A similar but less pronounced trend can be observed in the 500–2,000 and 4,000–10,000 population density categories. In contrast, the rest of the United States exhibited a gradual decline in travel activity among carless individuals over time, particularly in high-density areas. For instance, in the 50,000+ density category, the daily trips per person increased from 3.23 in 2001 to 3.42 in 2017, but this change is not statistically significant.

Another notable shift occurred in moderate-density areas (10,000–20,000 people per square mile), where daily trips in NYS metros increased statistically significantly from 2.84 in 2001 to 3.32 in 2017, whereas the rest of the United States remained relatively stable, increasing only slightly from 2.55 to 2.88 over the same period.

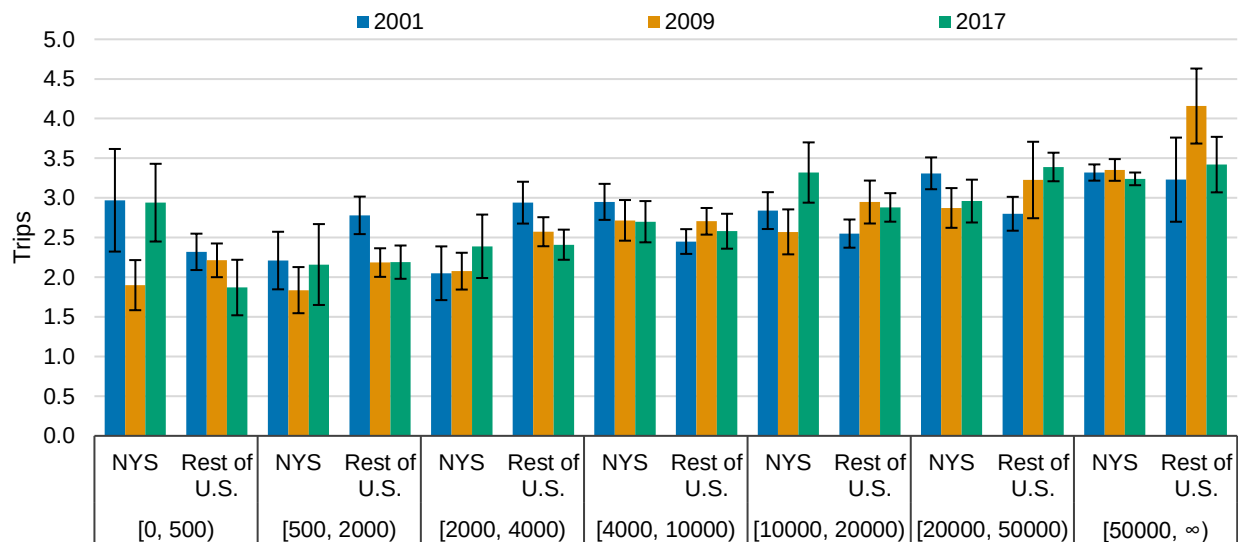


Figure 4-21. Average daily person trips of individuals without a vehicle by year, region, and population density.

The percentage of trips taken via public transit among individuals without a vehicle varies by population density and location, as shown in Figure 4-22. In 2001, NYS had a much higher share of trips taken by public transit among those without a vehicle compared with the rest of the United States, particularly in high-density areas. In census tracts with 50,000 or more persons per square mile, 34.8% of trips in NYS were made by transit, compared with 46.7% in the rest of the United States. Although the difference appears large, it was not statistically significant for this group. In the 20,000–50,000 persons per square mile category, transit accounted for 26.7% of trips in NYS and 33.9% in the rest of the United States. This difference is statistically significant, suggesting that in 2001, the rest of the United States had a higher reliance on transit in these high-density areas. However, at lower densities, such as 500–2,000 persons per square mile, NYS saw 20.6% of trips taken by transit compared with only 7.8% in the rest of the United States, with NYS having statistically significantly greater transit use in these lower-density regions.

In 2009, the percentage of trips taken by transit declined in many categories, with shifts between NYS and the rest of the United States. In areas with 50,000 or more persons per square mile, NYS saw a decrease to 31.3%, whereas the rest of the United States dropped statistically significantly to 22.1%. At 10,000–20,000 persons per square mile, NYS transit trips increased to 32.8%, whereas the rest of the United States declined to 26.1%, though that was not statistically significant. At lower densities, transit use fluctuated. In census tracts with 500–2,000 persons per square mile, NYS saw a modest decline to 19.3%, whereas the rest of the United States increased slightly to 9.6%, with the change not statistically significant. The biggest decline occurred in the 2,000–4,000 persons per square mile category, where NYS transit use dropped statistically significantly from 16.7% in 2001 to 5.1% in 2009, whereas the rest of the United States saw a statistically significant increase from 9.6% to 15.6%.

In 2017, transit use increased in NYS across most population densities, and trends were mixed in the rest of the United States. In the highest-density category of 50,000 or more persons per square mile, NYS transit use stabilized at 35.1%, whereas the rest of the United States increased slightly to 30.0%. In the 20,000–50,000 persons per square mile category, transit use in NYS declined slightly to 33.3%, whereas in the rest of the United States, it fell statistically significantly to 26.4%. In the 10,000–20,000 persons per square mile category, transit trips in NYS decreased slightly to 31.5%, whereas the rest of the United States saw a minor increase to 30.4%, though the change is not statistically significant. In the 4,000–10,000 persons per square mile category, NYS transit use increased to 23.5%, surpassing the rest of the United States at 18.8%, though this difference is not statistically significant. A similar trend can be observed in the 2,000–4,000 persons per square mile category, where NYS rebounded to 14.7%, whereas the rest of the United States increased to 20.7% with no significant difference. The most notable increases in transit use in 2017 occurred in lower-density areas. In census tracts with 500–2,000 persons per square mile, NYS transit use declined slightly to 14.7%, but in the rest of the United States, transit use nearly doubled to 19.7%, which is statistically significant. In the fewer than 500 persons per square mile category, NYS transit use increased statistically significantly to 11.9%, whereas the rest of the United States declined statistically significantly to 5.3%.

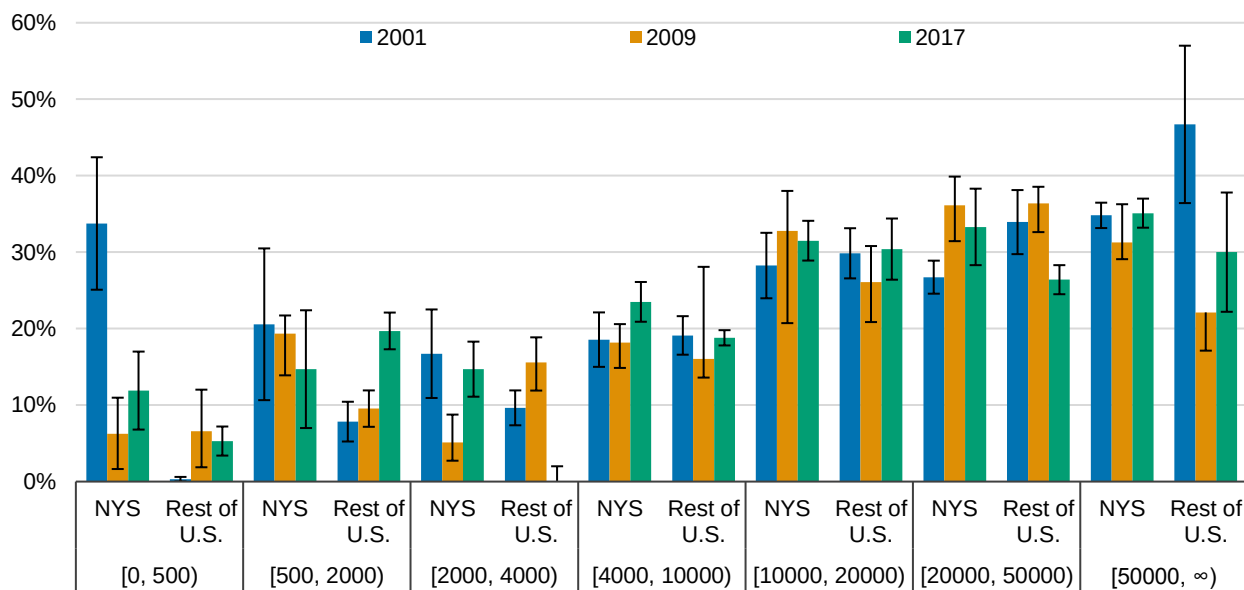


Figure 4-22. Percentage of trips taken via public transit by those without a vehicle by year, region, and population density.

Figure 4-23 shows the percentage of trips taken by walking by individuals without a vehicle. The population density is measured in persons per square mile based on census tract data. In 2001, NYS

exhibited a markedly higher percentage of walking trips among individuals without a vehicle compared with the rest of the United States, with 46.0% versus 15.4%, respectively, in regions with a population of fewer than 500 people per square mile. This substantial difference suggests that the higher walking percentage in NYS is statistically significant for that year. Across various population densities, NYS consistently outperformed the rest of the United States. For instance, in areas with a population density of 2,000–4,000 persons per square mile, NYS had 33.5% of trips taken by walking compared with 21.9% in the rest of the United States.

By 2009, the overall percentage of walking trips in NYS plummeted to 16.9%, whereas the rest of the United States saw an increase to 24.0% in regions with a population of fewer than 500 people per square mile. However, the decline in NYS and the rise in the rest of the United States were not statistically significant. Within specific population density categories, certain differences remain similar. For example, in the 2,000–4,000 persons per square mile category, NYS maintained a higher walking percentage (30.1%) compared with the rest of the United States (42.0%). However, this difference was not statistically significant.

In 2017, NYS rebounded to 42.4% in overall walking trips in regions with a population pf fewer than 500 people per square mile, surpassing the rest of the United States at 25.1%. NYS's increase was statistically significant, but the differences between NYS and the rest of the United States are not statistically significant. Nevertheless, within higher population densities, NYS continued to demonstrate higher walking percentages. For example, in areas with 2,000–4,000 persons per square mile, NYS reported 33.2%, compared with 45.8% in the rest of the United States. However, this difference is not statistically significant.

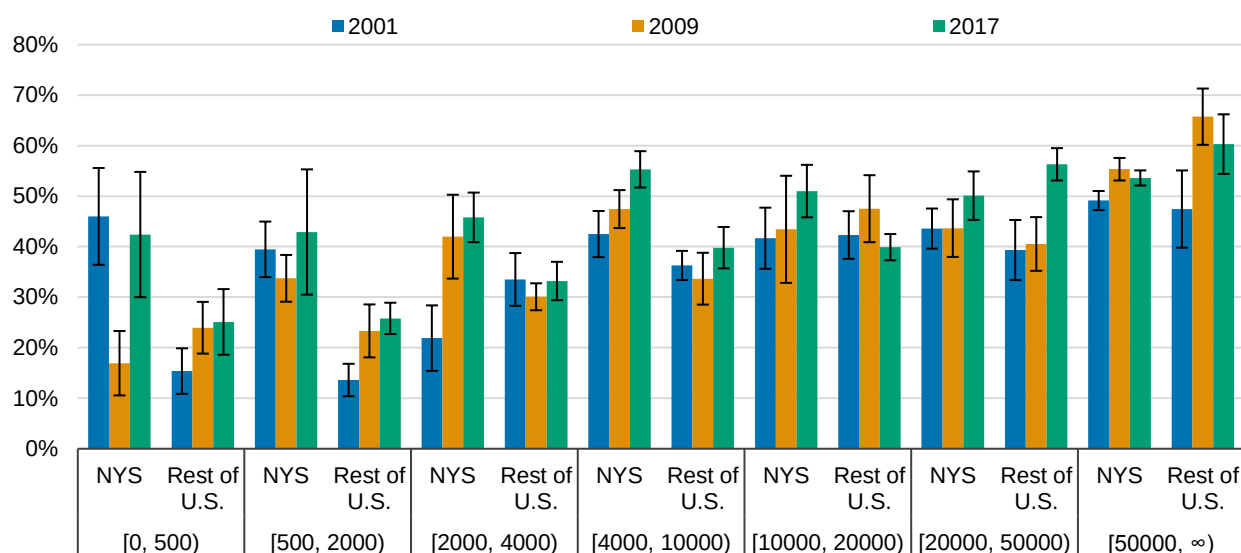


Figure 4-23. Percentage of trips taken via walking by those without a vehicle by year, region, and population density.

4.3 VEHICLE TRAVEL COMPARISONS

4.3.1 Driver Trip Rates

Table 4-13 provides insights into vehicle trips per driver and trip purposes across population density categories for NYS and the rest of the United States. A clear pattern emerges where the number of vehicle trips per driver decreases as population density increases in both regions. For areas with fewer than 500

people per square mile, NYS drivers average 2.64 trips per day, compared with 2.7 trips in the rest of the United States. However, in the highest-density areas (50,000+ people per square mile), NYS drivers take only 0.76 trips on average, which is statistically significantly lower than the 1.88 trips observed in the rest of the United States.

Looking at trip purposes, the share of work-related trips decreases with increasing density in both regions. For example, in the lowest-density areas (<500 people per square mile), NYS records 29.2% of trips in this category, which is statistically higher than the 26.1% observed in the rest of the United States. In the highest-density areas, work trips constitute 20.7% of trips in NYS, which is statistically significantly lower than the 27.3% observed in the rest of the United States. Trips for family and personal business consistently represent the largest share of trips across all density levels. In NYS, this category peaks at 45.8% in areas with 2,000–4,000 people per square mile, slightly higher than the 43.5% in the rest of the United States in the same density range. In the densest areas (50,000+ people per square mile), trips for family and personal business rise to 48.6% in NYS, which is statistically higher than the 34.9% observed in the rest of the United States. Social and recreational trips show a different trend, slightly decreasing in NYS as density increases. In the highest-density areas, 22.0% of NYS trips fall into this category, which is statistically lower than the 31.7% observed in the rest of the United States. This suggests a notable difference in how high-density living affects recreational and social travel between the two regions. Overall, the data reveal that NYS consistently sees fewer vehicle trips per driver and significant differences in trip purposes, highlighting the influence of urban density on transportation patterns.

Table 4-13. Share^a of vehicle trips by trip purpose in 2017 by region and population density.

Trip Purpose	Population Density (population per square mile)							
	[0, 500)		[500, 2000)		[2000, 4000)		[4000, 10000)	
	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.
Vehicle Trips per Driver	2.64	2.7	2.89	2.85	2.8	2.83	2.66	2.78
Earn a Living	29.2%	26.1%	25.3%	25.8%	22.7%	25.2%	25.2%	27.2%
Family & Personal Business	42.1%	43.1%	44.8%	43.2%	45.8%	43.5%	42.5%	42.8%
Civic, Educational, & Religious	4.1%	5.4%	4.6%	5.6%	4.5%	5.1%	4.3%	5.0%
Social & Recreational	21.8%	21.7%	22.1%	22.2%	23.3%	22.6%	24.9%	21.9%
Other	2.8%	3.7%	3.2%	3.1%	3.7%	3.6%	3.0%	3.1%
Unreported	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%

Trip Purpose	Population Density (population per square mile)					
	[10000, 20000)		[20000, 50000)		[50000, ∞)	
	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.
Vehicle Trips per Driver	2.32	2.4	1.39	1.88	0.76	1.48
Earn a Living	25.5%	28.2%	20.7%	28.4%	20.6%	27.3%
Family & Personal Business	44.8%	40.8%	45.2%	40.4%	48.6%	34.9%
Civic, Educational, & Religious	2.9%	5.2%	3.4%	5.4%	4.6%	0.9%
Social & Recreational	23.6%	22.1%	26.2%	22.2%	22.0%	21.3%
Other	3.2%	3.6%	4.5%	3.6%	4.1%	15.5%
Unreported	0.0%	0.1%	0.0%	0.1%	0.0%	0.0%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

VMT per driver decreases with increasing population density in both NYS and the rest of the United States (Table 4-14). For areas with fewer than 500 people per square mile, NYS drivers average 31.95 miles per day, whereas drivers in the rest of the United States average 34.25 miles. This difference is not statistically significant. In the densest areas (50,000+ people per square mile), NYS drivers average 8.14 miles per day, statistically significantly lower than the 19.57 miles in the rest of the United States.

The share of trips for work purposes decreases with increasing density in both NYS and the rest of the United States. In the lowest density areas (<500 people per square mile), NYS records 35.3% work-related trips, which is not statistically different from the 34.4% in the rest of the United States. However, in the densest areas (50,000+ people per square mile), work trips drop to 19.3% in NYS, statistically significantly lower than the 29.8% in the rest of the United States.

Trips for family and personal business dominate across most density levels and tend to increase slightly with population density, particularly in NYS. In areas with 20,000–50,000 people per square mile, family and personal business trips constitute 40.2% of all trips in NYS, which is significantly higher than the 27.1% observed in the rest of the United States. In the densest areas (50,000+), NYS continues to have a higher share (41.7%) compared with the rest of the United States (20.1%), with statistical significance.

Social and recreational trips show more variation, generally decreasing in NYS as density increases. For example, in the least dense areas (<500 people per square mile), NYS records 25.6% of trips in this category, compared with 23.1% in the rest of the United States, a difference that is not statistically significant. However, in the densest areas (50,000+), NYS has 22.5% of trips for social and recreational purposes, statistically lower than the 31.9% observed in the rest of the United States.

The average vehicle trip length decreases as population density increases in both NYS and the rest of the United States, reflecting shorter trips in more densely populated areas (Table 4-15). In areas with fewer than 500 people per square mile, NYS drivers average 12.76 miles per trip, whereas the rest of the United States averages 13.07 miles. There is no statistical significance between these values. In the densest areas (50,000+ people per square mile), NYS drivers average 10.17 miles, whereas the rest of the United States averages 11.34 miles. The difference is not statistically significant. Across all densities, NYS and the rest of the United States show a similar trend of decreasing trip lengths with increasing density, but with slightly shorter trips in NYS in most categories.

Average time spent on trips increases as population density increases, suggesting slower travel speeds in denser areas. In areas with fewer than 500 people per square mile, NYS drivers spend an average of 78.8 min on trips, whereas drivers in the rest of the United States spend 84.14 min. This difference is statistically significant, with NYS having lower average trip times. In the densest areas (50,000+ people per square mile), NYS drivers spend an average of 100.76 min on trips, whereas drivers in the rest of the United States spend 97.16 min. However, the difference has no statistical significance. For the 10,000–20,000 density category, NYS drivers spend 95.73 min, significantly more than the 75.9 min for the rest of the United States.

Table 4-14. Share^a of VMT by trip purpose in 2017 by region and population density.

Trip Purpose	Population Density (population per square mile)							
	[0, 500)		[500, 2000)		[2000, 4000)		[4000, 10000)	
	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.
VMT per Driver	31.95	34.25	31.24	30.16	26.17	26.77	22.52	26.82
Earn a Living	35.3%	34.4%	28.5%	34.2%	28.4%	32.8%	29.2%	31.9%
Family & Personal Business	31.0%	31.6%	42.2%	28.5%	33.9%	31.8%	27.6%	32.7%
Civic, Educational, & Religious	3.7%	4.8%	3.9%	6.3%	3.7%	4.6%	5.0%	7.3%
Social & Recreational	25.6%	23.1%	20.5%	25.7%	29.4%	25.1%	34.2%	23.4%
Other	4.4%	6.1%	4.8%	5.2%	4.7%	5.7%	3.9%	4.8%
Unreported	0.0%	0.0%	0.0%	0.2%	0.0%	0.0%	0.1%	0.0%

Trip Purpose	Population Density (population per square mile)					
	[10000, 20000)		[20000, 50000)		[50000, ∞)	
	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.
VMT per Driver	18.87	20.55	11.73	19.57	8.14	11.26
Earn a Living	38.1%	35.4%	24.9%	29.8%	19.3%	45.1%
Family & Personal Business	27.4%	27.1%	37.4%	40.2%	41.7%	20.1%
Civic, Educational, & Religious	1.5%	4.8%	1.2%	3.9%	2.8%	0.2%
Social & Recreational	26.7%	27.5%	29.4%	22.5%	31.9%	22.0%
Other	6.3%	5.0%	7.1%	3.5%	4.3%	12.7%
Unreported	0.0%	0.1%	0.0%	0.1%	0.0%	0.0%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 4-15. Average vehicle trip length and average driving time in 2017 by region and population density.

	Population Density (population per square mile)							
	[0, 500)		[500, 2000)		[2000, 4000)		[4000, 10000)	
	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.
Average Vehicle Trip Length (miles)	12.76	13.07	11.48	10.75	9.1	9.52	8.37	9.64
Average Time Spend Driving in a POV in a Typical Day (minutes)	78.8	84.14	76.73	78.95	72.74	76.55	80.74	75.9

	Population Density (population per square mile)					
	[10000, 20000)		[20000, 50000)		[50000, ∞)	
	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.
Average Vehicle Trip Length (miles)	7.8	8.71	8.79	10.17	11.34	12.1
Average Time Spend Driving in a POV in a Typical Day (minutes)	95.73	79.96	96.61	97.16	100.76	76.54

Figure 4-24 shows that daily vehicle trip rates decreased in both NYS and the rest of the United States between 2001 and 2017 across all population density categories. For example, in areas with fewer than 500 people per square mile, NYS daily trip rates dropped from 3.10 in 2001 to 2.64 in 2017, whereas the rest of the United States saw a decline from 3.35 to 2.7, which is statistically significant.

In low-density areas (<500 people per square mile), trip rates are consistently higher in the rest of the United States compared with those in NYS. In 2017, NYS had an average trip rate of 2.64, whereas the rest of the United States averaged 2.7. These differences are statistically significant. In medium-density areas (4,000–10,000 people per square mile) in 2001, NYS had an average trip rate of 3.33, slightly lower than the 3.61 observed in the rest of the United States. By 2017, trip rates declined to 2.78 in NYS and 2.83 in the rest of the United States, with no significant difference. In the densest areas (50,000+ people per square mile), trip rates are consistently lower for NYS compared with the rest of the United States. In 2017, NYS had a trip rate of 0.76, statistically significantly lower than the 1.88 for the rest of the United States.

NYS consistently exhibits lower trip rates compared with the rest of the United States in the densest areas (50,000+ people per square mile), suggesting greater reliance on nonvehicle travel modes in high-density areas. In medium- and low-density areas, NYS and the rest of the United States exhibit similar trip rates over time, with no statistically significant differences.

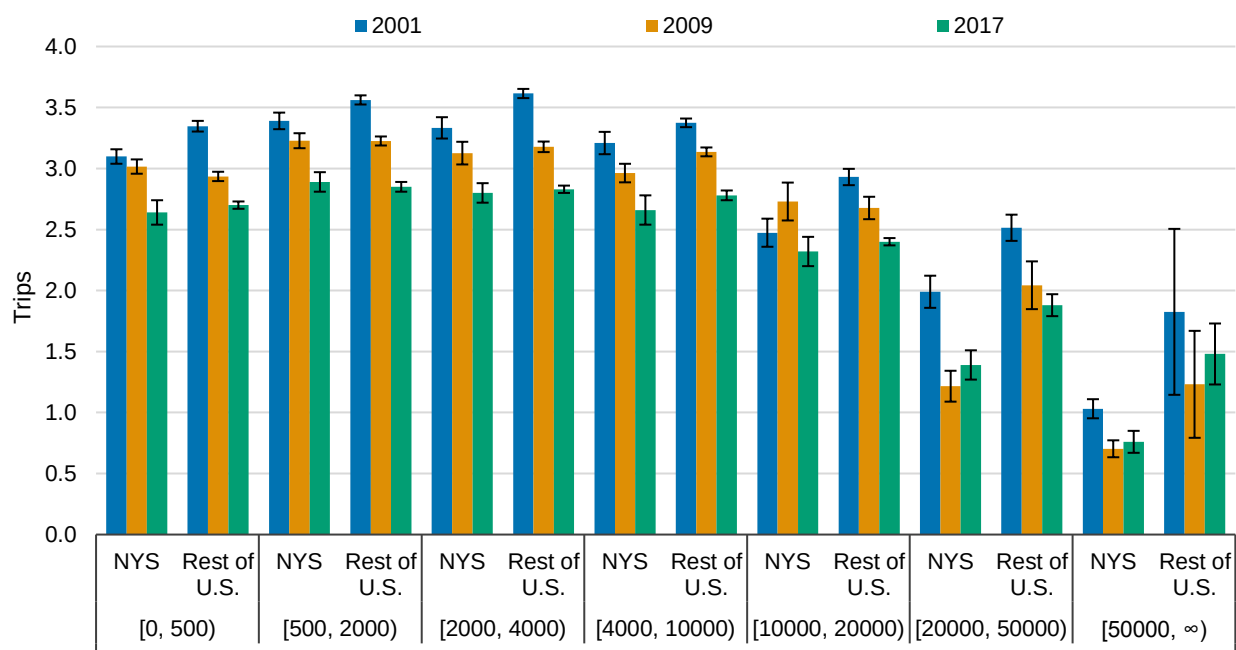


Figure 4-24. Daily vehicle trip rates statistics by year, region, and population density.

Figure 4-25 further highlights disparities in daily vehicle trip rates between NYS and the rest of the United States. Positive values indicate higher trip rates in the rest of the United States, whereas negative values suggest higher trip rates in NYS. The largest differences appear in high-density areas, especially for the 50,000+ category. In 2017, the gap widened further to 35.3% in the 20,000–50,000 category and 94.7% in the 50,000+ category, indicating far fewer trips in NYS compared with those in the rest of the United States. These differences have grown over time, with the most substantial gaps emerging in 2009 and persisting into 2017. In contrast, differences in low-density areas remain small, occasionally favoring NYS trip rates.

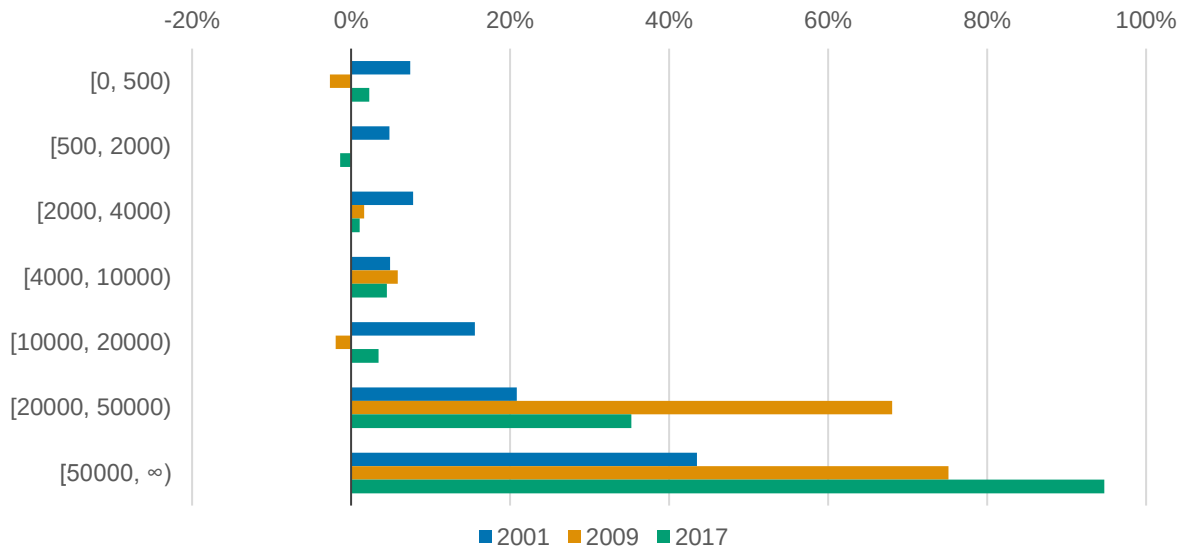


Figure 4-25. Percentage differences in daily vehicle trip rates between metros in NYS and in the rest of the United States by year and population density.

4.3.2 Driving Distances and Times

Figure 4-26 compares average driving distances (in miles) for drivers in NYS and the rest of the United States across population density categories for 2001, 2009, and 2017. Driving distance increased overall. Both NYS and the rest of the United States show an increase in average driving distances over time across all density categories. The trend is more pronounced in lower-density areas (<500 people per square mile). For instance, in the fewer than 500 people category, NYS increased statistically significantly from 10.95 miles in 2001 to 12.76 miles in 2017, whereas the rest of the United States increased statistically significantly from 11.42 to 13.07 miles. NYS drivers generally travel shorter distances in medium- and high-density areas compared with the rest of the United States. This is consistent across all years and reflects the effect of urban density on driving behavior, likely influenced by alternative transportation options in NYS metros (e.g., public transit or walkability). Driving distances remain shorter for NYS drivers in high-density areas, but the gap narrowed in 2017 as NYS drivers increased their average trip distance to 10.17 miles, compared with 11.34 miles in the rest of the United States.

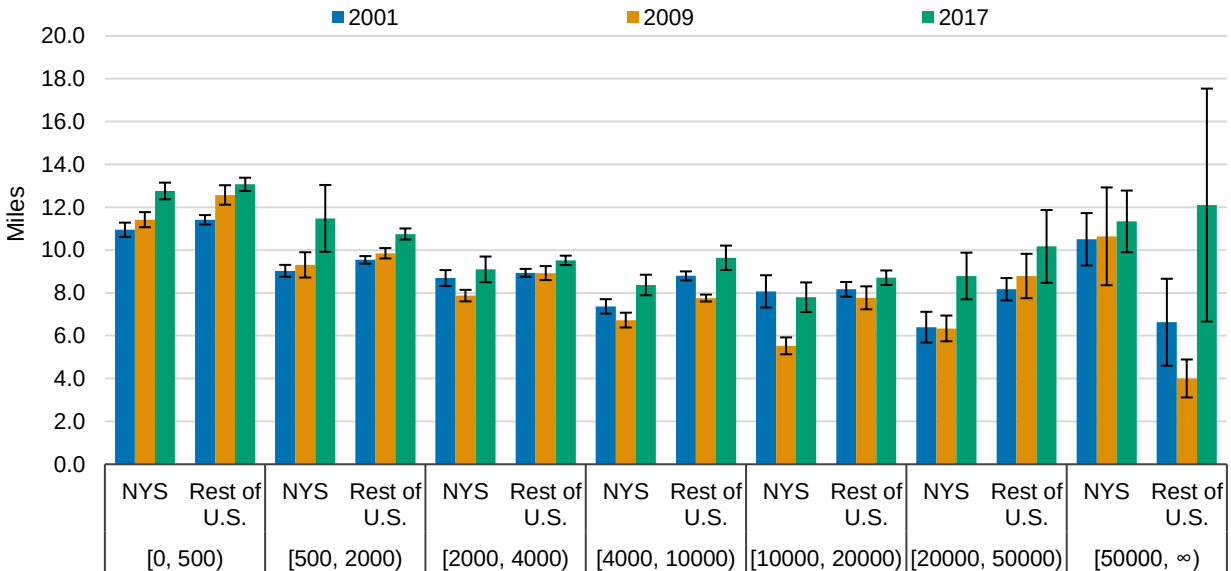


Figure 4-26. Average driving distances by year, region, and population density.

Figure 4-27 presents the average driving times (in minutes) for drivers in metro areas in NYS and in the rest of the United States, segmented by population density and measured for the years 2001, 2009, and 2017. In low-density areas (<500 people per square mile), average driving times are consistently higher for the rest of the United States compared with NYS across all years. In 2001, drivers in NYS averaged 78.76 min, whereas drivers in the rest of the United States averaged 84.75 min. The difference is statistically significant. By 2017, average driving times increased slightly in both regions to 78.8 min for NYS and 84.14 min for the rest of the United States.

In medium-density areas (500–4,000 people per square mile), driving times are comparable across NYS and the rest of the United States, with slight regional differences. For the 500–2,000 category, NYS drivers had an average driving time of 76.13 min in 2001, which remained relatively stable at 76.73 min in 2017. Similarly, the rest of the United States had an average time of 81.30 min in 2001, decreasing slightly to 78.95 min in 2017. In the 2,000–4,000 category, NYS driving times increased slightly from 74.22 min in 2001 to 75.9 min in 2017, whereas the rest of the United States remained largely unchanged, with an average of 80.74 min in 2017, which was not statistically significant.

In medium-high density areas (10,000–20,000 people per square mile), driving times increased more noticeably over time in NYS compared with the rest of the United States. In 2001, NYS drivers averaged 81.01 min, which rose significantly to 95.73 min in 2017. The rest of the United States, by contrast, experienced relatively smaller increases, rising from 77.59 min in 2001 to 79.96 min in 2017. This indicates a growing disparity in driving times between the two regions in medium- to high-density areas, which may reflect increased congestion or other regional differences in travel behavior.

In high-density areas (20,000+ people per square mile), driving times are significantly longer for both regions, with more pronounced increases in NYS over time. For areas with 20,000–50,000 people per square mile, NYS drivers in 2009 averaged 73.46 min, which rose statistically significantly to 96.61 min in 2017. The rest of the United States also saw increases, though less dramatic, averaging 96.17 min in 2017. In the densest areas (50,000+ people per square mile), NYS drivers averaged 81.16 min in 2001 and 100.76 min in 2017, whereas the rest of the United States statistically significantly increased from 97.22 min to 100.76 min over the same period. NYS experience larger increases over time.

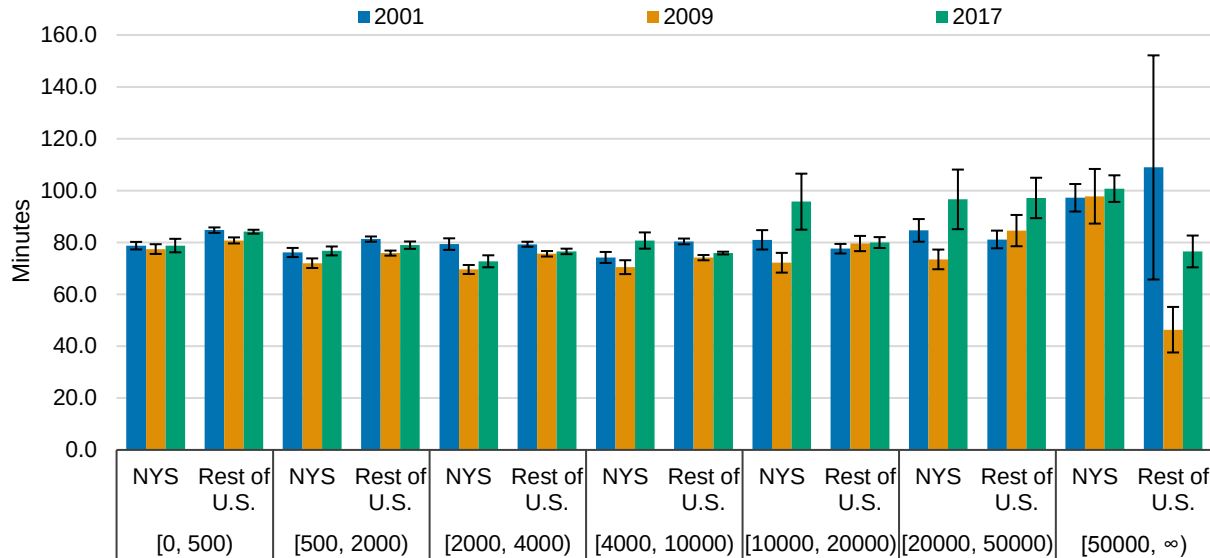


Figure 4-27. Average driving times by year, region, and population density.

4.4 COMMUTING COMPARISONS

4.4.1 Commute Modes

As population density increased, the share of commute trips using POVs decreased (Figure 4-28). For example, in areas with a population density of 10,000–20,000 people per square mile, the share of POV commute trips in NYS was 50%. In the next density bracket (20,000–50,000 people per square mile), this share dropped significantly to 22%. Conversely, the share of commute trips using public transportation modes or walking increased with higher population density. NYS and the rest of the United States showed a statistically significant difference in the use of subways or elevated rail for commute trips in densely populated areas. For instance, in areas with more than 50,000 people per square mile, 48% of commute trips in NYS were made by subway or elevated rail, compared with just 16% in the rest of the United States. In areas with a population density of fewer than 10,000 people per square mile, the difference between NYS and the rest of the United States was relatively minor.

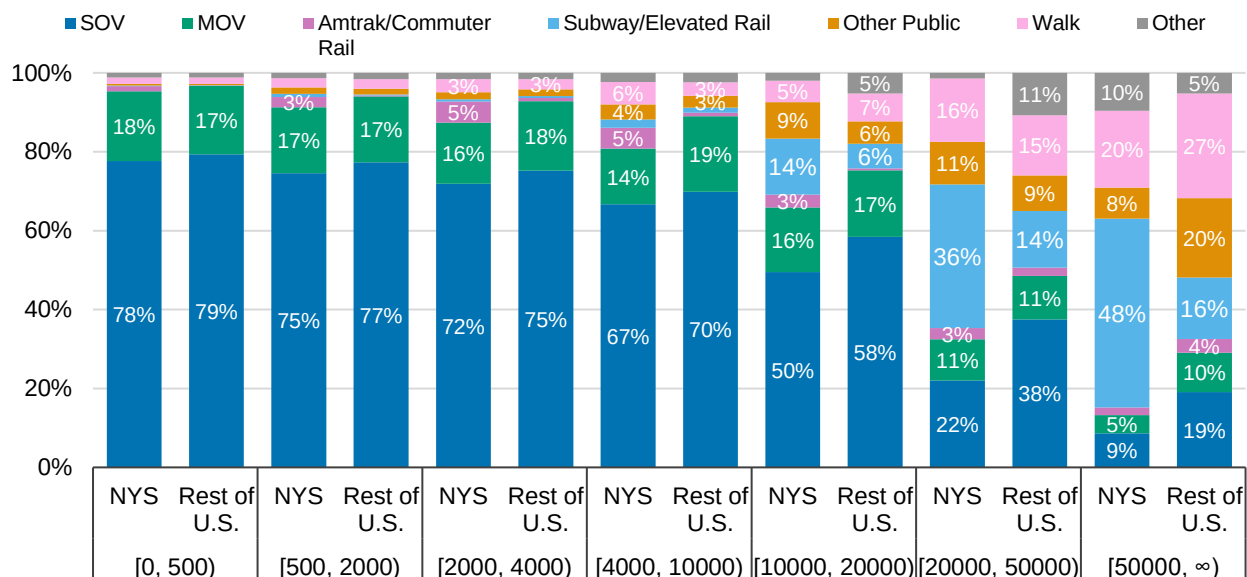


Figure 4-28. Commute trips by mode, region, and population density in 2017. Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number. Unlabeled segments represent values lower than 3%.

4.4.2 Commute Distances and Times

Table 4-16, Table 4-17, and Table 4-18 present an overview of commute statistics in NYS metropolitan areas and the rest of the United States metropolitan areas across different population density categories for the years 2001, 2009, and 2017. In terms of commute length, NYS consistently shows shorter private commute distances compared with the rest of the United States, particularly in areas with lower population density. However, this trend reverses in the highest density category (over 50,000 people per square mile), where NYS exhibits longer private commutes. Public transportation commute lengths in NYS were longer than those in the rest of the United States in almost all population density categories across the 3 years. Over time, the average commute lengths for private vehicles have remained relatively stable, whereas commute distances of public transportation modes in NYS have slightly increased. For commute travel time, NYS commuters generally experienced longer trips compared with their counterparts in the rest of the United States, especially for public transit users. The disparity is most pronounced in 2017 in the least populated areas (Figure 4-29), where public transit times often exceed an hour. Despite this, walking times in NYS tend to be similar to or slightly shorter than those in the rest of the United States in most population density categories.

Table 4-16. Daily commute statistics in 2017 by region and population density.

Mode	Population Density (population per square mile)													
	[0, 500)		[500, 2000)		[2000, 4000)		[4000, 10000)		[10000, 20000)		[20000, 50000)		[50000, ∞)	
	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.
Average Commute Length (miles)														
Private	14.0	15.5	10.1	12.6	10.7	10.6	8.6	10.0	10.0	9.8	8.0	8.8	11.7	14.9
Public	44.4	18.6	28.4	20.7	24.9	18.4	19.8	12.9	11.7	8.1	9.4	6.3	7.6	6.2
Walk	0.4	0.8	0.7	1.6	0.7	0.7	0.8	0.7	0.6	0.6	0.5	0.6	0.7	0.5
Other	16.7	37.7	5.3	25.8	2.7	14.2	5.9	6.1	9.3	4.0	6.1	9.8	4.0	2.6
Commute Travel Time (minutes)														
Private	23.3	26.5	20.9	23.7	23.0	24.0	21.2	23.4	28.2	26.1	32.1	26.4	54.4	35.3
Public	98.7	56.8	81.7	67.4	78.3	64.2	77.3	57.3	64.1	49.8	62.8	45.5	48.3	46.2
Walk	15.4	14.1	13.3	13.1	12.8	15.9	45.9	13.8	13.2	15.9	15.2	15.7	15.0	12.8
Other	26.5	49.8	30.4	46.4	17.9	24.2	25.2	23.9	67.3	24.2	20.6	21.2	28.1	20.4

Table 4-17. Daily commute statistics in 2009.

Mode	Population Density (population per square mile)													
	[0, 500)		[500, 2000)		[2000, 4000)		[4000, 10000)		[10000, 20000)		[20000, 50000)		[50000, ∞)	
	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.
Average Commute Length (miles)														
Private	14.4	15.2	11.8	12.9	11.3	10.9	9.9	10.1	8.9	9.8	12.1	10.4	10.6	4.7
Public	21.0	6.1	18.5	14.3	26.4	17.2	14.6	11.1	12.2	5.9	7.9	8.2	7.6	4.6
Walk	0.5	0.8	1.3	1.6	0.5	0.8	0.9	0.9	0.8	1.0	2.9	0.7	1.5	0.6
Other	50.6	21.1	8.2	21.1	7.1	14.2	5.4	10.3	28.5	13.6	5.1	4.9	7.0	1.4
Average Commute Travel Time (minutes)														
Private	24.0	25.1	22.9	23.4	23.1	22.5	24.2	22.1	24.1	25.4	34.6	26.9	33.4	17.9
Public	61.3	50.6	53.6	51.2	69.2	70.8	55.9	51.7	63.7	49.4	52.2	53.8	49.5	37.2
Walk	8.5	13.3	13.6	28.1	15.1	15.9	16.9	15.1	16.4	15.9	28.1	13.1	18.3	17.2
Other	31.8	25.7	26.5	26.3	19.1	33.9	26.0	27.8	41.0	26.9	43.2	21.6	26.9	22.2

Table 4-18. Daily commute statistics in 2001 by region and population density.

Mode	Population Density (population per square mile)													
	[0, 500)		[500, 2000)		[2000, 4000)		[4000, 10000)		[10000, 20000)		[20000, 50000)		[50000, ∞)	
	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.	NYS	Rest of U.S.
Average Commute Length (miles)														
Private	13.8	14.8	12.0	11.8	12.0	10.6	10.0	11.0	9.9	10.4	8.9	9.6	10.5	12.9
Public	20.0	22.6	29.9	23.2	27.0	14.8	21.9	12.8	14.0	9.2	11.0	6.1	7.7	9.1
Walk	2.9	0.8	0.9	0.8	0.9	1.2	1.9	0.9	1.3	0.9	1.2	0.8	1.2	0.1
Other	47.3	102.6	24.7	100.6	12.7	86.4	14.1	72.6	11.0	31.5	7.4	3.0	6.8	—
Commute Travel Time (minutes)														
Private	23.6	24.2	22.3	22.1	23.3	21.3	22.0	22.8	26.0	25.8	28.2	23.9	35.5	33.0
Public	58.0	106.7	74.5	61.7	71.2	56.3	77.5	59.5	72.0	48.4	58.0	43.9	51.1	54.7
Walk	12.6	8.9	33.2	15.0	14.5	13.7	12.9	13.2	17.7	14.5	20.3	15.2	19.8	3.2
Other	67.5	74.3	52.9	58.4	32.7	56.9	53.6	38.3	42.5	40.4	32.4	32.1	37.5	—

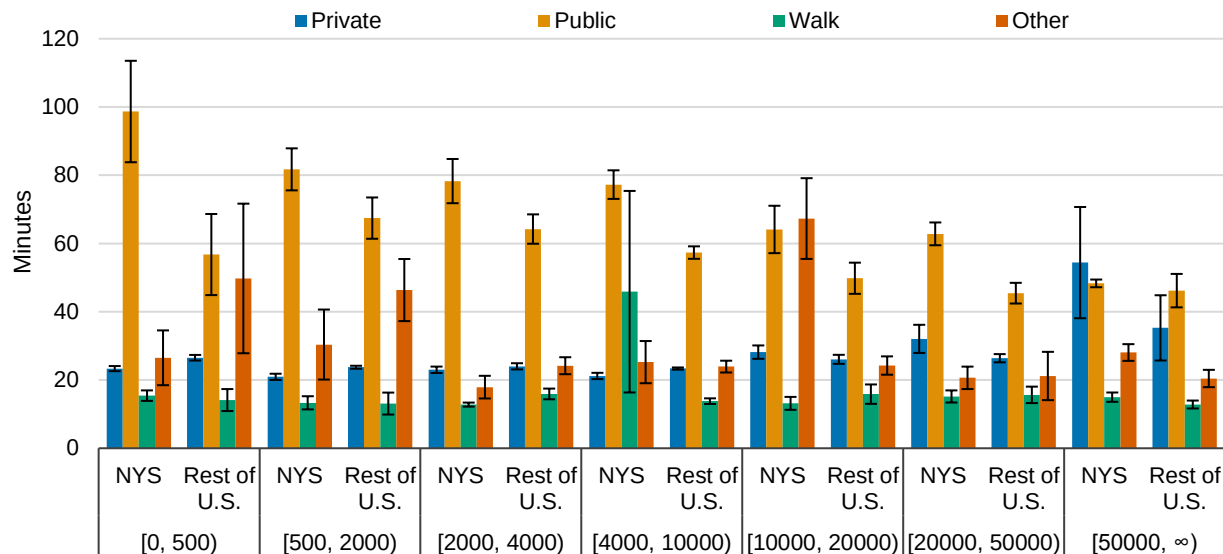


Figure 4-29. Commute travel time in 2017 by mode, region, and population density.

Figure 4-30 illustrates the average commute time across different transportation modes over the three NHTS survey years. In less densely populated areas, the differences in commute times between years were minor, indicating relative stability in travel patterns where lower population densities may result in fewer congestion-related delays and more consistent travel conditions. However, in areas with more than 10,000 people per square mile, more noticeable differences emerged. In NYS, commute travel time in 2017 showed a slight increase compared with 2009, but this change was not statistically significant, suggesting that although minor fluctuations may have occurred, overall commute durations remained relatively stable. In contrast, in the most densely populated areas of the rest of the United States, the 2017 average commute time was statistically significantly higher than the average in 2009. This observation suggests that for commuters in these high-density regions outside of NYS, travel conditions worsened over time, possibly owing to increasing congestion, growing demand for transit services, or declining transit efficiency.

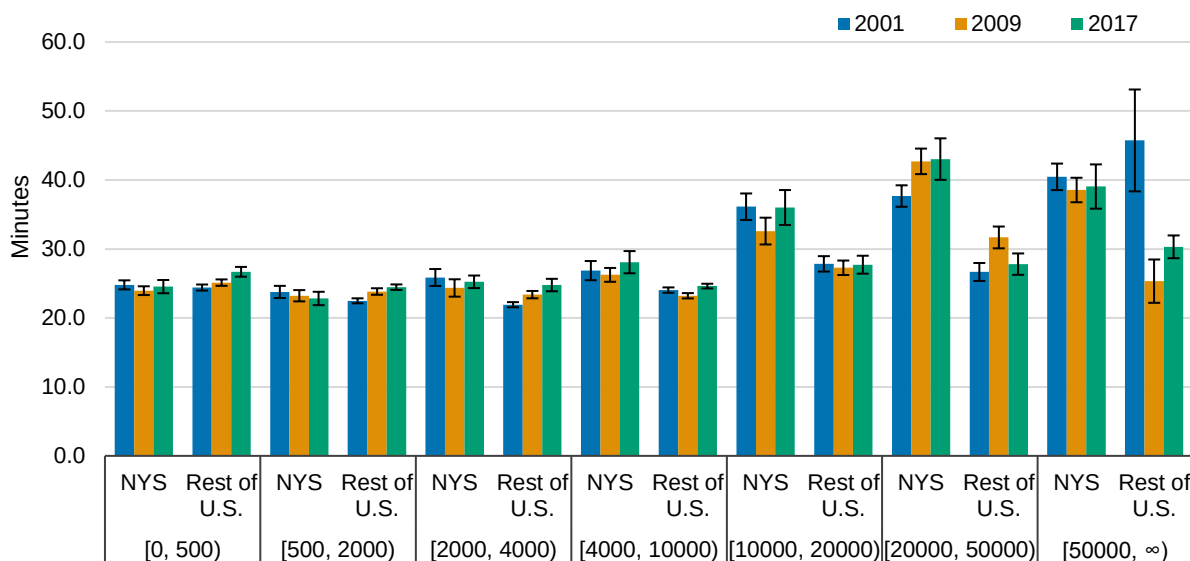


Figure 4-30. Trends in commute travel times by year, region, and population density.

Examining public transportation modes in detail, commute distance by public transit consistently decreased as population density increased (Figure 4-31). In low-density areas with fewer than 500 people per square mile, NYS had a significantly longer average public transit commute distance of 44 miles, compared with 19 miles in the rest of the United States—a statistically significant difference. However, as population density increased, the disparity between NYS and the rest of the United States gradually diminished, reflecting a shift toward shorter, more localized commutes in denser regions.

A similar trend emerged in commute travel time (Figure 4-32). In the lowest-density areas, the rest of the United States had significantly lower public transit travel times compared with those in NYS, with the longest commute times exceeding an hour and a half in NYS. This substantial difference suggests that in sparsely populated areas, NYS commuters using public transit must travel much longer distances to reach their destinations, potentially owing to limited transit infrastructure, fewer direct routes, or reliance on intercity transit options such as commuter rail. However, as population density increased, the travel time gap between NYS and the rest of the United States gradually narrowed. In the highest-density areas (50,000+ people per square mile), the average commute time was nearly identical: 48 min in NYS and 46 min in the rest of the United States, with no statistically significant difference.

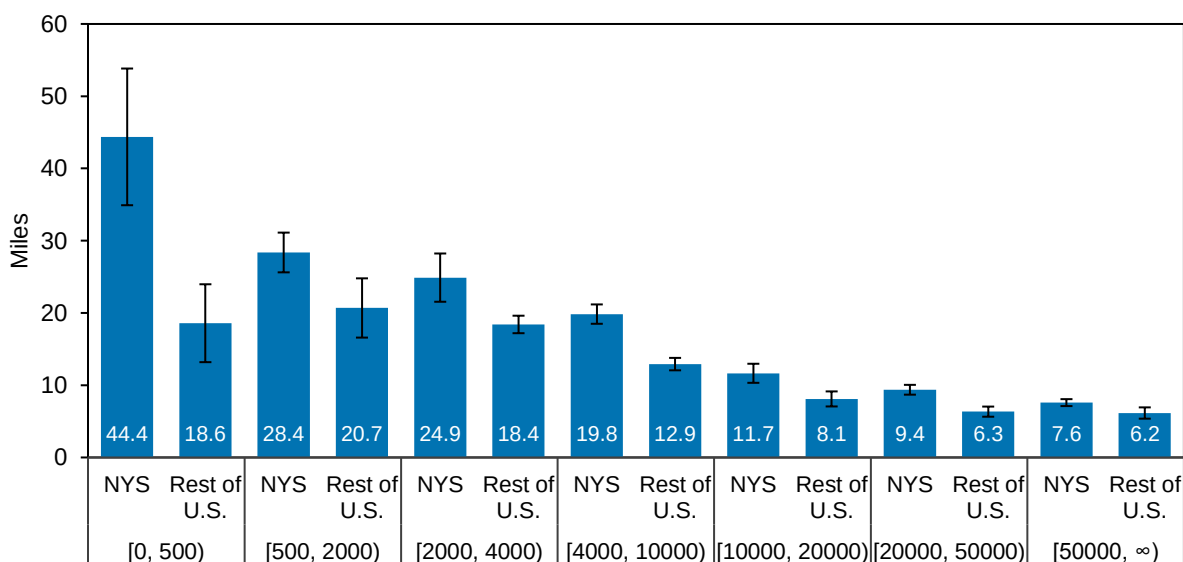


Figure 4-31. Metro area commute trip lengths of public transit in 2017 by region and population density.

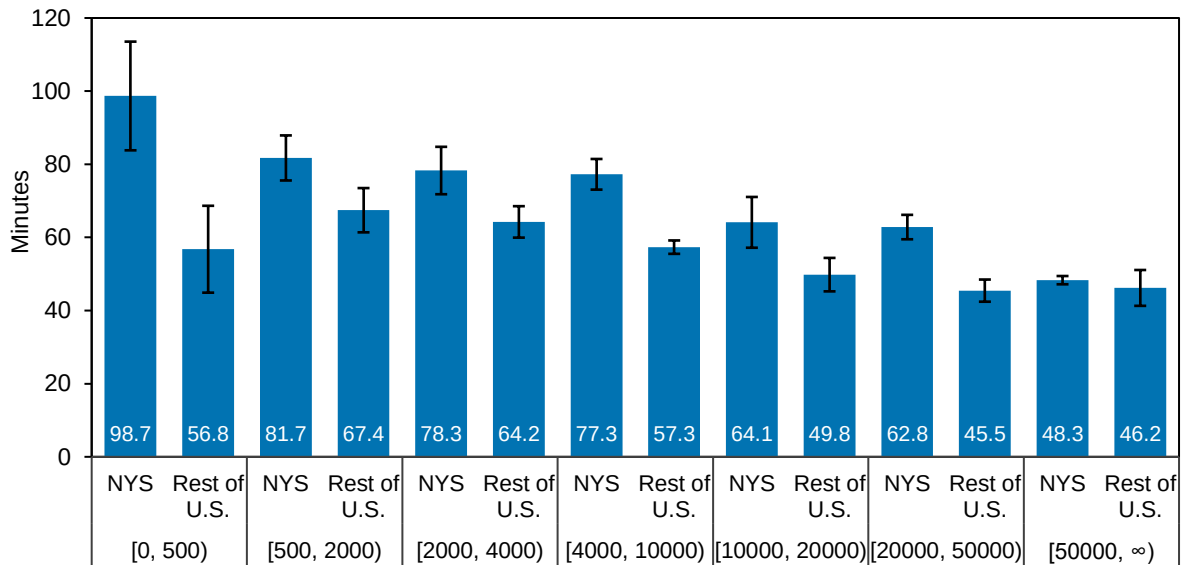


Figure 4-32. Metro area commute trip travel time of public transit in 2017 by region and population density.

4.5 TRAVEL BY MANHATTAN RESIDENTS

Approximately 90% of Manhattan households are in densely populated areas with a tract density exceeding 50,000 people per square mile, making it one of the most densely populated urban areas in the United States. Manhattan's unique combination of high residential density, extensive public transit infrastructure, and limited reliance on personal vehicles sets it apart from other urban areas. This raises the question of whether Manhattan residents travel differently compared with individuals living in similarly dense urban regions across the country.

In terms of the mode share (Figure 4-33), Manhattan residents have more walking trips. For example, in 2017, Manhattan residents have 55% of person trips using walking, with only 44% in the rest of NYC. Compared with 2001, 2009 walking share increased from 51% in 2001 to 54% to 2009 and further to 55% in 2017. Furthermore, the share of POV trips (12%) was almost half of that in the rest of NYC (22%) in 2017.

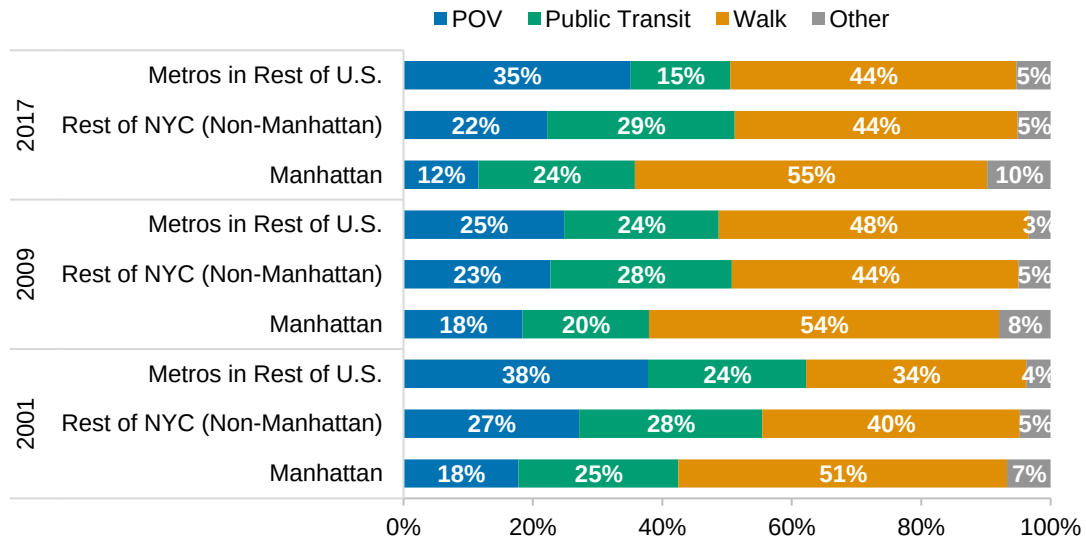


Figure 4-33. Mode choices statistics by year and region. Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number.

Over the years, travel and household characteristics in Manhattan, the rest of NYC, and other metropolitan areas have undergone shifts, reflecting changes in transportation choices, vehicle ownership, and commuting behavior (Table 4-19). Household vehicle ownership patterns highlight a stark contrast between Manhattan and the rest of NYC. In Manhattan, a significant majority of households remained car-free, with 73.3% having no vehicles in 2017, a slight decline from 77.4% in 2009. Meanwhile, the rest of NYC experienced a gradual increase in vehicle ownership, with fewer zero-vehicle households and a rise in single-vehicle households. This shift suggests a growing reliance on private transportation outside of Manhattan, possibly driven by accessibility improvements or reduced public transit dependence. Similarly, whereas the number of drivers per household remained relatively stable in the rest of NYC, Manhattan saw a slight drop, reinforcing its lower dependence on personal vehicles.

Travel behavior followed a similar pattern, with overall trip rates declining between 2001 and 2017 across all regions. Manhattan consistently had trip rates higher than those of the rest of NYC, but both areas saw a reduction in PMT, particularly Manhattan, where PMT per person dropped from 27.46 miles in 2001 to 10.56 miles in 2017. Walking remained the dominant mode of travel in Manhattan, with its share increasing from 50.8% in 2001 to 54.5% in 2017, indicating a continued reliance on pedestrian-friendly infrastructure. In contrast, public transit use declined slightly in Manhattan and more significantly in the rest of NYC, where it dropped from 28% in 2009 to 15.4% in 2017. This decline coincided with an increase in private vehicle use, particularly in less dense areas. Although vehicle trips per driver and VMT per driver remained lower in Manhattan, they showed slight increases in the rest of NYC, reinforcing the trend toward private vehicle dependency. These shifts suggest evolving urban mobility patterns shaped by infrastructure availability, transit reliability, and broader lifestyle changes, such as remote work and the growing prevalence of digital services.

Table 4-19. Household and travel characteristics of Manhattan, other NYC residents, and residents living in very dense metro areas by year.

	2001			2009			2017		
	Manhattan	Rest of NYC ^a	Metros in Rest of U.S.	Manhattan	Rest of NYC ^a	Metros in Rest of U.S.	Manhattan	Rest of NYC ^a	Metros in Rest of U.S.
Workers per Household	1.15	1.33	1.11	1.22	1.28	1.75	1.02	1.46	1.64
Drivers per Household	0.39	0.52	0.58	0.28	0.47	0.82	0.88	1.35	1.63
Vehicles per Driver	0.39	0.48	0.62	0.21	0.30	0.54	0.38	0.34	0.56
Vehicle Ownership (%) ^b									
0 vehicles	68.7%	56.8%	49.1%	77.4%	62.3%	27.0%	73.3%	59.9%	30.7%
1 vehicle	27.2%	35.9%	43.7%	18.1%	30.1%	65.5%	21.8%	34.2%	47.3%
2 vehicles	3.1%	6.5%	6.9%	3.7%	6.5%	6.2%	4.1%	5.3%	21.6%
3 vehicles	0.5%	0.6%	0.3%	0.6%	1.2%	1.4%	0.5%	0.5%	0.0%
4+ vehicles	0.5%	0.2%	0.0%	0.3%	0.0%	0.0%	0.3%	0.0%	0.4%
Driver Rate % (Drivers/16+)									
Average	61.4%	49.4%	64.8%	72.8%	66.1%	63.2%	59.4%	52.6%	78.5%
Male	73.3%	63.4%	67.3%	80.1%	75.0%	74.8%	70.7%	65.1%	81.7%
Female	51.0%	37.4%	58.5%	66.6%	58.2%	43.3%	49.4%	41.9%	75.7%
Person Trips ^b	3.91	3.15	3.60	3.96	3.34	3.23	3.81	3.12	4.28
POV	17.8%	27.1%	37.8%	18.4%	22.7%	24.8%	11.6%	22.2%	35.1%
Public	24.7%	28.2%	24.4%	19.5%	28.0%	23.9%	24.2%	29.0%	15.4%
Walk	50.8%	39.8%	34.0%	54.1%	44.3%	47.9%	54.5%	43.7%	44.2%
Other	6.7%	4.7%	3.8%	8.0%	4.9%	3.4%	9.8%	5.1%	5.3%
Unreported	0.1%	0.2%	0.1%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
PMT	27.46	14.18	24.00	18.22	14.69	7.10	40.06	20.17	32.32
Average Trip Length (miles)	7.59	5.32	7.75	4.84	4.98	2.59	10.56	6.47	7.55
Vehicle Trips	0.77	1.19	1.83	0.55	0.78	1.23	0.45	0.94	1.48
VMT	11.66	9.10	12.06	8.29	6.70	4.85	9.53	7.33	11.26

^a "Rest of NYC" excludes Manhattan.^b Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

5. COMPARISON OF TRAVEL PATTERNS AMONG NEW YORK STATE METROPOLITAN PLANNING AREAS

Previous chapters of this report compare the travel patterns of NYS residents with those in the rest of the nation. This chapter takes a closer look at how travel behavior has evolved across NYS's 13 MPO-based metropolitan areas over time, using data from the 2001, 2009, and 2017 NHTS. The analyzed metros are Albany, Binghamton, Watertown, Buffalo, Elmira, Glens Falls, Ithaca, Kingston, the New York Metropolitan Transportation Council (NYMTC) area, Newburgh, Poughkeepsie, Rochester, Syracuse, and Utica–Rome. Figure 5-1 illustrates population density distributions across NYS in 2017, with more detailed maps available in the Appendix for further geographic scrutiny at the census tract level.

As discussed in Chapter 4, each of these metropolitan areas corresponds to a NYS MPO region. However, owing to the nature of the NHTS sampling frame, which primarily covers the urban core counties of each MPO, some metropolitan areas may not be fully represented in every survey cycle. For consistency, these regions are referred to as metro areas or metros in this analysis. By examining data across 3 survey years, this chapter provides insights into changes in travel behavior over time, particularly in response to economic shifts, demographic trends, and policy changes.

To facilitate meaningful comparisons, the NYMTC region is further divided into three geographic components: (1) Nassau and Suffolk Counties; (2) the five boroughs of NYC (the Bronx, Kings/Brooklyn, Queens, New York/Manhattan, and Richmond/Staten Island); and (3) Putnam, Rockland, and Westchester counties. Within NYC, Manhattan is examined separately from the rest of NYC to account for its unique travel patterns. These metro areas and their subcomponents are grouped according to US Census-defined MSA size categories, allowing for comparisons among small, medium, and large metros. The classifications used in 2017 remain consistent with those in 2009 and 2001 to ensure continuity, with the exception of Kingston, which was newly added in 2009 and classified as a “Not in an MSA” area (see Table 5-1).

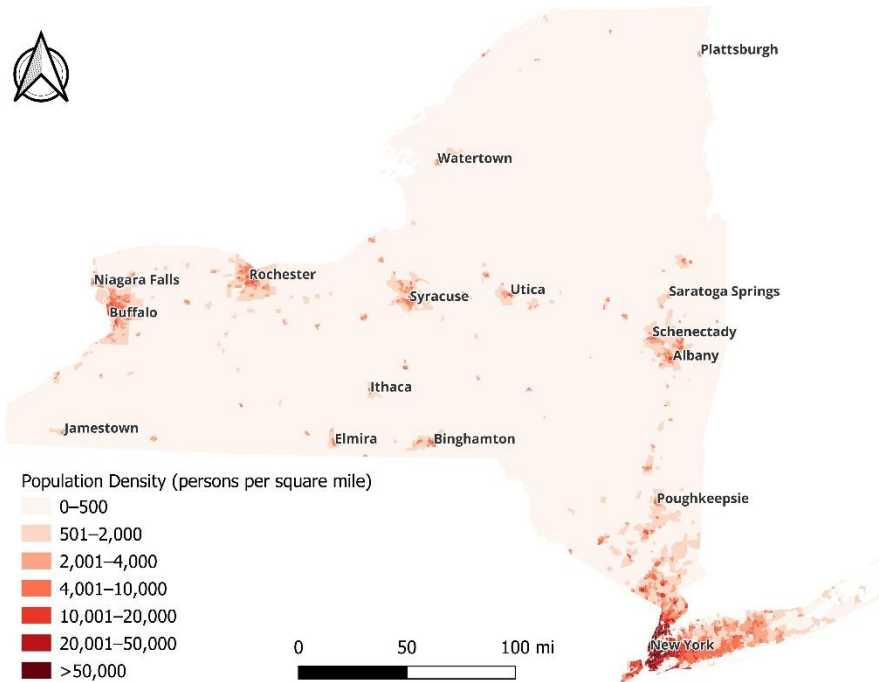


Figure 5-1. Population densities by 2017 census tracts in NYS.

Table 5-1. Population size categories and metropolitan areas in each.

MSA Size Class	NYS Metropolitan Areas
Not in MSA	• Ithaca • Kingston
<250,000	• Watertown • Elmira • Glens Falls
250,000–499,999	• Binghamton • Utica–Rome
500,000–999,999	• Albany • Syracuse
1 million–2.9 million	• Buffalo • Rochester
3+ million	• Newburgh • Poughkeepsie • Nassau/Suffolk • NYC (NYC’s 5 counties; Bronx, Kings, Queens, Manhattan, and Richmond) • Putnam/Rockland/Westchester

5.1 PROFILE OF NEW YORK STATE METRO AREA HOUSEHOLDS BY MSA SIZE

5.1.1 Workers Per Household, Vehicles Per Household, and Vehicles Per Driver

Table 5-2 and Figure 5-2 present data on household distribution across different MSA size classes in NYS for the years 2001, 2009, and 2017. The data reveal trends in household distribution across various MSA size classes over time. The number of households in the largest metropolitan areas (with populations of 3 million or more) saw a significant increase from 4.7 million in 2001 to 7.9 million in 2017. This indicates a growing concentration of households in the largest urban areas. Meanwhile, mid-sized metro areas with

populations between 1 and 2.9 million experienced modest growth, increasing from 769,000 households in 2001 to 822,000 in 2017. However, smaller metro areas, particularly those with populations below 500,000, saw relatively stagnant or declining household numbers. For instance, MSAs with populations between 250,000 and 499,999 declined from 220,000 households in 2001 to 189,000 in 2017.

Households in nonmetropolitan areas increased slightly over time, from 37,000 in 2001 to 115,000 in 2017, though they still represent an exceedingly small share of the total. The percentage distribution of households further reinforces these patterns. The share of households in the largest MSAs rose from 74% in 2001 to 82% in 2017, whereas mid-sized metro areas saw a decline in their overall share. Similarly, the percentage of households in metro areas with populations between 500,000 and 999,999 decreased from 8% in 2001 to 6% in 2017. These trends suggest a growing dominance of larger metro areas in NYS's household distribution, with smaller and mid-sized metro areas either experiencing slower growth or losing their share of households over time.

Table 5-2. Statistics for NYS metro area households.

	MSA Population Size Class					
	Not in MSA	<250,000	250,000–499,999	500,000–999,999	1 million–2.9 million	3 million+
2001	37	85	220	508	769	4698
2009	107	93	214	505	855	4661
2017	115	111	189	543	822	7863

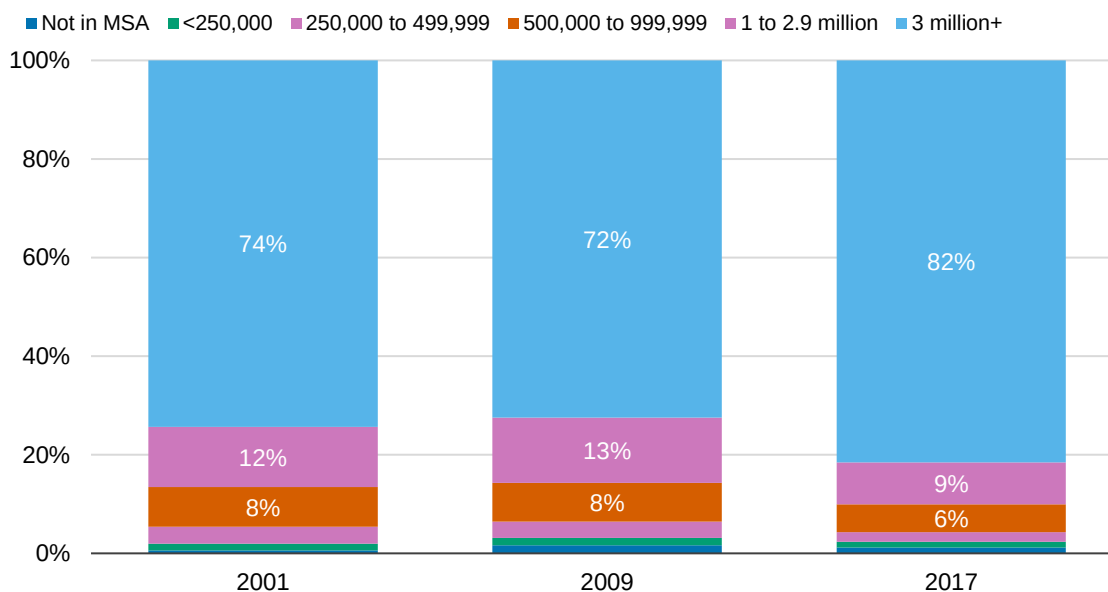


Figure 5-2. Statistics for NYS metro area households by year and MSA size. Note: Unlabeled segments represent values lower than 6%.

Table 5-3 provides estimates of household characteristics and vehicle ownership distributions across different MSA size classes in NYS in 2017. The data include metrics such as the number of households, drivers, workers, vehicles, and vehicle availability per household and per driver. Additionally, the data present the distribution of households by the number of vehicles owned. The number of workers per household varies across MSA size classes, with generally higher values in larger metro areas. In smaller MSAs and rural areas, the number of workers per household tends to be lower. Similarly, the number of

vehicles per household follows a pattern where larger MSAs, particularly those with over 3 million people, have fewer vehicles per household compared with smaller metro areas and non-MSA areas. The number of vehicles per driver is lower in larger MSAs, reflecting higher transit availability and lower car dependence in urban areas. The distribution of households by the number of vehicles owned shows variation by MSA size. The share of households with zero vehicles is highest in the largest metro areas, particularly in NYC, where more than half of households do not own a vehicle. In contrast, in smaller MSAs and rural areas, an exceedingly small proportion of households are without a vehicle. Households with multiple vehicles are more common in less densely populated areas, with the share of households owning three or more vehicles significantly higher in smaller metro areas and non-MSA regions compared with larger urban areas.

Table 5-4 shows data on household characteristics and vehicle ownership distributions across different MSA size classes in NYS in 2009. The number of workers per household varies across MSA size classes, with generally higher values in larger metro areas. In smaller MSAs and rural areas, the number of workers per household tends to be lower. Similarly, the number of vehicles per household follows a pattern where larger MSAs, particularly those with over 3 million people, have fewer vehicles per household compared with households in smaller metro areas and non-MSA areas. The number of vehicles per driver is lower in larger MSAs, reflecting higher transit availability and lower car dependency in urban areas. The distribution of households by the number of vehicles owned shows variation by MSA size. The share of households with zero vehicles is highest in the largest metro areas, particularly in NYC, where over half of households do not own a vehicle, and this figure is even higher in Manhattan. In contrast, in smaller MSAs and non-MSA areas, an exceedingly small proportion of households are without a vehicle. Households with multiple vehicles are more common in less densely populated areas, with the share of households owning three or more vehicles significantly higher in smaller metro areas and non-MSA regions compared with the households in larger urban areas.

Table 5-5 presents household characteristics and vehicle ownership patterns based on the 2001 NHTS data. The number of workers per household varies across MSA size classes. Smaller MSAs and non-MSA areas tend to have slightly fewer workers per household compared with larger metro areas, though the highest values are observed in midsized MSAs such as Newburgh and Poughkeepsie. The number of vehicles per household also follows a distinct pattern, with smaller metro areas and non-MSA areas having higher vehicle ownership rates, whereas the largest MSAs, particularly NYC, exhibit significantly lower rates. The number of vehicles per driver remains relatively consistent across most MSAs but is notably lower in NYC, especially in Manhattan, where public transit reliance is higher. The distribution of households by vehicle ownership further highlights differences in travel behavior across metro areas. The share of households with zero vehicles is highest in NYC, where more than half of households do not own a vehicle, and even higher in Manhattan. In contrast, smaller metro areas and non-MSA areas have extremely low shares of zero-vehicle households. Households with multiple vehicles are far more common in less densely populated areas, with the share of households owning three or more vehicles highest in smaller MSAs and non-MSA areas. These trends reflect the varying transportation needs and availability of transit options across different regions in NYS.

Table 5-3. Household characteristics and vehicle ownership patterns in 2017 by location (e.g., city, region) and MSA size.

MSA Size Class																		
Not in MSA			<250,000			250,000–499,999		500,000–999,999		1 million–2.9 million		3 million+						
Ithaca	Kingston		Watertown	Elmira	Glens Falls	Binghamton	Utica–Rome	Albany	Syracuse	Buffalo	Rochester	Newburgh	Poughkeepsie	NYMTC				
														Nass-Suff	Putn-Rock-West	NYC	Subcomponents of NYC	
																	Rest of NYC	Manhattan
Households	55	60	32	38	41	86	103	323	220	462	360	149	128	788	544	3127	2152	975
Drivers	70	134	55	48	97	140	178	627	331	718	569	240	211	2028	917	4246	3386	860
Workers	54	96	34	35	64	99	126	457	262	559	416	180	156	1461	685	4278	3291	988
Vehicles	86	110	48	59	78	135	174	575	378	741	615	299	261	1657	987	1968	1636	333
Workers per Household	0.97	1.6	1.07	0.93	1.56	1.15	1.23	1.42	1.19	1.21	1.15	1.21	1.22	1.86	1.26	1.37	1.53	1.01
Vehicles per Household	1.55	1.83	1.52	1.56	1.9	1.57	1.69	1.78	1.72	1.6	1.71	2.01	2.04	2.1	1.81	0.63	0.76	0.34
Vehicles per Driver	1.23	0.82	0.88	1.23	0.81	0.97	0.98	0.92	1.14	1.03	1.08	1.25	1.24	0.82	1.08	0.46	0.48	0.39
Distribution of Households by Household Vehicle Count (%) ^a																		
0 vehicles	12.9	6.6	9.6	14.7	7.3	14.8	15.5	9.3	12.4	15.4	12.1	12.1	6.7	5.1	10	54	46	12.9
1 vehicle	46.2	39.4	45.2	41.9	36.7	39.4	38.5	38.3	33.6	38.2	33.9	26.7	31.8	26.5	33.1	32.8	36.8	46.2
2 vehicles	26.6	30.9	34.7	26.8	31.8	29.2	26.6	30.3	35.1	27.1	34.5	31.4	32.1	39.1	34.9	10.6	13.7	26.6
3 vehicles	8.3	15.3	6.8	10.3	16	11.2	9.9	14.1	11	12.4	12.5	15.5	17.4	16.6	13.5	1.6	2.2	8.3
4+ vehicles	6	7.8	3.7	6.4	8.2	5.4	9.5	8.1	7.9	7	7	14.3	11.9	12.6	8.5	1	1.3	6

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 5-4. Household characteristics and vehicle ownership patterns in 2009 by location (e.g., city, region) and MSA size.

		MSA Size Class															
Not in MSA		<250,000		250,000–499,999		500,000–999,999		1 million–2.9 million		3 million+							
Ithaca	Kingston	Elmira	Glens Falls	Binghamton	Utica–Rome	Albany	Syracuse	Buffalo	Rochester	Newburgh	Poughkeepsie	NYMTC					
												Nass-Suff	Putn-Rock-West	NYC	Subcomponents of NYC		
															Rest of NYC	Manhattan	
Households	37	70	35	58	98	116	320	185	472	383	122	103	911	469	3055	2306	749
Drivers	81	138	63	110	181	207	587	329	807	712	256	218	2081	923	4836	3792	1044
Workers	67	103	45	81	125	149	428	236	586	486	194	160	1503	694	3978	3016	962
Vehicles	55	134	58	104	167	193	535	293	745	648	230	199	1784	783	1943	1722	221
Workers per Household	1.79	1.47	1.29	1.39	1.28	1.28	1.34	1.28	1.24	1.27	1.59	1.55	1.65	1.48	1.30	1.31	1.28
Vehicles per Household	1.47	1.91	1.66	1.78	1.71	1.66	1.67	1.58	1.58	1.69	1.88	1.92	1.96	1.67	0.64	0.75	0.29
Vehicles per Driver	0.69	0.97	0.91	0.95	0.92	0.93	0.91	0.89	0.92	0.91	0.90	0.91	0.86	0.85	0.40	0.45	0.21
Distribution of Households by Household Vehicle Count (%) ^a																	
0 vehicles	12.3	6.8	9.3	6.2	10.5	10.8	9.1	12.8	12.5	9.9	8.8	7.0	5.7	12.2	54.6	47.7	75.7
1 vehicle	44.4	33.3	36.4	36.0	34.9	36.6	35.9	36.5	37.4	33.9	28.0	30.9	26.9	35.0	31.4	35.1	20.2
2 vehicles	31.9	38.7	40.5	39.4	36.8	37.2	40.4	37.1	36.5	39.7	40.7	38.8	42.8	35.2	10.9	13.4	3.3
3 vehicles	8.0	11.2	8.4	12.5	12.0	10.7	10.6	9.3	9.4	11.6	14.8	14.2	18.9	11.7	2.2	2.7	0.6
4+ vehicles	3.4	10.0	5.4	5.9	5.8	4.8	4.0	4.2	4.2	5.0	7.6	9.0	5.8	5.9	0.9	1.1	0.3

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 5-5. Household characteristics and vehicle ownership patterns in 2001 by location (e.g., city, region) and MSA size.

MSA Size Class																
	Not in MSA	<250,000		250,000– 499,999		500,000– 999,999		1 million– 2.9 million		3 million+						
	Ithaca	Elmira	Glens Falls	Binghamton	Utica–Rome	Albany	Syracuse	Buffalo	Rochester	Newburgh	Poughkeepsie	NYMTC				
												Nass- Suff	Putn- Rock- West	NYC	Subcomponents of NYC	
															Rest of NYC	Manhattan
Households	37	36	49	102	118	324	184	477	292	117	101	933	471	3076	2324	752
Drivers	64	63	90	174	200	544	306	753	497	223	179	1860	883	3779	3018	761
Workers	49	45	61	118	145	416	219	589	364	173	142	1450	700	4045	3192	853
Vehicles	62	66	95	182	213	578	279	817	521	230	192	1755	811	2085	1795	290
Workers per Household	1.32	1.27	1.25	1.16	1.22	1.29	1.19	1.23	1.25	1.48	1.4	1.55	1.49	1.32	1.37	1.13
Vehicles per Household	1.67	1.85	1.93	1.78	1.8	1.78	1.52	1.71	1.79	1.97	1.89	1.88	1.72	0.68	0.77	0.39
Vehicles per Driver	0.96	1.04	1.05	1.05	1.06	1.06	0.91	1.08	1.05	1.03	1.07	0.94	0.92	0.55	0.59	0.38
Distribution of Households by Household Vehicle Count (%) ^a																
0 vehicles	9.41	7.6	7.52	11.14	8.32	9.06	12.62	10.24	7.19	5.43	5.61	4.77	10.19	51.59	45.9	69.16
1 vehicle	37.19	34.14	28.27	29.62	35.91	33.1	37.89	35.74	34.52	27.04	35.03	30.63	33.04	33.75	35.98	26.83
2 vehicles	37.02	35.18	39.27	39.46	33.69	38.32	37.92	36.29	39.94	43.13	35.6	43.13	38.33	11.35	14.03	3.07
3 vehicles	11.75	14.66	17	13.79	15.96	12.73	8.73	12.13	11.74	16.34	16.15	15.99	12.67	2.67	3.36	0.52
4+ vehicles	4.63	8.42	7.94	6	6.12	6.8	2.84	5.59	6.6	8.07	7.61	5.48	5.77	0.65	0.73	0.42

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest hundredth of a percent.

5.1.2 Vehicle Ownership

The vehicle ownership patterns in NYS metro areas in 2017 varied significantly by MSA size, reflecting differences in household composition, employment levels, and transportation preferences (Table 5-3). In general, smaller MSAs and non-MSA areas exhibited higher vehicle ownership rates, whereas larger metropolitan areas, particularly NYC, showed much lower vehicle availability per household.

In smaller metro areas and non-MSA regions, households had a relatively high number of vehicles per household, typically exceeding 1.5 vehicles. For instance, non-MSA areas had 1.55 vehicles per household, whereas MSAs with populations below 250,000 had even higher rates, such as 1.9 vehicles per household in Glens Falls and 1.83 in Kingston. These areas also had a relatively high percentage of households owning two or more vehicles, with more than 30% of households in many of these regions possessing at least two vehicles. Additionally, a small but notable proportion of households owned three or more vehicles, particularly in areas like Kingston, where 15.3% of households had three vehicles, and Glens Falls, where 8.2% of households had four or more vehicles. These patterns suggest a strong reliance on personal vehicles in smaller metro areas, likely due to lower transit accessibility and longer travel distances (Pan et al., 2024). Midsized MSAs, ranging from 250,000 to just under 3 million in population, showed a moderate vehicle ownership pattern. The average number of vehicles per household ranged from 1.6 to 1.81, with cities like Albany (1.78), Syracuse (1.72), and Buffalo (1.6) displaying values close to or slightly above the statewide average. Although one-vehicle households were still prevalent, a substantial portion of households owned two or more vehicles. In this category, the proportion of zero-vehicle households was lower than those in smaller MSAs but still significant, particularly in Binghamton (14.8%) and Buffalo (15.4%). The relatively high worker-to-household ratios in these midsized MSAs, often exceeding 1.2 workers per household, indicate a stronger demand for mobility, which may contribute to increased vehicle ownership.

In contrast, NYC and its surrounding areas exhibited dramatically different vehicle ownership patterns. The NYC metropolitan region, especially within the five boroughs, had the lowest vehicle ownership levels in the state. The total number of vehicles per household in NYC was just 0.63, with Manhattan exhibiting the lowest rate at 0.34 vehicles per household. The high percentage of zero-vehicle households was striking, with more than half (54%) of households in NYC reporting no personal vehicle. Even in the broader metropolitan area, including counties like Nassau, Suffolk, Putnam, Rockland, and Westchester, vehicle ownership remained relatively low, averaging 2.1 vehicles per household, which, whereas higher than in NYC proper, was still lower than ownership in most midsized MSAs. The percentage of zero-vehicle households in this category was also significantly lower (5.1%), reflecting the suburban nature of these regions where car ownership is more common but less essential than in rural areas. The differences in vehicle ownership across these regions are largely attributable to variations in transportation infrastructure and commuting patterns. In NYC, the extensive public transit network reduces the need for car ownership, whereas smaller metro areas and non-MSA regions, which typically lack robust public transportation, exhibit a heavier reliance on private vehicles.

5.1.3 Driving and Gender

Table 5-6 presents the percentage of the population that drives, categorized by gender, for various MSAs in NYS in 2001, 2009, and 2017. Between 2001 and 2017, the percentage of male and female drivers generally declined across most MSAs, particularly in larger metropolitan areas. In smaller MSAs and non-MSA regions, the changes were less pronounced, with some areas maintaining relatively high rates of drivers among both men and women. In 2001, male driving rates were high across all regions, with most areas reporting over 90% of males ages 16 and older as drivers. Female driving rates were slightly lower but still above 85% in most regions except for NYC, where only 50% of females were drivers compared

with 72% of males. The difference between genders was especially stark in NYC, indicating a much lower rate of female licensure or access to vehicles compared with men. By 2009, male and female driving rates remained relatively stable in most regions, with minor declines in some areas. Notably, driving rates in smaller MSAs such as Kingston and Binghamton saw slight drops, whereas some midsized MSAs like Albany and Rochester continued to report high driving rates. The gap between male and female drivers remained statistically significant, particularly in NYC, though a slight increase in female driving rates took place in NYC (from 50% to 65%). By 2017, a statistically significant decline in driving rates occurred, particularly among males, across most regions. The male driving rate dropped significantly in some areas, such as Ithaca (from 92% in 2001 to 83% in 2017) and Elmira (from 91% to 73%). Larger metropolitan areas like Albany, Buffalo, and Rochester maintained relatively high male driving rates, but NYC saw a continued decline, with only 70% of males driving. Female driving rates followed a similar trend, with statistically significant declines in regions such as Buffalo (from 92% in 2001 to 74% in 2017) and Elmira (from 89% to 71%). In contrast, some smaller MSAs, like Kingston, retained stable or slightly higher female driving rates.

Smaller MSAs (e.g., Ithaca, Kingston, Elmira, Watertown) exhibited higher rates of vehicle access for both men and women, with over 85% of each gender consistently reporting driving across the years. The highest rates were observed in non-MSAs and smaller metro areas, where public transportation options are limited, necessitating greater vehicle dependency. Midsized MSAs (250,000 to 2.9 million population)—such as Albany, Syracuse, and Buffalo—demonstrated slightly lower driving rates but remained above 80% for most years. These areas have more public transportation options than smaller metro areas but still rely heavily on personal vehicles for commuting. The largest MSA category, NYC (3 million+), consistently had the lowest driving rates, particularly among females. The disparity between male and female drivers was most significant in NYC, with only 50% of females being drivers in 2017 compared with 70% of males. This reflects the strong public transportation network in the city, which reduces the need for personal vehicle ownership, particularly among women.

Table 5-6. Percentage of population who drive by gender (drivers/16+) by location (e.g., city, region) and MSA size.

MSA Size Class																			
Not in MSA			<250,000			250,000–499,999		500,000–999,999		1 million–2.9 million		3 million+							
Ithaca	Kingston		Watertown	Elmira	Glens Falls	Binghamton	Utica–Rome	Albany	Syracuse	Buffalo	Rochester	Newburgh	Poughkeepsie	NYMTC					
														Nass-Suff	Putn-Rock-West	NYC	Subcomponents of NYC		
																	Rest of NYC	Manhattan	
2001																			
Male	92%	—	—	91%	96%	91%	96%	95%	91%	91%	93%	91%	91%	91%	94%	72%	71%	73%	
Female	91%	—	—	87%	92%	89%	86%	88%	90%	82%	92%	90%	88%	88%	84%	50%	49%	52%	
2009																			
Male	92%	95%	—	92%	93%	91%	88%	90%	94%	90%	92%	92%	93%	93%	92%	79%	78%	82%	
Female	92%	89%	—	87%	91%	89%	87%	88%	87%	86%	87%	85%	90%	87%	82%	65%	64%	68%	
2017																			
Male	83%	85%	85%	73%	90%	84%	79%	92%	82%	86%	92%	89%	90%	89%	89%	70%	70%	72%	
Female	84%	89%	92%	71%	85%	76%	75%	86%	84%	74%	82%	76%	86%	85%	81%	50%	50%	50%	

5.2 PERSONAL TRAVEL COMPARISONS

Table 5-7 presents personal travel statistics for NYS metro areas in 2017, segmented by MSA size class. It includes the number of person trips per person, average trip length, and the distribution of trips by purpose. The number of total daily trips per person varied across MSAs, with smaller metro areas generally having trip rates higher than those for larger ones. In non-MSA areas, the average was 4.00 trips per person, significantly higher than those in metro areas such as Buffalo (3.15 trips) and NYC (3.27 trips). Notably, Ithaca had the highest trip rate at 4.00 trips, which is significantly higher than many metro areas, including Albany (3.26 trips) and Rochester (3.41 trips). When looking at gender differences, men and women exhibited similar travel patterns overall, but with some variations by MSA. For example, in Kingston, men made 3.02 trips per person, whereas women made 3.23 trips with a statistically significant gap. Similarly, in Putnam–Rockland–Westchester, women traveled statistically significantly more frequently than men did (3.52 trips and 3.31 trips, respectively).

Trip length varied significantly across MSAs, with smaller areas tending to have longer average trips. The longest average trip length was observed in Poughkeepsie (13.48 miles), statistically significantly higher than in Syracuse (7.71 miles) and Manhattan (7.08 miles). Ithaca, a non-MSA area, also had a notably long average trip distance at 7.87 miles, but this was not significantly different from many other small metro areas like Glens Falls (9.87 miles). Larger MSAs tended to have shorter trips, particularly in NYC, where Manhattan had an average trip length of only 7.08 miles, which was statistically significantly shorter than trips in areas like Rochester (9.87 miles) and Albany (10.35 miles). The short travel distances in urban cores are likely due to greater availability of transit and nonmotorized travel options.

Trip purposes were divided into categories such as work-related travel; family and personal business; civic, educational, and religious trips; social and recreational travel; and other purposes. Across MSAs, “Family & Personal Business” was the dominant category, comprising around 30%–40% of all trips, with some variation. For instance, in Kingston, 36.5% of trips fell into this category, significantly higher than those in Binghamton (26.4%). The proportion of trips taken for work was relatively consistent across regions, generally ranging from 17% to 21%. However, Poughkeepsie had the highest work trip share at 20.6%, which is statistically significantly greater than work trips in Albany (17.6%). Social and recreational trips showed some of the widest variations across MSAs. In non-MSA areas such as Ithaca, these trips accounted for 29.0% of all travel, which is statistically higher than in Manhattan (20.3%). Meanwhile, smaller MSAs like Binghamton had 26.4% of trips in this category, statistically significantly higher than Putnam–Rockland–Westchester (21.1%).

For comparison with the 2017 statistics, Table 5-8 and Table 5-9 present personal travel statistics for NYS metro areas in 2009 and 2001, respectively, segmented by MSA size class

Table 5-7. Personal travel statistics in 2017 by location (e.g., city, region) and MSA size.

	MSA Size Class								
	Not in MSA		<250,000			250,000–499,999		500,000–999,999	
	Ithaca	Kingston	Watertown	Elmira	Glens Falls	Binghamton	Utica–Rome	Albany	Syracuse
Person Trips	4	3.13	3.22	2.86	3.26	3.26	3.47	3.44	3.15
Male	3.99	3.02	3.24	2.54	2.84	3.35	3.51	3.32	3.04
Female	4.01	3.23	3.2	3.23	3.64	3.18	3.42	3.56	3.26
Average Person Trip Length (miles)	7.87	12.88	10.23	7.32	12.29	9.87	8.21	13.15	10.33
Trip Purpose (%) ^a									
Earn a Living	17.5	19.6	17.1	17.7	17.0	18.4	16.2	19.8	19.6
Family & Personal Business	38.4	39.3	36.5	39.1	45.8	42.5	45.8	39.2	37.7
Civic, Educational, & Religious	10.2	6.6	14.2	13.2	6.2	9.4	6.6	8.7	13.1
Social & Recreational	29.0	31.4	29.7	26.5	26.5	26.1	28.3	28.6	26.0
Other	4.9	2.9	2.5	3.4	4.5	3.6	3.1	3.5	3.6
Unreported	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.2	0.0

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 5-7. Personal travel statistics in 2017 by location (e.g., city, region) and MSA size (continued).

	MSA Size Class								
	1 million– 2.9 million		3 million+						
	Buffalo	Rochester	Newburgh	Poughkeepsie	NYMTC				
					Nass-Suff	Putn- Rock-West	NYC	Subcomponents of NYC	
								Rest of NYC	Manhattan
Person Trips	3.41	3.24	3.16	3.35	3.42	3.46	3.23	3.1	3.76
Male	3.61	3.3	3.33	3.44	3.31	3.36	3.27	3.15	3.75
Female	3.22	3.19	2.99	3.27	3.52	3.55	3.19	3.05	3.77
Average Person Trip Length (miles)	7.71	9.87	11.97	12.11	13.48	10.74	7.76	7.08	10.06
Trip Purpose (%) ^a									
Earn a Living	18.6	20.2	20.8	20.6	17.5	18.0	20.1	20.3	19.3
Family & Personal Business	40.0	38.4	35.0	36.0	39.3	36.4	37.7	38.8	34.0
Civic, Educational, & Religious	10.8	13.1	19.7	11.1	8.7	11.2	9.0	9.8	6.5
Social & Recreational	26.9	24.5	20.8	28.0	30.4	30.5	29.4	27.5	35.7
Other	3.5	3.7	3.8	4.1	4.2	4.0	3.8	3.5	4.5
Unreported	0.2	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 5-8. Personal travel statistics in 2009 by location (e.g., city, region) and MSA size.

	MSA Size Class							
	Not in MSA		<250,000		250,000–499,999		500,000–999,999	
	Ithaca	Kingston	Elmira	Glens Falls	Binghamton	Utica–Rome	Albany	Syracuse
Person Trips	3.73	3.81	3.99	3.90	3.84	3.66	3.68	4.04
Male	3.58	3.77	3.82	3.79	3.94	3.54	3.73	4.08
Female	3.89	3.84	4.14	4.01	3.74	3.77	3.63	4.00
Average Person Trip Length (miles)	7.27	9.57	8.93	9.09	8.60	7.30	10.69	7.70
Trip Purpose (%) ^a								
Earn a Living	22.4	18.5	17.0	21.2	18.7	18.1	20.7	18.1
Family & Personal Business	38.1	47.5	46.5	41.6	44.6	43.5	41.4	42.1
Civic, Educational, & Religious	6.9	8.7	8.5	7.0	9.1	9.0	9.0	8.7
Social & Recreational	29.6	24.1	26.0	28.7	26.5	28.5	27.2	29.5
Other	1.7	0.7	0.7	0.6	0.6	0.2	0.7	0.8
Unreported	1.3	0.6	1.3	0.9	0.6	0.8	1.2	0.8

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 5-8. Personal travel statistics in 2009 by location (e.g., city, region) and MSA size (continued).

	MSA Size Class								
	1 million– 2.9 million		3 million+						
	Buffalo	Rochester	Newburgh	Poughkeepsie	NYMTC				
					Nass-Suff	Putn- Rock-West	NYC	Subcomponents of NYC	
								Rest of NYC	Manhattan
Person Trips	3.60	3.75	3.57	3.69	4.00	3.86	3.46	3.31	4.04
Male	3.56	3.79	3.50	3.66	3.94	3.88	3.46	3.26	4.28
Female	3.64	3.71	3.63	3.71	4.05	3.84	3.45	3.35	3.82
Average Person Trip Length (miles)	7.96	9.43	10.96	11.21	7.97	8.12	5.20	5.30	4.88
Trip Purpose (%) ^a									
Earn a Living	20.5	19.2	20.1	18.2	19.4	18.2	16.6	16.8	16.0
Family & Personal Business	40.5	41.5	40.9	42.9	43.3	43.2	44.1	44.2	43.4
Civic, Educational, & Religious	8.9	10.5	13.0	9.6	9.6	10.1	9.6	10.9	5.4
Social & Recreational	28.9	27.4	24.1	27.7	25.3	26.8	27.3	25.8	32.4
Other	0.3	0.6	0.8	0.7	1.8	0.7	1.4	1.2	1.7
Unreported	0.8	0.8	1.1	1.0	0.7	0.9	1.0	1.0	1.1

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 5-9. Personal travel statistics in 2001 by location (e.g., city, region) and MSA size.

		MSA Size Class					
		Not in MSA	<250,000		250,000–499,999		500,000–999,999
		Ithaca	Elmira	Glens Falls	Binghamton	Utica–Rome	Albany Syracuse
Person Trips	4.38	4.08	4.02	4.21	4.19	3.96	3.94
Male	4.41	4.10	3.92	4.26	4.30	3.95	4.09
Female	4.35	4.07	4.11	4.16	4.09	3.98	3.81
Average Person Trip Length (miles)	8.22	8.82	10.02	8.78	8.69	8.77	7.79
Trip Purpose (%) ^a							
Earn a Living	18.3	18.5	18.1	16.8	18.6	19.9	18.1
Family & Personal Business	42.5	46.3	46.0	45.9	44.3	42.9	45.3
Civic, Educational, & Religious	11.5	8.0	7.6	7.8	7.5	9.3	8.7
Social & Recreational	26.4	26.2	27.2	28.7	28.5	26.4	26.9
Other	1.2	0.8	0.6	0.6	0.9	1.0	0.5
Unreported	0.1	0.2	0.6	0.3	0.2	0.4	0.6

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 5-9. Personal travel statistics in 2001 by location (e.g., city, region) and MSA size (continued).

	MSA Size Class								
	1million– 2.9 million		3 million+						
	Buffalo	Rochester	Newburgh	Poughkeepsie	NYMTC				
					Nass-Suff	Putn- Rock-West	NYC	Subcomponents of NYC	
								Rest of NYC	Manhattan
Person Trips	4.03	4.00	3.94	3.83	3.79	3.97	3.44	3.34	3.86
Male	3.81	3.93	3.80	3.85	3.79	4.05	3.54	3.50	3.71
Female	4.23	4.07	4.08	3.81	3.80	3.91	3.35	3.20	3.99
Average Person Trip Length (miles)	7.38	8.48	11.68	10.38	8.65	9.51	6.82	6.30	8.61
Trip Purpose (%) ^a									
Earn a Living	16.9	18.0	19.1	18.8	19.9	19.2	20.1	20.0	20.6
Family & Personal Business	43.9	43.6	43.3	44.3	41.5	46.0	42.7	43.7	39.0
Civic, Educational, & Religious	9.4	9.8	9.5	10.5	8.9	7.9	11.1	11.5	9.6
Social & Recreational	28.6	27.3	26.4	25.1	28.0	25.2	24.6	23.4	28.9
Other	1.0	0.9	1.5	1.1	1.5	1.6	1.3	1.2	1.8
Unreported	0.3	0.4	0.2	0.3	0.3	0.3	0.3	0.3	0.2

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

5.2.1 Trip Frequencies

Figure 5-3 presents the average daily person trips in various NYS metro areas for the years 2001, 2009, and 2017, categorized by MSA size. The number of daily trips per person generally declines across nearly all metro areas between 2001 and 2017. This pattern is consistent with national trends indicating reduced trip-making, likely due to increased remote work, online shopping, and changing travel behaviors. Non-MSA areas and smaller metro areas started with higher trip rates in 2001 but saw significant declines over time. Ithaca, for example, dropped from 4.38 trips per person in 2001 to 4.00 in 2017, but this decrease is not statistically significant. However, the decline for Glens Falls from 4.08 to 3.26 is statistically significant. Similarly, Albany experienced a significant decline from 4.19 in 2001 to 3.26 in 2017. Larger metro areas experienced some of the most substantial declines in daily trip rates. NYC's trip rate decreased from 3.46 in 2009 to 3.23 in 2017, but this is not statistically significant. However, Putnam–Rockland–Westchester saw a statistically significant drop from 3.97 in 2001 to 3.46 in 2017. For midsized MSAs, some regions saw sharper declines than others. Rochester, for instance, dropped significantly from 4 trips in 2001 to 3.41 in 2017. Some small MSAs retained relatively stable trip rates. Kingston's decline from 3.81 to 3.13 was not statistically significant.

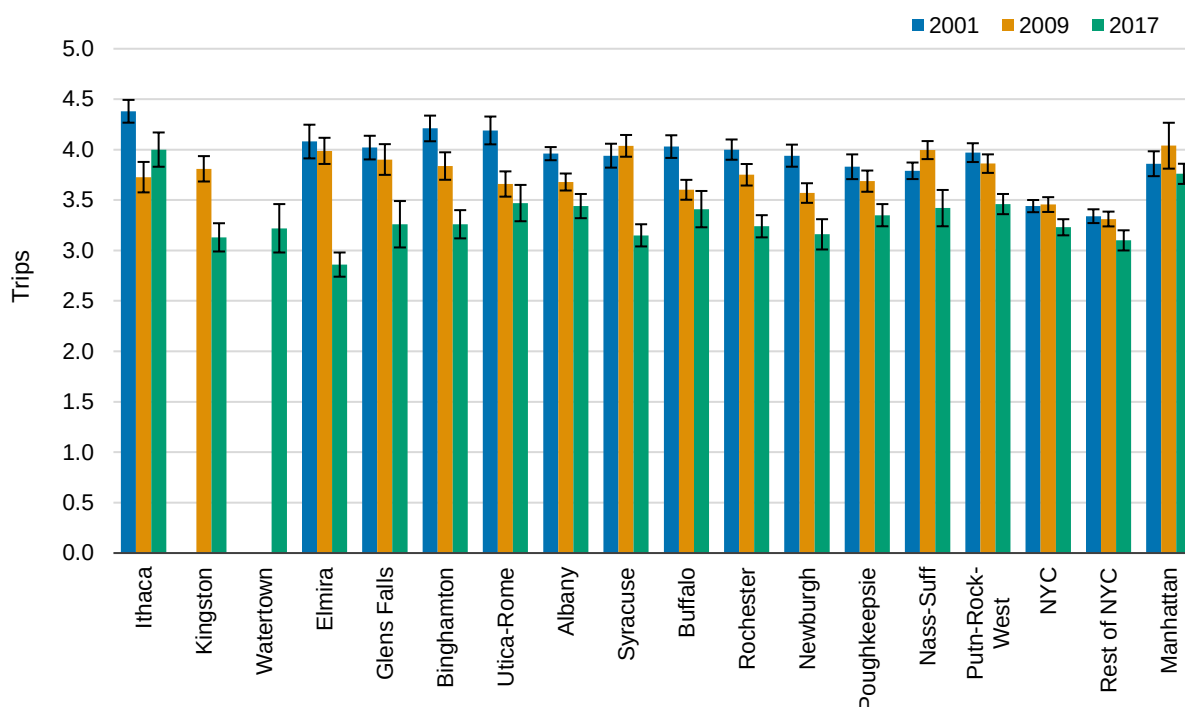


Figure 5-3. Person trips statistics by year and location (e.g., city, region).

5.2.2 Trip Lengths

Unlike trip frequency, which showed a general decline over time, trip distances increased in many areas between 2001 and 2017 (Figure 5-4). This suggests that although people are making fewer trips, the trips they do take tend to cover greater distances. For smaller metro areas and non-MSA regions, there is a clear increase in trip distances over time. For example, Kingston's trip distance rose statistically significantly from 9.57 miles in 2009 to 12.88 miles in 2017. Similarly, Albany saw a statistically significant increase from 10.69 miles in 2009 to 13.15 miles in 2017. In contrast, some regions have seen trip distances remain relatively stable. For instance, Syracuse's trip distance was 7.79 miles in 2001, 7.20

miles in 2009, and 7.71 miles in 2017. However, the change is not statistically significant. Larger MSAs have consistently shorter trip distances compared with trips in smaller MSAs. NYC had an average trip distance in 2001 of only 5.20 miles, which dropped to 5.30 in 2009 and increased slightly to 7.76 in 2017. Although the 2017 distance is higher, the increase is not statistically significant. Manhattan had the shortest trips in all years, rising from 6.30 miles in 2009 to 7.08 in 2017, but still remained much lower than trips in suburban areas. Among suburban and exurban areas, trip distances have grown more substantially. Poughkeepsie saw a statistically significant rise from 10.38 miles in 2001 to 12.11 miles in 2017. Nassau–Suffolk also saw a rise to 13.48 miles, making it one of the regions with the longest trip distances. These trends likely reflect increased commuting distances and suburban expansion.

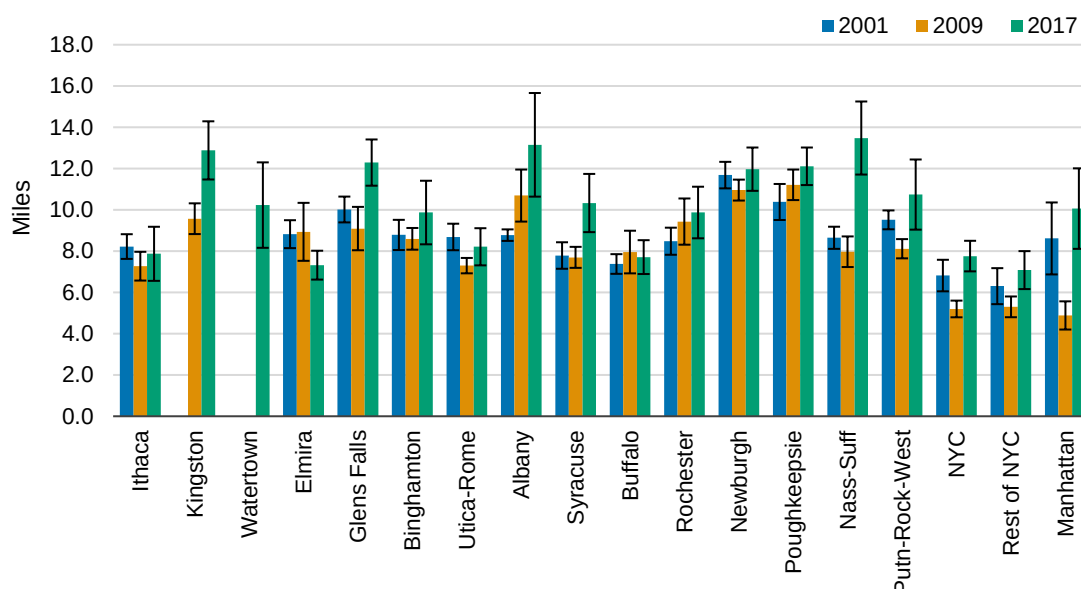


Figure 5-4. Average person trip distances by year and location (e.g., city, region).

5.2.3 Travel Purposes

Figure 5-5 through Figure 5-7 show the distribution of daily person trips by purpose in various NYS metro areas. Categories include trips for earning a living; family and personal business; civic, educational, and religious activities; social and recreational purposes; and other activities.

In 2017, trips for earning a living, including commuting, varied across regions but remained within a narrow range of 16%–21%. Smaller metro areas such as Utica–Rome and Binghamton reported the lowest share (16% and 18%, respectively), whereas larger metro areas like Syracuse and Albany had higher shares (20%). Newburgh and Poughkeepsie had the highest share of commuting trips at 21%, suggesting a higher reliance on work-related travel in these suburban regions. Interestingly, NYC’s share of trips for earning a living was 20.1%, slightly higher than that in many smaller metro areas but consistent with large urban centers like Albany and Syracuse. Manhattan’s share was 19%, similar to those of Buffalo and Rochester, showing that the dense urban core exhibits commuting behavior on par with other urban areas. Family and personal business trips made up the largest category across all regions, ranging from 35% to 46%. Smaller metro areas like Glens Falls, Albany, and Binghamton had the highest proportions at 46%, suggesting a strong dependence on travel for errands and household-related activities in these areas. Urban regions like NYC and Manhattan exhibited slightly lower shares of family and personal trips, at 38% and 34%, respectively. These lower shares may reflect higher access to services and facilities within walkable distances, reducing the need for longer errand-related travel. Social and recreational travel shares ranged from 21% to 36%, with some of the highest percentages in Manhattan

(36%) and Poughkeepsie (36%). The higher share of social and recreational trips in these areas may reflect their urban and suburban lifestyle characteristics, with greater access to recreational facilities and social venues. In contrast, smaller metro areas like Binghamton and Watertown had lower shares (26% and 30%, respectively). NYC overall reported 29%, consistent with its social and recreational opportunities, but slightly lower than Manhattan.

Over the years, daily person trip patterns in NYS metro areas have undergone noticeable shifts, reflecting broader societal, technological, and economic changes. One of the most evident trends is the slight decline in work-related travel across most regions. In 2001, work trips constituted a significant share of travel, particularly in urban areas like Manhattan, where 21% of trips were dedicated to commuting. By 2017, this number had dropped to 19%, a pattern also seen in Ithaca, where work-related travel declined from 22% in 2009 to 18% in 2017. In contrast, NYC overall saw a slight increase in work trips between 2001 and 2017, stabilizing around 20%. Despite fluctuations in work travel, family and personal business trips have consistently been the dominant trip category across all years. This category accounted for over 40% of all trips in many metro areas in 2001 and remained substantial in 2017. However, in some regions, such as Putnam–Rockland–Westchester, the share of these trips declined from 46% in 2001 to 36% in 2017, whereas Albany saw a slight drop from 41% to 39%. The decreasing share of errand-based travel in certain areas may be linked to increased online shopping, greater reliance on delivery services, and more efficient trip planning. Civic, educational, and religious travel, whereas a smaller proportion of total trips, has shown mixed trends across different metro areas. In suburban regions such as Newburgh, trips for civic, educational, and religious purposes grew substantially, from 11% in 2001 to 20% in 2017. This increase suggests greater participation in educational institutions, community services, and civic engagement. However, in Manhattan, the share of civic, educational, and religious travel declined from 11% in 2001 to just 7% in 2017. This shift may be due to changing education patterns, such as increased online learning, shorter commutes to local schools, or a stronger reliance on transit in the urban core.

One of the most notable changes over time has been the increase in social and recreational travel, particularly in urban centers. In Manhattan, these trips made up 29% of total travel in 2001 but increased significantly to 36% in 2017. A similar trend is seen in Poughkeepsie, where the share of social and recreational trips grew from 25% to 30% over the same period. This suggests a growing emphasis on leisure activities, entertainment, and social engagement in urban areas. In contrast, some suburban areas like Newburgh saw a slight decline in social and recreational travel, decreasing from 25% in 2001 to 21% in 2017. This divergence between urban and suburban areas could be attributed to the expansion of recreational amenities in cities and shifting lifestyle preferences among residents.

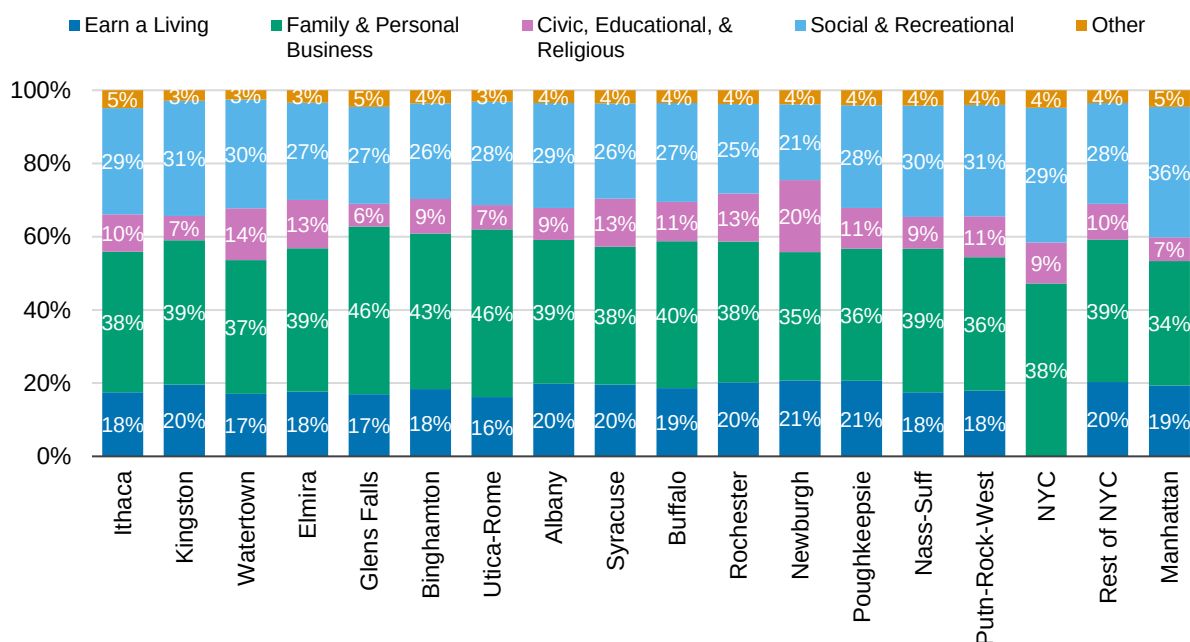


Figure 5-5. Daily person trips by purpose and location (e.g., city, region) in 2017. Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number.

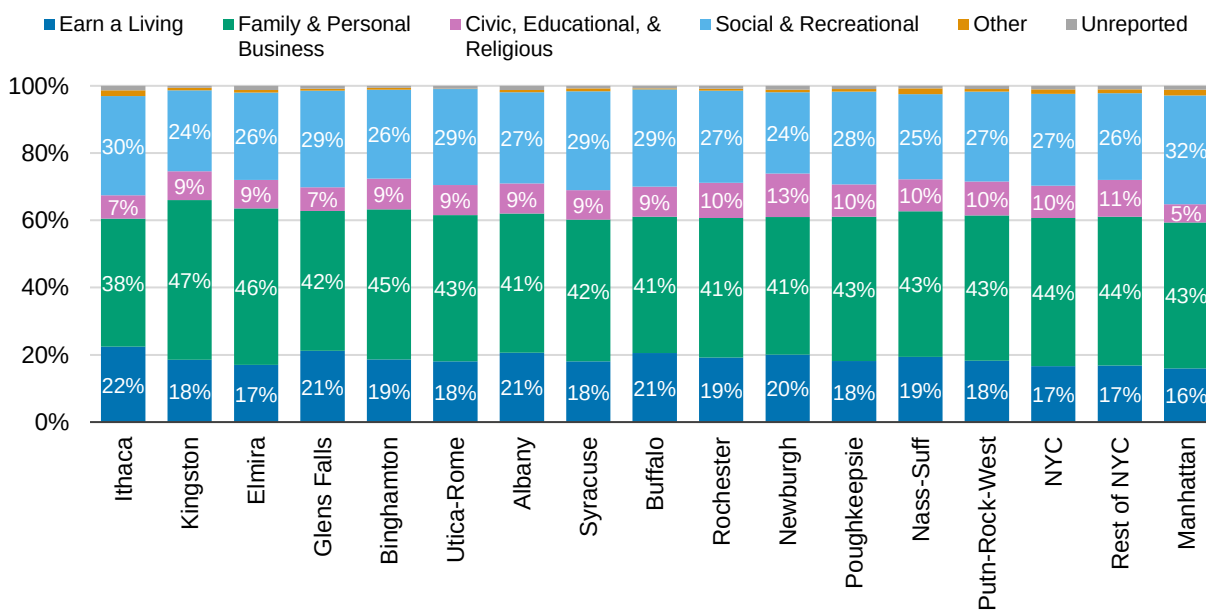


Figure 5-6. Daily person trips by purpose and location (e.g., city, region) in 2009. Notes: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number. Unlabeled segments represent values lower than 5%.

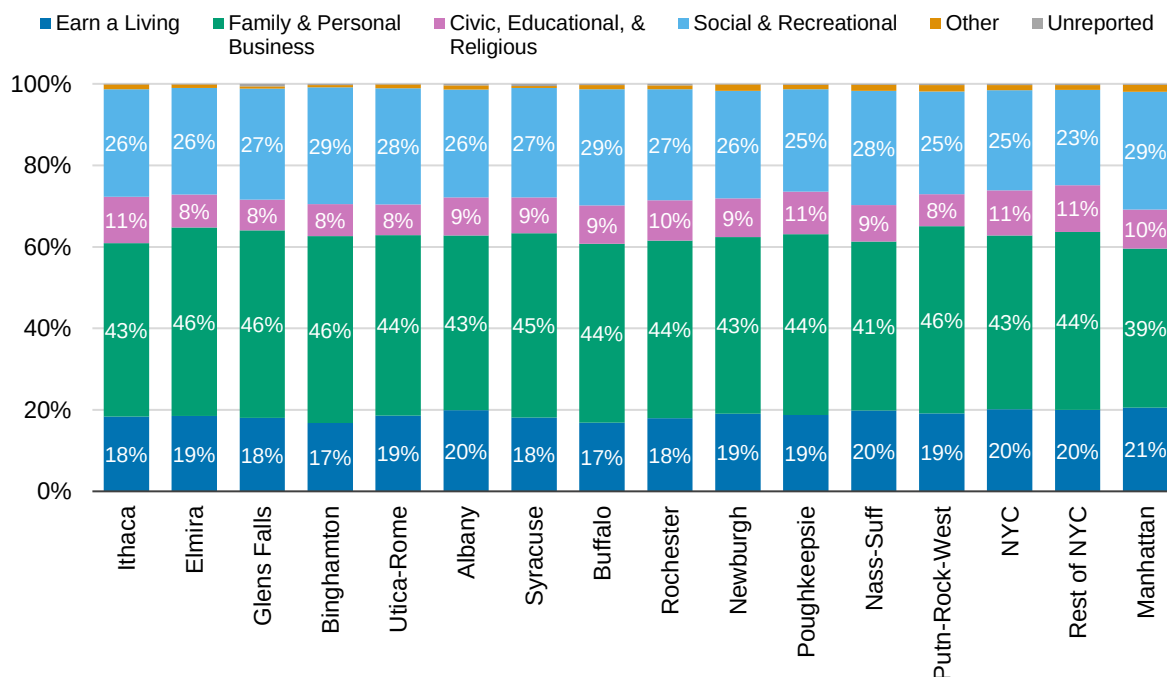


Figure 5-7. Daily person trips by purpose and location (e.g., city, region) in 2001. Note: Unlabeled segments represent values lower than 5%.

5.2.4 Public Transit, Walk, and Cycle Trips

Table 5-10 presents data on walking and biking trip activity in NYS metro areas for the years 2001, 2009, and 2017, broken down by MSA size class. The table includes the percentage of the population that did not take a walking or biking trip, as well as the average minutes spent walking and biking in a given day. The percentage of people not taking a walking trip has declined over time in most metro areas, indicating an overall increase in walking participation. In 2001, nonwalking rates were high across all areas, with Rochester at 38.5% and Buffalo at 32.5%. By 2017, these numbers had fallen slightly, though they remained high, at 31.0% for Glens Falls and 28.6% for Binghamton. The highest percentages of people not walking in 2017 were still in Buffalo (27.5%) and Albany (24.8%), whereas areas like Ithaca (16.6%) and Manhattan (15.2%) had the lowest rates, indicating the most walking activity. In terms of average minutes spent walking per day, in NYC, the average daily walking time rose slightly from 17.0 min in 2001 to 18.04 min in 2017. However, in some smaller metro areas, walking time remained relatively low. For example, Watertown reported just 4.56 min of walking per day in 2017, significantly lower than that in places like Ithaca (14.77 min). These differences likely reflect varying levels of pedestrian infrastructure and urban density.

Despite the growth in walking, biking activity remained limited across most metro areas and showed minimal change over time. The percentage of people not taking a biking trip was consistently high across all years, generally exceeding 85%–90% in most areas. In 2017, over 93% of people in areas like Albany, Syracuse, and Rochester reported not biking at all, a trend that remained stable compared with 2001 and 2009. Even in NYC, where biking infrastructure has expanded, the share of nonbikers remained around 88.5% in 2017. The average daily minutes spent biking was low across all years, with most metro areas reporting under 1.5 min per day on average. In 2017, Ithaca had the highest reported biking time at 2.96 min per day, a notable increase from 0.68 min in 2001. Other areas, such as Albany (0.27 min in 2017) and Buffalo (0.32 min), showed no substantial growth in biking activity over the years. This suggests that

while some cities have made strides in encouraging biking, overall participation remains low in many regions.

Looking at regional variations, NYC and Manhattan consistently had the lowest percentage of people not walking, reinforcing the city's reputation as a pedestrian-friendly environment. Meanwhile, upstate and suburban areas showed higher walking inactivity rates, particularly in places like Glens Falls and Rochester, where over 30% of residents reported not walking in 2017. These differences are statistically meaningful. For biking, the persistently high nonparticipation rates indicate no statistically significant growth in biking across most regions. Even in cities with substantial investment in bike infrastructure, such as NYC and Albany, the change in biking minutes per day from 2001 to 2017 is not statistically significant. The data suggest that walking participation has increased over time, particularly in dense urban areas like NYC and Ithaca, whereas biking activity remains low and largely unchanged across NYS metro areas.

Table 5-10. Walking and biking trip activity by year, location (e.g., city, region), and MSA size.

	MSA Size Class							
	Not in MSA		<250,000			250,000–499,999		500,000–999,999
	Ithaca	Kingston	Watertown	Elmira	Glens Falls	Binghamton	Utica–Rome	Albany Syracuse
2017								
Percent Not Taking Walk Trips in the Last Week	16.6%	20.8%	25.0%	27.0%	31.0%	28.6%	25.2%	24.8% 27.9%
Average Minutes Spent Walking Per Day	14.77	4.48	4.56	5.84	6.36	6.43	10.74	9.97 6.62
Percent Not Taking Bike Trips in the Last Week	83.9%	92.4%	90.2%	86.0%	88.5%	91.7%	87.5%	89.1% 88.1%
Average Minutes Spent Biking Per Day	2.96	0.19	0.09	1.03	1.11	1.14	0.91	0.27 0.32
2009								
Percent Not Taking Walk Trips in the Last Week	26.4%	33.5%	—	31.4%	30.5%	30.8%	34.3%	29.2% 32.5%
Average Minutes Spent Walking Per Day	9.62	4.58	—	4.74	6.35	3.41	5.41	6.56 4.93
Percent Not Taking Bike Trips in the Last Week	84.8%	88.4%	—	85.7%	85.2%	90.9%	88.6%	88.8% 89.5%
Average Minutes Spent Biking Per Day	0.49	0.47	—	0.62	1.08	0.28	1.00	0.82 1.31
2001								
Percent Not Taking Walk Trips in the Last Week	17.2%	—	—	28.6%	25.9%	28.0%	27.1%	26.8% 28.6%
Average Minutes Spent Walking Per Day	9.60	—	—	4.96	5.21	5.43	3.49	5.00 5.09
Percent Not Taking Bike Trips in the Last Week	90.0%	—	—	93.1%	91.5%	92.3%	91.9%	92.3% 92.1%
Average Minutes Spent Biking Per Day	0.68	—	—	1.39	0.76	0.87	0.31	1.18 0.63

Table 5-10. Walking and biking trip activity by year, location (e.g., city, region), and MSA size (continued).

	MSA Size Class								
	1 million– 2.9 million		3 million+						
	Buffalo	Rochester	Newburgh	Poughkeepsie	NYMTC				
					Nass- Suff	Putn- Rock- West	NYC	Subcomponents of NYC	
								Rest of NYC	Manhattan
2017									
Percent Not Taking Walk Trips in the Last Week	27.5%	29.1%	26.7%	22.8%	29.5%	22.3%	8.7%	10.1%	2.8%
Average Minutes Spent Walking Per Day	5.95	4.44	6.38	6.38	5.28	6.69	18.04	16.08	26.1
Percent Not Taking Bike Trips in the Last Week	87.2%	89.0%	93.3%	89.5%	89.6%	88.5%	88.4%	89.1%	85.3%
Average Minutes Spent Biking Per Day	0.57	0.8	0.1	0.64	0.37	0.27	2	1.62	3.59
2009									
Percent Not Taking Walk Trips in the Last Week	38.5%	36.7%	30.9%	31.8%	35.3%	27.7%	16.8%	18.7%	9.5%
Average Minutes Spent Walking Per Day	4.23	4.28	4.83	4.87	5.65	9.32	20.35	17.27	32.92
Percent Not Taking Bike Trips in the Last Week	87.1%	88.0%	89.5%	88.8%	89.8%	87.6%	93.4%	93.4%	93.2%
Average Minutes Spent Biking Per Day	1.04	0.35	0.44	0.62	0.27	0.59	0.38	0.38	0.37
2001									
Percent Not Taking Walk Trips in the Last Week	31.6%	31.3%	33.2%	26.6%	31.6%	28.7%	15.5%	17.4%	7.9%
Average Minutes Spent Walking Per Day	5.63	6.32	4.31	3.87	6.69	6.15	16.11	14.35	23.52
Percent Not Taking Bike Trips in the Last Week	91.2%	91.0%	93.6%	90.7%	91.9%	94.2%	94.4%	94.8%	92.7%
Average Minutes Spent Biking Per Day	0.78	0.83	0.66	0.81	0.57	0.34	0.65	0.67	0.55

5.2.5 Travel by Zero-Vehicle Households

Table 5-11 presents data on zero-vehicle households by income level across NYS metro areas in 2017, providing insights into the relationship between income and car ownership. The data reveal clear patterns, with lower-income households disproportionately representing zero-vehicle households, whereas higher-income groups show near-zero prevalence of carless households. As expected, lower-income households (those earning less than \$10,000 annually) make up a significant share of zero-vehicle households across all metro areas. However, this share varies significantly by region. In rural and small metro areas, the proportion is generally lower; for example, only 16.7% of zero-vehicle households in Ithaca and 12.9% in Kingston fell into this income category. In contrast, in larger metro areas and regions with more public transportation availability, such as Glens Falls (46.7%), Binghamton (51.7%), and Buffalo (31.3%), lower-income households made up the majority of carless households. NYC exhibits a similar pattern, with 20.7% of zero-vehicle households earning less than \$10,000 per year, whereas Manhattan had a significantly lower share at 11.6%. This suggests that, within NYC, zero-vehicle households are more distributed across lower-income outer boroughs than in high-density wealthier areas such as Manhattan. Households earning between \$10,000 and \$25,000 annually also represent a considerable share of zero-vehicle households. In Binghamton, 43.5% of carless households fell into this category, whereas in Rochester, it was 45.7%. However, in areas with stronger transit networks, such as NYC, this figure was slightly lower at 35%. The pattern suggests that although lower-income households in larger cities are more likely to forgo car ownership, those in mid-sized metro areas may also be forced into carlessness owing to affordability constraints.

For households earning more than \$50,000 annually, the proportion of zero-vehicle households drops sharply. Across most metro areas, households in this income range represent fewer than 5% of zero-vehicle households. In NYC, 11.2% of zero-vehicle households earned between \$50,000 and \$75,000, but in most other MSAs, this percentage was below 5%. Similarly, the share of zero-vehicle households earning over \$100,000 was negligible across all regions. The only notable exceptions were NYC and its surrounding suburban areas, where 6.9% of zero-vehicle households in Manhattan and 5.9% in Putnam–Rockland–Westchester fell into this income range. This suggests that in wealthier urban areas, some households choose to live without vehicles despite having the financial means, likely because of the convenience of transit and walkability.

The highest proportion of low-income zero-vehicle households is found in Binghamton (51.7%), Glens Falls (46.7%), and Rochester (45.7%), indicating strong economic constraints on car ownership in these areas. Meanwhile, suburban areas like Nassau–Suffolk and Putnam–Rockland–Westchester show much lower shares of low-income zero-vehicle households because car ownership is more necessary for mobility in these regions. In NYC, the distribution of zero-vehicle households by income differs by borough. Manhattan has a higher share of middle- and upper-income zero-vehicle households (6.9% earning over \$100,000) compared with the outer boroughs (3.5%), reflecting the greater reliance on public transit among wealthier Manhattan residents. The data indicate that zero-vehicle households are overwhelmingly concentrated among lower-income groups, particularly those earning less than \$25,000 per year. However, regional differences highlight important contextual factors. In smaller MSAs, economic hardship is a key driver of not having a vehicle, whereas in large urban areas like NYC and Manhattan, transit accessibility and lifestyle preferences also contribute to car-free living, even among higher-income households.

Table 5-11. Zero-vehicle households by income, location (e.g., city, region), and MSA size in 2017.

Household Income ^a	MSA Size Class								
	Not in MSA		<250,000			250,000–499,999		500,000–999,999	
	Ithaca	Kingston	Watertown	Elmira	Glens Falls	Binghamton	Utica–Rome	Albany	Syracuse
Less than \$10,000	20.0%	37.4%	79.8%	41.3%	46.7%	51.7%	43.5%	42.5%	40.6%
\$10,000 to \$14,999	16.7%	12.9%	11.0%	32.4%	24.7%	14.0%	18.3%	18.3%	34.3%
\$15,000 to \$24,999	27.6%	21.7%	6.6%	21.9%	13.4%	19.4%	25.5%	18.1%	10.5%
\$25,000 to \$34,999	19.2%	5.7%	2.6%	0.6%	10.7%	4.6%	1.1%	7.3%	8.2%
\$35,000 to \$49,999	6.0%	15.6%	0.0%	2.7%	0.0%	2.8%	4.1%	6.1%	0.0%
\$50,000 to \$74,999	6.4%	1.4%	0.0%	0.0%	0.0%	3.9%	0.0%	5.2%	1.3%
\$75,000 to \$99,999	0.8%	0.5%	0.0%	0.0%	0.0%	0.3%	0.0%	2.0%	0.0%.
\$100,000 to \$124,999	0.6%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.5%
\$125,000 to \$149,999	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%.
\$150,000 to \$199,999	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%.
\$200,000 or More	1.2%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%.
Unreported Households	1.4%	4.8%	0.0%	1.1%	4.5%	3.4%	7.5%	0.4%	4.5%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 5-11. Zero-vehicle households by income, location (e.g., city, region), and MSA size in 2017 (continued).

Household Income ^a	MSA Size Class								
	1 million– 2.9 million		3 million+						
	Buffalo	Rochester	Newburgh	Poughkeepsie	NYMTC			Subcomponents of NYC	
					Nass-Suff	Putn- Rock-West	NYC	Rest of NYC	Manhattan
Less than \$10,000	31.3%	45.7%	29.2%	23.8%	20.6%	15.0%	15.8%	20.7%	8.9%
\$10,000 to \$14,999	20.3%	11.4%	40.4%	13.3%	30.4%	35.0%	6.7%	10.2%	1.7%
\$15,000 to \$24,999	25.3%	8.2%	20.9%	42.0%	6.1%	12.2%	9.0%	11.8%	5.2%
\$25,000 to \$34,999	14.1%	21.2%	5.8%	11.7%	19.5%	12.4%	9.3%	9.9%	8.6%
\$35,000 to \$49,999	1.6%	0.0%	0.8%	5.5%	4.9%	14.4%	9.7%	11.2%	7.7%
\$50,000 to \$74,999	0.6%	2.2%	0.0%	0.0%	5.4%	2.6%	12.3%	11.2%	13.8%
\$75,000 to \$99,999	0.5%	0.0%	0.0%	0.0%	11.8%	3.4%	11.0%	8.3%	14.7%
\$100,000 to \$124,999	0.0%	0.5%	0.0%	0.4%	0.0%	1.9%	5.9%	4.4%	7.9%
\$125,000 to \$149,999	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	3.6%	2.0%	6.0%
\$150,000 to \$199,999	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	4.1%	2.6%	6.3%
\$200,000 or More	0.0%	0.0%	0.0%	0.0%	0.0%	0.8%	7.4%	1.7%	15.5%
Unreported Households	6.2%	10.8%	3.0%	3.3%	1.2%	2.3%	5.1%	6.0%	3.8%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Figure 5-8 presents personal travel trends of zero-vehicle households in various NYS metro areas from 2001 to 2017, measured by average daily person trips per household. Across most metro areas, the average number of daily trips per household remained stable or increased slightly between 2001 and 2017. This suggests that despite not owning a vehicle, households are still making regular trips, likely relying on public transit, walking, or shared mobility. The increase is particularly pronounced in certain metro areas, whereas other regions saw a minor decline or stagnation over the years. In NYC, zero-vehicle households maintained relatively high trip rates over time, decreasing from 3.30 trips in 2001 to 3.19 in 2017. This change is not statistically significant, indicating a relatively stable level of mobility among car-free households in the city. A similar pattern is observed in Manhattan, where the number of trips decreased slightly from 3.04 in 2001 to 2.96 in 2017. Meanwhile, several smaller metro areas experienced notable increases in trips per household over time. In Newburgh, the number of trips rose from 2.68 in 2001 to 3.66 in 2017, a statistically significant increase. Similarly, Glens Falls saw a statistically significant increase from 2.14 trips in 2001 to 3.49 in 2017, suggesting improved mobility options or greater travel demand in these areas. On the other hand, a few regions, particularly Elmira and Buffalo, experienced slight declines in trips per household. In Elmira, average daily trips dropped from 2.60 in 2001 to 1.53 in 2017, a decrease that is statistically significant.

The data suggest that zero-vehicle households continue to make a considerable number of daily trips, relying on nonauto modes of transport. Whereas NYC and larger metro areas exhibit relatively stable travel patterns, certain smaller MSAs, such as Newburgh and Glens Falls, have seen significant increases in mobility over time. In contrast, areas like Elmira have experienced a decline in travel activity among zero-vehicle households, which may indicate growing transportation barriers.

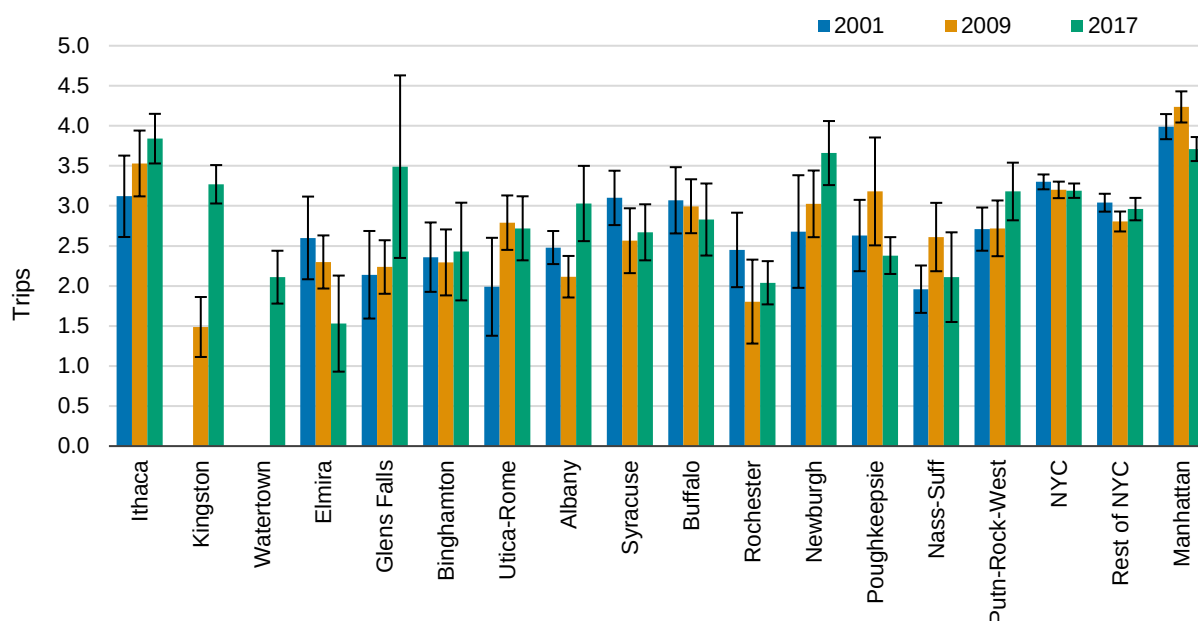


Figure 5-8. Average daily person trips of zero-vehicle households by year and location (e.g., city, region).

Figure 5-9 presents PMT per day for zero-vehicle households across NYS metro areas for the years 2001, 2009, and 2017. The data indicate that PMT for zero-vehicle households has fluctuated over time, with some areas experiencing significant increases or decreases in travel distance. In many regions, zero-vehicle households in 2017 traveled distances comparable to or greater than those of previous years, suggesting that mobility has remained stable or even improved despite the lack of personal vehicle access. The largest increases in PMT between 2001 and 2017 occurred in some smaller MSAs and suburban areas. Kingston experienced a high increase in PMT, from 7.24 miles in 2009 to 63.30 miles in 2017.

However, the margin of error for Kingston in 2017 is 53.41, meaning that this extreme value is likely unreliable and is not statistically significant. Similarly, Newburgh's PMT spiked from 16.90 miles in 2001 to 30.03 miles in 2009, before dropping to 5.08 miles in 2017. The differences are not statistically significant. Conversely, some urban and midsized metro areas exhibited more stable PMT trends over time. In NYC, PMT decreased from 17.87 miles in 2001 to 14.16 miles in 2017, remaining relatively stable. A similar pattern is observed in Manhattan, where PMT slightly declined from 17.11 miles in 2001 to 12.55 in 2017, though this change is not statistically significant. These trends indicate that zero-vehicle households in urban cores maintain relatively high travel distances, likely due to strong transit networks. Some regions experienced declining PMT over time, suggesting reduced mobility among zero-vehicle households. Elmira's PMT dropped statistically significantly from 4.74 in 2001 to just 3.15 in 2017. Similarly, Glens Falls' PMT decreased from 12.84 miles in 2001 to 10.04 in 2009, before recovering to 14.65 miles in 2017. The fluctuations in these areas may be attributed to changes in transit service availability or economic conditions affecting travel demand.

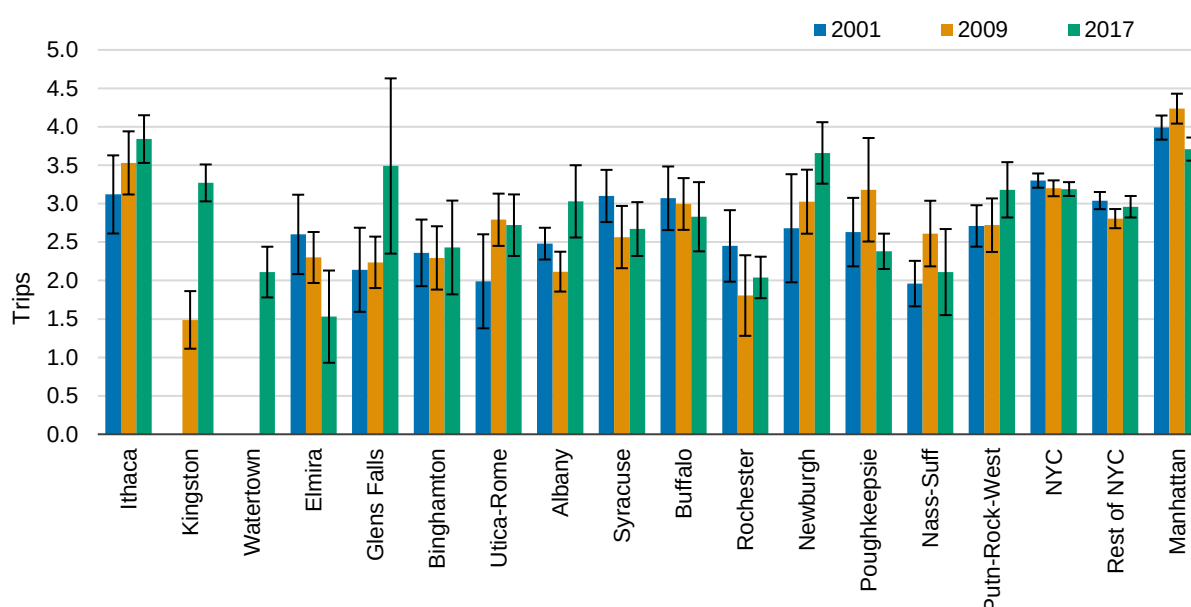


Figure 5-9. Average daily PMT of zero-vehicle households by year and location (e.g., city, region).

Table 5-12 presents person travel patterns for zero-vehicle households in NYS metro areas in 2017, broken down by mode of transportation and percentage of person trips and PMT by mode. Walking and public transit are the dominant travel modes for zero-vehicle households, but their usage varies significantly across different metro areas. Walking accounts for the highest share of trips in most regions, particularly in Newburgh (75.6%), Rochester (61.4%), and Buffalo (59.1%), where over half of all trips by carless households are on foot. In NYC, walking represents 52.3% of trips, with Manhattan showing a slightly lower share (49.0%), likely due to the widespread availability of public transit. Public transit use is highest in NYC, where it accounts for 28.3% of all person trips among zero-vehicle households. The share is even higher in Manhattan at 35.2%, underscoring the role of the city's extensive subway and bus network. However, in smaller MSAs, transit usage is considerably lower; for example, in Watertown, only 6.5% of trips are made by transit, and in Glens Falls, it is just 1.7%. This disparity suggests that outside of major metropolitan areas, transit access is insufficient for daily travel needs.

Private vehicle use among zero-vehicle households is notably higher in smaller metro areas. In Kingston, 43.2% of trips are made using private vehicles, the highest share among all areas, followed by Syracuse (23.1%) and Glens Falls (22.7%). This suggests that despite not owning a vehicle, many households in

these areas depend on private vehicles for travel. Regarding PMT, the reliance on different travel modes shifts slightly. Public transit accounts for the highest percentage of miles traveled in NYC (81.3%), followed by Manhattan (71.6%), reflecting the long-distance nature of many transit trips. In contrast, smaller metro areas have much lower transit PMT shares, with Watertown at just 10.2% and Elmira at 9.9%, further emphasizing the limited availability of transit options in these areas.

Even though walking represents the highest number of trips, it contributes a relatively low percentage to total miles traveled. For example, in NYC, walking makes up 52.3% of trips but only 10.3% of total PMT, suggesting that most walking trips are short distances. Similarly, in Syracuse, walking accounts for 49.8% of trips but only 6.2% of PMT. This reinforces the idea that while walking is crucial for local mobility, it is often supplemented by transit for longer journeys. Meanwhile, private vehicle PMT shares remain substantial in certain areas, particularly in Kingston (62.3%), suggesting that households without cars still travel considerable distances using private vehicles, likely via ride-hailing, taxis, or carpooling.

Table 5-13 presents person travel patterns for zero-vehicle households in NYS metro areas in 2009, categorized by mode of transportation and broken down into percentage of person trips and PMT by mode. These data provide insights into the transportation choices of carless households and how travel behavior differs across regions. Zero-vehicle households primarily relied on walking and public transit, but use patterns varied across different metro areas. Walking was the most common travel mode in most regions, particularly in Manhattan (61.5%), Newburgh (61.2%), and NYC overall (52.7%), highlighting the role of pedestrian-friendly infrastructure in dense urban areas. Similarly, smaller MSAs like Elmira (65.0%) and Ithaca (55.3%) also exhibited high walking shares, suggesting that in places where transit is less accessible, walking remains a dominant mode. In contrast, Glens Falls had only 39.4% of trips made by walking, indicating a higher dependency on other modes. Public transit played a significant role in large metro areas, especially NYC (32.5%) and its surrounding areas (37.5% in the rest of NYC and 23.7% in Manhattan). However, in smaller metro areas, transit usage was notably lower. For example, only 1.5% of trips in Glens Falls and 14.7% of trips in Binghamton were made using public transit. This highlights the lack of transit access in many smaller MSAs, where alternative transportation options are required. Private vehicle usage, despite zero-vehicle household status, was substantial in several regions. In Glens Falls, 52.5% of trips were made using private vehicles, the highest among all areas, followed by Newburgh (32.5%) and Kingston (27.1%). This suggests that in these areas, even households without a vehicle frequently rely on ridesharing, taxis, or borrowing cars for travel.

Looking at PMT, the reliance on different modes shifts significantly. Public transit accounted for the highest percentage of miles traveled in NYC (55.7%) and surrounding areas (60.4%), indicating that while transit may make up a smaller proportion of total trips compared with walking, it is the dominant mode for longer-distance travel. The same pattern is seen in Albany (55.5%) and Syracuse (32.1%), demonstrating the importance of transit in these midsized cities. Walking, whereas a significant share of total trips, contributed far less to total PMT. For example, in NYC, walking accounted for 52.7% of trips but only 16.4% of total miles traveled, indicating that most walking trips were short distances. Similarly, in Syracuse, walking made up 43.0% of trips but only 5.1% of total PMT, emphasizing the role of walking as a local travel mode rather than a means for longer-distance mobility. Private vehicles contributed significantly to PMT in suburban and smaller metro areas. In Glens Falls, private vehicles accounted for 90.5% of PMT, the highest among all areas, aligning with the high share of trips made using private vehicles. Even in smaller MSAs like Kingston (66.9%) and Utica–Rome (59.6%), private vehicle reliance remained dominant for longer-distance travel, underscoring the limited availability of public transit for zero-vehicle households in these regions.

Table 5-14 presents person travel patterns for zero-vehicle households in NYS metro areas in 2001, categorized by mode of transportation and broken down into percentage of person trips and PMT by mode. Zero-vehicle households primarily relied on walking and private vehicles, though their reliance on

different travel modes varied significantly by region. Walking accounted for the highest share of trips in many areas, particularly in NYC (47.2%) and Manhattan (55.4%), reflecting the pedestrian-friendly infrastructure and high-density urban environment. Smaller metro areas such as Ithaca (53.6%) and Newburgh (47.0%) also had a high share of walking trips, suggesting that in places where transit is less accessible, walking plays a key role in mobility. In contrast, private vehicle usage was notably high in smaller MSAs, particularly in Poughkeepsie (49.5%) and Elmira (46.8%), indicating that even among zero-vehicle households, alternative transportation modes such as taxis, ridesharing, or carpooling were heavily used. Public transit usage was highest in NYC (32.7%) and Rochester (38.6%), with Manhattan reaching 35.2%, reinforcing the reliance on subways and buses for longer-distance travel. However, in smaller metro areas, public transit accounted for a much lower share of trips. For instance, transit usage was only 14.2% in Glens Falls and 14.8% in Binghamton. This reflects the more limited transit networks available outside of major metropolitan centers.

When examining PMT, which measures the total distance traveled per mode, the distribution shifts significantly. Private vehicle PMT was highest in suburban and smaller metro areas. Private vehicles accounted for 93.2% of miles traveled in Poughkeepsie and 88.6% of miles traveled in Glens Falls, reflecting that even though these households did not own cars, they still traveled long distances using private vehicle options such as carpooling, taxis, or borrowing vehicles. In contrast, in NYC, private vehicle PMT was just 16.8%, and in Manhattan, it was 13.5%, showing that transit and walking play a much larger role in daily mobility. Public transit dominated PMT in urban regions, particularly in NYC (55.1%) and Rochester (69.6%), indicating that while transit may account for fewer trips compared with walking, it is the primary mode for longer-distance travel. This trend is particularly evident in larger metro areas where bus and rail systems provide reliable options for commuting and other essential travel. Walking, despite being the most frequent mode of travel in many areas, accounted for a much smaller share of total miles traveled. In Manhattan, walking made up 55.4% of trips but only 7.6% of PMT, reinforcing the idea that most walking trips are short distances. Similarly, in Syracuse, walking made up 38.3% of trips but only 6.2% of PMT, demonstrating that individuals depend on other modes for longer-distance trips.

Comparing 2001 data with 2017 trends, notable shifts in travel behavior among zero-vehicle households have emerged, particularly in the decline of private vehicle dependence and the increasing reliance on walking and public transit in certain regions. In 2001, private vehicles accounted for 40.8% of trips in Ithaca and 46.8% in Elmira, but by 2017, these numbers had dropped to 8.9% and 3.2%, respectively, signaling a significant reduction in car dependency among zero-vehicle households in smaller MSAs. Similarly, in NYC, private vehicle use declined from 13.3% in 2001 to just 9.5% in 2017, further reinforcing the shift toward public transit and active transportation. Public transit use has also evolved, especially in urban areas. In NYC, public transit trips increased from 32.7% in 2001 to 28.3% in 2017, showing slight fluctuations but overall maintaining their dominant role in long-distance mobility. However, in mid-sized metro areas such as Albany and Syracuse, public transit saw modest increases over time, indicating some improvements in transit accessibility or ridership trends. Walking has consistently grown as a dominant mode for zero-vehicle households, with its share rising in NYC, where it increased from 47.2% in 2001 to 52.3% in 2017, and in places like Rochester, which saw an increase from 40.3% to 61.4%. This suggests a growing trend in pedestrian mobility, potentially influenced by urban densification, improved walkability, and changing lifestyle preferences.

Table 5-12. Person travel for zero-vehicle households by mode, location (e.g., city, region), and MSA size in 2017.

Mode ^a	MSA Size Class							
	Not in MSA		<250,000			250,000–499,999		500,000–999,999
	Ithaca	Kingston	Watertown	Elmira	Glens Falls	Binghamton	Utica–Rome	Albany Syracuse
Person Trips (%)								
Private	8.9	43.2	13.0	20.1	22.7	22.6	15.7	16.7 23.1
Public	24.9	8.6	6.5	10.7	1.7	17.1	11.7	25.0 17.9
Walk	58.8	32.0	61.6	45.6	57.6	47.5	55.1	48.4 49.8
Other	7.3	16.2	18.9	23.6	18.0	12.9	17.5	9.8 9.2
PMT (%)								
Private	18.1	91.9	23.6	40.9	51.9	73.1	28.9	28.9 55.1
Public	45.7	4.5	10.0	25.1	1.3	14.0	13.8	43.6 25.2
Walk	23.8	0.9	37.0	10.2	29.6	4.1	31.4	5.7 8.7
Other	12.4	2.7	29.4	23.9	17.2	8.8	26.0	21.9 11.0

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 5-12. Person travel for zero-vehicle households by mode, location (e.g., city, region), and MSA size in 2017 (continued).

Mode ^a	MSA Size Class								
	1 million– 2.9 million		3 million+						
	Buffalo	Rochester	Newburgh	Poughkeepsie	NYMTC				
					Nass-Suff	Putn- Rock-West	NYC	Subcomponents of NYC	
Rest of NYC	Manhattan								
Person Trips (%)									
Private	8.6	10.6	12.9	10.4	20.1	9.5	4.3	3.8	5.4
Public	22.0	31.4	5.2	6.6	17.3	28.3	35.2	40.6	25.6
Walk	59.1	25.5	61.4	75.6	43.0	51.8	52.6	49.0	59.1
Other	10.4	32.5	20.6	7.5	19.7	10.3	7.8	6.6	10.0
PMT (%)									
Private	14.2	13.7	42.9	19.8	28.7	6.6	18.7	15.6	23.6
Public	44.7	35.1	8.0	11.1	49.8	81.3	55.2	71.6	29.3
Walk	13.4	4.4	20.4	45.0	10.0	5.3	5.7	5.4	6.3
Other	27.7	46.8	28.7	24.1	11.5	6.8	20.3	7.4	40.7

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 5-13. Person travel for zero-vehicle households by mode, location (e.g., city, region), and MSA size in 2009.

Mode ^a	MSA Size Class							
	Not in MSA		<250,000		250,000–499,999		500,000–999,999	
	Ithaca	Kingston	Elmira	Glens Falls	Binghamton	Utica–Rome	Albany	Syracuse
Person Trip (%)								
Private	12.1	27.1	16.2	52.5	26.3	20.5	21.8	9.9
Public	27.2	16.5	9.1	1.5	14.7	11.2	32.4	16.2
Walk	55.3	40.5	65.0	39.4	29.5	38.4	37.1	43.0
Other	5.2	15.9	9.6	6.6	29.5	24.0	8.8	25.3
Unreported	0.2	0.0	0.0	0.0	0.0	6.0	0.0	5.5
PMT (%)								
Private	54.6	66.9	35.2	90.5	59.6	28.3	38.5	22.6
Public	36.6	10.7	14.6	2.1	17.6	24.4	55.5	32.1
Walk	6.7	14.5	20.1	3.4	2.1	16.2	5.1	17.2
Other	1.8	7.9	30.1	4.0	20.8	21.1	0.9	16.2
Unreported	0.3	0.0	0.0	0.0	0.0	10.0	0.0	12.0

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 5-13. Person travel for zero-vehicle households by mode, location (e.g., city, region), and MSA size in 2009 (continued).

Mode ^a	MSA Size Class							
	1 million– 2.9 million		3 million+					
	Buffalo	Rochester	Newburgh	Poughkeepsie	NYMTC			
					Nass-Suff	Putn- Rock-West	NYC	Subcomponents of NYC
								Rest of NYC Manhattan
Person Trip (%)								
Private	28.1	34.5	13.3	32.5	18.1	17.5	7.4	7.9 6.5
Public	17.1	12.6	5.3	9.2	29.1	23.5	32.5	37.5 23.7
Walk	29.6	34.3	61.2	48.9	40.0	50.0	52.7	47.6 61.5
Other	25.2	18.7	20.2	9.4	9.2	8.9	7.3	6.8 8.3
Unreported	0.0	0.0	0.0	0.0	3.7	0.0	0.1	0.2 0.0
PMT (%)								
Private	35.2	76.3	42.1	54.7	26.7	21.7	20.1	15.8 27.2
Public	25.6	9.2	41.6	18.7	40.0	44.7	55.7	60.1 48.2
Walk	6.7	6.8	3.5	18.7	6.8	16.4	13.3	11.5 16.4
Other	32.6	7.6	12.9	7.9	5.9	17.3	10.9	12.5 8.2
Unreported	0.0	0.0	0.0	0.0	20.6	0.0	0.1	0.1 0.0

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 5-14. Person travel for zero-vehicle households by mode, location (e.g., city, region), and MSA size in 2001.

Mode ^a	MSA Size Class						
	Not in MSA	<250,000		250,000–499,999		500,000–999,999	
	Ithaca	Elmira	Glens Falls	Binghamton	Utica–Rome	Albany	Syracuse
Person Trips (%)							
Private	40.8	46.8	39.3	30.0	48.8	32.6	34.9
Public	3.9	15.3	14.2	14.8	11.4	17.1	14.4
Walk	53.6	31.7	43.5	45.1	39.8	42.3	38.3
Other	1.7	6.2	3.0	10.1		7.9	12.5
Unreported	0.0	0.0	0.0	0.0	0.0	0.0	0.0
PMT (%)							
Private	77.6	71.8	88.6	53.7	76.0	63.3	63.5
Public	11.3	6.3	7.3	22.1	13.2	22.5	17.2
Walk	9.7	14.8	3.9	13.3	10.8	6.5	6.2
Other	1.4	7.1	0.2	10.9		7.7	13.1
Unreported	0.0	0.0	0.0	0.0	0.0	0.0	0.0

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 5-14. Person travel for zero-vehicle households by mode, location (e.g., city, region), and MSA size in 2001 (continued).

Mode ^a	MSA Size Class							
	1 million– 2.9 million		3 million+					
	Buffalo	Rochester	Newburgh	Poughkeepsie	NYMTC			
					Nass-Suff	Putn- Rock-West	NYC	Subcomponents of NYC
								Rest of NYC Manhattan
Person Trips (%)								
Private	28.5	17.2	32.1	49.5	41.9	31.7	13.3	15.0 9.9
Public	11.9	38.6	6.5	3.1	1.4	24.0	32.7	35.2 27.5
Walk	33.8	40.3	47.0	36.9	37.7	35.4	47.2	43.2 55.4
Other	25.8	3.9	14.4	10.5	19.0	7.9	6.6	6.4 7.0
Unreported	0.0	0.0	0.0	0.0	0.0	0.9	0.2	0.2 0.1
PMT (%)								
Private	56.5	21.9	75.4	93.2	56.4	32.1	16.8	13.5 24.5
Public	13.4	69.6	7.7	0.1	1.6	55.1	31.5	33.4 27.1
Walk	8.2	6.3	4.1	1.7	17.2	8.4	5.7	4.9 7.6
Other	21.9	2.2	12.9	5.1	24.8	4.4	45.9	48.1 40.8
Unreported	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1 0.0

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

5.3 VEHICLE TRAVEL COMPARISONS

5.3.1 Driving Frequencies, Distances, and Times

Table 5-15 reveals distinct regional trends in daily vehicle trips per driver, average trip length, and daily VMT per driver across different MSAs in NYS from 2001 to 2017. A key finding is the decline in daily vehicle trips per driver across most regions (Figure 5-10). In 2001, drivers in smaller MSAs (e.g., Ithaca, Watertown, and Elmira) made over 3.2 trips per day, but by 2017, these numbers had dropped to around 2.6–2.8 trips per day. The decline is also observed in larger metro areas, including Buffalo, Rochester, and Albany, where trip rates fell below three per day. This decline is statistically significant in several regions. In contrast, in some suburban areas like Putnam–Rockland and Nassau–Suffolk, trip rates remained relatively stable, and the decline in these regions was not statistically significant. Although trip rates declined, average trip lengths increased significantly over time (Figure 5-11). In 2001, trip lengths ranged from approximately 8 to 10 miles in most MSAs, but by 2017, they had risen to between 9.8 and 13.2 miles. This increase is particularly notable in suburban and midsized metro areas such as Utica–Rome and Binghamton, where trip lengths in 2017 were statistically significantly higher than those in 2001. Conversely, in NYC and Manhattan, where transit use is more prevalent, trip lengths remained relatively stable, with no notable change.

Daily VMT had variations across different regions (Figure 5-12). In smaller MSAs such as Ithaca and Kingston, daily VMT dropped significantly from 2001 to 2009 before stabilizing or slightly increasing in 2017. In contrast, in larger metro areas like Buffalo, Rochester, and Newburgh, daily VMT remained consistent or even increased slightly over time. The increase in Buffalo and Rochester is statistically significant. Regional differences are particularly pronounced between urban centers and suburban areas. NYC and Manhattan consistently exhibit significantly lower vehicle trip rates and daily VMT compared with suburban regions like Putnam–Rockland and Nassau–Suffolk. This contrast is expected, given NYC’s extensive public transportation network and higher population density, which reduces reliance on private vehicles.

Table 5-15. Daily vehicle travel statistics by year, location (e.g., city, region), and MSA size.

MSA Size Class																		
Not in MSA			<250,000			250,000–499,999		500,000–999,999		1 million–2.9 million				3 million+				
Ithaca	Kingston		Watertown	Elmira	Glens Falls	Binghamton	Utica–Rome	Albany	Syracuse	Buffalo	Rochester	Newburgh	Poughkeepsie	NYMTC				
														Nass–Suff	Putn–Rock–West	NYC	Subcomponents of NYC	
																	Rest of NYC	Manhattan
Vehicle Trips																		
2001	3.25	—	—	3.36	3.24	3.41	3.63	3.27	3.21	3.43	3.43	3.32	3.35	3.07	3.17	1.57	1.77	0.75
2009	2.35	2.99	—	3.12	3.04	3.20	3.08	2.95	3.18	3.02	3.21	2.97	2.95	3.18	2.90	1.13	1.29	0.53
2017	2.63	2.57	2.76	2.68	2.65	2.78	3.06	2.72	2.71	3.05	2.89	2.38	2.70	2.74	2.64	1.12	1.29	0.45
Average Trip Length (miles)																		
2001	8.30	—	—	8.91	10.04	8.73	8.64	8.75	8.47	7.85	8.26	12.08	9.85	8.59	9.40	7.91	7.10	15.59
2009	7.53	9.22	—	7.58	9.02	8.66	8.13	9.17	8.57	6.94	9.10	11.98	11.76	7.89	8.70	7.55	6.65	15.48
2017	8.29	11.76	13.23	8.41	10.75	10.00	8.68	12.90	9.80	7.74	10.55	13.96	11.82	9.27	10.22	9.34	7.99	19.84
VMT																		
2001	26.51	—	—	29.91	32.34	29.07	31.31	28.22	26.85	26.69	28.07	39.93	32.66	25.73	29.16	11.86	12.06	11.10
2009	17.62	27.24	—	23.59	27.01	27.62	24.93	26.89	27.09	20.74	28.81	35.07	34.08	24.83	24.72	8.13	8.20	7.89
2017	21.05	28.88	27.57	22.67	27.55	27.30	26.54	32.98	26.28	24.05	30.11	32.60	31.13	25.92	25.04	10.11	10.38	9.06

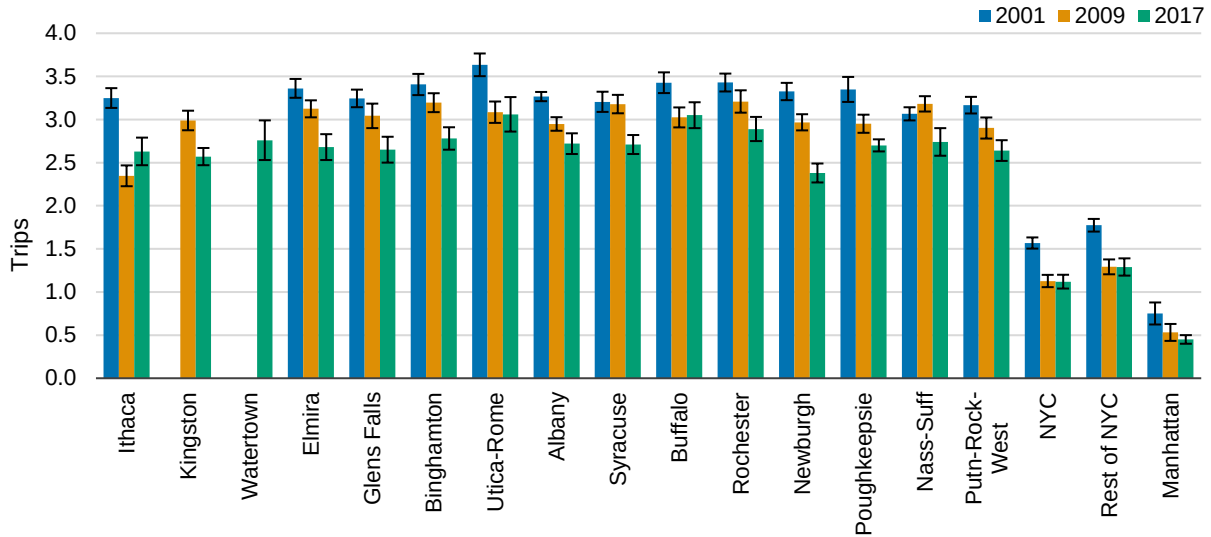


Figure 5-10. Vehicle trips statistics by year and location (e.g., city, region).

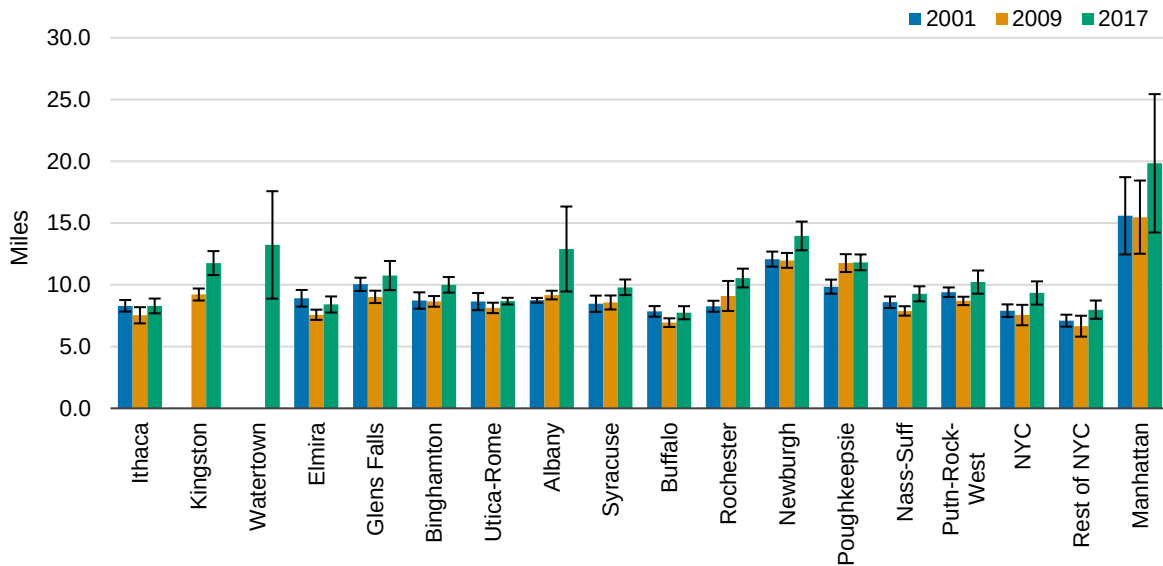


Figure 5-11. Average trip length (miles) statistics by year and location (e.g., city, region).

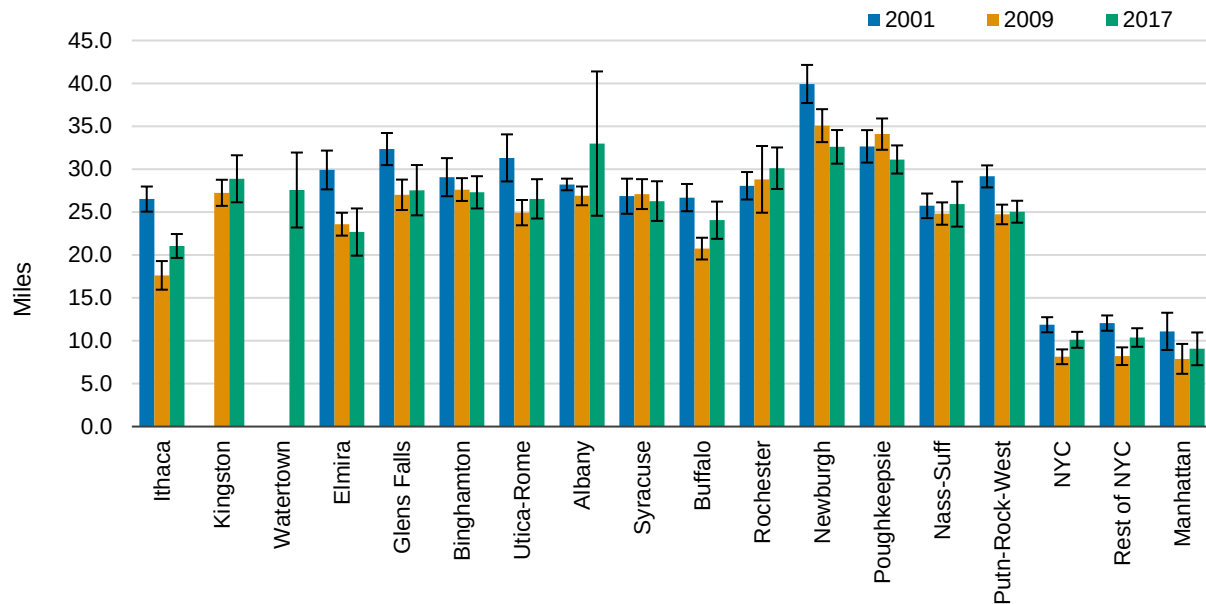


Figure 5-12. VMT statistics by year and location (e.g., city, region).

Figure 5-13 analyzes the average time spent driving a private vehicle across multiple regions for the years 2001, 2009, and 2017. Over the years, there have been observable trends in driving time. In many regions, the average driving time decreased between 2001 and 2009 but slightly rebounded by 2017. For instance, in Ithaca, the driving time declined from 72.08 min in 2001 to 63.24 min in 2009 before increasing again to 69.03 min in 2017. The differences are not statistically significant. A similar trend is shown in Glens Falls, which dropped statistically significantly from 80.62 min in 2001 to 69.78 in 2009 before rising slightly to 75.13 min in 2017. However, some regions, such as Watertown and Kingston, saw a more consistent increase in driving time over the years. NYC and its surrounding areas display distinct trends. Manhattan has the highest average driving time across all years, with 85.09 min in 2001, dropping slightly to 77.16 min in 2009, and then increasing to 96.19 min in 2017. This trend suggests increasing congestion or changes in travel behavior. In contrast, the rest of NYC follows a similar pattern but remains lower than Manhattan, with 88.31 min in 2001, 81.66 in 2009, and 98.19 in 2017.

Comparing regions, urban centers generally exhibit driving times longer than those in smaller cities and towns. Albany, Syracuse, and Rochester consistently report lower driving times than those reported in Manhattan, NYC, and Rockland–Westchester. Additionally, some upstate cities like Buffalo and Rochester have relatively moderate fluctuations over the years, suggesting more stable travel patterns than those in downstate areas.

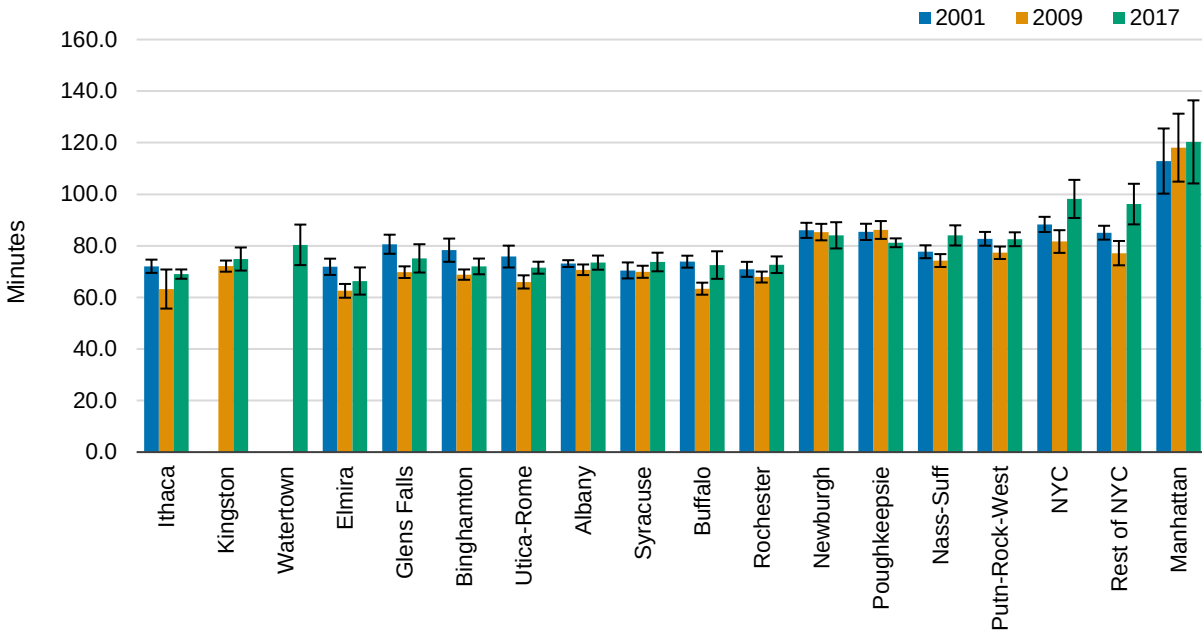


Figure 5-13. Average driving time for private vehicles by year and location (e.g., city, region).

Table 5-16 provides a detailed breakdown of vehicle trip statistics by purpose in 2017. A clear trend emerges in work-related travel, where smaller metro areas and non-MSA regions exhibit higher shares of trips for earning a living (Figure 5-14). Cities such as Watertown (26.2%), Elmira (25.8%), and Syracuse (26.9%) have some of the highest shares, reflecting the greater reliance on private vehicles for commuting. In contrast, NYC (22.9%) and particularly Manhattan (9.0%) show statistically significantly lower shares, highlighting the widespread use of public transit and alternative commuting options. Among all regions, Newburgh–Poughkeepsie stands out with the highest share of work-related travel at 28.9%, whereas Manhattan remains the lowest, reinforcing the idea that dense urban cores rely on modes of transport other than private vehicles. Family and personal travel account for the largest share of vehicle trips across all regions, though the distribution varies. In smaller metropolitan areas such as Ithaca (50.0%), Glens Falls (48.0%), and Elmira (47.2%), these trips make up nearly half of all travel, indicating a strong dependence on personal vehicles for daily life. Family and personal vehicle trips remain a substantial portion even in densely populated areas like NYC (45.2%) and its subcomponents (45.4%), suggesting that although public transportation plays a major role in commuting, private vehicles are still widely used for personal errands and family-related travel. Civic, educational, and religious trips account for a relatively small proportion of vehicle travel across all regions. Areas such as Watertown (5.5%) and Elmira (5.8%) report slightly higher shares, likely due to the necessity of private vehicles for school commutes and community activities. Social and recreational travel constitutes a notable portion of vehicle use, especially in smaller metro areas and NYC. Cities like Watertown (25.4%), Kingston (25.0%), and Glens Falls (22.5%) show relatively high shares, indicating that driving remains a key mode of transportation for leisure activities. Interestingly, Manhattan stands out with the highest proportion of social and recreational trips at 41.3%, suggesting that although work-related trips by car are minimal, discretionary travel remains significant. This trend reflects the urban lifestyle, where social activities and entertainment play a vital role in mobility patterns.

When considering VMT in 2017, the trends largely mirror trip distributions but offer additional insight into trip lengths (Figure 5-15). Work-related travel accounts for a higher VMT share in larger metropolitan areas, with Newburgh–Poughkeepsie (44.4%) and Nassau–Suffolk (38.8%) leading in long-distance commuting. In contrast, NYC (26.5%) and especially Manhattan (6.4%) report significantly

lower VMT shares for work, confirming that even when work-related trips occur, they tend to be shorter or involve alternative transportation modes. Family and personal travel contribute the highest VMT percentages in many regions, with Albany (50.6%) and Ithaca (39.7%) at the top. In contrast, Manhattan reports a relatively lower VMT percentage (46.1%), reinforcing the idea that although many personal trips still occur, they are often shorter because of the dense urban environment. Civic, educational, and religious trips account for a small fraction of VMT across all regions, with Buffalo (5.7%) and Nassau–Suffolk (3.3%) showing slightly higher numbers, whereas Manhattan (1.7%) again has the lowest share. Social and recreational travel represents a major component of VMT, particularly in urban areas. Manhattan stands out once again, with 43.9% of its total VMT dedicated to social and recreational purposes. This aligns with the earlier finding that social trips make up the highest percentage of vehicle trips in Manhattan, further emphasizing that although fewer people drive for work, recreational travel by private vehicle remains substantial. Smaller metropolitan areas such as Kingston (41.3%) and Glens Falls (29.6%) also show high VMT shares for social and recreational travel, indicating that such trips are often longer in these regions.

Overall, the 2017 data show that smaller MSAs exhibit a greater reliance on private vehicles for work and family-related trips, leading to higher VMT in these categories. In contrast, larger metro areas like NYC see a lower share of work-related driving but a higher share of social and recreational trips, particularly in Manhattan, where vehicle use for discretionary activities dominates. Suburban areas such as Nassau–Suffolk and Rockland–Westchester display a balanced mix of trip purposes.

Table 5-17 and Table 5-18 show daily vehicle travel statistics by trip purpose across different MSA sizes for the years 2001 and 2009. One of the most notable shifts between 2001 and 2009 is the decline in work-related vehicle trips, particularly in larger metropolitan areas like NYC and its surrounding suburbs. In 2001, work-related trips accounted for 25.6% of vehicle trips in NYC and 30.6% in Manhattan, but by 2009, these figures had declined to 18.2% and 13.8%, respectively. This trend suggests a shift away from private vehicle commuting, likely due to expanded transit options, increased telecommuting, or changing economic conditions during this period. In smaller metropolitan areas and non-MSA regions, the decline in work-related trips was less pronounced, with some areas like Newburgh–Poughkeepsie even seeing slight increases. Family and personal trips, on the other hand, maintained their dominant share across all regions, increasing slightly in urban areas between 2001 and 2009. For example, the percentage of family and personal trips rose from 50.3% in 2001 to 55.5% in 2009 in NYC and from 39.8% to 47.8% in Manhattan in the same period. This shift suggests that although work-related vehicle trips declined, personal vehicle use remained steady for household and errand-related travel. In suburban and smaller metro areas, family and personal trips remained relatively stable, hovering between 45% and 52% of total vehicle trips. Social and recreational travel saw interesting fluctuations. While the share of social and recreational travel remained steady in most regions, a significant increase occurred in Manhattan, rising from 28.6% in 2001 to 36.5% in 2009. In contrast, in many suburban and small metro areas, the share of social and recreational trips remained around 20%–23%, indicating that vehicle use for leisure activities remained relatively stable.

A similar pattern emerges when examining VMT by trip purpose. Work-related VMT decreased in urban areas like NYC, where it dropped from 34.5% in 2001 to 24.1% in 2009, reinforcing the idea that car commuting became less dominant. However, in smaller metro areas such as Ithaca and Glens Falls, VMT for work-related travel remained high, even increasing in some places (e.g., Glens Falls saw an increase from 34.0% to 42.0%). This suggests that while urban centers were reducing their dependence on private vehicles for work, smaller cities and rural areas continued to rely on them for commuting. Family and personal VMT remained a substantial portion of total miles traveled across all regions, though it slightly increased in suburban and urban regions. In NYC, it rose from 35.8% in 2001 to 34.5% in 2009, with similar increases in nearby areas such as Nassau–Suffolk and Rockland–Westchester. This suggests that although work trips declined, household and errand-related trips became longer or more frequent. Social

and recreational VMT saw an increase in urban areas, particularly in Manhattan, where it grew from 23.8% in 2001 to 33.5% in 2009. This aligns with the earlier observation that private vehicle use in Manhattan has increasingly shifted from work to leisure activities. In contrast, smaller metropolitan areas like Binghamton and Utica–Rome saw a slight decline in social and recreational VMT.

When looking at the entire period from 2001 to 2017, a few key trends emerge. First, work-related vehicle trips have consistently declined in major metropolitan areas, particularly in NYC and Manhattan, where they accounted for over 25% of trips in 2001 but dropped to 18.2% in 2009 and even lower by 2017 (9.0% in Manhattan). This suggests a sustained shift over time toward public transit, walking, biking, or remote work. Second, family and personal trips have remained the dominant category of vehicle travel, with their share increasing slightly in urban regions over time. This could be due to greater suburbanization, longer distances between households and services, or changes in shopping and errand patterns. Third, social and recreational trips have gained a more prominent share, particularly in urban areas, where they have offset declines in work-related travel. Manhattan, in particular, has seen one of the largest shifts toward leisure-related travel, with social and recreational VMT increasing significantly over the years. Last, the distribution of VMT aligns with these trends; as urban centers see less work-related travel but longer leisure trips, suburban and rural areas maintain steady or increasing VMT for work and family purposes. The increase in social and recreational VMT in urban areas suggests that, despite improvements in transit infrastructure, cars remain a crucial mode for nonwork trips, particularly for discretionary travel.

Table 5-16. Daily vehicle travel statistics by trip purpose, location (e.g., city, region), and MSA size in 2017.

Trip Purpose ^a	MSA Size Class							
	Not in MSA		<250,000			250,000–499,999		500,000–999,999
	Ithaca	Kingston	Watertown	Elmira	Glens Falls	Binghamton	Utica–Rome	Albany Syracuse
Vehicle Trips (%)								
Earn a Living	20.5%	25.5%	26.2%	25.8%	23.5%	25.3%	25.2%	25.9% 26.9%
Family & Personal Business	50.0%	42.7%	41.0%	47.2%	48.0%	45.1%	45.0%	43.6% 43.1%
Civic, Educational, & Religious	4.2%	3.9%	5.5%	5.8%	3.2%	3.7%	3.3%	4.0% 5.7%
Social & Recreational	21.3%	25.0%	25.4%	18.3%	22.5%	23.1%	23.5%	23.3% 21.8%
Other	4.0%	2.7%	1.9%	2.9%	2.8%	2.7%	3.0%	3.1% 2.5%
Unreported	0.0%	0.1%	0.0%	0.0%	0.0%	0.0%	0.0%	0.2% 0.0%
VMT (%)								
Earn a Living	20.0%	32.3%	20.5%	31.7%	28.4%	29.1%	34.5%	22.4% 31.6%
Family & Personal Business	39.7%	35.5%	29.6%	40.2%	35.3%	36.4%	30.3%	50.6% 28.3%
Civic, Educational, & Religious	4.2%	4.3%	3.8%	5.6%	1.5%	1.9%	3.6%	3.5% 5.2%
Social & Recreational	30.0%	23.3%	41.3%	19.7%	29.6%	27.4%	26.9%	19.2% 30.1%
Other	6.1%	4.5%	4.8%	2.7%	5.1%	5.2%	4.7%	4.1% 4.8%
Unreported	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.2% 0.0%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 5-16. Daily vehicle travel statistics by trip purpose, location (e.g., city, region), and MSA size in 2017 (continued).

Trip Purpose ^a	MSA Size Class								
	1 million– 2.9 million		3 million+						
	Buffalo	Rochester	Newburgh	Poughkeepsie	NYMTC				
					Nass– Suff	Putn– Rock– West	NYC	Subcomponents of NYC	
								Rest of NYC	Manhattan
Vehicle Trips (%)									
Earn a Living	25.3%	29.3%	28.7%	28.9%	23.2%	22.9%	22.9%	24.1%	9.0%
Family & Personal Business	43.2%	41.9%	42.2%	41.5%	45.4%	44.9%	45.2%	45.4%	42.2%
Civic, Educational, & Religious	4.9%	4.9%	6.3%	4.5%	3.6%	4.8%	3.5%	3.7%	1.0%
Social & Recreational	24.1%	20.7%	19.8%	22.2%	24.0%	23.9%	24.5%	23.0%	41.3%
Other	2.5%	3.2%	3.0%	3.0%	3.7%	3.5%	4.0%	3.7%	6.6%
Unreported	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
VMT (%)									
Earn a Living	31.9%	32.8%	44.4%	38.8%	28.3%	31.4%	26.5%	31.0%	6.4%
Family & Personal Business	29.7%	32.1%	25.4%	29.9%	31.0%	34.7%	36.3%	34.1%	46.1%
Civic, Educational, & Religious	4.3%	5.7%	4.2%	3.7%	3.3%	4.1%	1.8%	2.2%	0.4%
Social & Recreational	31.1%	24.9%	20.5%	21.9%	33.0%	24.2%	29.6%	26.6%	43.3%
Other	3.1%	4.5%	5.4%	5.6%	4.5%	5.6%	5.7%	6.1%	3.9%
Unreported	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 5-17. Daily vehicle travel statistics by trip purpose, location (e.g., city, region), and MSA size in 2009.

Trip Purpose ^a	MSA Size Class							
	Not in MSA		<250,000		250,000–499,999		500,000–999,999	
	Ithaca	Kingston	Elmira	Glens Falls	Binghamton	Utica–Rome	Albany	Syracuse
Vehicle Trips (%)								
Earn a Living	29.2%	23.9%	24.2%	29.2%	23.9%	24.9%	28.4%	25.6%
Family & Personal Business	42.7%	52.8%	50.9%	44.0%	48.2%	46.7%	46.0%	46.9%
Civic, Educational, & Religious	3.6%	3.9%	4.7%	3.0%	5.2%	5.3%	4.9%	4.0%
Social & Recreational	22.1%	18.7%	19.4%	22.7%	21.8%	22.2%	19.5%	22.7%
Other	1.0%	0.2%	0.1%	0.3%	0.3%	0.0%	0.2%	0.1%
Unreported	1.4%	0.5%	0.7%	0.9%	0.6%	1.0%	1.0%	0.8%
VMT (%)								
Earn a Living	35.6%	34.8%	28.1%	42.0%	31.8%	30.4%	35.9%	33.8%
Family & Personal Business	31.5%	37.3%	38.1%	31.9%	38.8%	39.2%	31.3%	31.3%
Civic, Educational, & Religious	3.2%	3.5%	5.3%	1.8%	4.3%	3.5%	4.7%	3.5%
Social & Recreational	24.1%	22.4%	27.1%	21.4%	22.4%	23.8%	23.4%	30.2%
Other	1.6%	0.1%	0.1%	1.1%	0.8%	0.0%	0.4%	0.1%
Unreported	4.0%	1.9%	1.3%	1.9%	2.0%	3.1%	4.4%	1.2%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 5-17. Daily vehicle travel statistics by trip purpose, location (e.g., city, region), and MSA size in 2009 (continued).

Trip Purpose ^a	MSA Size Class								
	1 million– 2.9 million		3 million+						
	Buffalo	Rochester	Newburgh	Poughkeepsie	NYMTC				
					Nass– Suff	Putn– Rock– West	NYC	Subcomponents of NYC	
								Rest of NYC	Manhattan
Vehicle Trips (%)									
Earn a Living	27.1%	26.5%	28.0%	23.8%	25.4%	23.9%	18.2%	18.7%	13.8%
Family & Personal Business	46.6%	45.2%	48.1%	47.7%	48.5%	49.9%	55.5%	56.3%	47.8%
Civic, Educational, & Religious	4.6%	4.4%	3.4%	4.5%	5.0%	5.9%	3.4%	3.8%	0.0%
Social & Recreational	21.0%	23.2%	19.1%	22.6%	20.0%	19.4%	21.3%	19.5%	36.5%
Other	0.1%	0.1%	0.5%	0.3%	0.6%	0.3%	0.2%	0.2%	0.4%
Unreported	0.7%	0.7%	0.8%	1.1%	0.5%	0.6%	1.5%	1.5%	1.6%
VMT (%)									
Earn a Living	32.7%	29.6%	44.4%	34.1%	39.6%	37.3%	24.1%	25.3%	19.6%
Family & Personal Business	32.4%	30.0%	31.8%	32.5%	34.3%	31.3%	34.5%	36.8%	26.1%
Civic, Educational, & Religious	4.5%	3.0%	2.0%	2.8%	3.7%	4.6%	2.3%	2.9%	0.0%
Social & Recreational	28.9%	21.8%	20.1%	25.5%	20.2%	24.0%	36.8%	33.4%	50.0%
Other	0.2%	0.3%	0.4%	0.7%	0.6%	0.5%	0.4%	0.3%	0.0%
Unreported	1.4%	15.2%	1.2%	4.3%	1.5%	2.3%	1.9%	1.4%	0.4%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 5-18. Daily vehicle travel statistics by trip purpose, location (e.g., city, region), and MSA size in 2001.

Trip Purpose ^a	MSA Size Class					
	Not in MSA	<250,000		250,000–499,999		500,000–999,999
	Ithaca	Elmira	Glens Falls	Binghamton	Utica-Rome	Albany Syracuse
Vehicle Trips (%)						
Earn a Living	24.3%	25.7%	24.8%	23.3%	25.7%	27.6% 23.6%
Family & Personal Business	47.9%	48.9%	50.9%	51.4%	46.4%	47.3% 51.4%
Civic, Educational, & Religious	6.1%	4.1%	3.1%	3.9%	3.6%	4.4% 4.1%
Social & Recreational	20.9%	20.8%	20.4%	21.1%	23.6%	20.0% 20.4%
Other	0.7%	0.3%	0.4%	0.2%	0.6%	0.6% 0.3%
Unreported	0.1%	0.2%	0.4%	0.2%	0.1%	0.2% 0.2%
VMT (%)						
Earn a Living	29.2%	29.7%	34.0%	27.8%	40.7%	36.4% 31.5%
Family & Personal Business	39.6%	39.3%	38.5%	47.8%	34.5%	34.0% 31.6%
Civic, Educational, & Religious	5.9%	3.4%	3.8%	3.2%	2.6%	3.6% 2.4%
Social & Recreational	24.5%	26.3%	22.4%	20.9%	20.9%	24.5% 34.0%
Other	0.5%	0.7%	0.8%	0.2%	1.0%	0.9% 0.4%
Unreported	0.2%	0.5%	0.6%	0.1%	0.4%	0.5% 0.1%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 5-18. Daily vehicle travel statistics by trip purpose, location (e.g., city, region), and MSA size in 2001 (continued).

Trip Purpose ^a	MSA Size Class								
	1 million– 2.9 million		3 million+						
	Buffalo	Rochester	Newburgh	Poughkeepsie	NYMTC				
					Nass– Suff	Putn– Rock– West	NYC	Subcomponents of NYC	
								Rest of NYC	Manhattan
Vehicle Trips (%)									
Earn a Living	24.4%	24.7%	27.0%	25.8%	25.7%	24.0%	25.6%	25.1%	30.6%
Family & Personal Business	49.3%	48.0%	50.8%	49.3%	46.9%	50.7%	50.3%	51.5%	39.8%
Civic, Educational, & Religious	4.7%	5.0%	3.2%	4.6%	4.6%	3.8%	3.8%	4.2%	0.1%
Social & Recreational	21.2%	22.0%	18.2%	19.5%	21.9%	20.8%	19.1%	18.1%	28.6%
Other	0.5%	0.3%	0.7%	0.7%	0.8%	0.5%	1.0%	1.1%	0.9%
Unreported	0.0%	0.0%	0.1%	0.1%	0.2%	0.3%	0.1%	0.1%	0.0%
VMT (%)									
Earn a Living	31.0%	32.3%	38.1%	38.9%	38.2%	37.5%	34.5%	37.0%	23.8%
Family & Personal Business	39.4%	36.1%	34.0%	32.7%	29.6%	31.0%	35.8%	35.5%	37.2%
Civic, Educational, & Religious	3.8%	4.6%	2.5%	4.8%	2.8%	2.7%	3.8%	4.7%	0.1%
Social & Recreational	25.2%	24.8%	22.4%	22.5%	28.9%	27.8%	25.1%	22.0%	38.4%
Other	0.6%	1.0%	2.8%	1.1%	0.5%	0.8%	0.6%	0.6%	0.5%
Unreported	0.0%	1.3%	0.2%	0.0%	0.1%	0.2%	0.2%	0.3%	0.0%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

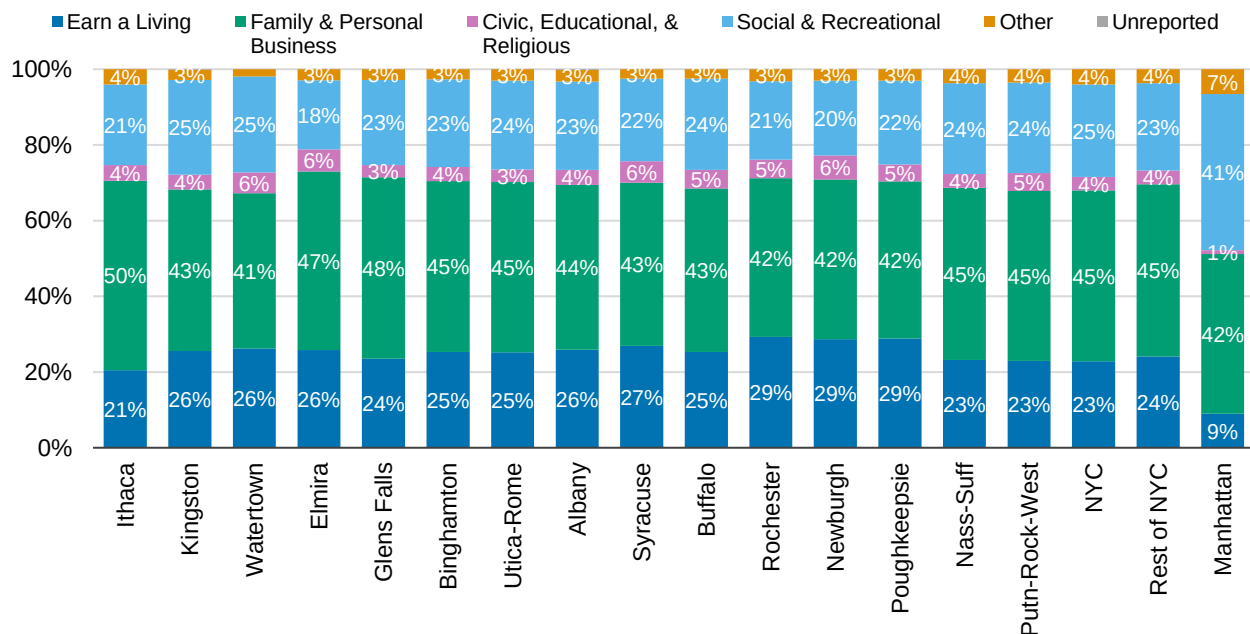


Figure 5-14. Distribution of vehicle trips by trip purpose and location (e.g., city, region) in 2017. Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number. Unlabeled segments represent values lower than 3%.

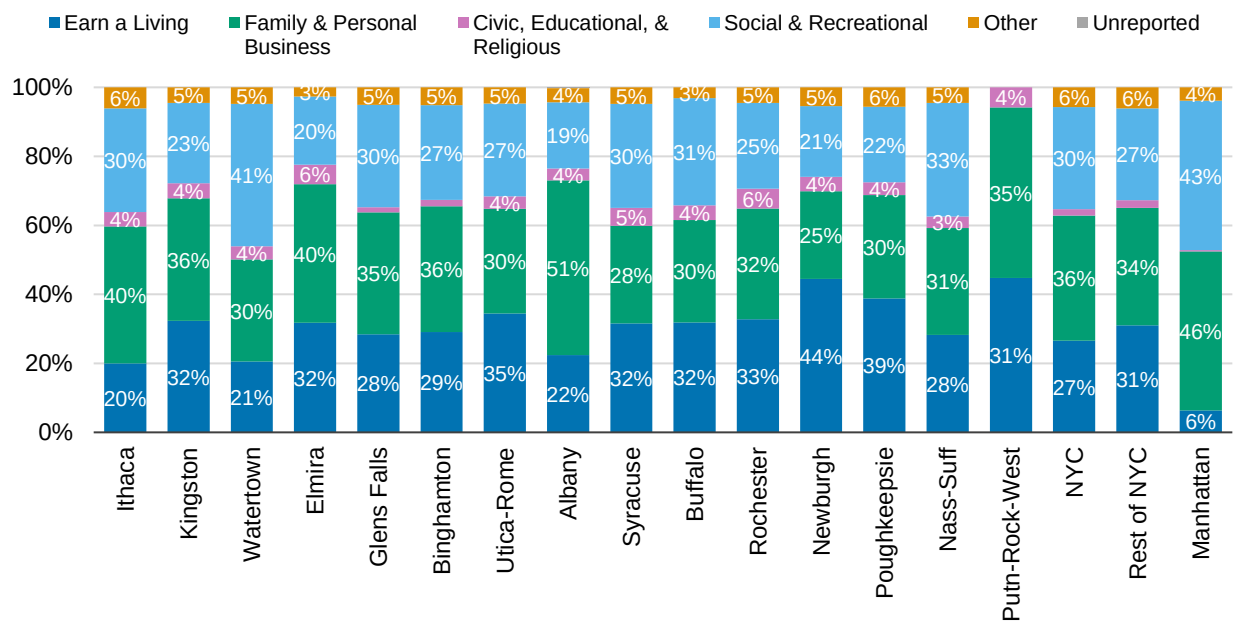


Figure 5-15. Distribution of VMT by trip purpose and location (e.g., city, region) in 2017. Note: Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number. Unlabeled segments represent values lower than 3%.

Figure 5-16 shows vehicle trips per day for various regions, differentiating between weekday and weekend travel patterns. It provides insights into travel behavior across different urban and suburban areas in NYS. Overall, most regions exhibit higher vehicle trips on weekdays compared with weekends, except for a few areas where weekend travel is comparable or slightly higher. For instance, in Ithaca, the number of trips per day is 2.65 on weekdays and 2.63 on weekends, but the difference is not statistically significant. Similarly, Kingston (2.55 weekday vs. 2.64 weekend) and Watertown (2.82 weekday vs. 2.65 weekend) also show minimal variation, suggesting travel patterns remain stable across the week. In contrast, some regions show notably fewer trips on weekends compared with weekdays, particularly in more urbanized areas. For example, in Buffalo, the average number of trips drops statistically significantly from 3.22 on weekdays to 2.63 on weekends. A similar pattern is observed in Rochester (3.02 to 2.59) and Albany (3.02 to 1.99), where the weekend decrease is statistically significant.

NYC and Manhattan exhibit the lowest vehicle trips per day, particularly on weekends, reinforcing the role of public transit and alternative modes of travel. In Manhattan, weekday trips average just 1.50, decreasing statistically significantly to 0.65 on weekends. In contrast, suburban and upstate areas rely more on vehicle travel, with consistently higher trips per day across both weekdays and weekends. For instance, in Glens Falls, trips per day remain high (2.50 on weekdays, 3.05 on weekends), whereas Utica–Rome shows a small weekend increase (3.08 to 3.02,); neither difference is statistically significant.

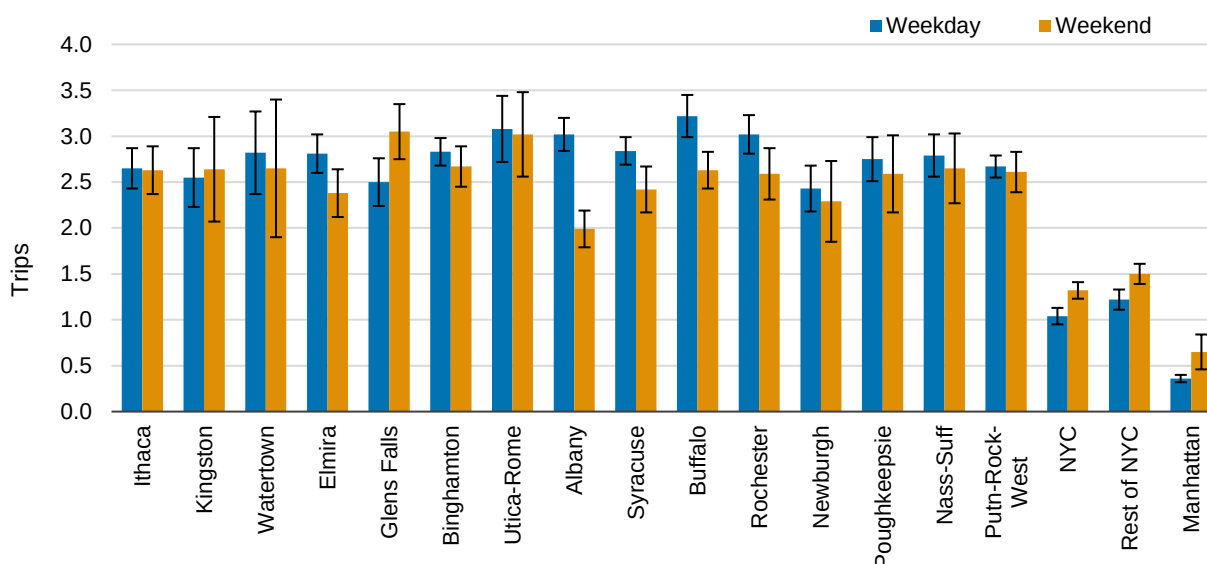


Figure 5-16. Vehicle trips by day and location (e.g., city, region) in 2017.

Note: Weekend trips = Saturday and Sunday trips.

5.3.2 Types and Ages of Vehicles Driven

Table 5-19 presents household in-use vehicle fleet percentages across NYS's metro areas for the years 2001, 2009, and 2017, categorized by different vehicle types, including automobiles, vans, SUVs, pickups, and motorcycles. The share of traditional automobiles (cars and sedans) has declined across all regions, reflecting a broader national trend toward SUVs and light trucks. In 2001, autos represented 66.0% of the household fleet in Ithaca, 58.4% in Elmira, and 62.5% in Manhattan, but by 2017, these figures had fallen to 56.1%, 47.7%, and 54.6%, respectively. The decline in automobile share is seen across all metro sizes, including both small towns and major metropolitan areas, as SUVs and pickups gained popularity. SUVs saw significant growth, particularly in suburban and rural areas. In 2001, SUV ownership was relatively low, with shares around 10%–15% in smaller metro areas and under 20% in larger regions like Buffalo and NYC. However, by 2017, SUV ownership surged, reaching over 30% in

many regions, including 30.4% in Watertown, 31.4% in Syracuse, and 30.5% in NYC. The sharp rise in SUV ownership aligns with changing consumer preferences, improvements in fuel efficiency, and a growing demand for larger, more versatile vehicles.

Pickup truck ownership, traditionally higher in rural and smaller metro areas, has remained relatively stable. In non-MSA areas like Ithaca and Glens Falls, pickup ownership hovered around 13%–16% in 2001 and remained within that range in 2017. However, in urban areas such as NYC and its suburbs, pickup ownership has stayed consistently low, typically below 5%. Van ownership has declined over time across nearly all metro areas. In 2001, vans accounted for approximately 10% of household fleets in many upstate regions, but by 2017, their share had fallen to around 5%–6%. This decline reflects the shift away from traditional minivans toward SUVs, which offer similar passenger capacity with perceived benefits in styling, performance, and versatility. Motorcycles have remained a small but steady segment of the vehicle fleet, with slightly higher shares in rural and less urbanized areas. In regions like Elmira and Glens Falls, motorcycle ownership increased slightly from a range of 2%–3% in 2001 to 3%–5% in 2017, whereas in denser areas like NYC and Manhattan, it has consistently remained below 2%.

Vehicle ownership varies significantly by urbanization level. In rural and smaller metro areas, the reliance on pickups and SUVs is more pronounced, reflecting the need for all-weather, utility-oriented vehicles in these areas. For instance, in 2017, SUVs made up 30.4% of the fleet in Watertown, compared with just 26.0% in Ithaca. Similarly, pickup trucks represent around 15% of the fleet in Glens Falls but only 5% in NYC. In contrast, urban areas like NYC and Manhattan exhibit a stronger preference for traditional automobiles and lower overall vehicle ownership rates. In 2017, cars still accounted for over 54.6% of the fleet in Manhattan, despite the national trend toward SUVs and light trucks. Additionally, pickup truck and motorcycle ownership remains minimal in these high-density urban cores, where public transit, biking, and walking are more prevalent. Suburban regions, such as Nassau–Suffolk and Rockland–Westchester, represent a middle ground between urban and rural vehicle preferences. These areas have seen a major increase in SUV ownership, with SUVs making up over 30% of household fleets in 2017, compared with less than 20% in 2001. However, these regions still maintain relatively low pickup truck ownership rates, reinforcing the idea that pickups remain more common in rural and small-town settings.

The data from 2001, 2009, and 2017 illustrate a long-term transition toward SUVs and a gradual decline in sedans and vans. This trend reflects shifting consumer preferences, advancements in fuel efficiency, and changing perceptions of vehicle utility and safety. The decline of van ownership suggests that families who once relied on minivans are now opting for SUVs, whereas the stability in pickup truck ownership indicates their continued relevance in rural and utility-based settings. Urban areas, particularly NYC and Manhattan, have resisted the SUV boom to some extent, maintaining a higher share of traditional automobiles than those in other regions. However, even in these dense environments, SUVs have gained ground, with ownership rising from under 20% in 2001 to over 30% in 2017.

Table 5-19. Household in-use vehicle fleet percentages by vehicle type, year, location (e.g., city, region), and MSA size.

Vehicle Type ^a	MSA Size Class								
	Not in MSA		<250,000			250,000–499,999		500,000–999,999	
	Ithaca	Kingston	Watertown	Elmira	Glens Falls	Binghamton	Utica–Rome	Albany	Syracuse
2017									
Auto	56.1%	52.2%	44.0%	47.7%	45.3%	50.2%	48.0%	57.6%	48.5%
Van	7.5%	5.2%	10.5%	11.3%	3.5%	8.3%	7.2%	5.1%	6.1%
Sport Utility	22.6%	27.6%	30.4%	21.9%	29.3%	24.8%	23.0%	23.8%	31.4%
Pickup	10.0%	11.2%	11.4%	14.0%	16.0%	13.4%	15.9%	9.1%	9.9%
Motorcycle	2.5%	2.3%	2.8%	3.8%	4.9%	2.0%	5.0%	3.1%	3.1%
2009									
Auto	58.5%	55.3%	—	51.5%	45.9%	53.4%	54.2%	56.8%	56.0%
Van	9.5%	8.1%	—	8.9%	7.4%	11.0%	8.8%	8.0%	9.9%
Sport Utility	18.5%	18.3%	—	20.6%	22.4%	18.2%	18.1%	22.4%	21.7%
Pickup	10.6%	13.0%	—	14.7%	19.8%	14.3%	15.8%	9.9%	9.7%
Motorcycle	2.5%	4.8%	—	3.4%	3.7%	2.2%	2.8%	2.4%	2.4%
2001									
Auto	66.0%	—	—	58.4%	53.1%	60.9%	58.5%	62.4%	66.5%
Van	8.0%	—	—	10.3%	8.5%	10.0%	9.6%	10.1%	9.7%
Sport Utility	9.8%	—	—	10.4%	12.9%	10.8%	11.8%	11.8%	11.2%
Pickup	13.4%	—	—	16.6%	21.0%	14.7%	14.5%	12.2%	10.4%
Motorcycle	1.9%	—	—	2.9%	3.3%	2.8%	4.7%	2.5%	1.3%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 5-19. Household in-use vehicle fleet percentages by vehicle type, year, location (e.g., city, region), and MSA size(continued).

Vehicle Type ^a	MSA Size Class									
	1 million– 2.9 million		3 million+							
	Buffalo	Rochester	Newburgh	Poughkeepsie	NYMTC				Subcomponents of NYC	
					Nass–Suff	Putn– Rock– West	NYC	Rest of NYC	Manhattan	
2017										
Auto	48.7%	58.9%	55.5%	53.9%	54.1%	54.6%	59.8%	59.2%	62.5%	
Van	7.1%	6.5%	5.4%	6.8%	6.0%	6.8%	6.3%	7.0%	2.5%	
Sport Utility	30.1%	24.2%	24.9%	26.0%	32.0%	32.5%	30.5%	30.5%	30.6%	
Pickup	10.9%	7.6%	10.8%	8.9%	5.2%	3.3%	1.4%	1.3%	1.9%	
Motorcycle	2.4%	2.4%	2.5%	3.5%	1.7%	1.8%	1.6%	1.7%	1.3%	
2009										
Auto	61.0%	57.3%	55.3%	58.1%	59.6%	59.9%	64.7%	64.6%	65.3%	
Van	10.0%	11.8%	9.1%	8.7%	8.0%	8.4%	9.9%	10.5%	5.6%	
Sport Utility	17.2%	17.3%	20.8%	20.0%	25.3%	25.0%	22.6%	22.1%	26.3%	
Pickup	9.2%	11.2%	11.2%	8.3%	4.6%	4.9%	1.2%	1.3%	0.4%	
Motorcycle	2.2%	1.6%	3.0%	4.1%	2.0%	1.5%	1.6%	1.5%	2.4%	
2001										
Auto	64.3%	63.3%	65.3%	64.1%	66.8%	73.3%	75.7%	75.3%	78.2%	
Van	10.7%	11.3%	9.6%	8.8%	11.2%	8.9%	9.3%	9.7%	6.4%	
Sport Utility	10.1%	11.7%	11.6%	12.7%	14.6%	13.1%	12.4%	12.5%	11.7%	
Pickup	11.9%	10.7%	10.7%	9.1%	5.6%	3.6%	1.0%	1.0%	1.3%	
Motorcycle	2.2%	2.0%	2.1%	4.2%	1.7%	0.9%	1.3%	1.1%	2.5%	

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Looking at the annual VMT (Figure 5-17), vehicle mileage varied significantly across regions and years. Examining the estimates, Glens Falls, Rochester, and Newburgh–Poughkeepsie recorded some of the highest vehicle mileages in 2001, whereas NYC and Manhattan had the lowest. Between 2001 and 2009, estimated mileage in Elmira, Binghamton, and Rochester declined, but these decreases are not statistically significant. However, Kingston saw a statistically significant increase in vehicle mileage from 2001 to 2009. From 2009 to 2017, Watertown experienced a statistically significant increase in vehicle mileage. Conversely, Manhattan saw a statistically significant decrease over the same period. Meanwhile, the decline in NYC’s vehicle mileage between 2001 and 2009 is not statistically significant. Although some regions, such as Albany and Syracuse, showed fluctuations over the years, none of these changes are statistically significant. Similarly, the drop in Binghamton’s mileage between 2001 and 2009 is not statistically significant.

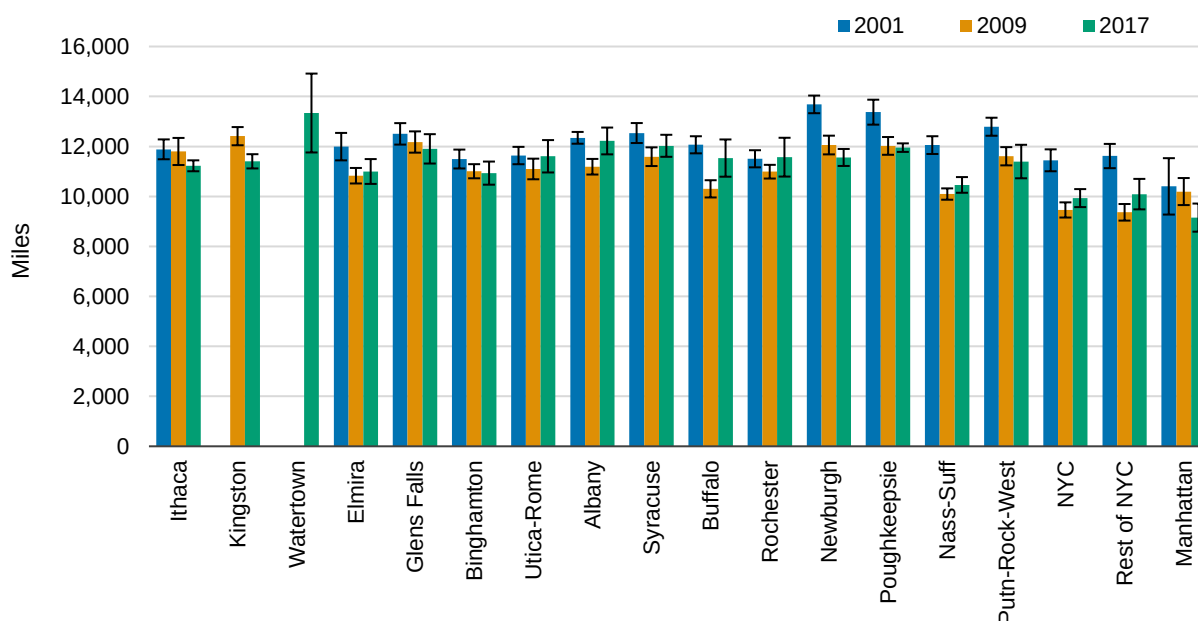


Figure 5-17. Annual estimated vehicle miles driven by year and location (e.g., city, region).

5.4 COMMUTING COMPARISONS

5.4.1 Commuting Trip Frequencies and Miles of Travel by Mode

Table 5-20 shows the daily commuting patterns across different MSAs in NYS, categorized by population size. Smaller MSAs, particularly those with populations under 250,000, exhibit a significantly higher reliance than that of larger metropolitan areas on SOVs. For instance, Kingston and Watertown have SOV shares exceeding 72%, whereas places like Ithaca, which is not in an MSA, report a substantially lower SOV share at 44.5% (Figure 5-18). The difference in SOV use between these areas is statistically significant, reflecting potential differences in urban form, transit availability, and travel behavior. Conversely, MOV use remains fairly consistent in smaller regions, ranging from approximately 18.9% to 24.2%, with no statistically significant differences between many of these MSAs. Larger MSAs, especially those with populations exceeding 1 million, display a marked shift toward public transit. In NYC, for instance, SOV use drops dramatically to just 16.9%, a statistically significant reduction compared with smaller MSAs. Within NYC, further disaggregation reveals even lower SOV use in Manhattan at just 1.1%, whereas subway and elevated rail account for 46.5% of commutes, a statistically significant increase relative to other MSAs. In contrast, Nassau–Suffolk and the northern Putnam–

Rockland–Westchester regions, as part of the greater NYC metropolitan area, exhibit higher SOV shares at 72.7% and 56.6%, respectively, suggesting substantial variation within the NYC metro area itself.

Public transit use varies widely, with subway/elevated rail, Amtrak/commuter rail, and other public transportation modes playing a more significant role in larger MSAs. In regions like NYC, the subway share reaches 41.2%, a statistically significant difference compared with the shares of all other MSAs. The Amtrak/commuter rail share, whereas generally low, is notably higher in areas with strong regional rail networks, such as Nassau–Suffolk (7.8%) and Rockland–Westchester (11.4%), where it provides crucial commuter connections to NYC. These shares are significantly higher compared with smaller MSAs, where commuter rail is either nonexistent or negligible. Walking mode share also follows a distinct pattern, with substantially higher values in dense urban areas. For example, Manhattan sees 27.7% of commutes completed on foot, which is a statistically significant increase over all other MSAs, reflecting the area’s walkability and high population density. In contrast, smaller regions, particularly those with populations under 500,000, typically report walk mode shares between 2% and 7%, with no statistically significant differences among them.

Examining PMT provides additional insight into commute distances (Figure 5-19). In smaller MSAs like Ithaca and Kingston, daily PMT is relatively low, suggesting shorter average commute distances. Conversely, in larger metro areas such as NYC, PMT varies significantly by subregion. Manhattan, with its extensive public transit use and high walk mode share, records a notably low PMT of 4.8, whereas suburban components of NYC, such as Nassau–Suffolk, show significantly higher PMT values, reflecting the longer commutes that are often associated with suburban living and reliance on personal vehicles or commuter rail.

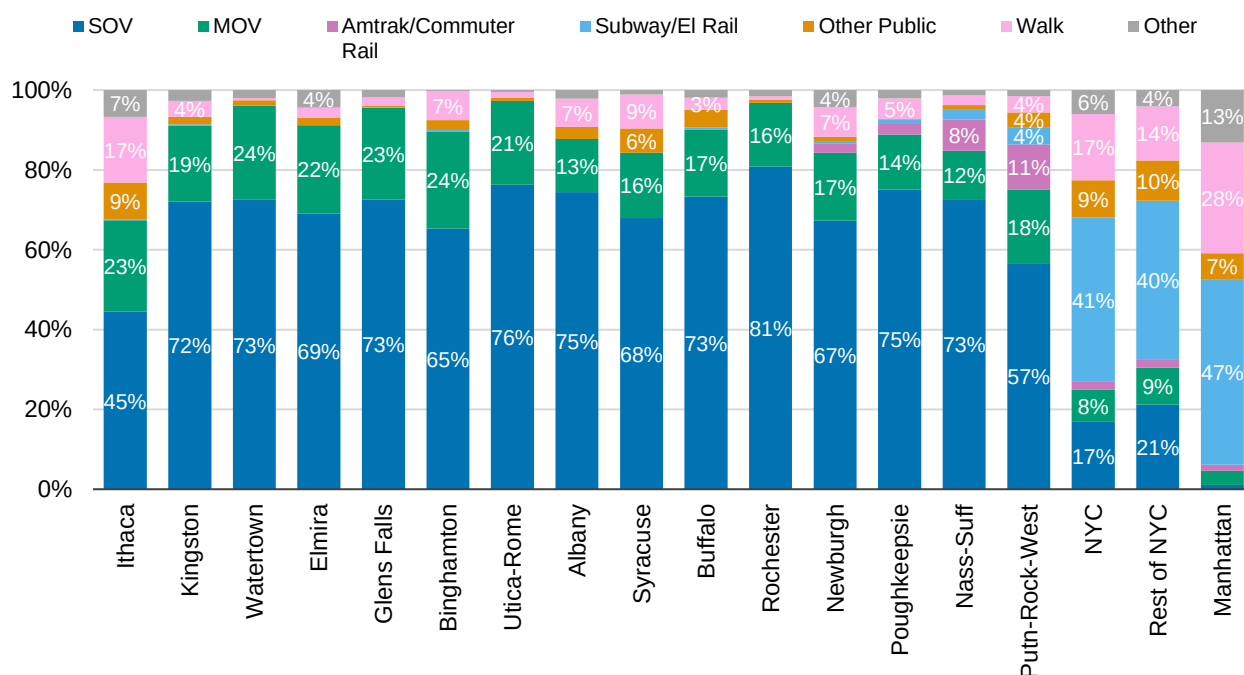


Figure 5-18. Daily commute trips by mode of transportation by year and location (e.g., city, region) in 2017.
 Notes: El Rail = elevated rail. Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number. Unlabeled segments represent values lower than 3%.

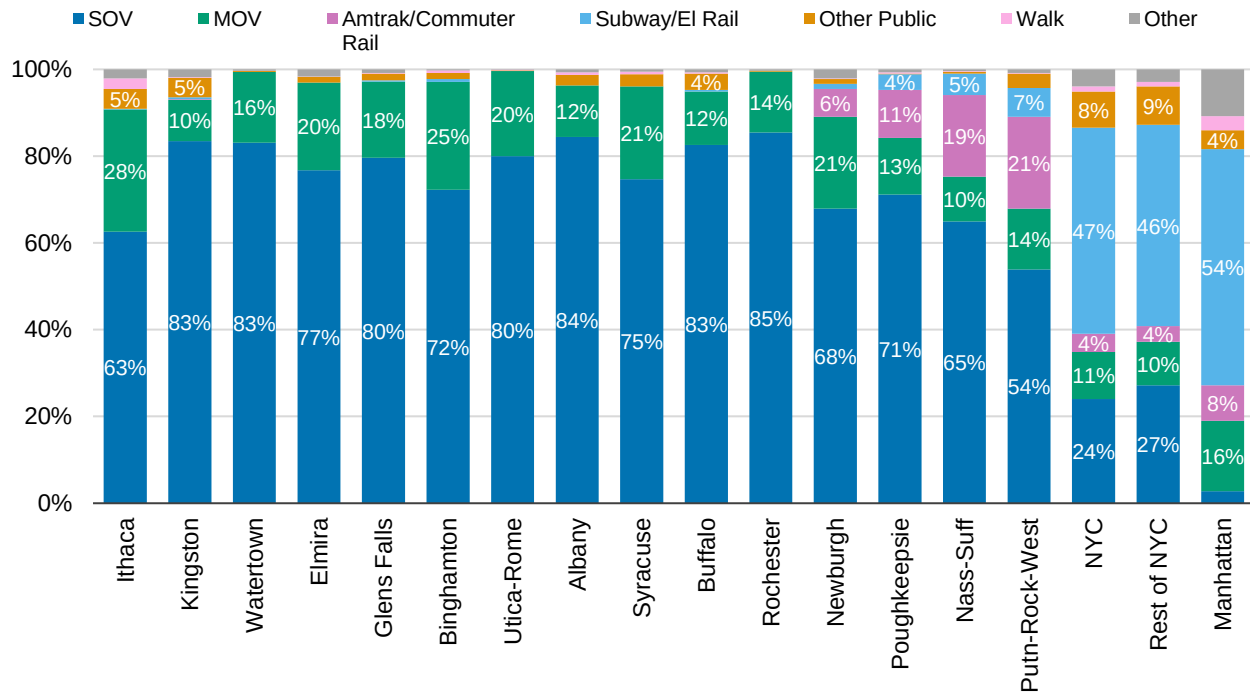


Figure 5-19. Daily commute PMT by mode of transportation by year and location (e.g., city, region) in 2017.
 Notes: El Rail = elevated rail. Totals may not equal exactly 100% because some quantities have been rounded to the nearest whole number. Unlabeled segments represent values lower than 4%.

Table 5-20. Share of commute trips and PMT by mode of transportation, location (e.g., city, region), and MSA size in 2017.

Mode ^a	MSA Size Class								
	Not in MSA		<250,000			250,000–499,999		500,000–999,999	
	Ithaca	Kingston	Watertown	Elmira	Glens Falls	Binghamton	Utica–Rome	Albany	Syracuse
Daily Commute Trips per Worker	1.1	1.0	1.2	1.1	1.0	1.1	1.1	1.1	1.1
SOV	44.5%	72.2%	72.5%	69.1%	72.6%	65.4%	76.2%	74.5%	68.0%
MOV	22.9%	18.9%	23.6%	22.1%	22.9%	24.2%	21.0%	13.3%	16.4%
Amtrak/Commuter Rail	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Subway/Elevated Rail	0.1%	0.4%	0.0%	0.0%	0.1%	0.5%	0.0%	0.0%	0.0%
Other Public	9.3%	1.9%	1.3%	1.9%	0.4%	2.5%	0.8%	3.1%	6.1%
Walk	16.5%	4.0%	0.6%	2.5%	2.2%	7.2%	1.4%	7.0%	8.5%
Other	6.7%	2.7%	2.0%	4.4%	1.8%	0.3%	0.5%	2.2%	1.1%
Daily Commute PMT per Worker	6.1	18.2	8.8	9.1	12.0	10.1	12.3	10.3	10.7
SOV	62.5%	83.4%	83.1%	76.7%	79.7%	72.3%	80.0%	84.4%	74.7%
MOV	28.2%	9.5%	16.3%	20.2%	17.6%	24.9%	19.6%	11.9%	21.4%
Amtrak/Commuter Rail	0.0%	0.1%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Subway/Elevated Rail	0.1%	0.5%	0.0%	0.0%	0.3%	0.7%	0.0%	0.0%	0.0%
Other Public	4.6%	4.5%	0.2%	1.3%	1.5%	1.4%	0.2%	2.4%	2.8%
Walk	2.4%	0.1%	0.1%	0.1%	0.2%	0.6%	0.1%	0.6%	0.6%
Other	2.1%	1.8%	0.3%	1.6%	0.8%	0.2%	0.1%	0.7%	0.5%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Table 5-20. Share of commute trips and PMT by mode of transportation, location (e.g., city, region), and MSA size in 2017 (continued).

Mode ^a	MSA Size Class								
	1 million– 2.9 million		3 million+						
	Buffalo	Rochester	Newburgh	Poughkeepsie	NYMTC				
					Nass–Suff	Putn– Rock– West	NYC	Subcomponents of NYC	
								Rest of NYC	Manhattan
Daily Commute Trips per Worker	1.2	1.1	1.1	1.1	1.0	1.1	1.1	1.1	1.0
SOV	73.4%	80.8%	67.4%	75.1%	72.7%	56.6%	16.9%	21.2%	1.1%
MOV	16.8%	15.9%	17.0%	13.8%	12.1%	18.3%	8.0%	9.2%	3.5%
Amtrak/Commuter Rail	0.0%	0.0%	2.2%	2.6%	7.8%	11.4%	2.0%	2.2%	1.5%
Subway/Elevated Rail	0.5%	0.0%	0.5%	1.2%	2.4%	4.3%	41.2%	39.7%	46.5%
Other Public	4.4%	0.7%	1.3%	0.0%	1.2%	3.7%	9.3%	10.1%	6.5%
Walk	3.1%	1.0%	7.4%	5.2%	2.5%	4.1%	16.6%	13.5%	27.7%
Other	1.9%	1.5%	4.3%	2.1%	1.3%	1.5%	6.0%	4.1%	13.1%
Daily Commute PMT per Worker	10.2	12.1	20.7	17.7	12.9	15.3	8.5	9.7	4.8
SOV	82.6%	85.4%	67.8%	71.2%	64.9%	53.9%	24.0%	27.2%	2.8%
MOV	12.3%	14.0%	21.2%	13.1%	10.3%	14.0%	10.8%	10.0%	16.2%
Amtrak/Commuter Rail	0.0%	0.0%	6.4%	11.1%	18.8%	21.2%	4.2%	3.6%	8.2%
Subway/Elevated Rail	0.4%	0.0%	1.2%	3.6%	4.9%	6.6%	47.4%	46.4%	54.4%
Other Public	3.7%	0.2%	1.1%	0.1%	0.5%	3.3%	8.3%	8.9%	4.4%
Walk	0.3%	0.0%	0.1%	0.3%	0.1%	0.2%	1.2%	1.0%	3.2%
Other	0.7%	0.3%	2.1%	0.7%	0.4%	0.8%	3.9%	2.9%	10.8%

^a Totals may not equal exactly 100% because some quantities have been rounded to the nearest tenth of a percent.

Figure 5-20 presents daily commute person trips across NYS metro areas for the years 2001, 2009, and 2017. Looking at the statewide trends, most metro areas exhibit relatively stable commute trip rates over the years, with some minor fluctuations. In 2001, daily commute trips per person ranged between 1.07 (Newburgh–Poughkeepsie) and 1.20 (Syracuse and Utica–Rome). By 2009, many metro areas experienced slight declines, with NYC seeing the lowest commute trip rate at 0.95, statistically significantly lower than those of all other regions. However, by 2017, most regions showed a slight rebound or remained stable, with Watertown (1.18) and Buffalo (1.15) showing some of the highest trip rates. Some metro areas exhibit consistent stability over time. For example, Albany, Rochester, and Newburgh–Poughkeepsie maintain similar daily commute trip rates throughout the years. Overall, the data suggest relatively stable commuting behavior across NYS metro areas over the years, with some regions experiencing slight fluctuations that are not always statistically significant. However, Utica–Rome and Watertown show clear upward trends in commute trip rates from 2009 to 2017, suggesting potential changes in travel demand or commuting patterns in those areas.

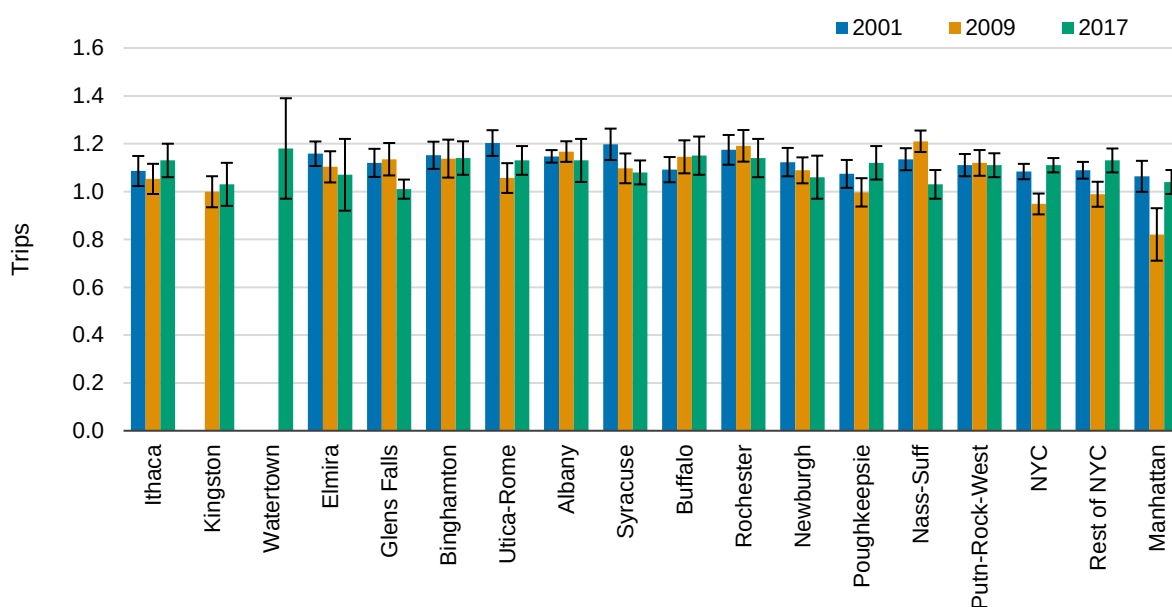


Figure 5-20. Daily commute person trips statistics by year and location (e.g., city, region).

Figure 5-21 presents daily commute travel time in various MSAs of NYS across 2001, 2009, and 2017. In general, commute times have increased over the years in most MSAs. In smaller regions such as Ithaca, Kingston, and Elmira, travel times have remained relatively stable, with only minor increases between 2001 and 2017. For example, in Ithaca, the commute time increased slightly from 17.18 min in 2001 to 17.46 min in 2017, a change that is not statistically significant. Similarly, Elmira saw an increase from 17.83 min to 18.34 min, again not statistically significant. In contrast, larger metro areas and suburban regions surrounding NYC have experienced more substantial increases. Commute times in Nassau–Suffolk, the northern Putnam–Rockland–Westchester region, and Newburgh–Poughkeepsie have all risen notably. For example, in Nassau–Suffolk, commute times increased from 29.54 min in 2001 to 29.19 min in 2017. NYC and its subcomponents have consistently exhibited the longest commute times, with steady increases. The overall NYC commute time rose from 40.21 min in 2001 to 41.14 min in 2017, whereas Manhattan saw an increase from 36.96 to 44.98 min. The rise in Manhattan’s commute duration is statistically significant. This suggests increasing congestion and possibly longer travel distances despite extensive public transit infrastructure. Overall, these data suggest a growing disparity in commute times between smaller and larger MSAs. Whereas rural and small metro areas have experienced relatively stable travel times, suburban and urban commuters, particularly in and around NYC, face increasing

commute durations, likely driven by population growth, congestion, and evolving transportation dynamics.

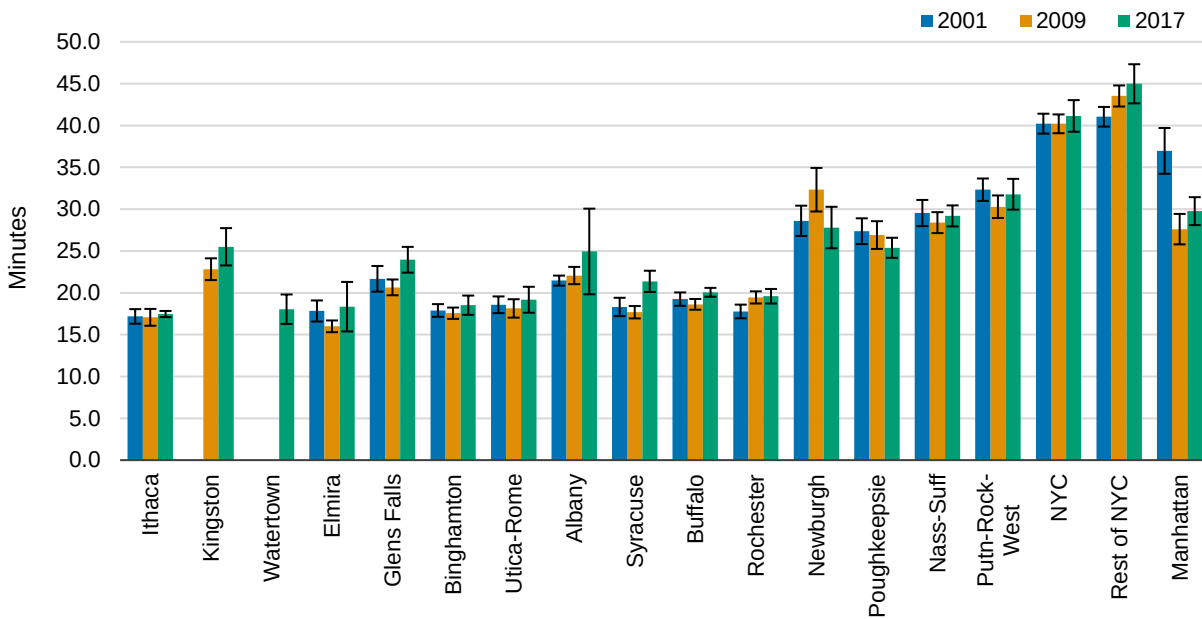


Figure 5-21. Daily commute travel time statistics by year and location (e.g., city, region).

6. SUMMARY

This report provides a comprehensive analysis of transportation patterns, vehicle ownership trends, and travel behaviors across NYS and the rest of the United States. The study examines key factors influencing mobility, including regional differences in vehicle ownership, travel frequency, transportation mode choices, and environmental impacts over multiple NHTS years. NHTS data from 2001, 2009, 2017, and 2022 were used. A significant focus is placed on differences in travel by geographical characteristics and shifting personal, vehicle, and commute patterns. Findings highlight how transportation habits have evolved over time, with declines in vehicle use, increases in alternative transportation modes, and variations based on household characteristics, income levels, and geographic location.

In Chapter 2, the analysis reveals distinct patterns in vehicle ownership, transportation habits, and environmental impacts between NYS and the rest of the United States. Across all observed years, NYS had a significantly higher share of zero-vehicle households, particularly in NYC, where a robust public transportation system reduces the necessity of owning a car. In 2017, 29% of households in NYS did not own a vehicle, compared with only 8% of households nationwide. Households with two or more vehicles were more common in the rest of the United States, whereas single-vehicle households were similarly prevalent across regions. Regarding travel behavior, person trips in NYS declined from 3.7 trips per person in 2001 to 3.3 in 2017. Although the NYS rate was initially lower than the national rate, the difference narrowed over time, with the rest of the United States averaging 3.4 trips in 2017. Vehicle ownership also affected travel frequency; zero-vehicle households made fewer person trips than those with vehicles across all years, a trend more pronounced among low-income households. In terms of travel modes, private vehicles accounted for a sizable portion of person trips in both regions, although this share was notably lower in NYS (59%) than the rate in the rest of the United States (84%). Walking trips were substantially higher in NYS, making up 24% of person trips in 2017, compared with just 10% nationally. In NYS, the average number of daily vehicle trips per driver declined from 2001 to 2017. Social and recreational trips increased from 21% in 2009 to 23% in 2017, whereas family and personal business vehicle trips decreased from 49% to 45% during the same period. VMT in NYS remained relatively stable, decreasing slightly from 19.7 miles per driver in 2009 to 19.3 in 2017. However, work-related trip VMT declined significantly, dropping from 34% in 2009 to 31% in 2017. Across all 3 years, the 35-to-54-year-old age group recorded the highest VMT in both NYS and the rest of the United States. Males consistently had significantly higher VMT than that of females; in 2017, male drivers in NYS averaged 26 miles, 52% higher than the 17 miles average for female drivers.

Chapter 3 examined key transportation and travel trends across NYC, the rest of NYS, and the rest of the United States from 2001 to 2017. Despite accounting for 43% of NYS households, NYC has significantly lower vehicle ownership, with only 21% of the state's vehicles and a high proportion (80%) of zero-vehicle households. This reflects the city's reliance on public transit, walking, and cycling, supported by its dense infrastructure and extensive transit network. NYC has a larger share of lower-income households than the shares of the rest of NYS and the rest of the United States. In 2017, 10% of NYC households earned less than \$10,000 annually, a statistically higher share than the 7% in the rest of NYS. However, NYC also has a greater proportion of high-income households earning over \$200,000 annually, particularly in Manhattan. The driver's licensing rate in NYC is consistently lower than the rates in the rest of NYS and the rest of the United States, with a notable gender gap—71% of men and only 50% of women in NYC held a license in 2017. Vehicle ownership remains significantly lower in NYC, where 54% of households have no vehicle, compared with just 15% in the rest of NYS. Additionally, NYC has a much lower share of multivehicle households. Average daily person trips per person declined across all regions between 2001 and 2017. However, average trip lengths increased, particularly in NYC, where they rose from 5.2 miles in 2009 to 7.8 miles in 2017. Walking was the most common travel mode in NYC (41% of trips), whereas single-occupancy vehicle trips represented a 10% share in 2017,

significantly lower than the shares in the rest of NYS and the rest of the United States. Zero-vehicle households in NYC take more daily trips compared with their counterparts elsewhere, with a significantly higher reliance on public transit (36% of trips) and walking (53%). In contrast, zero-vehicle households in the rest of NYS and the United States rely more on ride-hailing, taxis, and carpooling. Daily vehicle trips per driver declined across all regions, particularly in NYC, where the rate dropped to 1.1 trips in 2017—far lower than the 2.7 trips in the rest of NYS and the rest of the United States. Although work-related vehicle trips declined, social and recreational trips increased. NYC has seen a rise in transit commuting, with subway ridership increasing from 29% in 2009 to 42% in 2017. In contrast, single-occupancy vehicle commuting declined to 17% in NYC, whereas it remained the dominant mode in the rest of NYS and the United States. Commute distances decreased slightly across all regions, but commute times increased, particularly for private vehicle users in NYC, reflecting worsening congestion.

The analysis of travel patterns across population densities in Chapter 4 reveals distinct trends in personal, vehicle, and commute behaviors. Overall, trip rates have declined from 2001 to 2017 across both NYS and the rest of the United States, with more pronounced decreases in lower-density areas. However, in high-density areas (20,000+ people per square mile), the rest of the United States experienced a statistically significant increase in trip rates, diverging from the more stable patterns observed in NYS. This shift may reflect regional differences in urbanization, transit infrastructure, and commuting behavior, particularly as trip lengths increased dramatically in dense urban areas over time. In 2017, NYS saw increases in trip lengths in mid-density areas, whereas the rest of the United States experienced most increases in trip lengths in the highest-density areas, suggesting the growth of urban sprawl or economic decentralization. Vehicle travel patterns also evolved, with VMT per driver decreasing as density increased, but at a slower rate in the rest of the United States. Public transit remained dominant in high-density NYS, whereas the rest of the United States saw only modest gains. Walking also became more prevalent with density, accounting for nearly 48% of trips in NYS's densest areas. Commute distances and travel times varied by region, with NYS consistently experiencing longer public transit commutes in low-density areas but shorter private vehicle commutes compared with the rest of the United States. The longest public transit commutes (exceeding 90 min) were observed in NYS's least populated areas, indicating challenges in transit accessibility for rural and suburban residents. Meanwhile, the gap in public transit commute times narrowed in the highest-density areas between NYS and the rest of the United States, reflecting more efficient transit services in dense urban centers. These findings highlight shifting mobility patterns, the growing role of urban transit, and the need for targeted infrastructure investments to improve accessibility and efficiency in both urban and rural settings.

In Chapter 5, household and travel patterns are analyzed across different MSAs. Household and vehicle ownership trends reveal distinct differences based on MSA size. Large metro areas with populations exceeding 3 million saw a significant increase in households, growing from 4.7 million in 2001 to 7.9 million in 2017, whereas smaller MSAs with populations under 500,000 experienced a decline. Vehicle ownership patterns also varied by MSA size: larger MSAs, particularly NYC, had fewer vehicles per household, whereas smaller MSAs and non-MSA areas exhibited higher vehicle ownership rates. Zero-vehicle households were most prevalent in NYC, reaching approximately 54% in 2017, especially in Manhattan. In contrast, rural areas had near-universal vehicle ownership. Non-MSA areas and smaller MSAs showed higher rates of multivehicle ownership, reflecting a greater dependence on personal vehicles owing to limited transit access. In terms of personal travel, daily trips per person declined across nearly all MSAs between 2001 and 2017, aligning with national trends. However, average trip distances increased, particularly in suburban and mid-sized MSAs, suggesting longer commutes and continued suburbanization. Work-related trips saw a notable decrease, especially in NYC, driven by a shift toward remote work, increased transit use, and reduced reliance on personal vehicles. Conversely, social and recreational trips increased, particularly in urban areas like Manhattan, indicating a shift in travel behavior toward leisure activities. Walking rates also rose, with NYC and Ithaca leading in pedestrian participation, whereas biking remained low across the state with minimal growth. Zero-vehicle

households exhibited distinct characteristics. Lower-income households made up a significant share of zero-vehicle households, particularly in Buffalo, Binghamton, and Glens Falls. Meanwhile, higher-income zero-vehicle households were concentrated in Manhattan, reflecting transit accessibility and urban lifestyle preferences. Public transit use was highest in NYC, where 29% of trips were transit-based, whereas in smaller MSAs, transit use remained minimal. Interestingly, even among zero-vehicle households in smaller MSAs, private vehicle use was still common, likely owing to reliance on ride-hailing, taxis, and carpooling as well as limited availability of public transportation. Vehicle travel patterns also changed significantly between 2001 and 2017. Daily vehicle trips per driver declined across most MSAs, particularly in urban areas, whereas average trip lengths increased, especially in suburban and mid-sized metro areas. In contrast, trip lengths in urban areas like NYC remained stable. Trends in VMT varied: smaller MSAs experienced declines, whereas suburban areas such as Buffalo and Rochester saw statistically significant increases. Work-related vehicle trips decreased, particularly in large MSAs like NYC, whereas family and personal trips remained dominant. Social and recreational travel by car increased, particularly in urban areas, suggesting a rise in discretionary vehicle use.

Future research should explore the effect of emerging transportation technologies on travel behavior and environmental outcomes. Additionally, studying the long-term effects of remote work and evolving urban development patterns on vehicle ownership and transit demand will provide valuable insights for policymakers and planners. Expanding data collection efforts to include more recent trends, particularly post-pandemic travel shifts, will further enhance our understanding of mobility changes and inform future transportation strategies.

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APPENDIX A. GLOSSARY OF NHTS TERMS

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This glossary provides the most commonly used terms in the National Household Travel Survey (NHTS) and the terms used in this report, along with definitions. These definitions are provided to assist the user in interpreting the NHTS data.

Adult	For NHTS, this is defined as a person 18 years or older.
Block Group	A subdivision of a census tract that contains 600–3,000 people. The source used for the 2017 NHTS was the United States Census Bureau 2014 TIGER/Line Shapefiles (derived from Census 2010 definition).
Census Tract	A small subdivision of a county, generally having a population size between 1,200 and 8,000 people, with an optimum size of 4,000 people. The geographic size of the tract may vary considerably, depending on population density. Tracts were designed to be homogeneous regarding population characteristics, economic status, and living conditions when they were first delineated. Because the first tracts were delineated for the 1890 Census, current tracts may be far from homogeneous. The source used for the 2017 NHTS was the United States Census Bureau 2014 TIGER/Line Shapefiles (derived from Census 2010 definition).
Destination	For travel day trips, the destination is the point at which there is a break in travel, except if the break is only to change vehicles or means of transport.
Driver	A driver is a person who operates a motorized vehicle. If more than one person drives the vehicle on a single trip, the person who drives the most miles is classified as the principal driver.
Elderly	A person age 65 years or older.
Elderly Household	Households with one or more persons age 65 years or older.
Employed	A person is considered employed if that person worked for pay, either full time or part time, during the week before the interview. This includes persons who work at home and/or have more than one job.
Household	A group of persons whose usual place of residence is a specific housing unit; these persons may or may not be related to each other. The total of all US households represents the total civilian noninstitutionalized population. A household does not include group quarters (i.e., 10 or more persons living together, none of whom are related).
Household Income	Household income is the money earned by all family members in a household, including those temporarily absent. Annual income consisted of the income earned 12 months preceding the interview. Household income includes monies from all sources, such as wages and salary, commissions, tips, cash bonuses, income from a business or farm, pensions, dividends, interest, unemployment or worker's compensation, social security, veterans' payments, rent received from owned property (minus the operating costs), public assistance payments, regular gifts of money from friends or relatives not living in the household, alimony, child support, and other kinds of periodic money income other than earnings.

	Household income excludes in-kind income such as room and board, insurance payments, lump-sum inheritances, occasional gifts of money from persons not living in the same household, withdrawal of savings from banks, tax refunds, and the proceeds of the sale of one's house, car, or other personal property.
Household Members	Household members include all people, whether present or temporarily absent, whose usual place of residence is in the sample unit. Household members also include people staying in the sample unit who have no other usual place of residence elsewhere.
Household Vehicle	A household vehicle is a motorized vehicle that is owned, leased, rented, or company-owned and available to be used regularly by household members during the 2 week travel period. Household vehicles include vehicles used solely for business purposes or business-owned vehicles if they are driven home and can be used for the home-to-work trip (e.g., taxicabs and police cars). Household vehicles include all vehicles that were owned or available for use by members of the household during the travel period, even though a vehicle may have been sold before the interview. Vehicles excluded from household vehicles are (1) those that were not working and were not expected to be working within 60 days and (2) those that were purchased or received after the designated travel day.
Journey-to-Work Trips (Commute Trips)	Includes travel to and from a place where one reports for work. Does not include any other work-related travel. Does not include trips for persons who work at home.
Means of Transportation	A mode of travel used for going from one place (origin) to another (destination). A means of transportation includes private and public transit modes, as well as walking. The following transportation modes, grouped by major mode, are included in the NHTS data. Active modes—include modes by which a person must actively move from one place to the next. • Walk: This category includes walking and jogging. • Bicycle: This category includes bicycles of all speeds and sizes that do not have a motor. Private vehicle—a stipulation for being a private vehicle is that the vehicle is privately operated, including rental cars. • Car: Includes cars and station wagons. Leased and rented cars are included if privately operated and not used for picking up passengers in return for fare. • Sport utility vehicle: Includes vehicles that are a hybrid of design elements from a van, pickup truck, and station wagon. Examples include a Ford Explorer, Jeep Cherokee, or Nissan Pathfinder. • Van: Includes vans or minivans designed to carry 5–13 passengers or to haul cargo. • Pickup truck: Includes vehicles with an enclosed cab that usually accommodates two to three passengers and has an open cargo area in the rear. Late-model pickups often have a back seat that allows for total seating of four to six passengers. Pickup trucks usually have a wheelbase

the same size as that of a full-size station wagon. This category also includes pickups with campers.

- Motorcycle/moped: This category includes large, medium, and small motorcycles and mopeds.
- RV (motor home, ATV, snowmobile): An RV or motor home includes a self-powered recreational vehicle that is operated as a unit without being towed by another vehicle (e.g., a Winnebago motor home). This category also includes ATVs and snowmobiles.

Rental car—Includes Zipcar and Car2Go, in addition to commercially rented cars for private use.

- Golf cart/Segway: This includes all electric or gas-operated vehicles designed for use on a golf course, but whose use has recently extended to within smaller, often gated, communities. NOTE: Travel taken via golf cart is NOT included in vehicle trips and vehicle miles of travel.

Public transportation—as used in Federal Highway Administration publications and analysis of NHTS data, typically includes the following (indicated in *italics* in the following text): public or commuter bus, commuter rail, and subway/elevated rail/light rail/streetcar.

Bus—This category includes:

- *Public or commuter buses*: local public transit buses available to the public and buses used for short-distance public transport purposes (e.g., city bus or public bus), school buses
- Private/charter/tour/shuttle buses: private buses operating on a fixed schedule between population centers and buses that shuttle passengers from one fixed place to another (e.g., airport shuttles)
- City-to-city buses: buses that run from one urban center to another (e.g., Greyhound)

Train—This category includes:

- *Amtrak/commuter rail*: trains that run from one urban center to another
- *Subway/elevated rail/light rail/streetcar* (also known as rail rapid transit): a high-capacity system operated on a fixed-rail or guideway system on a private right of way, and vehicles that run on a fixed-rail system powered by electricity obtained from an overhead power distribution system

Other modes—This category includes:

- School buses
- Paratransit/dial-a-ride
- Taxi/limo: Use of a taxicab by a passenger for fare, including limousines. In 2017, this category also includes ridesharing such as Uber and Lyft.
- Airplane: includes commercial airplanes and smaller planes available for use by the public in exchange for a fare. Private and corporate planes and helicopters are also included.
- Boat/ferry/water taxi: includes travel by passenger line ferries
- Something else: includes any type of transportation not previously listed (e.g., skateboards, roller blades, sailboats, cruise ships)

**Metropolitan
Statistical Area
(MSA)**

Except in the New England states, a metropolitan statistical area is a county or group of contiguous counties that contains at least one city of 50,000 inhabitants or more, or “twin cities” with a combined population of at least 50,000. In addition, contiguous counties are included in an MSA if, according to certain

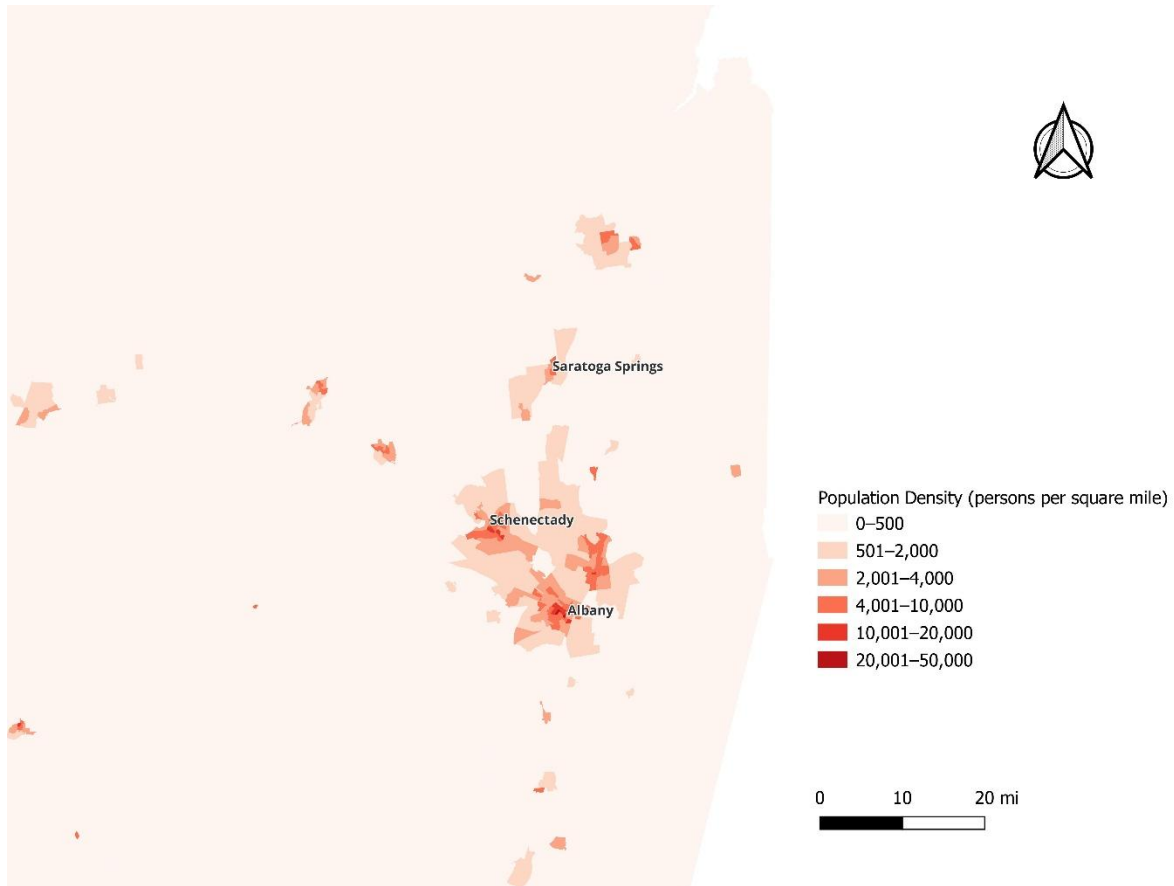
	criteria, they are socially and economically integrated with the central city. In the New England states, MSAs consist of towns and cities instead of counties. The source used for the 2017 NHTS was the United States Census Bureau 2014 TIGER/Line Shapefiles (derived from the Census 2010 definition).
Motorized Vehicle	Motorized vehicles include all vehicles licensed for highway driving. Snow mobiles and minibikes are specifically excluded.
New York City (NYC)	New York City is defined in this report as the five-county area: Bronx, Kings (Brooklyn), Queens, New York (Manhattan), and Richmond (Staten Island).
Occupancy	Occupancy is the number of persons, including driver and passenger(s), in a vehicle.
Occupancy Rate	NHTS occupancy rates are generally defined as the mileage-weighted averages of the number of persons on a vehicle trip.
Origin	Origin is the starting point of a trip.
Passenger	For a specific trip, a passenger is any occupant of a motorized vehicle, other than the driver.
Person Miles of Travel (PMT)	PMT is a primary measure of person travel. When one person travels 1 mile, 1 person mile of travel results. When two or more persons travel together in the same vehicle, each person makes the same number of person miles as the vehicle miles. Therefore, four persons traveling 5 miles in the same vehicle results in 20 person miles ($4 \times 5 = 20$).
Person Trip	A person trip is a trip by one or more persons in any mode of transportation. Each person is considered as making one person trip. For example, four persons traveling together in one auto are counted as four person trips.
Population Density	Category of population density (persons per square mile) in the Census Block Group of the household's home location: 0–99, 100–499, 500–999, 1,000–1,999, 2,000–3,999, 4,000–9,999, 10,000–24,999, and 25,000–999,999.
Privately Owned Vehicle (POV)	A privately owned or privately operated vehicle. In either case, the vehicle is not available to the public for a fee (i.e., it is not a bus, subway, taxi, or other vehicle for which a fee is charged for use).
Travel Day	A travel day is a 24 h period from 4:00 a.m. to 3:59 a.m. designated as the reference period for studying trips and travel by members of a sampled household.
Travel Day Trip	A travel day trip is defined as any time the respondent went from one location to another by private motor vehicle, public transportation, bicycle, walking, or other means during the NHTS-assigned reporting travel day. However, a separate trip is not counted in two instances: 1. When the sole purpose for the trip is to get to another vehicle or mode of transportation to continue to the destination. 2. Travel within a shopping center, mall, or shopping areas of four to five blocks is to be considered as travel to one destination.

Vehicle	In the 2017 NHTS, the term <i>vehicle</i> includes autos, passenger vans, sport utility vehicles, pickups and other light trucks, RVs, motorcycles, and mopeds owned or available to the household.
Vehicle Miles of Travel (VMT)	VMT is a unit to measure vehicle travel made by a private vehicle, such as an automobile, van, pickup truck, or motorcycle. Each mile traveled is counted as one vehicle mile regardless of the number of persons in the vehicle.
Vehicle Occupancy	Vehicle occupancy is the number of persons, including driver and passenger(s) in a vehicle; it also includes persons who did not complete a whole trip.
Vehicle Occupancy Rate	NHTS vehicle occupancy rates are generally calculated as person miles divided by vehicle miles.
Vehicle Trip	A trip by a single POV, regardless of the number of persons in the vehicle.
Work-Related Travel	Trips related to business activities except travel to the place of work; for example, a plumber drives to a wholesale dealer to purchase supplies for his business or a company executive travels from his office to another building to attend a business meeting. Business, out-of-town trips, and professional conventions are also included.
Worker	See “Employed.”

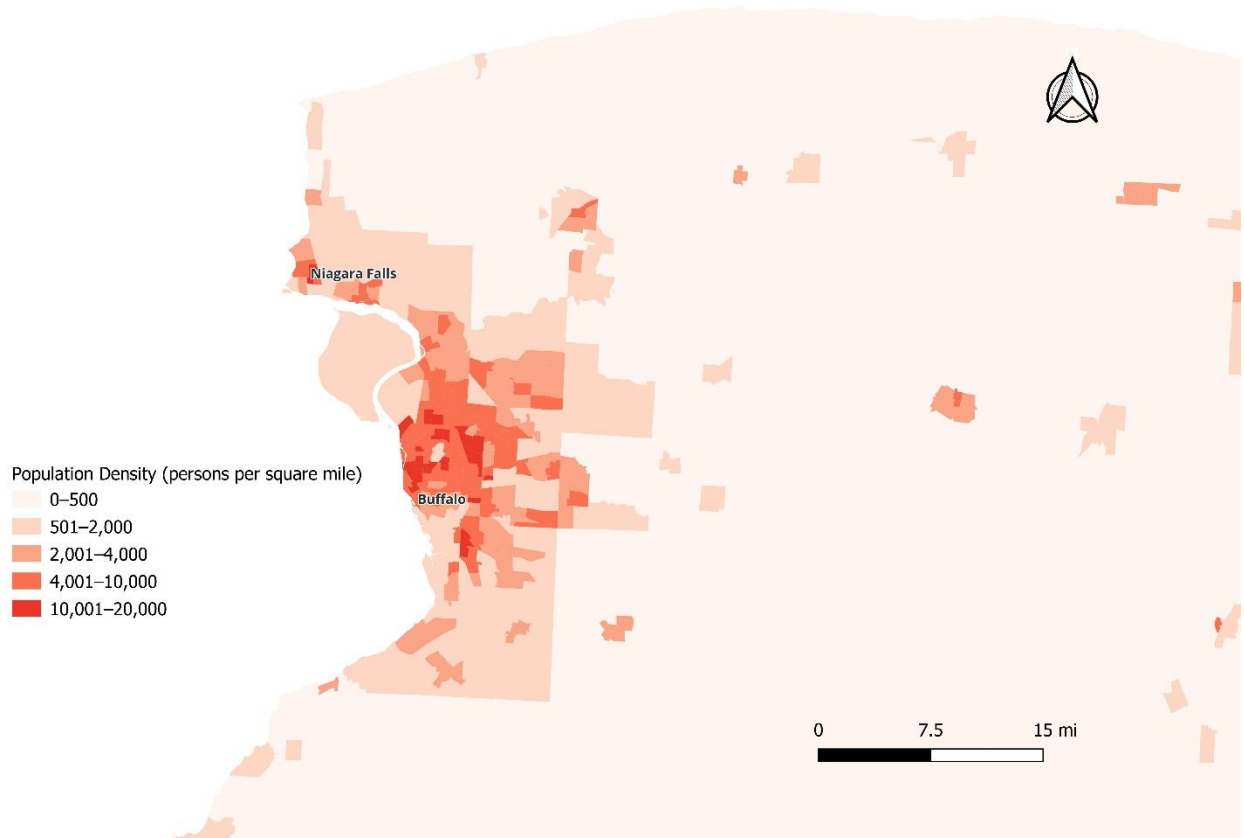
APPENDIX B. MAPS OF NYS AREAS

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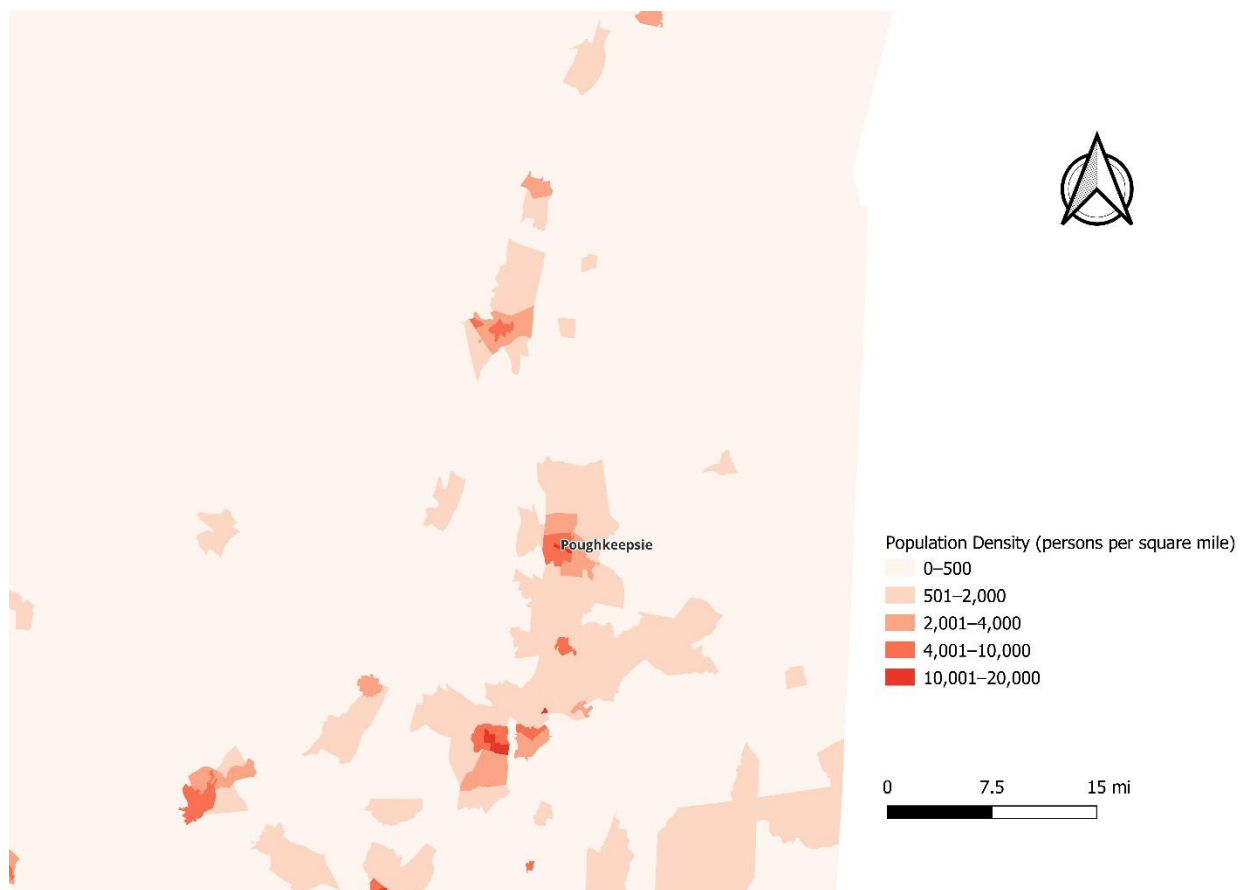
Albany/Capital District and Glens Falls Area



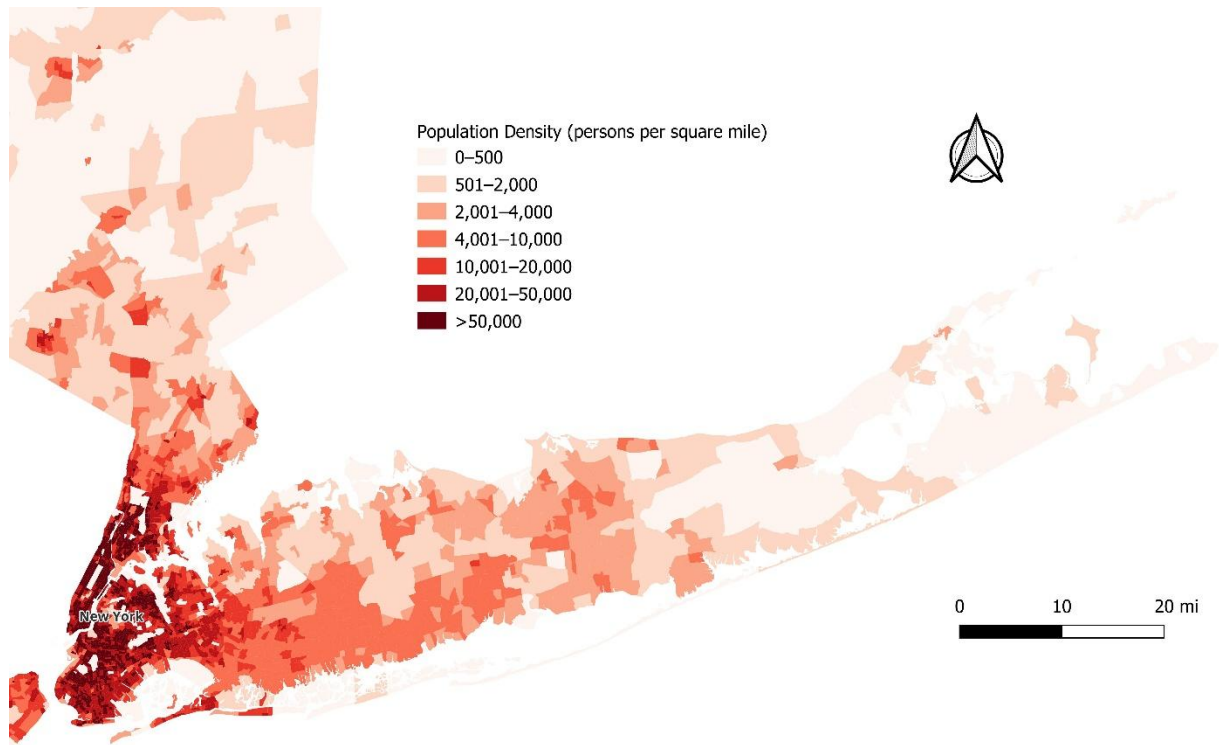
Buffalo Area



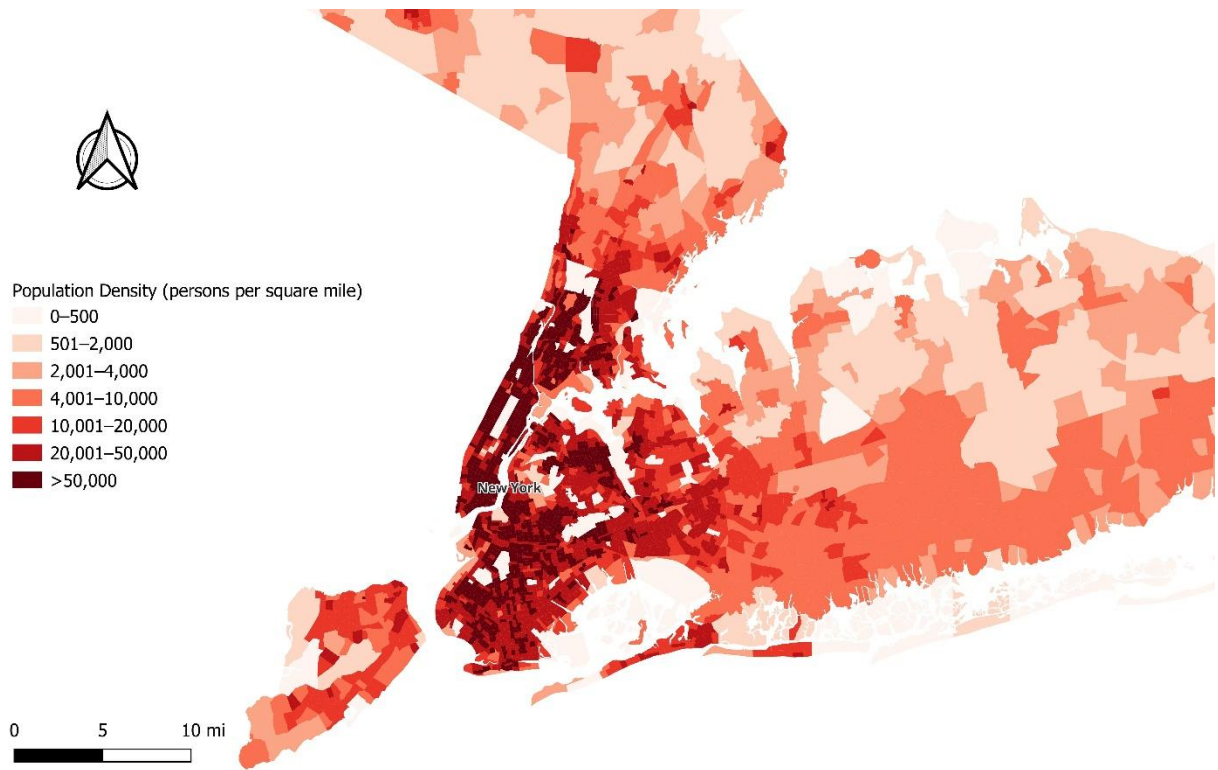
Kingston and Poughkeepsie Area



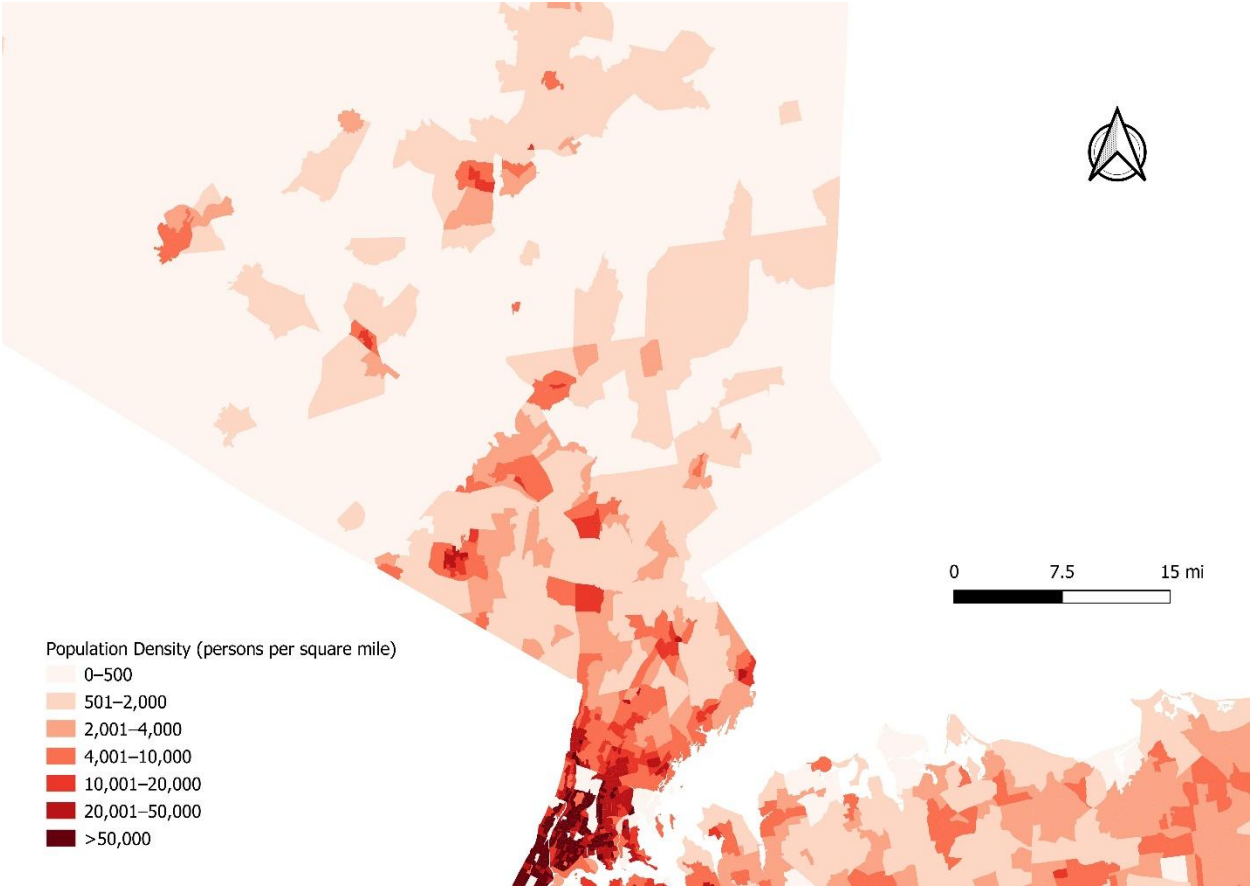
Nassau-Suffolk Area



NYC Area



Putnam–Rockland–Westchester Area

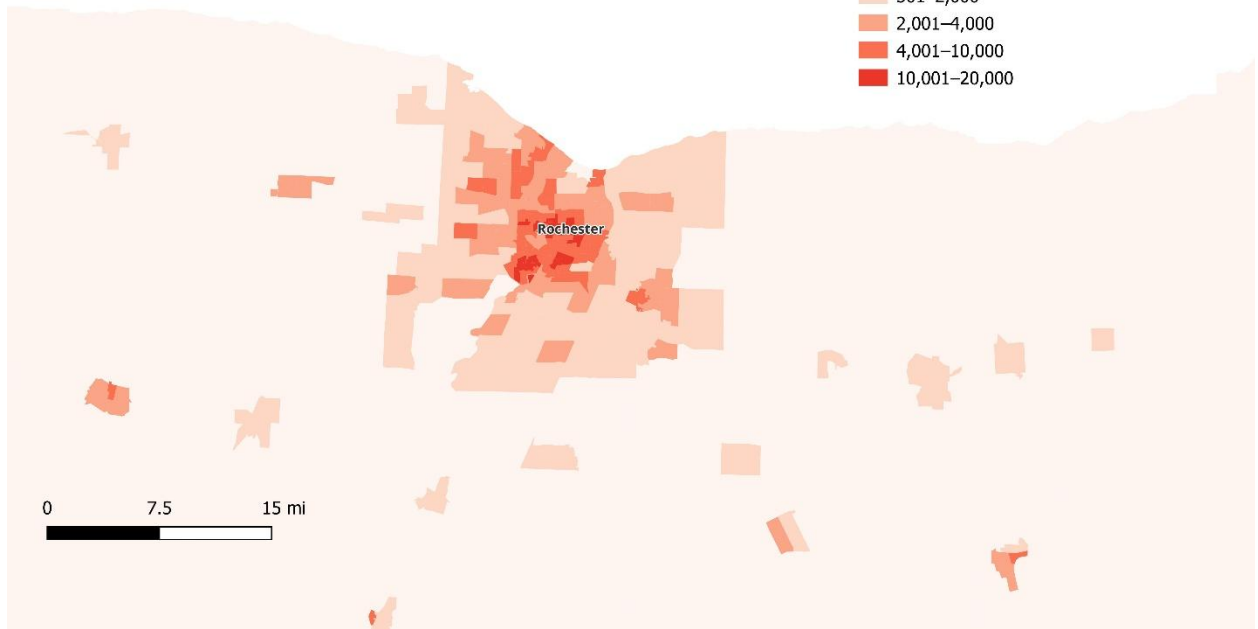


Rochester Area



Population Density (persons per square mile)

- 0–500
- 501–2,000
- 2,001–4,000
- 4,001–10,000
- 10,001–20,000



Syracuse Area

